TRIBHUVAN UNIVERSITY

Institute of Engineering

PULCHOWK CAMPUS

Lalitpur

A

REPORT

ON

SURVEY CAMP 2082

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Preface

This survey report is based on 10 days' field work conducted by the Survey Instruction Committee, **Pulchowk Campus** from **Mangsir 6th to 15th** for the partial fulfillment of the requirement for the Bachelor's Degree in Civil Engineering as per the syllabus of Institute of Engineering, Tribhuvan University in second year second part. This surveying has been able to impart us the great opportunity to consolidate and review the practical and theoretical knowledge on surveying, which we gained in previous years. This survey camp has also helped in the team co-ordination for the long term team work with the friendly environments. This practice of team work is the great achievement for the future professional work in the practical life. This survey camp also developed the sense of individual responsibility towards the steps taken by the group.

We have been able to achieve the true objectives of survey and upgrade the knowledge as handling of the instrument, working procedure, problem solving and field booking precisely. This survey camp gave us the practical knowledge of overcoming the technical difficulties and developing a skill in tackling it. It encouraged us to cope with the team members, as the surveying involved all the members equally during the field procedures, calculations and plotting and report preparations.

In conclusion, those 10 days were very much fruitful days and we are sure that these learning and experiences gained during the camp period as a real field engineer will helps us for shaping our future career as Civil Engineer.

Sincerely,
GROUP 25

Acknowledgement

We would like to express our sincere gratitude towards the Department of Civil Engineering, Pulchowk Campus, T.U, for initiating and facilitating the survey camp to enhance our knowledge of surveying and its applications.

We would like to thank our teachers Mr. Bharat Bahadur Dhakal, Mr. Narayan Basnet, Mr. Abhimanyu Lal Singh, Mr. Jaya Ram Maharjan, Mr. Birendra Kumar Shrestha, Mr. Roshan Karki, Mr. Sanjay Kumar Shah and Mr. Thaman Bahadur Khadka for providing insightful lectures and crucial understanding of survey techniques and all the working staffs and store keepers for their prompt assistance with the equipment and technical issues we encountered throughout the camp. Also, we are thankful to T.U. administrator who provided us the working space and homely environment. Your collective efforts have not only enhanced our academic knowledge but also inspired us to pursue excellence in our future endeavors. We are truly grateful for the opportunity to learn from such dedicated and experienced professionals. We would also want to show our genuine appreciation for the management committee of our college for creating healthy and creative environment for learning and handling the problems that arise in field.

Lastly, we would also like to thank the TU staff who helped us during the 10 days of survey camp by being supportive and providing the friendly and familiar environment to us and everyone who helped us directly or indirectly throughout the survey camp and in the preparation of this report.

SURVEY CAMP 2082
GROUP 25

Abstract

This report has been prepared as per the syllabus of Bachelor in Civil Engineering for Second Year Second Part (CE 604). We were taken for the Survey Camp in Tribhuvan University, Kirtipur for 10 days with the prescheduled tasks to be fulfilled as per the course requirement.

With the accomplishment of the objective of gaining experience of surveying on a larger area for the preparation of topographic map, learning to select a suitable bridge axis and select the road alignment along with the preparation of longitudinal and cross section of the road, this report is presented as the outcome of the 10-days Camp. This report gives a brief information on the working procedures and techniques along with the observations, calculations and the required maps. Starting off with the contents on topographic survey with all of its observations, calculations and corresponding drawings, it is then followed by bridge site survey and road alignment.

Despite our best efforts on preparing and compiling this report, errors might have made their way along. We will be pleased to hear from anyone who brings it to light and helps us with some constructive criticisms and feedbacks.

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List of acronyms

AP	Apex Point
BC	Beginning of Curve
BM	Bench Mark
BS	Back Sight
TP	Turning Point
EC	End of Curve
EDM	Electronic Distance Measurement
Er.	Engineer
FS	Fore Sight
HI	Height of Instrument
IP	Intersection Point
IS	Intermediate Sight
Recce	Reconnaissance
RL	Reduced Level
TBM	Temporary Bench Mark
TP	Turning Point
HFL	High Flood Level
NFL	Normal Flow Level
CFL	Center Flow Level
SW	Side Walk
GP	Ground Point

Chapter 1 : INTRODUCTION

1.1 Introduction to Survey

Surveying is the branch of engineering that deals with the art and science of determining the relative positions of distinctive features on or beneath the surface of the earth, by measurements of distances, directions and elevations. It is the most important subject matter before and during all engineering works like civil engineering works such as designing and construction of highways and transportation engineering, bridges, water supply systems, irrigation projects, commercial and residential buildings etc.

Surveying is the main root for the execution of any civil engineering projects. The science of surveying has been developing since the initial stage of human civilization according to their requirements. The art of surveying preparation of maps has been practiced from the ancient times and the further advanced until present. In the absence of the map, it is impossible to layout the alignments of road, canals tunnels, transmission power line, and microwave or television relaying towers and so on. Detailed map of the sites of engineering projects are necessary for the precision establishment of sophisticated instruments. Surveying is the first step for the execution of any project. As the success of any engineering is based upon the accurate and complete survey work, an engineer must therefore be thoroughly familiar with the principle and different methods of surveying and mapping.

For the purpose of water supply-sanitary system, irrigation system, highway designing, the relative altitudes are required, which is ascertained by the process of leveling. The details of the enclosed area and the ground nature can also be portrayed in the combined form of a topographic map. Not only this, the whole land can be surveyed as different areas and can be plotted into a single map, the main thing is not to violate the basic survey principles viz. working from whole to part, consistency in work, accuracy required according to scale and independent check.

The B.E. Survey Camp 2082, Kirtipur organized by the Survey Instruction Committee, Pulchowk Campus is a part of the four-year Bachelor's degree in Civil Engineering course, second year second semester, carrying a total of 100 marks. The total duration of the survey camp was 10 days,6th Mangshir to 16th Mangshir**.**

This is a detail report of the task, which were performed by group number 25, during the camp period. It briefly explains the working procedures and technique used by this group during the camp period. In addition, it contains observations, calculations, methods of adjustment of error, main problem faced during work and their solution, results of all calculations and their assessments with some comments presented in a concise form.

1.2 Objectives of survey camp

The main objectives of the survey camp are as follows:

- To become familiar with the surveying problems that are arise during the field works.
- To became familiar with the parts of the instruments, their functions and handling the surveying instruments for its use in surveying.
- To become familiar with the spirit and importance of teamwork, as surveying is not a single person work.
- To complete the given project in scheduled time and thus knows the value of time.
- To collect required data in the field in systematic ways.
- To compute and manipulate the observed data in the required accuracy and present it in diagrammatic and tabular form in order to understand by other engineers and related personnel easily.
- To tackle the mistake and incomplete data from the field while in office work.
- To know the complete method of report preparation.

1.3 Camping Schedule:

Survey camp was scheduled for 10 working days, starting from **6th Mangsir to 15th Mangsir**. Field work in the camp starts from 6:00am to 6:00 pm.Viva were also held as per requirement.

1.4 Project area

The typical features related to the site are as follows:

1.4.1 Location and Accessibility:

It is accessible place motor able road from Ringroad. The details about the area are:

- **Country:** Nepal
- **Province: Bagmati**
- **District:** Kathmandu
- **Municipality:** Kirtipur Municipality
- **Location:** TU Premises

1.4.2 Rainfall, Climate and Vegetation:

Tribhuvan University has a warm-temperate/subtropical climate with hot summers and cool winters. The area receives about 1,400 mm of rainfall yearly, mostly during the monsoon (June–September). July is the wettest month, while November is the driest. Vegetation includes mixed subtropical–temperate species like pines, oaks, and ferns, growing rapidly in monsoon. Urbanization and development have reduced natural greenery, and declining rainfall may further affect vegetation.



Figure 1 : Location TU Premises

Chapter 2 :TOPOGRAPHICAL SURVEY

Topographical surveying is the process of determining the positions of natural and artificial features of the locality by means of conventional signs up on a topographical map. Topographic surveys are three-dimensional; they provide the techniques of plane surveying and other special techniques to establish both horizontal and vertical control.
Hence the fieldwork in a topographical surveying consists of three parts.

1. Establishing both horizontal and vertical control.
2. Locating the contours.
3. Locating the details such as rivers, streams, lakes, roads, railways, houses, and trees etc.

2.1 Objectives:

The main Objective is to prepare the topographic map of the given area with horizontal control and vertical control with required accuracy. This also includes the calculations and diagrammatic representation of the area with the help of the co-ordinates in the paper with gridlines.

2.2 Brief Description of the area:

The area, where surveying was performed, is situated at Tribhuwan University Premises. The major traverse was run throughout the training centre area, which cover the whole compound area. Our objective was to prepare a topographic map of the given small area, which is a part of the large area.

2.3 Norms (Technical Specification):

- Conduct reconnaissance survey of the given area. Form a close traverse (major and minor) around the perimeter of the area by making traverse station. In the selection of the traverse station maintain the ratio of maximum traverse leg to minimum traverse leg less than 2 for major **(i.e. 1:2)** and less than 3 for minor **(i.e 1:3).**
- Measure the traverse legs in the forward and reverse directions by means of a tape calibrated against the standard length provided in the field, note that discrepancy between forward and backward measurements should be better than 1:5000.
- Measure traverse angle on two sets of reading by Theodolite. Note that difference between the mean angles of two sets reading should be within the square root of no of Stations times least count of the instrument.
- Determine the R.L. of traverse stations by fly leveling from the given B.M. Perform two peg test before the start of fly leveling. Note that collimation error should be less than 1:10000. Maintain equal foresight and back sight distances to eliminate collimation error.

  The Permissible error for fly leveling is $\pm 24\sqrt{k}$ mm where k is loop distance in km.
- Balance the traverse. The permissible angular error for the sum of interior angles of the traverse should be less than **$\pm C\sqrt{N}$, where C=30"** for Major Traverse and **$\pm C\sqrt{N}$ where 1'** for Minor Traverse (N = no of traverse station). For major and minor traverse the relative closing error should be less than **1: 5000** and **1: 3000** respectively.
- Plot the traverse stations by coordinate method in appropriate scale, i.e. **1:1000** for major traverse and **1:500** for minor traverses.
- Carry out the detail survey of the given area by tachometric method with reference to the major and minor traverse stations, which have been already plotted. Use conventional symbols for plotting.

2.4 Instruments and Accessories:

Different survey instrument were used for different purpose in the survey camp. Some of them are listed below:

- Theodolite
- Leveling staffs
- Ranging Rods
- Measuring Tapes 30m & 50m
- Leveling Instruments
- Plumb Bob
- Pegs
- Marker Pen

In the camp we also used Total station (combination of Theodolite and the EDM). This is very advanced surveying instrument which has capacity to measure the horizontal and zenithal angle and also measure horizontal distance. It can also store the field data in its memory and can be directly transfer to the computer. In the camp by using this instrument we measure the horizontal distance between the difficult station and compared with the measured horizontal distance between by the tape and if high difference was observed we had re-measure that stations distance by the Tape again.

2.5 Methodology:

The methodology of surveying is based on the principle of surveying. They are as follows:

i. Working from whole to a part
ii. Independent check
iii. Consistency of work
iv. Location of a point with respect to two control points.
v. Required accuracy

The different methodologies were used in surveying to solve the problems arise in the field. These methodologies are as follows:

2.5.1 Reconnaissance:

Reconnaissance means the exploration or scouting of an area. In survey, it involves walking around the survey area and roughly planning the number of stations and the position of the traverse stations. Recci is primarily done to get an overall idea of the site. This helps to make the necessary observations regarding the total area, type of land, topography, vegetation, climate, geology and inter visibility conditions that help in detailed planning. The following points have to be taken into consideration for fixing traverse stations:

- The adjacent stations should be clearly inter-visible.
- The whole area should include the least number of stations possible.

- The steep slopes and badly broken ground should be avoided as far as possible, which may cause inaccuracy in taping.
- The traverse station should maintain the ratio of maximum traverse leg to minimum traverse leg less than 2:1 for Major Traverse and 3:1 for Minor Traverse.
- The traverse line of sight should not be near the ground level to avoid the refraction the stations should provide minimum level surface required for setting up the instrument.
- The traverse line of sight should not be near the ground level to avoid the refraction
- Thinking the above given points into consideration, the traverse stations were fixed. Then two-way taping was done for each traverse leg. Thus, permanent fixing of the control points completes Recci.

2.5.2 Traversing:

Traversing is a type of surveying in which a number of connected survey lines form the framework. It is also a method of control surveying. The survey consists of the measurement of:

- Angles between successive lines or bearings of each line
- The length of each line

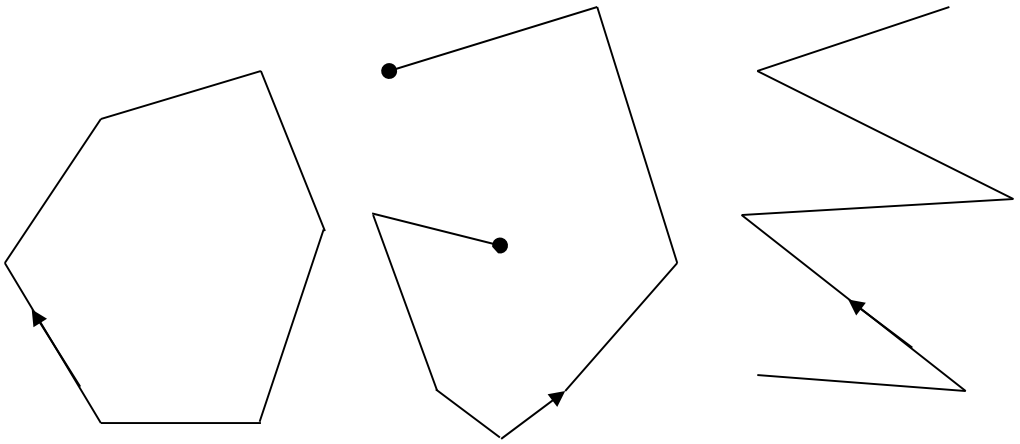
There are two types of traverse. They are as follows:

i. Closed traverse:

If the figure formed by the lines closes at a station i.e. if they form a polygon or it starts and finishes at the points of known co-ordinates then the traverse is called closed traverse.

ii. Open traverse:

If a traverse starts and finishes at points other than the starting point or point of unknown coordinates, then the traverse is called open traverse.



(a) Closed Loop Traverse **(b) Closed Traverse** **(c) Open Traverse**

Figure 2 Types of Traverses

Theodolite traversing is defined as the course taken when measuring a connected series of straight lines, each line joining two points on the ground. These points are called traverse station. The straight line between two consecutive traverse stations is called a traverse leg. The angle at any station between two consecutive traverse legs is known as traverse angle.

The directions and the lengths of the survey lines are measured with the help of an angle measuring instrument such as Theodolite and a tape. If the co-ordinates of the first station and the bearing of the first line are known, the co-ordinates of all successive points can be computed as follows:

$X_B = X_A + L*\cos\theta$

$Y_B = Y_A + L*\sin\theta$

where,

L=Length of traverse leg

2.5.2.1 Measurement of Traverse Length:

After completion of reconnaissance survey, taping of the major traverse was performed with the help of tapes. The distances between the adjacent control points were measured accurately as far as possible for the accuracy of the whole traverse. To attain the accuracy required, a two-way taping was done independently so that the length from each measurement was found within specified range.

To measure the horizontal distances accurately on the slopping ground, the short length was measured at a time so that the tape could be pulled horizontally without sagging. For this ranging was done accurately to divide the length into shorter length. Finally, all the lengths were added to obtain the whole length, which is also, called **stepping method.** For accuracy, traverse legs may be checked by electronic distance measuring instrument (EDM).

2.5.2.2 Major Traverse:

The skeleton of lines joining those control points, which covers the whole entire area, is called Major Traverse. Work on Major traverse must be precise. So two-set of reading should be taken for Major Traverse. For convenience, the readings are taken by setting the Theodolite at $0^0$0’0” for one set and $90^0$00’00” for the second.

In the Kirtipur Survey Camp, two traverses - major and minor had to be established. The major traverse had 21 control stations including two given control points. The control stations were named as M1, M2 and so on along with CP1 and CP2 (the two given control points) .The leg ratio of maximum traverse leg to minimum traverse leg was maintained within 1:2. The discrepancy in length between the forward measurements and the backward measurements of all the traverse legs was within 1:5000. Two sets of theodolite readings were taken for measuring the horizontal traverse angles. The difference between the mean angles of two sets of readings was within a minute for all the angles.

2.5.2.3 Minor Traverse:

A closed frame-work made within the major traverse for the ease and comfort to carry out detailed survey or detailing is known as minor traverse. The entire vertical as well as the horizontal controls is transferred from the major traverse. Minor traverse legs are fixed and stretched in and out the area to be surveyed. Minor traverse stations are fixed in such a way that it covers the maximum details which can be surveyed in the time frame with less effort and much ease. The leg ratio of maximum traverse leg to minimum traverse leg was maintained within 3:1. The discrepancy in length between the forward measurements and the backward measurements of all the traverse legs was within 1:3000.

2.5.2.4 Measurement of the horizontal and vertical angle:

Two set of horizontal angle was measured at each station and one set of vertical angle. And it was done in the following way:-

2.5.2.4.1 One the face left temporary adjustment was done.

2.5.2.4.2 After setting zero to the first station the second station was sighted by unclamping the upper screw.

2.5.2.4.3 .For better accuracy and exact bisection horizontal angle was measured at the bottom of the arrow.

2.5.2.4.4 And on the same setting or same face vertical angle at both the station was taken.

2.5.2.4.5 Now again changing the face the horizontal angle was taken and vertical angle too.

2.5.2.4.6 Now setting the reading to ninety at the first station again one set of horizontal angle was taken but the vertical angle is enough, taken earlier.

2.5.2.4.7 Before shifting the instrument to the next station the height of instrument was taken.

2.5.2.4.8 Similarly, the instrument was shifted to other station and in each station one set of vertical angle and two set of horizontal angle and height of instrument was measured.

2.5.2.4.9 For comparison of the tape distance and the Tachometric distance the stadia reading (top, mid, bottom) was taken at each station and for the calculation of the reduce level of each station we need to read mid reading which can be compared with the level transferred using auto level.

2.5.2.5 Balancing the traverse:

The process of adjusting the consecutive co-ordinates by applying the correction to the latitudes & departures of each of the traverse legs such that their algebraic sum is equal to zero is called balancing the traverse or balancing the consecutive co-ordinates.

A closed traverse can be balanced by any one of the following methods.

1. Bowditch's method
2. Transit rule
3. Graphical method
4. Axis method

a. Bowditch's Method

The method is based on the assumption that errors in the linear measurement are proportional to √L and the errors in the angular measurements are inversely proportional to √L where 'L' is the length of a line. The method is applicable when both the n
linear as well as angular measurements are of equal precision.

The Bowditch rule is:

Correction to latitude (or departure) of any side

$$= \text{Total error in lattitude or departure} * \frac{\text{Length of that side}}{\text{Perimeter of that traverse}}$$

$$C_L = \sum Lat * \frac{L}{\sum L}$$

$$C_D = \sum Dep * \frac{D}{\sum D}$$

where,

C_L = Correction to latitude of any side

C_D = Correction to departure of any side

Σ_{Lat} = Total error in latitude

Σ_{Dep} = Total error in departure

ΣL = Total perimeter of traverse

L = Length of any side

b. Transit Method

The method is most applicable when angular measurements are of more precision than linear measurement. According to this rule, the total error in latitude and in departure is distributed in proportion to the latitude and departure of the sides. The angles are less affected by the corrections applied by this method than by the Bowditch method. The Transit rule is:

Correction in Latitude (or Departure) of any side

$$= \text{Total error in latitude or departure} * \frac{\text{Latitude (or Departure) of that line}}{\text{Arithmetic sum of Latitudes (Departure)}}$$

$$C_L = \sum L * \frac{L}{L_T}$$

$$C_D = \sum D * \frac{D}{D_T}$$

where, C_L= Correction to latitude of any side

C_D = Correction to departure of any side

L = Latitude of any line

D = Departure of any line

L_T = Arithmetic sum of latitudes

D_T = Arithmetic sum of departures

Plotting of major traverse stations

After the computation and correction of the coordinates of the major traverse stations, the traverse stations were plotted in the grid sheet. The entire grid should be checked diagonally to avoid the plotting error. The major traverse stations were plotted in the scale of **1:1000** to the grid paper. Paper management is done so that the drawn traverse lies in the center of the grid sheet which also comforts the detail drawing or the preparation of the topographic map.

Minor Traverse

A closed frame-work made within the major traverse for the ease and comfort to carry out detailed survey or detailing is known as minor traverse. The entire vertical as well as the horizontal controls is transferred from the major traverse. Minor traverse legs are fixed and stretched in and out the area to be surveyed. Minor traverse stations are fixed in such a way that it covers the maximum details which can be surveyed in the time frame with less effort and much ease.

2.5.2.6 Procedure

- **Reconnaissance**

The whole area at the survey camp was divided into plots in kirtipur. As in the case of major traverse, reconnaissance was carried out before the selection of the minor control points or traverse stations. Minor stations were fixed such that there were 2-5 stations in a loop. One or two loops were formed as per the requirement and ease in detailing. The stations were fixed in such a way that maximum number of details could be controlled from a single minor station.

- **Marking and fixing of control points**

After the completion of reconnaissance, 1 minor loop was formed. Altogether 2 minor control points were fixed at suitable places considering all the required criteria.

- **Measurement of traverse legs**

As in the case of the major traverse, two-way measurements of all the traverse legs were carried out. The accuracy required for two-way measurement in the case of minor traverse is **1:3000**. The leg ratio should be within **1:3**.

- **Measurement of interior angles**

Only one set of horizontal angle observation is sufficient for the minor traverse. As in the case of major traverse, the difference of the observed angle in each observation should not exceed **1′**. In the same way, 0º0'0" was set at the preceding station and the telescope was turned in the clock-wise direction for the required horizontal angle.

Permissible Angular Error for the closed traverse **= 30"**$\sqrt{N}$

Where,

N = no. of traverse leg

For the closed traverse,

Sum of interior angles = (2n – 4) ×90°

Closing error = (2n – 4) ×90° - $\sum$ **Observed sum of internal angles**

If the angular error is within the permissible value of **30"**$\sqrt{N}$, then the error in the sum of internal angles is not equally distributed to all the angles as in the case of major traverse. Here, the major angle cannot be corrected or given correction. Correction is provided for the angles included by minor traverse legs.

- **Computation of bearing of the traverse legs**

As in the case of the major traverse, the bearing of the entire minor traverse legs are obtained from the bearing of the preceding leg (which has already been calculated in the major traverse) and the measured horizontal traverse angle. Prior to computation of bearing, correction for angular misclosure is applied as stated earlier.

- **Computation of coordinates of minor control points**

Using the co-ordinates of the major traverse which are already defined or computed, the coordinates of the minor control points are calculated. The co-ordinates of the minor traverse stations are calculated using the bearings and the average length of the minor traverse legs using their latitudes and departures.

- **Plotting of minor traverse stations**

As in the case of plotting of the major traverse, minor traverse is plotted in the grid sheet. The grid should be checked diagonally in order to avoid the plotting error. The minor traverse is plotted in the scale of **1:500**.

- **Checking for Orientation**

The cross check of the orientation of the traverse was done using plane table and its accessories. The traverse plot was placed coinciding the corresponding station and its orientation was checked by resection method.

2.5.3 Computation of the co-ordinates

According to the accuracy aimed and the nature of the ground, the lengths of traverse legs are measured directly on the ground either by chaining or taping. The traverse angles are measured with a theodolite by setting up the instrument at each station in turn and the vertical angle at each station measured will help to find the tacheometric distance and reduce level of that computation point. And the bearing of the any one of the traverse leg measured and the entire traverse angle measured, the bearing of all the legs can be calculated by:

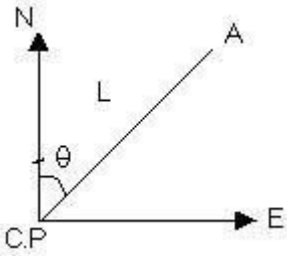


Figure 3 coordinate computation

Bearing of a line = (bearing of previous line +included angle) $\pm$ (180) or (540)

If θ is the bearing of line (c.p,A say), and l be the length of the line and provided that coordinate of the control point(c.p) is known then the co-ordinate of the point 'A' can be calculated as follow-:

X-coordinate of A=x-coordinate of control point (c.p) +l*sinθ

Y-coordinate of A=y-coordinate of control point (c.p) +l*cosθ

R.L or z-coordinate of A=R.L of point (c.p) +H.I $\pm$ H*Tanθ-Height of signal.

Where, H.I=Height of instrument

H=horizontal distance

- **Balancing of the consecutive co-ordinate:**

The process of adjusting consecutive co-ordinates of each line by applying correction to them in such a way that each algebraic sum of the latitude and departure of a close circuit is equal to zero i.e. the sum of the northing should be exactly equal to the sum of the southing and sum of the easting should be exactly equal to the sum of the westing.

The closing error however is distributed throughout the whole traverse stations such that its effect is not apparent on the plotted location of the station. And the error can be distributed among the stations if the closing error is within the permissible limit, which is given by-:

Precision = $\sqrt{(\Delta X^2+\Delta Y^2)}$ /P = e/P

This should be greater than 1:2000

- **Demonstration of EDM:**

Electronic Distance Measurement (EDM) is one of the modern surveying equipments, which is very accurate and hence very popular. As its name suggests, the EDM is used for measuring the horizontal distance between two places - the instrument station and the target. Due to the ease of use of the instrument, the EDM has in many places, replaced the conventional methods of distance measurement like chaining, taping, etc.

The basic principle of EDM is that the distance between any two points can be known once the time light takes to travel the distance and back and the velocity of light is known. Then the following relation, which is already programmed in the memory of the instrument along with other correction factors, calculates the required horizontal distance and is displayed on the LCD screen.

Distance (d) = velocity (v) * time (T/2)

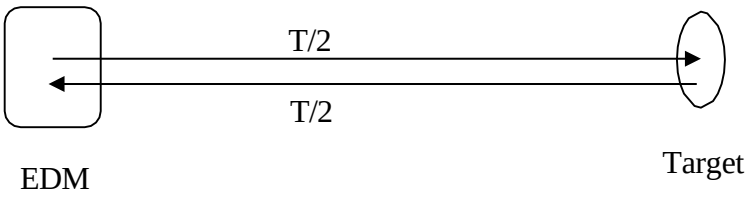


Figure 4 Workig of EDM

The battery powered EDM instrument used was Topkon and it had to be used in addition to the Topkon theodolite. There were a number of buttons on the instrument and an LCD screen. Another important part of the instrument was the target panel, which was kept at the target. With the proper use of the instrument and the target panel, the distance between two points can be obtained with great precision. We were told that the instrument, which was demonstrated to us, was also outdated among its kind and that there were very accurate EDMs available in the market nowadays.

- **Theodolite:**

The theodolite is the most precise instrument designed for the measurement of horizontal and vertical angles and has wide applicability in surveying such as laying off horizontal angles, locating points on line, prolonging survey lines, establishing grades, determining difference in elevation, setting out curves etc.

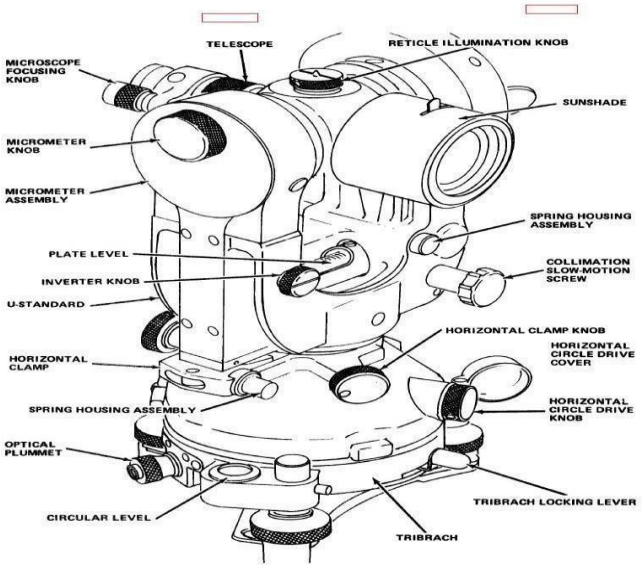


Figure 5 Theodolite

2.5.4 Levelling:

Leveling is a branch of surveying, the objectives of which are:

- To find the elevation of given points with respect to a given or assumed datum.
- To establish points at a given elevation or at different elevations with respects to a given or assumed datum.
- Two types of leveling are used in general Engineering practices, namely direct leveling (spirit leveling) and indirect leveling (trigonometric leveling).

2.5.4.1 Direct levelling:

It is the branch of leveling in which the vertical distances with respect to a horizontal line (perpendicular to the direction of gravity) may be used to determine the relative difference in elevation between two adjacent points. A level provides horizontal line of sight, i.e. a line tangential to a level surface at the point where the instrument stands. The difference in elevation between two points is the vertical distance between two level lines. With a level set up at any place, the difference in elevation between any two points within proper lengths of sight is given by the difference between the staff readings taken on these points. By a succession of instrument stations and related readings, the difference in elevation between widely separated points is thus obtained.

Following are some special methods of direct (spirit) leveling:

2.5.4.1.1 Differential Leveling:

It is the method of direct leveling the objective of which is solely to determine the difference in elevation of two points regardless of the horizontal positions of the points with respect of each other. This type of leveling is also known as fly leveling.

2.5.4.1.2 Profile Leveling:

It is the method of direct leveling the objective of which is to determine the elevations of points at measured intervals along a given line in order to obtain a profile of the surface along that line.

2.5.4.1.3 Cross Sectioning:

Cross-sectioning or cross leveling is the process of taking levels on each side of main line at right angles to that line, in order to determine a vertical cross-section of the surface of the ground, or of underlying strata, or of both.

2.5.4.1.4 Reciprocal Leveling:

It is the method of leveling in which the difference in elevation between two points is accurately determined by two sets of reciprocal observations when it is not possible to set up the level between the two points.

2.5.4.2 Indirect Leveling:

Indirect method or trigonometric leveling is the process of leveling in which the elevations of points are computed from the vertical angles and horizontal distances measured in the field, just as the length of any side in any triangle can be computed from proper trigonometric relations.

The first operation is required to enable the works to be designed while the second operation is required in the setting out of all kinds of engineering works. Leveling deals with measurements in a vertical plane.

Temporary Adjustment of Level:

The temporary adjustment for a level consists of the following:

- **Setting up the level:** The operation of setting up includes fixing the instrument on the stand and leveling the instrument approximately.
- **Leveling up:** Accurate leveling is done with the help of foot screws and with reference to the plate levels. The purpose of leveling is to make the vertical axis truly vertical and horizontal line of sight truly horizontal.
- **Removal of parallax:** Parallax is a condition when the image formed by the objective is not in the plane of the cross hairs. Parallax is eliminated by focusing the eyepiece for distinct

vision of the cross hairs and by focusing the objective to bring the image of the object in the plane of cross hairs.

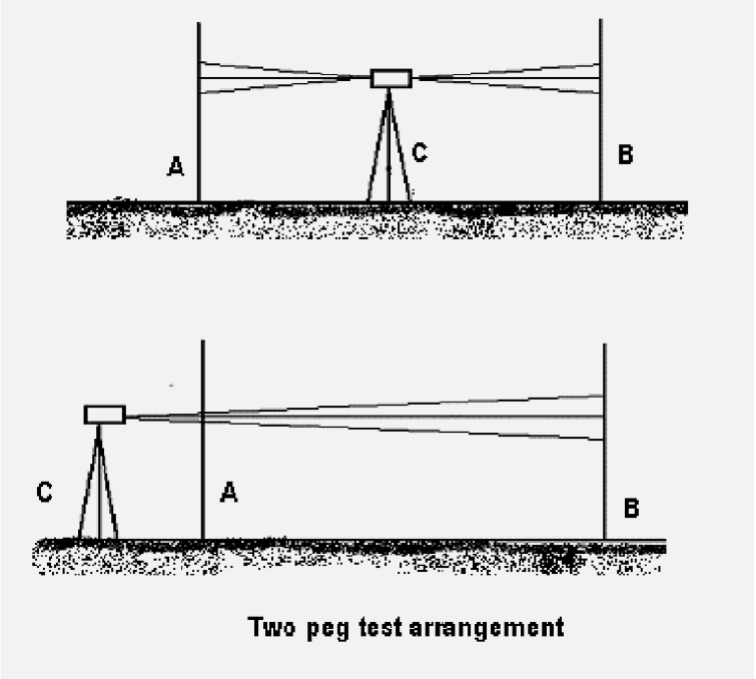


Figure 6 Two peg test arrangement

Permanent adjustments of level:

To check for the permanent adjustments of level **two-peg test** method should be performed. Two staffs were placed at A and B of known length (about 60 m). First the instrument was setup on the line near B and both staff readings (Top, Middle, and Bottom) were taken. Then, the instrument was setup at the middle C on the line and again both staff readings on A and B was taken. Then computation was done in order to check whether the adjustment was within the required accuracy or not.

The error obtained wasn't within the given permissible error. So, the permanent adjustment was done.

Booking and reducing levels:

There are two methods of booking and reducing the elevation of points from the observed staff reading:

- Height of the Instrument method
  - ✞ Arithmetic Check: ∑BS – ∑F.S. = Last R.L. – First R.L.

- Rise and Fall method
  - ✞ Arithmetic Check: ∑ BS – ∑ F.S. = ∑ Rise – ∑fall = Last R.L. – First R.L.

Fly Leveling:

The RL of Given TBM1 point was found by transferring the level from Known BM by the process of fly leveling. In this method auto level was used and the level was transferred directly by taking BS and FS at every Turning Point.

Level Transfer to Major and Minor Traverse Stations:

The R. L of the temporary benchmark was then transferred to the control stations of the major and minor traverse. The closing error was found to be within the permissible limits. The misclosure was adjusted in each leg of the leveling path by using the following formula:

Permissible error = $\pm 25\sqrt{K}$ mm.

Where k is perimeter in Km

Actual Error (e) = $\sum$BS – $\sum$F.S. = Last R.L. – First R.L.

Correction i^{th} leg=-(e x (L_1 + L_2 +….+ L_i)/P

Where L_1, $L_{2...}L_i$are Length of 1st 2nd,.....i^{th} leg.

P is perimeter

Relative Precision= 1/(p/e)

2.5.5 Detailing:

Detailing means locating and plotting relief in a topographic map. Detailing can be done by either plane table surveying or tachometric surveying. Plane tabling needs less office work than tachometric survey. Nevertheless, during our camp, we used the tachometric method.

Tacheometry

Tacheometry is a branch of angular surveying in which the horizontal and vertical distances of points are obtained by optical means. Though it only has accuracy about 1/300 to 1/500, it is faster and convenient than the measurements by tape or chain. It is very suitable for steep or broken ground, deep ravines, and stretches of water or swamp where taping is impossible and unreliable.

The objective of the tachometric survey is to prepare of contour maps or plans with both horizontal and vertical controls. For the survey of high accuracy, it provides a check on the distances measured by tape.

The formula for the horizontal distance is

$$H = 100 \times S \times \cos^2\theta$$

The formula for the vertical distance is

$$V = 100 \times S \times (\sin 2\theta)/2$$

Where, S = staff intercept; θ = Vertical Angle

If the angle used is zenithal angle then

$$H = 100 \times S \times \sin^2\theta$$

$$V = 100 \times S \times (\sin 2\theta)/2$$

Where, θ=zenithal angle.

Contouring:

A contour is an imaginary line, which passes through the points of equal elevation. It is a line in which the surface of ground is intersected by a level surface. Every fifth contour lines must be made darken. While drawing the contour lines, the characteristics of the contours should be approached.

The characteristics are as follows:

- Two contours of different elevations do not cross each other except in the case of an overhanging cliff.
- Contours of different elevations do not unite to form one contour except in the case of a vertical cliff.

- Contours drawn closer depict a steep slope and if drawn apart, represent a gentle slope.
- Contours equally spaced depict a uniform slope. When contours are parallel, equidistant and straight, these represent an inclined plane surface.
- Contour at any point is perpendicular to the line of the steepest slope at the point.
- A contour line must close itself but need not be necessarily within the limits of the map itself.
- A set ring contours with higher values inside depict a hill whereas a set of ring contours with lower values inside depict a pond or a depression without an outlet.
- When contours cross a ridge or V-shaped valley, they form sharp V-shapes across them. Contours represent a ridgeline, if the concavity of higher value contour lies towards the next lower value contour and on the other hand these represent a valley if the concavity of the lower value contour, lies toward the higher value contours.
- The same contour must appear on both the sides of a ridge or a valley.
- Contours do not have sharp turnings.

Methods of contouring:

Taking the reading at the change point on the ground does the indirect method of locating contours. The interpolation method is used to draw the contour lines. Interpolation of contours is done by estimation, by arithmetic calculations or by graphical method. The eye estimation method is extremely rough and is used for small-scale work only.

There are two methods of locating contour:

i) The Direct Method:

In this method, the points of equal elevations are found directly on the field. The horizontal control of the point is found by the help of plane table.

ii) The Indirect Method:

In this method, some suitable guide points need not necessarily be on the contour. There are some of the indirect methods of location the ground points:

a) Square Method
b) Cross- Section Method
c) Tachometric Method

Interpolation is the process of spacing the contours proportionately between the slopes of the ground between the two points is uniform. The interpolation of contour cans be done on following three ways:

i. Estimation:

In this method, we just draw the line between two RL points and estimate the contour point on the line.

ii. Arithmetic Calculation:

Generally, arithmetic calculation method of interpolation is used to draw the contour lines and is performed as follows:

$$Dist.of\ ContourPoint = \frac{(Diff.\ in\ RL\ of\ one\ point\ and\ req\ point)*(Horizontal\ DIst.\ in\ scale)}{Difference\ in\ RL\ of\ two\ known\ points}$$.

iii. Graphical Method:

Generally, we use arithmetic method of interpolation to draw the contour line, and the graphical methods are as follow:

2.5.6 Computation and plotting:

For the calculations as well as plotting, we applied the coordinate method (latitude and departure method). In this method, two terms latitude and departure are used for calculation. Latitude of a survey line may be defined as its coordinate lengths measured parallel to an assumed meridian direction. The latitude (L) of a line is positive when measured towards north, and termed Northing and it is negative when measured towards south, and termed Southing. The departure (D) of a line is positive when measured towards east, and termed Easting and it is negative when measured towards south, and termed westing. The latitude and departures of each control station can be calculated using the relation:

Latitude = L Cosθ

Departure = L Sinθ where,

L=distance of the traverse legs

θ=Reduced bearing

If a closed traverse is plotted according to the field measurements, the end of the traverse will not coincide exactly with the starting point. Such and error is known as closing error.

Mathematically,

Closing error (e) = $\sqrt{\{(\Sigma L)\ 2 + (\Sigma D)\ 2\}}$

The relative error of closure = e / p

The error (e) in a closed traverse due to bearing may be determined by comparing the two bearings of the last line as observed at the first and last stations of traverse. If the closed traverse, has N number of sides then,

Correction for the first line = e/N

Correction for the second line = 2e/N

And similarly, correction for the last line = Ne/N = e

In a closed traverse, by geometry, the sum of the interior angles should be equal to(2n4) x 90˚ where n is the number of traverse sides. If the angles are measured with the same degree of precision, the error in the sum of the angles may be distributed equally among each angle of the traverse.

Mathematically,

a) Correction in departure of a side of traverse

= - (Total departure misclosure / traverse perimeter) x length of that side

b) Correction in latitude of a side of traverse

= - (Total latitude misclosure / traverse perimeter) x length of that side

In the case of length, the difference in values obtained by forward and backward taping is called discrepancy. In addition, the reciprocal of the discrepancy divided by the mean of the two measurements is called precision. Both the discrepancy and the precision for each traverse leg should be within the given limits.

Mathematically,

Discrepancy = | Forward length - Backward length |

Linear precision = 1 / (Mean length / Discrepancy)

The coordinates of common points CP1 & CP2 are given.

2.5.7 Comments and Conclusion

The site for survey camping was the TU premises in Kirtipur, Kathmandu. The pattern (area) was very suitable because all the facilities for engineering work were available with a good environment for doing fieldwork. TU Kirtipur campus provides ample space, open grounds, and access to basic infrastructure which supports surveying and practical field exercises.

The arrangements of the survey instruments were appreciable although there were some faulty instruments that made the fieldwork time-consuming. Some instruments like Theodolite, levels etc. did not give accurate readings. We hope that there will be a sufficient number of instruments for the next survey camp. The stationary accessories should be managed inside the campus area because it is difficult to take all the stationary goods from outside and there is no such stationary shop near TU premises. Some other problems during the fieldwork occurred during fly leveling while transferring the R.L. from a given benchmark due to traffic disturbances being around the TU premises on the way to nearby areas.

The given Topography survey camp work was finished within the given span of time. The subject survey needs practice as much as possible. For surveying, theory can only be taken as an introduction but if there is practice, there will be much gain of knowledge about the techniques of surveying. Thus, this camp helps us by practicing the survey work to gain essential knowledge as far as possible. It provides confidence to perform surveys and apply techniques to any type of problem faced during actual work in a future career.

All groups prepared their topographic map of the given area of the TU premises in Kirtipur in the same scale. The whole area was divided in such a way that the area allocated for one group contains some part of the area allocated for another group. One traverse leg is also common to all groups and hence the combination of all groups' effort will provide a perfect and complete topographic map of TU premises after combining it.

Chapter 3 :BRIDGE SITE SURVEY

3.1 Objectives:

The main objective of the bridge site survey is to give the students the preliminary knowledge on selection and planning of possible bridge site and axis for the future construction of the bridge. The purpose of the bridge site survey was not only to prepare plan and layout of the bridge site but also from the engineering point of view, the purpose is to collect the preliminary data about the site such as normal water flow level, high flood level, geological features of the ground for planning and designing of the bridge from the details taken during the surveying. Moreover, bridge construction is an important aspect in the development of transportation network. Surveying is required for topographical mapping, knowledge of longitudinal sections of the river and cross sections at both the upstream and in downstream side of the river for the construction of a bridge.

The following are the main objectives of the bridge site survey.

a. To develop an idea of proper selection of the site for bridges such that the bridge axis should be as short as possible and should be stable, safe and economic.

b. To prepare the topographical map for the river site by carrying out topographical survey and hence draw the longitudinal and cross sections of the rivers at required u/s and d/s of the river.

c. To depict the nature of river flow.

3.2 Brief Description of the Area:

The bridge site survey was conducted in the Tribhuvan University area, Kirtipur. The site consists of mild vegetation. The gradient of the nearby water channels is low, and the banks are mostly composed of soil. The water discharge varies with the season but does not dry up, indicating that the source of water is not only rainfall but also spring water. The site is suitable for a bridge, though the foundation will need to be constructed with imported or engineered materials, as natural rocks or boulders are not available. The surrounding area consists mainly of sandy soil and clay.

3.3 Hydrology, Geology and Soil:

The site is surrounded with steep hill, which is covered with shrubs. The width of stream is not so big but high flood level covers large area. Water scoured marks on the side show the highest flood level. The terrace was on the right side of the bank while on the the left side there is steep slope.

3.4 Norms (Technical Specification):

- Reconnaissance was conducted in order to establish triangulation points for determining Bridge Axis Length, as well as horizontal and vertical control of the area. Triangles need to be well conditioned.
- Measurement of distance of Base Line in triangulation in accuracy of 1:2000.
- Measurement of the apex angle of triangulation on two sets of horizontal circle reading by Theodolite with discrepancy of one minute.
- Computation of average distance of the proposed bridge axis by two adjacent triangulation.
- Fly leveling was conducted to transfer the RL from given BM to the nearest end point of the bridge axis and error of closure was checked by making circuit close.
- Reciprocal leveling was done to transfer level from one bank to another. RL of the other triangulation stations are determined by fly leveling from the end point of the bridge axis.

- Prepare a topographic map by tachometric surveying indicating contour lines at suitable contour interval. Interpolate the contour lines with the help of guide points and draw longitudinal (along the river bed up to 150m U/S and 50m D/S) and cross-section (at 25m interval and one at the bridge axis) profile of the area. The scale for plotting is as follows:
  Scale of topographic map = 1:200

Scale of L-Section	**Scale of Cross-section**
Horizontal scale = 1:500	Horizontal scale =1:200
Vertical scale =1:50	Vertical scale =1:200

3.5 Equipments:

The equipments used in the survey during the preparation of topographic map are as follows:

- Total Stations
- Staffs
- Prisms
- Ranging rods
- Tapes
- Leveling
- Pegs
- Marker
- Hammer

3.6 Methodology:

The various methods performed during the bridge site survey were triangulation, leveling, tacheometry, and cross section, L-section etc. The brief descriptions of these methodologies were given below:

3.6.1 Site Selection:

There are various factors for the selection of bridge site such as geological condition, socio-economical and ecological aspect etc. Therefore, the sites was chosen such that it should be laid on the very stable rocks at the bed of river as far as possible and not affect the ecological balance of the flora and fauna of the site area. The bridge axis should be so located hat it should be fairly perpendicular to the flow direction and at the same time, the river width should be narrow from the economical point of view and the free board should be at least 4m. The starting point of bridge axis should not in any way lie or touch the curve of the road.

The site selected for the bridge axis was near the curve of the river with no community but with the temples and the shed house nearby. For the purpose of the shortest span, the stations were set perpendicular to the river flow direction. The riverbanks were not eroded and were suitable for bridge construction. The chance of change of direction of river on the selected axis line was nominal.

3.6.2 Topographic Survey:

For the topographic survey of the bridge site triangulation was done. First the bridge axis was set and horizontal control stations were fixed on either side for detailing. Distances between stations on the same sides of river i.e. base line were measured with tape precisely. Then the interconnecting triangles were formed and angles were measured with Theodolite. The bridge axis length or span was calculated by solving the triangles using the sine rule. Thus the horizontal control was set out.

For vertical control, the level was transferred from the BM to preceding IP A of the road and was transferred to the stations on the next bank by reciprocal leveling. For the same bank direct level transfer method was used.

3.6.3 Longitudinal Section

Profile leveling is the process of determining the elevations of the points at the short intervals along a fixed line or alignment such as the center line of the railway, highway, canal or sewer. The fixed line may be a single straight line or it may be composed of a succession of straight lines or of a series of straight lines connected by curves. It is also known as longitudinal leveling. By means of such sections the engineer is able to study the relationship between the existing ground surface and the levels of the proposed construction in the direction of its length. The profile is usually plotted on specially prepared profile paper, on which the vertical scale is much larger than the horizontal.

Profile leveling, like differential leveling, requires the establishment of turning points on which both back and fore sights is taken. In addition, any number of intermediate sights may be taken on the points along the line from each set up of the instrument. It is generally best to set up the instrument to one side of the profile line to avoid too short sights on the points near the instrument. For each set up, intermediate sights should be taken after the fore sight on the next turning point has been taken. The position of the intermediate points on the profile is simultaneously located by chaining along the profile and noting their distances from the point of commencement. For the longitudinal section of the river, it was drawn from the contour map of the site with first chainage at 10m then at every 25m chainage by interpolation.

3.6.4 Cross-Section:

Cross-sections are run at right angles to the longitudinal profile and on the either side of it for the purpose of lateral outline of the ground surface. They provide the data for estimating quantities of earth work and for other purposes. The cross-sections are numbered consecutively from the commencement of the centre line and are set out at right angles to the main line of section with the chain and tape. Cross-sections may be taken at each chain. The length of crosssection depends upon the nature of the work. The longitudinal and cross-sections may be worked together or separately as per the requirement. In the case of bridge site survey, crosssections are taken at 10m from the bridge axis then at every 25m interval. The spot heights were taken where the change of the slope of the ground was abrupt. Cross-sections must be taken at normal flood level, high flood level, and river banks etc. Total Station was used for this purpose as a tachometer.

3.6.5 Leveling:

Transferring R.L. from B.M. to control points:

The R.L of benchmark T.B.M.2 = was given and was transferred to the triangular stations from the B.M. by fly leveling along the road turning points by taking the back sight reading to the bench mark which should be within the given accuracy. The R.L. was transferred to the opposite bank of the river by reciprocal leveling.

Reciprocal Leveling:

For transferring the RL across the bridge, reciprocal leveling was performed. It is the method of leveling in which the difference in elevation between two points is accurately determined by two sets of reciprocal observations when it is not possible to set up the level between the two points. For transferring the R.L. across the bridge axis, reciprocal leveling was done.

Reciprocal leveling must be used to obtain accuracy and to eliminate the following errors due to focusing, collimation, earth's curvature and refraction of atmosphere etc

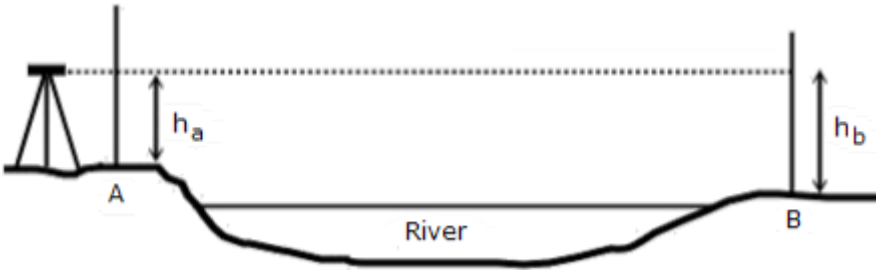


Figure 7 Reciprocal Levelling (a)

Fig (a)

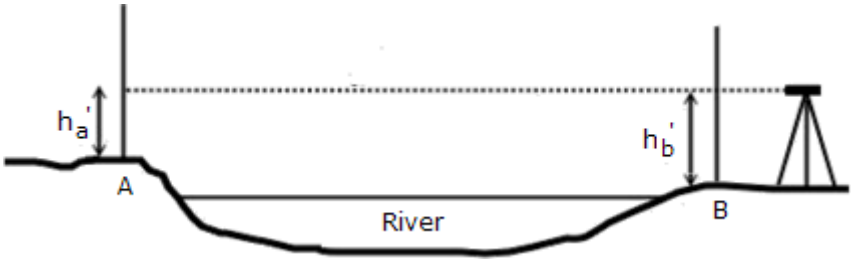


Figure 8 Reciprocal Levelling (b)

True difference in elevation between A and B = *H = ha- (hb-e)*
Also the true difference in elevation = *H = (ha'- e) - hb'*
Taking the average of the two differences we get the difference in elevation between A and B.

3.6.6 Detailing:

The detailing was done with the help of total station using electronic data recording.

3.6.7 Triangulation

Triangulation was performed for determination of the approximate span of the bridge. The triangulation station also serves as control points for detailing. Two points on either bank of the river were fixed as control points and side was assumed as the bridge axis. Then two triangles from each bank were fixed.

The bank line was measured accurately by two way tapping as well as tachometry was done and interior angles were measured by taking two sets of reading. The accurate span of bridge was computed by applying sine rule. To minimize the plotting error well-conditioned triangles

were tried to construct i.e. the angles greater than 30 degree. The best triangle is equilateral triangle.

The topographic map, the longitudinal section and the cross section were plotted on the respective scales after the completion of calculations. Control stations were plotted accurately on Grid Sheet. Then all hard details as well as contours were plotted with reference to the control stations by the method of angle and distances.

3.7 Comments and Conclusions:

The bridge axis should be design such that the span length should be minimum and in safe location. That means the bridge axis should not be below the flood level so that during course of monsoon it is affected by floods of flow. The result of the computations of the triangulation gave the axis span of **70.108.**

During the selection of the site all the considerations like geological, socio-economical and topographical considerations were made and the best site was selected. The inspection of the area showed that no springs, streams and sewer were discharged into the river up to the 100-m upstream and 50m downstream of the axis site. The flow in river was normal and showed no danger of changing its direction of flow for the design period of the bridge.

Chapter 4 :ROAD ALIGNMENT AND GEOMETRIC DESIGN

4.1 General:

Road alignment and bridge site survey includes both the works - to run a road between two far distance points and to carry a survey for the bridge construction along the route. This specific job is essential for an engineer combating with the mountainous topography of Nepal. Road alignment and bridge site survey includes both the works to run a road between two terminals and to carry a survey for the bridge construction along the route. This specific job is essential for an engineer combating with the mountainous topography of Nepal. The starting point of the route was in the southern west direction of solar panel, beneath the canteen. The site is surrounded with steep hill, which is covered with densely planted shrubs. The maximum allowable grade is 12%. There are several rise and fall along the route needing lots of cutting, and filling.

4.2 Hydrology and Geology:

The study of the hydrology in road survey is of great importance to drain the water from surface run-off or seepage from the road periphery. So the hydrology of the area affects the design of the road elements such as drainage arrangements for surface runoff and sub-surface drainage, design of cross drainage structures etc.

The geology of the area is the most important factor for the selection of the road alignment. Generally, the road alignment is avoided to cross the area such as faults, fold landslides marshy and muddy area etc. After selecting the alignment, the soil investigation should be carried out for the following purposes.

- To determine the nature and physical properties of soil to be used in the embankment.
- To facilitate the design of the embankment and cuts.
- To determine the construction techniques for handling the earthwork.
- To classify earthwork(ordinary soil, hard soil, soft rock, hard rock etc)to enable estimation of cost and planning for blasting operation and excavation technique.
- To design the pavement thickness and specifications.

4.3 Norms (Technical Specifications):

Reconnaissance alignment selection was carried out of the road corridor considering permissible gradient, obligatory points, bridge site and geometry of tentative horizontal and vertical curves. The road setting horizontal curve, cross sectional detail in 15m interval and longitudinal profile were prepared.

The topographic map (scale 1:1000) of road corridor was prepared. Geometric curves, road formation width, right of way, crossings and other details were shown in the map.

While performing the road alignment survey, the following norms were strictly followed:

- The road had to be designed starting at teacher hostel building to the final point behind the bamboo trees. If the external deflection angle at the I.P. of the road is less than 3°, curves need not be fitted.
- Simple horizontal curves had to be laid out where the road changed its direction, determining and pegging three points on the curve - the beginning of the curve, the middle point of the curve and the end of the curve along the centerline of the road.
- The radius of the curve had to be chosen such that it was convenient and safe.

- The gradient of the road had to be maintained below 7 %.
- Cross sections had to be taken at 20 m intervals and at the beginning, middle and end of the curve, along the centerline of the road - observations being taken for at least 10 m on either side of the centerline.
- Plan of the road had to be prepared on a scale of 1:1000
- L-Section of the road had to be plotted on a scale of 1:1000 horizontally and 1:100 vertically.
- The cross section of the road had to be plotted on a scale of 1:100 (both vertical and horizontal).
- The amount of cutting and filling required for the road construction had to be determined from the L-Section and the cross sections. However, the volume of cutting had to be roughly equal to the volume of filling.

4.4 Equipments:

The equipments used in the survey during the preparation of topographic map are as follows:

- Total Station
- Leveling Staffs
- Ranging rods
- Prisms
- Measuring Tapes 30m
- Leveling instruments
- Pegs
- Marker

4.5 Methodology:

4.5.1 Horizontal Alignment:

Horizontal alignment is done for fixing the road direction in horizontal plane. For this, the bearing of initial line connecting two initial stations was measured using compass. The interior angles were observed using 20" Theodolite at each IP and then deflection angles were calculated.

Deflection angle, d = 180 - interior angle

If +ve, the survey line deflects right (clockwise) with the prolongation of preceding line and deflects left if –ve (anti-clockwise). The radius was assumed according to the deflection angle.

Then the tangent length, EC, BC, apex distance along with their Chainage were found by using following formulae,

Tangent length (T L) = R x tan (d/2)

Length of curve (L.C) = 3.142 x R x d/180

Apex distance = R x 1/ (Cos (d/2)-1)

Chainage of BC = Chainage of IP – TL

Chainage of MC = Chainage of BC +LC/2

Chainage of EC = Chainage of MC + LC/2

The BC and EC points were located along the line by measuring the tangent length from the apex and the points were marked distinctly. The radius was chosen such that the tangent does

not overlap. The apex was fixed at the length of apex distance from IP along the line bisecting the interior angle.

4.5.2 Leveling:

The method of fly leveling was applied in transferring the level from the given B.M. to all the IPs, beginnings, mid points and ends of the curves as well as to the points along the center line of the road where the cross sections were taken. After completing the work of one way leveling on the entire length of the road, fly leveling was continued back to the B.M. making a closed loop for check and adjustment. The difference in the R.L. of the B.M. before and after forming the loops should be less than $25\sqrt{k}$ mm, where k is the total distance in km.

4.5.3 Longitudinal Section:

The L-Section of the road is required to give the road engineer an idea about the nature of the ground and the variation in the elevations of the different points along the length of the road and also to determine the amount of cutting and filling required at the road site for maintaining a gentle slope. In order to obtain the data for L-Section, staff readings were taken at points at 15m intervals along the centerline of the road with the help of a level by the method of fly leveling. Thus after performing the necessary calculations, the level was transferred to all those points with respect to the R.L. of the given B.M. Then finally the L-Section of the road was plotted on a graph paper on a vertical scale of 1:100 and a horizontal scale of 1:1000. The staff readings at BC, EC and apex were also taken. The RL of each point were calculated.

4.5.4 Cross–section:

Cross sections at different points are drawn perpendicular to the longitudinal section of the road on either side of its centerline in order to present the lateral outline of the ground. Cross sections are also equally useful in determining the amount of cut and fill required for the road construction. Cross sections were taken at 15m intervals along the centerline of the road and at points where there was a sharp change in the elevation. While doing so, the horizontal distances of the different points from the centerline were measured with the help of a tape and the vertical heights with a measuring staff. The R.L. was transferred to all the points by performing the necessary calculations and finally, the cross sections at different sections were plotted on a graph paper on a scale of both vertical and 1:100 - horizontal.

Topographic Survey of Road Corridor:

Topographic survey of road corridor was done by taking the deflection angle at each point where two straight roads meet. The Chainage of intersection point, tangent point and middle points were also taken by taping and applying formula. The staff readings of each of these points were also taken. The R.L was also transferred to find out the elevation and plot it in a map.

Chapter -5 : CURVE SETTING

5.1 General:

Curves are generally used on highways and railways where it is necessary to change the direction of motion. A curve may be circular, parabola or spiral and is always tangential to two straight directions.

5.2 Curve:

A curve may be circular, parabolic or spiral and is always tangential to the two straight directions commonly known as tangents.

Curves which are generally used on highways are as follows:

i. Simple Circular Curve

ii. Transition Curve

iii. Vertical Curve

5.2.1 Simple Circular Curve

A simple circular curve is the one which consists of a single arc of a circle. It is tangential to both of the straight lines namely tangents. During the road survey, it is always kept in mind that the radius of the simple circular curve should not be less than 12m. As far as possible, flat circular curves are preferred to that of the sharp one. Flat curves are comfortable to the passengers and there is less possibility of accident. Before setting out the curve, its elements are essential to be computed. Some essential elements of simple circular curve are as follows:

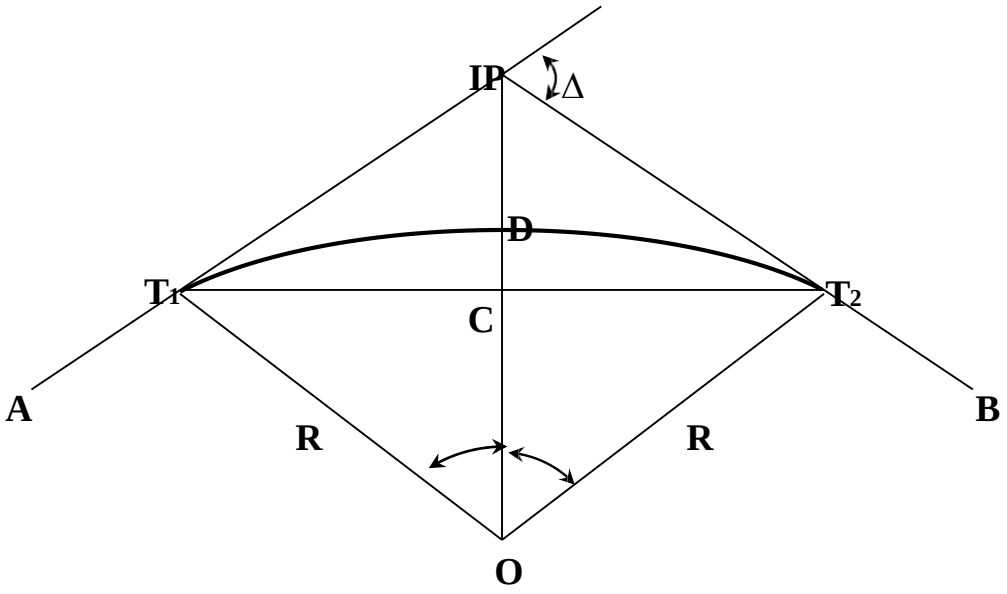


Figure 9 Simple Circular Curve

- Length of Tangent= $R*\tan\frac{\Delta}{2}$ Where, R= radius of simple circular curve
- Δ = deflection angle
- Length of long chord= $2R\sin\frac{\Delta}{2}$

- Apex distance= $R*\sec(\frac{\Delta}{2}-1)$
- Mid ordinate= $R*(1-\cos\frac{\Delta}{2})$
- Length of curve=$\frac{\pi R\Delta}{180^0}$
- Chainage of T_1= Chainage of IP -$R\tan\frac{\Delta}{2}$
- Chainage of T_2= Chainage of $T_2+\frac{\pi R\Delta}{180^0}$

Setting Out of Simple Circular Curves

A simple circular curve can be set in the field by various linear and angular methods which are listed as follows:

a. **Linear method:** Linear method is defined as the method of setting curve in which only chain or tape is used, i.e. no angular instruments are used to set the curve. This method is preferable where high accuracy is not required and the length of the curve to be set is short. Some common linear methods of setting of the simple circular curve are as follows:
   - By ordinates from the long chord
   - By perpendicular offset from tangents
   - By radial offset from tangents
   - By offset from the chords produced
   - By successive bisection of the curves

b. **Angular method**: Angular method is the one in which both angles and the distances are used to set the curve in the field. Generally, tangential deflection angle is observed with the help of Theodolite and the distance is made to be measured by making use of tape provided. Some of the most common angular methods of setting out of simple circular curve are as follows:

- Rankine's method of tangential angles
- Two Theodolite method
- Tachometric method

Setting out:

Setting of curves can be done by two methods depending upon the instrument used.

a. Linear method:-

In this method, only a chain or tape is used. Linear methods are used when a high degree of accuracy is not required and with or without chain or tape.

Before a curve is set out, it is essential to locate the tangents, point of intersection, point of curves and points of tangents.

The linear method adopted in field was Rankine's method.

Ordinate from long chord

Mid-ordinate can be determined by the relation

$O_o=R-\sqrt{(R^2-(L/2)^2)}$

To set out the curve, the long chord is divided into an even number of equal parts. Offsets are calculated from the relation

$O_x=\sqrt{(R^2-X^2)}-(R-O_o)$

Here R=Radius of the curve

O_o=mid- ordinate

O_x= ordinate at distance x from the mid point of the chord
T1 and T2 = tangents points
L= length of the long chord actually measured on the ground
Then the curve set out at each of these points.

b. Rankine's method:

In Rankine's method, we assume that the length of the curve and the chord length are equal for small chords. The deflection angle to any point on the chord from the point of contact to that point.this method is based on the principle that the deflection angle to any point on a circular curve is measured by one half the angle subtended by the arc on P.C. to that point. The angle subtended by each chord is given by the formula $\delta = 1718.9C/R$ if δ_1 $\delta_2\delta n$ are the tangential angles or the angles made by successive chords.
$\Delta 1$, $\Delta 2$, Δn are the total tangential angles or the deflection angles and C1,C2,C3…
Cn are the lengths of the chords
Then, for the first chord,
$\Delta 1 = \delta_1$
And for the second chord,
$\Delta 2 = \delta_1 + \delta_2$
$= \Delta 1 + \delta_1$
Similarly,
$\Delta n = \Delta n\text{-}1 + \delta n$

Field procedure:

1. The instrument was set at T1 and zero set at PI
2. Then the Theodolite was set to an angle of 0º00'000".
3. The tape was sung with one end at T1 and another end towards the right of the Theodolite.
4. The arrow was marked at the intersection of the tape with crosshairs.
5. Then another angle δ_2 was set on the Theodolite and with one end of the tape at 2 m from forward tangent it was again intersected by cross hair.
6. Using all the above statements, all the points were located and the curve was done.

5.2.2 Transition Curve

A transition curve is a curve of varying radius introduced between a straight and a circular curve, or between two branches of a compound curve or reverse curve. The functions of a transition curve are as follows:

5.2.2.1 To accomplish gradually the transition from the tangent to the circular curve, so that the curve is increased gradually from zero to a specified value.

5.2.2.2 To provide a medium for the gradual introduction or change of the required super-elevation. A transition curve introduced between the tangent and the circular curve should fulfill the following conditions:

5.2.2.2.1 It should be tangential to the straight.
5.2.2.2.2 It should meet the circular curve tangentially.
5.2.2.2.3 Its curvature should be zero at the origin on straight.
5.2.2.2.4 Its curvature at the junction with the circular curve should

be the same as that of the circular curve.

5.2.2.2.5 The rate of increase of curvature along the transition should be the same as that of increase of cant or super-elevation.

5.2.2.2.6 Its length should be such that full cant or super-elevation is attained at the junction with the circular curve.

Super-elevation

When a pavement or a track is sloped upwards the outside of the curve, it is termed as banked or super elevated. Thus, super-elevation or can't is the amount by which the outer end of the road or outer rail is raised above the inner one.

The length of transition curve should be such that the required super-elevation or can't is provided at a suitable rate. There are three methods for determining its length:

1. By an Arbitrary Gradient
2. By the Time Rate
3. By the Rate of Change of Radial Acceleration

Elements of Transition Curves

- *Length of Tangent* $= R + Stan\frac{\Delta}{2} + \frac{L}{2}$
- *Where,* R *= Radius of simple circular curve joining transition curve*
  S *= Shift*
  L *= Length of Transition curve*
- *Shift(S)* $= \frac{L^2}{24R}$
- *Spiral Angle* $(\Delta_s) = \frac{180L}{2\pi R}$
- *Central Circular Angle* $(\Delta_c) = \Delta - 2\Delta_s$
- *Length of the circular curve* $= \frac{\pi R(\Delta - 2\Delta_s)}{180^0}$
- *Length of the combined curve* $= \frac{\pi R(\Delta - 2\Delta_s)}{180^0} + 2L$

5.2.3. Vertical Curve

A vertical curve is used to join two intersecting grade lines of railroads, highways or other routes to smooth out the changes in the vertical motion. An abrupt change in the rate of the grade could otherwise subject a vehicle passing over it to an impact that would be either injurious or dangerous. The vertical curve, thus, contributes to the safety, comfort and appearance.

A grade which is expressed as percentage or **1** vertical is to **n** horizontal, is said to be upgrade or positive grade when the elevation along the road alignment increases, while it is termed as downward grade or negative grade when the elevation decreases along the direction of the motion.

5.3 Leveling

The method of differential leveling was applied in transferring the level from the given B.M. to all the I.P.s as well as other components of the curve. Along with the transfer of the level to the chainage at the interval of 20m and the components of the curve level was also transferred to the cross-section up to the distance of 10m on either side of the chainage and the components of the curve.

5.3.1 Profile Leveling (Longitudinal Sectioning)

Profile leveling is the process of determining the elevations of the points at the short-measured intervals along a fix line or alignment such as the center line of the railway, highway, canal or sewer. The fixed line may be a single straight line or it may be composed of a succession of straight lines or of a series of straight lines connected by curves. It is also known as longitudinal sectioning. By means of such sections the engineer is able to study the relationship between the existing ground surface and the levels of the proposed construction in the direction of its length. The profile is usually plotted on specially prepared profile paper, on which the vertical scale is much larger than the horizontal, of costs are made.

Profile leveling, like differential leveling, requires the establishment of turning points on which both back and fore sights is taken. In addition, any number of intermediate sights may be taken on the points along the line from each set up of the instrument. It is generally best to set up the instrument to one side of the profile line to avoid too short sights on the points near the instrument. For each set up, intermediate sights should be taken after the fore sight on the next turning point has been taken. The position of the intermediate points on the profile is simultaneously located by chaining along the profile and noting their distances from the point of commencement.

For the longitudinal section of the road, the staff reading was taken at the interval of every 20m along the center line of the road. Beside this, staff readings at beginning of the curve, ending of the curve and the apex of the curve were also taken. The R.L. of each point was calculated. The profile was plotted on the graph paper at the horizontal scale of **1:1000** and the vertical scale of **1:100**; chainage of each point along the horizontal direction and R.L. in the vertical direction.

5.3.2 Cross Sectioning

Cross-sections are run at right angles to the longitudinal profile and on the either side of it for the purpose of lateral outline of the ground surface. They provide the data for estimating quantities of earth work and for other purposes. The cross-sections are numbered consecutively from the commencement of the centre line and are set out at right angles to the main line of section with the chain and tape. Cross-sections may be taken at each chain. The length of crossection depends upon the nature of the work.The longitudinal and cross-sections may be worked together or separately as per the requirement.Cross-section was plotted on the graph paper both the horizontal as well as vertical scale of **1:100**.

5.4. Tachometry

Tachometry is the branch of angular surveying in which the horizontal and vertical distances of the points are determined or obtained by optical means. The method is very rapid, convenient. The primary object of tachometry is the preparation of the contoured maps or plans requiring both horizontal as well as vertical control. In tachometry, tachometer is used. A tachometer is an ordinary transit Theodolite fitted with a stadia diaphragm. The stadia diaphragm essentially consists of one stadia hair above and below at equal distance of the horizontal cross hair.

The tachometric process is applied to determine the elevation of the points at the cross-section. As for cross-section, horizontal control is not needed to define as it is in right angle to the road alignment; only vertical control is to defined or determined which is enabled with the aid of

tachometric surveying of cross section points. Fly leveling is carried out to define the elevation of the IPs.

5.5 Structures

The main structures provided for the road construction are retaining structures, cross drains, side drains, bio-engineering structures, etc. Retaining structures are provided where slope is critical. Gabion structure, dry masonry structures are the example. The cross drainage is provided at the road mostly at the valley and wherever necessary. Causeways, culverts, and bridges are the example of cross drainage. The side drain is the channel by which the pavement can be protected from the surface water. It is usually constructed along the road just below the cut slope. The collected water is drained off by the means of cross drainage.

5.6. Comments and Conclusions:

Survey of the road alignment is done to make most economical, comfortable, and durable. Extra case is taken to avoid any soil erosion and any other ecological damage. Vertical and horizontal curves are set according to Road Design Standards for comfort and other factors. While setting the road alignment, it should be kept in mind that the minimum IP points should be taken as far as possible and deflection angles should be minimum as far as possible. The task was challengeable and tough due to the route high altitude.

Chapter-6 : TWO PEG TEST

6.1 Introduction

Two Peg test is also known as collimation test. This test is carried out to test whether the line of collimation is parallel to the axis of bubble tube or not. It is applied for the adjustment of the line of collimation.

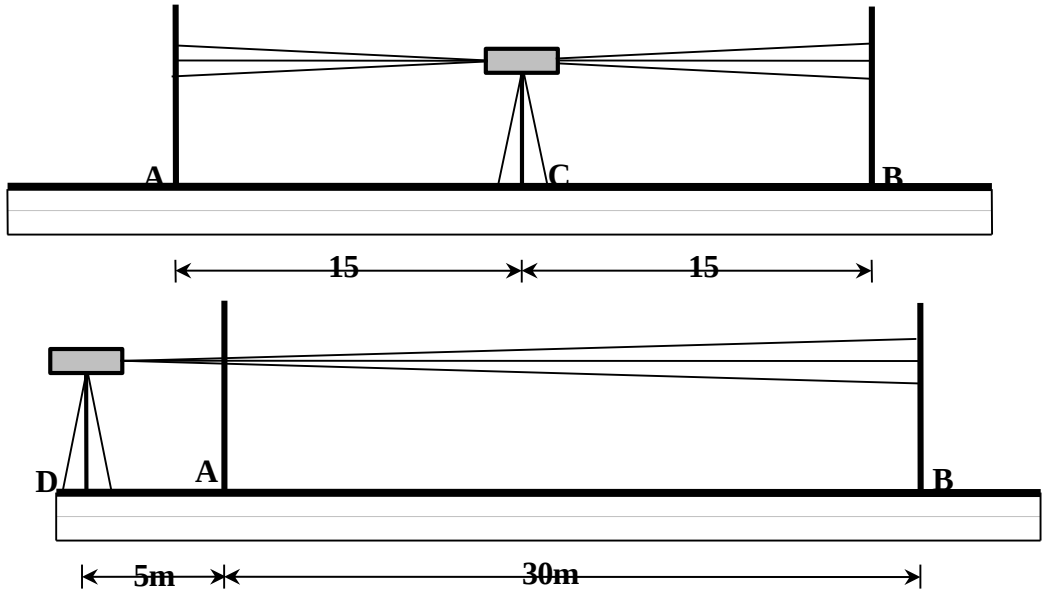


Figure 10 Two peg Test

6.2 Equipments

1. Auto Level
2. Staff

6.3 Methodology

1. Two points A and B were chosen on a fairly leveled ground at a distance of 30m. Instrument was set at C which was exactly at the midway of A and B.
2. Staffs were kept at points A and B and three wire readings were taken on the staff when the bubble was exactly centered.
3. Difference in elevation was calculated between two points, i.e. A & B. The difference in two staff readings give the correct difference in elevation even if the line of sight is inclined as balancing of back sight and fore sight is well carried out.
4. The level machine was shifted to point D about 5m from A and three wire readings were observed on both the staffs kept at A &B.
5. The level was shifted to another point E about 5m from B and three wire readings were observed on both the staffs kept at A &B.
6. Again the differences in elevations were carried out. If the level difference obtained previously is equal to level difference obtained, line of collimation is parallel to the axis of bubble tube. In this case, the collimation error should be less than **1:10000**.
7. If collimation error is greater than **1:10000**, permanent adjustment of the level instrument should be carried out.

6.4 Comments and Conclusion

The accuracy obtained was within the permissible range of **1:10000**. Collimation error occurs due to the following reasons:

- The ground was not well leveled.
- The focusing power of the instrument was just satisfactory.
- There were some observational errors as the eye power of different observers is not same.
- The staff graduation was poor as it was old.
- There may be errors while measuring the distance between the two staff stations.
- As the observation was carried out in the morning, atmospheric conditions also play a vital role.

ANNEX A1

OBSERVATION AND CALCULATION

(TWO PEG TEST &
LEVELING)

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Department of Civil Engineering

Two Peg Test

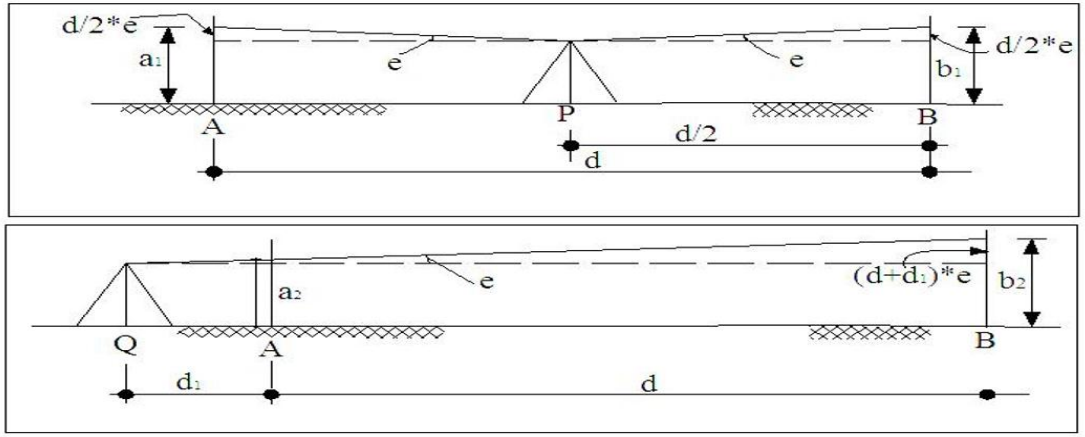
When instrument is at midway of two pegs:

Instrument at	Sighted to	Staff Readings (m)			Mean Value = {(T+M+B) ÷ 3}	True Difference; h1=(MidA-MidB)	Remarks
		Top	Mid	Bottom			
P	A	1.414	1.314	1.214	1.314	0.099	
	B	1.315	1.215	1.115	1.215		

When Instrument is at near one Peg:

Instrument at	Sighted to	Staff Readings (m)			Mean Value = {(T+M+B) ÷ 3}	Apparent Difference; h2=(MidA-MidB	Remarks
		Top	Mid	Bottom			
Q	A	1.49	1.31	1.13	1.31	0.1	
	B	1.23	1.21	1.19	1.21		

Note: Compare mean value with middle wire readings which should be within tolerance ± 1mm



Distance between two pegs (m) =	40
True difference, h1 = (a1-b1) =	0.099
Apparent difference, h2 = (a2-b2) =	0.1
Collimation error, e = |[(h1-h2)]|=	0.001
Precision = 1/d/e ≥	1/ 10000
Obtained precision = 1/d/e =	1/ 31999.9999999893

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Institute of Engineering
Central Campus, Pulchowk
Department of Civil Engineering

Fly Levelling

RL of TBM 2 = 1298.434

Forward Loop (TBM 2 to TBM 3)																
Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	S=(S1 +S2)	Hz. Distance (m)	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
TBM2	1.263	1.202	1.140	1.202	0.123								1298.434	0.123	12.3	TBM2
1	1.619	1.565	1.512	1.565	0.107	0.826	0.769	0.713	0.769	0.113	0.433		1298.867	0.22	22	
2	1.744	1.709	1.674	1.709	0.07	0.802	0.748	0.693	0.748	0.109	0.817		1299.684	0.179	17.9	
3	1.720	1.655	1.588	1.654	0.132	0.760	0.722	0.685	0.722	0.075	0.987		1300.671	0.207	20.7	
4	1.450	1.401	1.353	1.401	0.097	1.454	1.387	1.320	1.387	0.134	0.267		1300.938	0.231	23.1	
5	1.827	1.798	1.770	1.798	0.057	0.900	0.857	0.813	0.857	0.087	0.544		1301.482	0.144	14.4	
6	1.842	1.791	1.740	1.791	0.102	0.922	0.891	0.860	0.891	0.062	0.907		1302.389	0.164	16.4	
7	1.520	1.440	1.360	1.440	0.16	1.213	1.166	1.120	1.166	0.093	0.625		1303.014	0.253	25.3	
8	1.507	1.463	1.419	1.463	0.088	1.231	1.153	1.074	1.153	0.157	0.287		1303.301	0.245	24.5	
9	1.890	1.866	1.844	1.867	0.046	0.845	0.804	0.763	0.804	0.082	0.659		1303.96	0.128	12.8	
10	1.592	1.564	1.538	1.565	0.054	0.661	0.640	0.619	0.64	0.042	1.227		1305.187	0.096	9.6	
11	1.466	1.452	1.440	1.453	0.026	0.732	0.706	0.681	0.706	0.051	0.859		1306.046	0.077	7.7	
12	1.940	1.911	1.882	1.911	0.058	0.662	0.649	0.636	0.649	0.026	0.804		1306.85	0.084	8.4	
13	1.966	1.950	1.933	1.950	0.033	0.634	0.603	0.571	0.603	0.063	1.308		1308.158	0.096	9.6	
14	1.926	1.906	1.887	1.906	0.039	0.654	0.633	0.611	0.633	0.043	1.317		1309.475	0.082	8.2	
15	1.938	1.898	1.861	1.899	0.077	0.705	0.682	0.660	0.682	0.045	1.224		1310.699	0.122	12.2	
16	1.905	1.888	1.871	1.888	0.034	0.863	0.824	0.785	0.824	0.078	1.075		1311.774	0.112	11.2	

TBM3						0.765	0.744	0.722	0.744	0.043	1.144		1312.918	0.043	4.3	TBM3
Sum				28.46					13.978		14.484				260.6	

Arithmetic Check:	
|∑BS-∑FS|=	14.48
∑Rise-∑Fall|=	14.48
t RL - First R	14.48
Distance(k1)=	260.6

Backward Loop (TBM 3 to TBM 2)

Stations Chainage	**BS**			**Mean BS**	**S1**	**FS**			**Mean FS**	**S2**	**Rise +**	**Fall -**	**Elevation**	**S=(S1 +S2)**	**Hz. Distance (m)**	**Remarks**
	Top	**Mid**	**Bot**		**(T-B)**	**Top**	**Mid**	**Bot**		**(T-B)**						
TBM3	0.765	0.744	0.722	0.744	0.043								1312.913	0.043	4.3	TBM3
1	0.655	0.622	0.590	0.622	0.065	1.905	1.888	1.871	1.888	0.034		1.14	1311.769	0.099	9.9	
2	1.159	1.103	1.047	1.103	0.112	1.494	1.461	1.427	1.461	0.067		0.84	1310.93	0.179	17.9	
3	0.754	0.714	0.673	0.714	0.081	1.927	1.871	1.816	1.871	0.111		0.77	1310.162	0.192	19.2	
4	0.772	0.751	0.730	0.751	0.042	1.960	1.921	1.882	1.921	0.078		1.21	1308.955	0.12	12	
5	0.643	0.604	0.565	0.604	0.078	1.802	1.779	1.756	1.779	0.046		1.03	1307.927	0.124	12.4	
6	0.712	0.690	0.667	0.690	0.045	1.937	1.900	1.863	1.9	0.074		1.3	1306.631	0.119	11.9	
7	0.783	0.770	0.757	0.770	0.026	1.960	1.940	1.921	1.94	0.039		1.25	1305.381	0.065	6.5	
8	1.460	1.431	1.401	1.431	0.059	1.830	1.815	1.801	1.815	0.029		1.05	1304.336	0.088	8.8	
9	0.795	0.776	0.758	0.776	0.037	1.885	1.856	1.826	1.856	0.059		0.43	1303.911	0.096	9.6	
10	1.317	1.260	1.201	1.259	0.116	1.584	1.561	1.538	1.561	0.046		0.79	1303.126	0.162	16.2	
11	1.551	1.504	1.455	1.503	0.096	1.617	1.555	1.493	1.555	0.124		0.3	1302.83	0.22	22	
12	1.723	1.656	1.590	1.656	0.133	1.755	1.704	1.654	1.704	0.101		0.2	1302.629	0.234	23.4	
13	0.729	0.695	0.661	0.695	0.068	1.693	1.630	1.566	1.63	0.127	0.026		1302.655	0.195	19.5	
14	1.080	1.050	1.020	1.050	0.06	1.638	1.606	1.572	1.605	0.066		0.91	1301.745	0.126	12.6	

15	0.695	0.673	0.650	0.673	0.045	1.845	1.811	1.778	1.811	0.067		0.760	1300.984	0.112	11.200	
16	0.927	0.872	0.817	0.872	0.110	1.950	1.930	1.910	1.930	0.040		1.257	1299.727	0.150	15.000	
17	0.935	0.872	0.810	0.872	0.125	1.720	1.670	1.610	1.667	0.110		0.795	1298.932	0.235	23.500	
TBM2						1.432	1.370	1.306	1.369	0.126		0.497	1298.435	0.126	12.600	TBM2
Sum				16.79					31.26		0.026	14.50			268.50	

Arithmetic Check:		
|∑BS-∑FS|=		14.48
|∑Rise-∑Fall|=		14.48
|Last RL - First RL| =		14.48
	Distance(k1)=	268.5

Error Computation:	
Error of Closure (m) (mm)=	6
Loop Distance in (km) =	0.5291
Permissible Error = 24√k =	17.457
RL of TBM3 (m) =	1312.918

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Fly Levelling

Observer: Group 25
Recorder: Group 25
Instrument: Auto-Level

Forward Loop from TBM4 to 25M9

Station chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise(m)	Fall(m)	RL	S=(S1+S2)	Hz. Dst. (m)	Remark
	T	M	B		(T-B)	T	M	B		(T-B)	+	-				
TBM4	1.731	1.651	1.571	1.651	0.160								1322.676	0.160	16.000	TBM4
St. 1	1.736	1.681	1.626	1.681	0.110	1.154	1.078	1.003	1.078	0.151	0.573		1323.249	0.261	26.100	
St.2	1.832	1.777	1.722	1.777	0.110	1.215	1.160	1.105	1.160	0.110	0.521		1323.770	0.220	22.000	
St.3	1.928	1.889	1.850	1.889	0.078	1.085	1.025	0.966	1.025	0.119	0.752		1324.521	0.197	19.700	
St.4	1.840	1.789	1.739	1.789	0.101	0.871	0.830	0.788	0.830	0.083	1.059		1325.581	0.184	18.400	
St.5	1.689	1.643	1.599	1.644	0.090	0.784	0.732	0.680	0.732	0.104	1.057		1326.638	0.194	19.400	
St.6	1.629	1.593	1.558	1.593	0.071	0.897	0.852	0.807	0.852	0.090	0.792		1327.430	0.161	16.100	
25M9						1.002	0.966	0.927	0.965	0.075	0.628		1328.058	0.075	7.500	25M9
SUM				12.024					6.642		5.382				145.200	

Arithmetic check:	
∑BS-∑FS=	5.382
∑Rise-∑Fall=	5.382
Last RL-First RL	5.382
Distance(k1)	145.200

Backward Loop from TBM4 to 25M9

Station chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise(m)	Fall(m)	RL	S=(S1+S2	Hz. Dst. (m)	Remark
	T	M	B		(T-B)	T	M	B		(T-B)	+	-				
25M9	1.002	0.966	0.927	0.965	0.075								1328.058	0.075	7.500	25M9
St.7	1.223	1.186	1.150	1.186	0.073	1.629	1.593	1.558	1.593	0.071		0.628	1327.430	0.144	14.400	
St.8	1.068	1.027	0.987	1.027	0.081	1.820	1.786	1.752	1.786	0.068		0.600	1326.830	0.149	14.900	
St.9	1.076	1.033	0.990	1.033	0.086	1.868	1.831	1.795	1.831	0.073		0.804	1326.026	0.159	15.900	
St.10	0.846	0.799	0.752	0.799	0.094	1.922	1.881	1.840	1.881	0.082		0.848	1325.178	0.176	17.600	
St.11	0.872	0.799	0.723	0.798	0.149	1.832	1.784	1.734	1.783	0.098		0.984	1324.194	0.247	24.700	
St.12	1.034	0.948	0.861	0.948	0.173	1.777	1.700	1.624	1.700	0.153		0.902	1323.291	0.326	32.600	
TBM4						1.646	1.564	1.480	1.563	0.166		0.615	1322.676	0.166	16.600	TBM4
SUM				6.757					12.139			5.381			144.200	

Arithmetic check:	
∑BS-∑FS=	-5.382
∑Rise-∑Fall=	-5.382
Last RL-First RL	-5.382
Distance(k2)	144.200

Error Computation:	
Error of Closure (mm)=	0
Loop Distance in(km)=	0.2894
Permissible Error=24√k=	12.91102
RL of 25M9=	1328.058

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Loop Levelling

Loop Levelling (LIBRARY)

RL of known station	M9	1328.06

Staff Stations	BS	FS	Rise +	Fall -	Reduced Level (m)	Correction	Corrected RL	Remarks
25M9	1.062				1328.06		1328.06	25M9
1	1.968	1.739		0.677	1327.38	0.001	1327.38	TP1
2	1.028	1.292	0.676		1328.06	0.002	1328.06	TP2
3	1.548	2.050		1.022	1327.04	0.003	1327.04	25ml1
4	1.848	1.123	0.425		1327.46	0.004	1327.46	TP3
5	1.495	1.351	0.497		1327.96	0.005	1327.96	24ml2
6	1.996	0.750	0.745		1328.7	0.006	1328.71	TP4
7	1.444	0.832	1.164		1329.87	0.008	1329.87	25ml3
8	1.669	0.655	0.789		1330.66	0.009	1330.66	TP5
9	1.054	1.212	0.457		1331.11	0.01	1331.12	25M10
10	0.933	1.460		0.406	1330.71	0.011	1330.72	TP6
11	0.835	1.788		0.855	1329.85	0.012	1329.86	TP7
12	1.205	2.171		1.336	1328.52	0.013	1328.53	TP8
25M9		1.676		0.471	1328.04	0.014	1328.06	25M9
Sum	18.085	18.099	4.753	4.767				

Arithmetic Check:	
$\lvert\sum BS-\sum FS\rvert=$	0.014
$\lvert\sum Rise-\sum Fall\rvert=$	0.014

Error Calculation:	
Error (mm)(-)	14
oop Distance k (km)	0.34936
rmissble = 24√k mm	14.186
Correction =	+

Loop Levelling (CEDA)

RL of known station 25M10 = 1331.12

Staff Stations	BS	FS	Rise +	Fall -	R L (m)	Cor.	Corrected RL	Remarks
25M10	1.036				1331.12		1331.122	25M10
1	1.042	2.161		1.125	1330	0.001	1329.998	TP1
2	0.730	2.304		1.262	1328.74	0.002	1328.737	TP2
3	1.020	1.585		0.855	1327.88	0.002	1327.882	25mc5
4	2.306	1.595		0.575	1327.31	0.003	1327.308	TP3
5	2.468	0.735	1.571		1328.88	0.003	1328.879	TP4
6	1.156	1.409	1.059		1329.94	0.004	1329.939	25mc4
7	1.845	1.610		0.454	1329.48	0.004	1329.485	TP5
8	0.714	1.471	0.374		1329.86	0.005	1329.86	25mc3
9	0.774	2.876		2.162	1327.69	0.005	1327.698	TP6
10	0.725	2.825		2.051	1325.64	0.006	1325.648	TP7
11	1.206	2.646		1.921	1323.72	0.006	1323.727	TP8
12	0.889	2.639		1.433	1322.29	0.007	1322.295	TP9
13	1.128	2.465		1.576	1320.71	0.008	1320.72	TP10
14	1.892	1.793		0.665	1320.05	0.008	1320.055	TP11
15	1.964	2.623		0.731	1319.32	0.009	1319.325	TP12
16	2.548	1.107	0.857		1320.17	0.009	1320.182	TP13
17	1.785	0.725	1.823		1322	0.01	1322.006	TP14
18	1.782	1.782	0.003		1322	0.01	1322.009	25mc2
19	2.014	1.750	0.032		1322.03	0.011	1322.042	TP15
20	2.034	0.975	1.039		1323.07	0.011	1323.081	TP16
21	2.205	1.095	0.939		1324.01	0.012	1324.021	25mc1
22	2.239	0.935	1.27		1325.28	0.012	1325.291	TP17
23	2.041	0.720	1.519		1326.8	0.013	1326.811	TP18
24	2.648	0.860	1.181		1327.98	0.013	1327.992	TP19
25	2.193	0.645	2.003		1329.98	0.014	1329.996	TP20
25M10		1.067	1.126		1331.11	0.014	1331.122	M10
Sum	42.384	42.398	14.796	14.81				

Arithmetic Check:	
|∑BS-∑FS|=	0.014
|∑Rise-∑Fall|=	0.014

Error Calculation:	
Error (mm)(-)	14
oop Distance k (km)	0.56778
rmissble = 24√k mm	18.084
Correction =	+

ANNEX A2

OBSERVATION AND CALCULATION

(TOPOGRAPHIC SURVEY)

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Department of Civil Engineering

Horizontal Angle Measurement (Major Traverse)

Inst St.	Sighted To	Face	Set I					Set II					Mean HZ. Angle	Hz. Distance	Remarks
			HCR Observation			Face Angle	Mean	HCR Observation			Face Angle	Mean			
			D	M	S			D	M	S					
CP1	25M11	L	0	0	0	96° 46' 48"	96° 46' 56"	90	0	0	96° 47' 7"	96° 47' 12"	96° 47' 4"	57.874	
	CP2	L	96	46	48			186	47	7					
	25M11	R	179	59	54	96° 47' 4"		269	59	47	96° 47' 17"			79.087	
	CP2	R	276	46	58			6	47	4					
CP2	CP1	L	0	0	0	145° 50' 26"	145° 50' 33"	90	0	0	145° 50' 32"	145° 50' 35"	145° 50' 34"	79.092	
	25M12	L	145	50	26			235	50	32					
	CP1	R	179	59	54	145° 50' 40"		270	0	5	145° 50' 37"			77.314	
	25M12	R	325	50	34			55	50	42					
25M12	CP2	L	0	0	0	211° 36' 17"	211° 36' 18"	90	0	0	211° 36' 26"	211° 36' 24"	211° 36' 21"	77.308	
	25M13	L	211	36	17			301	36	26					
	CP2	R	180	0	5	211° 36' 19"		270	0	2	211° 36' 21"			91.965	
	25M13	R	31	36	24			121	36	23					
25M13	25M12	L	0	0	0	119° 42' 36"	119° 42' 39"	90	0	0	119° 42' 32"	119° 42' 29"	119° 42' 34"	91.953	
	25M14	L	119	42	36			209	42	32					
	25M12	R	179	59	57	119° 42' 42"		270	0	19	119° 42' 26"			75.601	
	25M14	R	299	42	39			29	42	45					
25M14	25M13	L	0	0	0	145° 37' 23"	145° 37' 20"	90	0	0	145° 37' 26"	145° 37' 24"	145° 37' 22"	75.609	
	25M15	L	145	37	23			235	37	26					
	25M13	R	180	0	12	145° 37' 17"		270	0	18	145° 37' 22"			122.548	
	25M15	R	325	37	29			55	37	40					

25M15	25M14	L	0	0	0	201° 48' 15"	201° 48' 17"	90	0	0	201° 48' 27"	201° 48' 33"	201° 48' 25"	122.555	
	25M16	L	201	48	15			291	48	27					
	25M14	R	179	59	45	201° 48' 19"		269	59	55	201° 48' 39"			80.814	
	25M16	R	21	48	4			111	48	34					
25M16	25M15	L	0	0	0	124° 13' 3"	124° 13' 9"	90	0	0	124° 13' 13"	124° 13' 19"	124° 13' 14"	80.802	
	25M17	L	124	13	3			214	13	13					
	25M15	R	179	59	53	124° 13' 15"		270	0	5	124° 13' 25"			68.882	
	25M17	R	304	13	8			34	13	30					
25M17	25M16	L	0	0	0	160° 19' 30"	160° 19' 21"	90	0	0	160° 19' 28"	160° 19' 37"	160° 19' 29"	68.895	
	25M18	L	160	19	30			250	19	28					
	25M16	R	180	0	7	160° 19' 12"		269	59	56	160° 19' 46"			71.640	
	25M18	R	340	19	19			70	19	42					
25M18	25M17	L	0	0	0	236° 13' 51"	236° 13' 57"	90	0	0	236° 14' 2"	236° 14' 7"	236° 14' 2"	71.637	
	25M19	L	236	13	51			326	14	2					
	25M17	R	179	59	43	236° 14' 3"		269	59	47	236° 14' 12"			91.957	
	25M19	R	56	13	46			146	13	59					
25M19	25M18	L	0	0	0	43° 16' 49"	43° 16' 58"	90	0	0	43° 17' 3"	43° 17' 10"	43° 17' 4"	91.944	
	25M1	L	43	16	49			133	17	3					
	25M18	R	179	59	48	43° 17' 7"		270	0	4	43° 17' 17"			75.347	
	25M1	R	223	16	55			313	17	21					
25M1	25M19	L	0	0	0	227° 25' 43"	227° 25' 52"	90	0	0	227° 26' 4"	227° 26' 12"	227° 26' 2"	75.361	
	25M2	L	227	25	43			317	26	4					
	25M19	R	180	0	16	227° 26' 1"		269	59	43	227° 26' 20"			94.089	
	25M2	R	47	26	17			137	26	3					
25M2	25M1	L	0	0	0	233° 11' 51"	233° 11' 57"	90	0	0	233° 12' 9"	233° 12' 17"	233° 12' 7"	94.095	
	25M3	L	233	11	51			323	12	9					
	25M1	R	179	59	43	233° 12' 3"		269	59	58	233° 12' 25"			84.409	
	25M3	R	53	11	46			143	12	23					
25M3	25M2	L	0	0	0	152° 21' 28"	152° 21' 37"	90	0	0	152° 21' 41"	152° 21' 49"	152° 21' 43"	84.419	
	25M4	L	152	21	28			242	21	41					
	25M2	R	179	59	46	152° 21' 46"		269	59	41	152° 21' 57"			68.838	
	25M4	R	332	21	32			62	21	38					

25M4	25M3	L	0	0	0	138° 53' 1"	138° 52' 56"	90	0	0	138° 53' 3"	138° 53' 8"	138° 53' 2"	68.848
	25M5	L	138	53	1			228	53	3				
	25M3	R	179	59	53	138° 52' 51"		269	43	19	138° 53' 13"			66.157
	25M5	R	318	52	44			48	36	32				
25M5	25M4	L	0	0	0	117° 27' 28"	117° 27' 37"	90	0	0	117° 27' 47"	117° 27' 53"	117° 27' 45"	66.162
	25M6	L	117	27	28			207	27	47				
	25M4	R	180	0	4	117° 27' 46"		270	0	4	117° 27' 59"			97.013
	25M6	R	297	27	50			27	28	3				
25M6	25M5	L	0	0	0	136° 7' 54"	136° 7' 56"	90	0	0	136° 8' 2"	136° 8' 8"	136° 8' 2"	97.029
	25M7	L	136	7	54			226	8	2				
	25M5	R	179	59	52	136° 7' 58"		269	59	43	136° 8' 14"			82.693
	25M7	R	316	7	50			46	7	57				
25M7	25M6	L	0	0	0	236° 2' 37"	236° 2' 43"	90	0	0	236° 3' 2"	236° 2' 56"	236° 2' 49"	82.691
	25M8	L	236	2	37			326	3	2				
	25M6	R	179	59	43	236° 2' 49"		269	59	54	236° 2' 50"			105.869
	25M8	R	56	2	32			146	2	44				
25M8	25M7	L	0	0	0	160° 23' 11"	160° 23' 7"	90	0	0	160° 23' 18"	160° 23' 25"	160° 23' 16"	105.857
	25M9	L	160	23	11			250	23	18				
	25M7	R	179	59	54	160° 23' 3"		270	0	3	160° 23' 32"			125.362
	25M9	R	340	22	57			70	23	35				
25M9	25M8	L	0	0	0	133° 36' 54"	133° 37' 2"	90	0	0	133° 37' 6"	133° 37' 14"	133° 37' 8"	125.348
	25M10	L	133	36	54			223	37	6				
	25M8	R	179	59	57	133° 37' 10"		269	59	53	133° 37' 22"			86.637
	25M10	R	313	37	7			43	37	15				
25M10	25M9	L	0	0	0	129° 35' 42"	129° 35' 48"	90	0	0	129° 35' 56"	129° 36' 4"	129° 35' 56"	86.651
	25M11	L	129	35	42			219	35	56				
	25M9	R	180	0	2	129° 35' 54"		269	59	53	129° 36' 12"			97.634
	25M11	R	309	35	56			39	36	5				
25M11	25M10	L	0	0	0	269° 27' 3"	269° 27' 7"	90	0	0	269° 26' 43"	269° 26' 51"	269° 26' 59"	97.646
	CP1	L	269	27	3			359	26	43				
	25M10	R	180	0	6	269° 27' 11"		269	59	52	269° 26' 59"			57.866
	CP1	R	89	27	17			179	26	51				

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Linear Distance Measurement (Major Traverse)

S.No.	Line		Distance (m)		Average	Discrepancy	Precision ≥1/5000	Remarks
	From	To	Forward (FW)	Backward (BW)				
1	CP1	CP2	79.087	79.092	79.090	0.005	1/ 15817.9	
2	CP2	M12	77.314	77.308	77.311	0.006	1/ 12885.167	
3	M12	M13	91.965	91.953	91.959	0.012	1/ 7663.25	
4	M13	M14	75.601	75.609	75.605	0.008	1/ 9450.625	
5	M14	M15	122.548	122.555	122.552	0.007	1/ 17507.357	
6	M15	M16	80.814	80.802	80.808	0.012	1/ 6734	
7	M16	M17	68.882	68.895	68.889	0.013	1/ 5299.115	
8	M17	M18	71.640	71.637	71.639	0.003	1/ 23879.5	
9	M18	M19	91.957	91.944	91.951	0.013	1/ 7073.115	
10	M19	M1	75.347	75.361	75.354	0.014	1/ 5382.429	
11	M1	M2	94.089	94.095	94.092	0.006	1/ 15682	
12	M2	M3	84.409	84.419	84.414	0.01	1/ 8441.4	
13	M3	M4	68.838	68.848	68.843	0.01	1/ 6884.3	
14	M4	M5	66.157	66.162	66.160	0.005	1/ 13231.9	
15	M5	M6	97.013	97.029	97.021	0.016	1/ 6063.813	
16	M6	M7	82.693	82.691	82.692	0.002	1/ 41346	
17	M7	M8	105.869	105.857	105.863	0.012	1/ 8821.917	
18	M8	M9	125.362	125.348	125.355	0.014	1/ 8953.929	
19	M9	M10	86.637	86.651	86.644	0.014	1/ 6188.857	
20	M10	M11	97.634	97.646	97.640	0.012	1/ 8136.667	
21	M11	CP1	57.866	57.874	57.870	0.008	1/ 7233.75	

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Major Traverse Computation (Gale's Table)

Bearing	D	M	S
CP1-CP2	184	53	10

Refrence Coordinates	N	E
CP1	3063148.96	626865.9897

Stn.	Line	Length	Observed Angle			Cor.	Correccted Aangle			Bearing			Consecutive Coordinates		Correction		Corrected Cons. Cor.		Independent Coordinates		Adjusted Length	Adjusted Bearing		
			D	M	S	S	D	M	S	D	M	S	L	D	L	D	L	D	Northing	Easting		D	M	S
CP1																			3063148.959	626865.979				
CP2	CP1CP2	79.090	145	50	34	-3	145	50	31	184	53	10	-78.802	-6.736	0	0	-78.802	-6.736	3063070.157	626859.243	79.0895	184	53	10
M12	CP2M12	77.311	211	36	21	-3	211	36	18	150	43	41	-67.439	37.802	-0.002	0.005	-67.441	37.806	3063002.716	626897.049	77.3149	150	43	32
M13	M12M13	91.959	119	42	34	-2	119	42	32	182	19	59	-91.882	-3.743	-0.002	0.006	-91.885	-3.738	3062910.831	626893.311	91.9605	182	19	46
M14	M13M14	75.605	145	37	22	-3	145	37	19	122	2	31	-40.112	64.087	-0.002	0.005	-40.113	64.092	3062870.718	626957.403	75.6099	122	2	29
M15	M14M15	122.552	201	48	25	-3	201	48	22	87	39	50	4.995	122.450	-0.003	0.007	4.992	122.458	3062875.710	627079.861	122.5593	87	39	56
M16	M15M16	80.808	124	13	14	-2	124	13	12	109	28	12	-26.934	76.187	-0.002	0.005	-26.936	76.192	3062848.774	627156.053	80.8133	109	28	13
M17	M16M17	68.889	160	19	29	-3	160	19	26	53	41	24	40.793	55.512	-0.002	0.004	40.791	55.516	3062889.565	627211.569	68.8909	53	41	35
M18	M17M18	71.639	236	14	2	-3	236	13	59	34	0	50	59.381	40.074	-0.002	0.004	59.379	40.079	3062948.944	627251.648	71.6395	34	1	4
M19	M18M19	91.951	43	17	4	-2	43	17	2	90	14	49	-0.396	91.950	-0.002	0.006	-0.399	91.955	3062948.546	627343.603	91.9561	90	14	54
M1	M19M1	75.354	227	26	2	-3	227	25	59	313	31	52	51.900	-54.632	-0.002	0.005	51.898	-54.627	3063000.444	627288.976	75.3494	313	31	57
M2	M1M2	94.092	233	12	7	-3	233	12	4	0	57	50	94.079	1.583	-0.002	0.006	94.076	1.589	3063094.520	627290.564	94.0898	0	58	3
M3	M2M3	84.414	152	21	43	-3	152	21	40	54	9	54	49.420	68.435	-0.002	0.005	49.418	68.440	3063143.938	627359.004	84.4170	54	10	5
M4	M3M4	68.843	138	53	2	-3	138	52	59	26	31	34	61.596	30.746	-0.002	0.004	61.594	30.750	3063205.533	627389.754	68.8434	26	31	47
M5	M4M5	66.160	117	27	45	-2	117	27	43	345	24	32	64.026	-16.667	-0.002	0.004	64.024	-16.663	3063269.557	627373.091	66.1569	345	24	43
M6	M5M6	97.021	136	8	2	-3	136	7	59	282	52	16	21.612	-94.583	-0.002	0.006	21.610	-94.577	3063291.167	627278.514	97.0147	282	52	13
M7	M6M7	82.692	236	2	49	-3	236	2	46	239	0	14	-42.585	-70.884	-0.002	0.005	-42.587	-70.879	3063248.580	627207.635	82.6887	239	0	4
M8	M7M8	105.863	160	23	16	-3	160	23	13	295	3	0	44.823	-95.905	-0.003	0.006	44.821	-95.899	3063293.401	627111.736	105.8561	295	3	1
M9	M8M9	125.355	133	37	8	-3	133	37	5	275	26	13	11.878	-124.791	-0.003	0.008	11.874	-124.783	3063305.275	626986.953	125.3471	275	26	9
M10	M9M10	86.644	129	35	56	-3	129	35	53	229	3	18	-56.781	-65.446	-0.002	0.005	-56.783	-65.440	3063248.492	626921.513	86.6414	229	3	6
M11	M10M11	97.640	269	26	59	-3	269	26	56	178	39	11	-97.613	2.295	-0.002	0.006	-97.615	2.301	3063150.877	626923.814	97.6425	178	38	58
CP1	M11CP1	57.870	96	47	4	-2	96	47	2	268	6	7	-1.917	-57.838	-0.001	0.004	-1.918	-57.835	3063148.959	626865.979	57.8665	268	6	2
SUM		1801.749	3420	0	58								0.042	-0.105										

Observed Sum of Horizontal Angle	3240˚0'58"
Theoretical Sum of Horizontal Angle	3240˚
Angular Error	0˚0'58"
Permissible Value of Angular Error(30"√N)	0˚2'17"
Total Correction	-58"

Error in Latitude	0.042
Error in Departure	-0.105
Closing Error	0.113
Perimeter for Correction	1722.660
Relative Precision	1/15289

Tribhuvan University Institute of Engineering
Pulchowk Campus
Department of Civil Engineering
Survey Instruction Committee
Detailing by Total Station

Observer:Group 25
Recorder:Group 25
Instrument:Total Station

Points	Northing	Easting	Elevation	**Remarks**
1	63148.959	26865.979	1311.734	CP1
2	63070.173	26859.254	1311.466	CP2
3	63150.876	26923.828	1308.208	M11
4	63248.5	26921.533	1307.565	M10
5	63302.971	26988.927	1304.232	M9
6	63246.248	27033.828	1303.188	MI1
101	63267.042	27011.623	1304.296	GP
102	63268.547	27010.845	1304.145	GP
103	63264.689	27005.023	1304.178	B C
104	63253.854	27001.106	1305.523	B C
105	63264.451	26971.889	1305.197	B C
106	63273.055	26975.423	1305.306	GP
107	63273.208	26978.186	1305.061	GP
108	63276.243	26977.057	1305.029	GP
109	63277.229	26985.432	1304.27	GP
110	63276.462	26993.563	1304.219	GP
111	63285.082	26993.82	1303.614	GP
112	63285.179	26998.872	1303.672	GP
113	63287.925	27005.724	1303.402	GP
114	63287.9	27005.728	1303.401	GP
115	63303.924	27026.944	1302.252	GP
116	63279.453	27005.666	1303.773	GP
117	63300.325	27023.713	1302.288	GP
118	63286.594	26979.945	1304.837	GP
119	63302.505	27028.415	1302.21	POLE
120	63278.322	26974.045	1305.297	POLE
121	63265.195	26985.609	1304.646	POLE
122	63309.512	27029.64	1302.258	ROAD
123	63310.855	27024.842	1302.442	ROAD
124	63309.002	27015.251	1303.024	ROAD
125	63308.028	27008.281	1303.326	ROAD
126	63304.479	26992.382	1304.023	ROAD
127	63267.747	26974.154	1305.17	GP
128	63274.337	26971.695	1305.465	GP
129	63281.301	26972.979	1305.089	GP

1	63148.959	26865.979	1311.734	CP1
130	63285.843	26975.981	1304.624	GP
131	63292.457	26990.921	1304.22	GP
132	63294.427	26994.266	1303.635	GP
133	63286.931	26980.689	1304.57	TRE
134	63280.997	26973.503	1305.042	TRE
135	63280.619	26971.936	1305.213	TRE
136	63272.881	26975.549	1305.203	TRE
137	63269.94	26992.094	1304.467	TRE
138	63272.755	27001.124	1303.638	TRE
139	63267.028	27011.773	1304.112	TRE
140	63265.5	27016.088	1304.235	TRE
141	63303.088	27006.203	1303.355	ROAD
142	63299.979	26992.434	1303.979	ROAD
143	63296.945	26984.674	1304.444	ROAD
144	63299.471	26980.086	1304.71	ROAD
145	63289.145	26972.126	1305.079	ROAD
146	63290.714	26966.642	1305.307	ROAD
147	63279.472	26960.726	1305.664	ROAD
148	63277.519	26952.158	1306.045	ROAD
149	63266.637	26947.714	1306.231	ROAD
150	63262.426	26938.182	1306.676	ROAD
151	63250.819	26934.638	1306.767	ROAD
152	63244.73	26923.255	1307.102	ROAD
153	63253.855	26925.814	1307.591	BW
154	63275.258	26946.262	1306.276	BW
155	63296.072	26966.26	1305.037	BW
156	63283.905	26953.976	1305.861	BW
157	63299.853	26971.075	1305.027	BW
158	63272.449	26944.513	1306.325	POLE
159	63292.361	26963.085	1305.406	POLE
160	63290.104	26961.828	1305.546	POLE
161	63307.043	26993.326	1303.969	POLE
162	63309.906	27003.054	1303.593	POLE
163	63310.196	27008.773	1303.278	POLE
164	63312.32	27024.409	1302.513	BW
165	63310.595	27005.886	1303.459	BW
166	63312.703	27030.601	1302.091	BW
167	63305.464	26985.112	1304.405	BW
7	63325.237	26949.854	1304.236	C5
8	63374.465	26892.04	1306.737	C4
9	63375.096	26892.258	1306.335	C4
201	63369.517	26897.861	1306.004	GP
202	63369.561	26903.195	1305.493	GP
203	63367.846	26904.728	1304.714	GP
204	63367.312	26905.476	1304.353	GP
205	63366.309	26905.887	1304.081	GP
206	63364.962	26907.359	1303.524	GP
207	63365.077	26906.081	1303.514	PS
208	63324.971	26945.174	1303.532	PS

1	63148.959	26865.979	1311.734	CP1
209	63300.061	26970.402	1305.2	FP
210	63299.362	26969.211	1305.193	CRD
211	63297.169	26965.748	1305.191	CRD
212	63295.747	26963.055	1305.051	GP
213	63293.917	26961.637	1305.312	GP
214	63289.566	26956.027	1305.327	GP
215	63286.898	26950.668	1305.459	GP
216	63281.406	26941.559	1305.777	GP
217	63279.334	26936.181	1306.052	GP
218	63318.105	26948.231	1304.05	GP
219	63314.46	26946.284	1304.637	GP
220	63310.501	26942.788	1305.662	GP
221	63308.44	26940.632	1305.988	GP
222	63306.796	26941.27	1306.093	TRE
223	63324.135	26944.377	1303.925	TRE
224	63310.722	26935.252	1306.228	TRE
225	63317.351	26944.178	1304.861	TRE
226	63313.418	26947.807	1304.921	TRE
227	63331.85	26925.699	1305.078	TRE
228	63338.821	26919.364	1305.327	TRE
229	63333.462	26924.937	1304.881	TRE
230	63345.411	26914.959	1305.21	TRE
231	63347.856	26913.386	1305.117	TRE
232	63355.92	26908.44	1304.866	TRE
233	63359.701	26904.942	1305.043	TRE
234	63370.343	26902.707	1305.926	TRE
235	63341.48	26939.889	1303.836	BRW
236	63344.714	26944.203	1303.251	BRW
237	63331.491	26953.951	1303.793	BRW
238	63323.468	26953.112	1303.705	BRW
239	63319.894	26956.251	1304.46	FP
240	63314.611	26960.028	1304.326	FP
241	63334.318	26962.493	1303.415	FP
242	63331.9	26957.286	1303.739	FP
243	63327.119	26948.15	1303.535	CRD
10	63174.645	26999.563	1304.215	MI2
301	63252.161	27024.147	1304.174	B-C
302	63259.859	27020.182	1304.126	B-C
303	63237.487	27018.816	1304.121	B-C
304	63257.345	27024.782	1304.201	B-C
305	63258.905	27019.92	1304.123	B-C
306	63227.195	27022.978	1303.414	B-C
307	63251.703	27024.05	1304.139	B-C
308	63251.859	27023.284	1304.196	B-C
309	63237.127	27019.923	1304.031	B-C
310	63229.22	27016.384	1303.417	B-C
311	63228.056	27018.913	1305.232	B-C
312	63215.073	27015.122	1304.875	B-C
313	63225.84	27024.941	1303.512	S TR

1	63148.959	26865.979	1311.734	CP1
314	63225.579	27024.414	1303.665	S TR
315	63226.411	27022.81	1303.463	S TR
316	63213.702	27021.093	1303.5	S TR
317	63214.061	27020.908	1303.666	S TR
318	63226.115	27022.653	1303.641	S TR
319	63214.183	27020.439	1303.668	S TR
320	63214.363	27020.518	1303.81	S TR
321	63216.92	27021.333	1303.976	S TR
322	63217.178	27020.704	1304.298	S TR
323	63219.909	27024.486	1303.467	S-T
324	63219.54	27025.511	1303.434	S-T
325	63218.853	27024.076	1303.469	S-T
326	63224.586	27022.05	1304.456	S TR
327	63224.727	27021.711	1304.485	S TR
328	63217.58	27019.476	1304.477	S TR
329	63216.223	27019.454	1304.471	S TR
330	63222.025	27028.545	1303.401	S TR
331	63221.106	27031.793	1302.304	S TR
332	63264.451	27036.55	1303.144	GP
333	63261.244	27039.605	1303.47	GP
334	63264.895	27035.524	1302.817	GP
335	63262.444	27035.271	1302.342	GP
336	63253.594	27029.924	1301.819	GP
337	63259.964	27034.766	1302.211	GP
338	63258.445	27035.202	1302.94	GP
339	63258.284	27036.69	1303.422	GP
340	63258.077	27035.44	1303.26	GP
341	63255.743	27042.391	1303.945	GP
342	63253.955	27040.997	1303.399	GP
343	63252.125	27040.956	1303.715	GP
344	63249.872	27042.163	1304.074	GP
345	63244.678	27042.956	1304.078	GP
346	63241.404	27038.054	1303.723	GP
347	63239.17	27039.08	1303.998	GP
348	63242.493	27037.726	1303.307	GP
349	63234.044	27036.529	1303.614	GP
350	63236.919	27038.018	1303.949	GP
351	63233.778	27034.712	1302.98	GP
352	63237.437	27032.847	1303.059	GP
353	63234.057	27033.43	1303.213	GP
354	63239.244	27029.411	1303.03	GP
355	63239.934	27026.698	1302.85	GP
356	63250.596	27028.483	1301.871	GP
357	63252.759	27029.198	1301.837	GP
358	63254.873	27029.642	1301.951	GP
359	63257.029	27030.658	1302.024	GP
360	63259.294	27031.174	1302.032	GP
361	63261.289	27031.935	1302.052	GP
362	63263.742	27032.73	1302.167	GP

1	63148.959	26865.979	1311.734	CP1
363	63220.392	27045.286	1303.419	TRE
364	63220.166	27042.436	1303.257	TRE
365	63218.677	27029.461	1302.098	TRE
366	63220.59	27026.749	1302.133	TRE
367	63253.819	27044.227	1304.743	TRE
368	63199.639	27000.753	1300.872	TRE
369	63227.355	27016.059	1301.769	TRE
370	63236.361	27015.827	1301.882	ROAD
371	63249.952	27023.356	1302.073	ROAD
372	63235.121	27018.203	1301.899	ROAD
373	63271.934	27026.98	1302.139	ROAD
374	63271.46	27030.669	1302.249	ROAD
11	63410.98	26946.327	1306.276	C3
400	63388.28	26914.905	1305.857	B C
401	63385.435	26914.209	1305.948	B C
402	63386.501	26914.084	1305.941	B C
403	63381.958	26916.561	1306.159	B C
404	63387.208	26914.237	1305.901	B C
405	63379.267	26911.812	1306.18	B C
406	63376.579	26912.464	1305.663	S TAIRS
407	63376.93	26913.531	1305.551	S TAIRS
408	63376.708	26913.817	1305.539	S TAIRS
409	63375.963	26912.834	1305.513	S TAIRS
410	63372.667	26915.526	1304.277	S TAIRS
411	63370.525	26916.9	1304.233	S TAIRS
412	63370.33	26917.138	1304.123	S TAIRS
413	63371.166	26917.889	1304.37	S TAIRS
414	63370.953	26918.058	1304.272	S TAIRS
415				S TAIRS
416	63355.202	26928.203	1303.883	S TAIRS
417	63355.777	26929.206	1303.853	S TAIRS
418	63355.987	26929.048	1303.757	S TAIRS
419	63355.432	26928.03	1303.772	S TAIRS
420				S TAIRS
421	63371.379	26912.729	1304.176	PS
422	63371.016	26912.307	1304.129	PS
423	63371.543	26910.614	1304.652	PS
424	63373.787	26909.25	1305.592	PS
425	63374.319	26908.838	1305.64	PS
426	63372.896	26908.964	1305.376	PS
427	63374.881	26908.336	1306.124	GP
428	63371.805	26905.819	1305.66	GP
429	63370.525	26906.699	1305.089	GP
430	63373.091	26908.771	1305.349	GP
431	63370.002	26908.246	1304.662	GP
432	63370.689	26910.451	1304.258	GP
433	63368.42	26905.104	1304.795	GP
434	63366.575	26905.42	1304.203	GP
435	63366.197	26900.637	1305.291	GP

1	63148.959	26865.979	1311.734	CP1
436	63368.753	26898.231	1306.075	GP
437	63365.067	26898.836	1305.892	GP
438	63366.838	26889.586	1306.435	GP
439	63364.964	26885.776	1306.582	GP
440	63360.743	26889.588	1306.673	GP
441	63366.675	26886.69	1306.55	GP
442	63360.047	26898.454	1306.209	GP
443	63370.025	26889.325	1306.379	GP
444	63372.211	26891.221	1306.309	GP
445	63364.297	26900.659	1305.469	GP
446	63363.01	26900.247	1305.841	GP
447	63360.229	26903.257	1305.473	GP
448	63357.231	26906.039	1305.311	GP
449	63357.442	26904.571	1305.702	GP
450	63358.037	26908.127	1304.444	GP
451	63353.439	26908.346	1305.446	GP
452	63351.411	26912.867	1304.478	GP
453	63383.884	26906.809	1306.211	GP
454	63382.918	26903.159	1306.17	GP
455	63398.093	26921.533	1306.056	GP
456	63378.477	26903.307	1306.133	GP
457	63399.859	26929.786	1306.235	GP
458	63395.452	26926.328	1306.178	GP
459	63403.673	26935.029	1306.302	GP
460	63401.38	26936.768	1306.366	GP
461	63407.51	26942.042	1306.208	GP
462	63407.678	26938.678	1306.245	TRE
463	63380.703	26906.926	1306.239	TRE
464	63371.281	26901.728	1306.158	TRE
465	63375.923	26896.182	1306.228	TRE
466	63368.465	26895.713	1306.218	TRE
467	63370.896	26893.393	1306.152	TRE
468	63361.817	26902.811	1305.286	TRE
469	63358.282	26903.155	1306.043	TRE
470	63373.457	26890.291	1306.312	TRE
471	63361.184	26892.445	1306.505	TRE
472	63358.977	26888.474	1306.779	TRE
473	63361.545	26887.273	1306.73	TRE
474	63370.702	26888.548	1306.307	TRE
475	63367.909	26880.979	1307.071	TRE
476	63379.136	26896.133	1306.166	TRE
12	63204.747	26959.961	1306.282	MI3
490	63181.325	27006.163	1304.163	TRE
491	63181.671	27004.772	1304.186	TRE
492	63181.647	27003.64	1304.178	TRE
493	63181.992	27002.169	1304.222	TRE
494	63182.537	27000.011	1304.233	TRE
495	63182.77	26998.589	1304.263	TRE
496	63182.709	26997.385	1304.258	TRE

1	63148.959	26865.979	1311.734	CP1
497	63183.754	26993.896	1304.303	TRE
498	63184.025	26992.33	1304.293	TRE
499	63184.517	26990.022	1304.314	TRE
500	63184.695	26988.758	1304.3	TRE
501	63185.066	26987.431	1304.321	TRE
502	63185.511	26986.051	1304.377	TRE
503	63185.374	26985.193	1304.316	TRE
504	63177.675	26991.693	1304.385	TRE
505	63176.199	26997.344	1304.705	TRE
506	63197.494	27009.967	1304.293	TRE
507	63174.699	27002.51	1304.777	TRE
508	63203.327	27012.038	1304.743	TRE
509	63203.019	27010.707	1304.189	GP
510	63199.811	27010.418	1304.219	GP
511	63197.344	27007.461	1304.195	GP
512	63189.427	27005.788	1304.266	GP
513	63173.061	27005.859	1303.925	GP
514	63172.277	27007.382	1303.638	GP
515	63181.834	26999.607	1304.159	GP
516	63179.392	27008.054	1303.774	GP
517	63179.613	26999.487	1304.142	GP
518	63176.833	26999.346	1304.127	GP
519	63184.396	27010.995	1303.691	GP
520	63186.6	26989.519	1304.29	GP
521	63186.932	27010.054	1303.951	GP
522	63188.61	26982.158	1304.569	GP
523	63190.33	26979.81	1304.936	GP
524	63190.238	26979.128	1305.234	GP
525	63191.488	26978.179	1305.49	GP
526	63194.392	26979.003	1305.339	GP
527	63177.937	26984.862	1304.697	GP
528	63178.599	26985.08	1304.591	GP
529	63183.553	26987.987	1304.255	GP
530	63193.27	26982.81	1304.386	B C
531	63193.048	26983.218	1304.296	B C
532	63193.974	26983.568	1304.236	B C
533	63188.673	26997.298	1304.228	B C
534	63188.512	26997.853	1304.184	B C
535	63189.428	26998.272	1304.189	B C
536	63187.95	27003.034	1304.177	B C
537	63192.714	27004.604	1304.204	B C
538	63192.429	27005.517	1304.19	B C
539	63194.204	26999.829	1304.811	B C
540	63170.761	27008.977	1303.791	FP
541	63186.116	27014.679	1303.596	FP
542	63170.306	27010.099	1303.786	FP
543	63207.555	27019.558	1303.542	FP
544	63208.285	27017.217	1303.54	FP
545	63210.52	27017.564	1303.564	FP

1	63148.959	26865.979	1311.734	CP1
546	63209.511	27021.496	1303.49	FP
547	63181.472	27025.208	1303.163	ROAD
548	63181.365	27028.151	1303.046	ROAD
549	63178.07	27024.143	1303.277	ROAD
550	63178.315	27027.191	1303.202	ROAD
551	63170.936	27024.066	1303.516	ROAD
552	63171.191	27020.613	1303.386	ROAD
553	63167.817	27021.211	1303.614	ROAD
554	63169.703	27018.593	1303.503	ROAD
555	63166.395	27017.333	1303.731	ROAD
556	63168.74	27015.201	1303.624	ROAD
557	63166.07	27012.88	1303.83	ROAD
558	63168.856	27011.805	1303.789	ROAD
559	63168.98	26998.335	1304.145	ROAD
560	63170.84	26989.847	1304.442	ROAD
561	63174.128	26993.181	1304.495	ROAD
562	63175.945	26973.188	1305.362	ROAD
563	63179.677	26971.852	1305.564	ROAD
564	63174.983	26964.175	1305.923	DR
565	63168.928	26990.006	1304.427	DR
566	63167.495	26989.623	1304.713	DR
567	63172.579	26974.216	1305.571	DR
568	63173.281	26974.578	1305.641	DR
569	63174.641	26991.2	1304.691	EPOLE
570	63169.888	27027.92	1303.083	EPOLE
571	63171.505	27015.45	1303.411	SL
572	63197.106	27028.347	1302.59	SL
573	63156.256	26994.184	1304.354	GATE
574	63157.196	26989.979	1304.131	GATE
575	63171.173	27007.824	1303.809	F ENCE
576	63172.098	27005.463	1303.79	F ENCE
577	63172.458	27003.259	1304.02	F ENCE
578	63173.142	27000.82	1304.112	F ENCE
579	63173.834	26998.345	1304.151	F ENCE
580	63174.655	26995.852	1304.299	F ENCE
581	63174.953	26993.486	1304.436	F ENCE
582	63175.737	26991.155	1304.492	F ENCE
583	63177.02	26986.031	1304.796	F ENCE
584	63178.707	26981.587	1304.888	F ENCE
13	63359.441	27062.395	1298.472	C2
600	63407.916	26944.675	1306.258	GP
601	63405.228	26946.511	1306.076	GP
602	63403.787	26947.309	1305.795	GP
603	63352.721	26955.002	1303.131	B C
604	63353.556	26957.08	1303.072	B C
605	63359.621	26958.037	1303.18	B C
606	63360.391	26958.993	1303.151	B C
607	63398.713	26926.842	1306.28	GP
608	63359.956	26959.435	1303.139	B C

1	63148.959	26865.979	1311.734	CP1
609	63362.581	26962.916	1303.177	B C
610	63361.621	26969.086	1303.124	B C
611	63358.031	26971.775	1303.163	B C
612	63359.621	26975.042	1303.203	B C
613	63358.25	26972.195	1303.176	B C
614	63359.7	26975.198	1303.934	B C
615	63362.56	26978.907	1303.767	B C
616	63360.17	26975.501	1303.747	B C
617	63361.625	26977.771	1303.961	B C
618	63362.609	26979.617	1304.01	B C
619	63378.595	26967.995	1304.024	B C
620	63365.709	26974.86	1303.725	B C
621	63364.09	26948.4	1304.017	B C
622	63364.121	26971.91	1303.949	B C
623	63361.952	26949.969	1303.995	B C
624	63365.763	26973.895	1303.94	B C
625	63376.721	26969.342	1304.002	B C
626	63362.721	26952.333	1303.752	B C
627	63361.477	26950.762	1303.755	B C
628	63361.062	26953.581	1303.743	B C
629	63364.392	26948.482	1302.956	B C
630	63368.339	26945.757	1302.905	B C
631	63378.548	26967.598	1302.909	B C
632	63382.246	26964.653	1302.898	B C
633	63383.696	26966.635	1302.94	B C
634	63383.206	26967.218	1302.888	B C
635	63383.25	26967.744	1302.875	B C
636	63382.795	26970.314	1302.89	B C
637	63383.079	26970.873	1302.867	B C
638	63363.892	26985.135	1302.831	B C
639	63360.297	26980.947	1302.972	B C
640	63362.367	26979.598	1302.894	B C
641	63372.025	26944.092	1300.091	B C
642	63381.528	26962.679	1300.157	B C
643	63385.411	26959.84	1300.05	B C
644	63388.317	26963.777	1298.591	B C
645	63384.893	26933.997	1303.298	FP
646	63372.083	26943.467	1303.309	FP
647	63371.755	26942.949	1303.315	FP
648	63372.014	26942.688	1303.296	FP
649	63371.821	26942.468	1303.275	FP
650	63372.88	26945.207	1300.002	FP
651	63372.526	26945.194	1300.011	FP
652	63373.042	26945.645	1299.997	FP
653	63373.551	26945.495	1299.971	FP
654	63382.959	26958.037	1300.018	FP
655	63366.799	26969.966	1302.973	FP
656	63359.749	26975.032	1303.181	FP
657	63392.267	26955.392	1300.081	FP

1	63148.959	26865.979	1311.734	CP1
658	63390.29	26952.802	1300.064	FP
659	63387.046	26934.834	1302.879	FP
660	63385.379	26932.614	1302.959	FP
661	63393.789	26934.674	1305.792	GP
662	63394.236	26936.168	1305.428	GP
663	63392.781	26934.882	1305.241	GP
664	63392.558	26935.291	1304.97	GP
665	63392.15	26935.245	1304.821	GP
666	63393.169	26938.076	1304.487	GP
667	63391.162	26936.192	1304.287	GP
668	63392.325	26938.857	1303.927	GP
669	63390.084	26940.181	1303.104	GP
670	63388.508	26940.568	1302.345	GP
671	63387.781	26941.739	1301.736	GP
672	63388.46	26944.211	1301.084	GP
673	63404.249	26943.858	1306.206	GP
674	63404.262	26944.133	1306.303	GP
675	63406.888	26945.476	1306.186	GP
676	63405.769	26948.513	1305.889	GP
677	63412.407	26944.465	1306.015	GP
678	63413.647	26949.928	1305.182	GP
679	63414.444	26952.569	1304.119	GP
680	63415.439	26955.747	1303.046	GP
681	63413.017	26958.934	1302.523	GP
682	63413.483	26959.94	1302.098	GP
683	63413.065	26961.585	1301.277	GP
684	63412.453	26962.722	1300.804	GP
685	63410.039	26968.246	1298.839	GP
686	63406.887	26971.483	1298.471	GP
687	63403.614	26975.034	1298.431	GP
688	63400.851	26978.285	1298.505	GP
689	63401.442	26982.751	1297.534	GP
690	63400.898	26985.595	1297.065	GP
691	63398.019	26990.653	1296.878	GP
692	63394.497	26996.95	1296.652	GP
693	63392.276	26994.674	1297.004	GP
694	63401.002	26994.927	1296.901	GP
695	63402.372	26995.186	1296.744	GP
696	63403.317	26993.272	1296.954	GP
697	63399.723	26996.558	1296.636	GP
698	63398.281	26995.71	1296.718	GP
699	63388.393	26989.24	1297.974	GP
700	63386.619	26987.926	1298.247	GP
701	63386.57	26987.9	1298.255	GP
14	63306.918	27065.62	1300.632	C1
710	63359.068	27059.499	1298.414	GP
711	63357.679	27058.675	1298.541	GP
712	63355.821	27058.437	1298.513	GP
713	63354.746	27056.346	1298.511	GP

1	63148.959	26865.979	1311.734	CP1
714	63352.649	27054.827	1298.6	GP
715	63351.972	27053.712	1298.611	GP
716	63351.239	27052.542	1298.62	GP
717	63348.929	27049.055	1298.585	GP
718	63347.127	27050.684	1298.459	GP
719	63347.759	27052.827	1298.488	GP
720	63345.44	27053.017	1298.51	GP
721	63342.858	27052.842	1298.486	GP
722	63340.768	27051.044	1298.417	GP
723	63340.985	27049.508	1298.467	GP
724	63339.621	27046.675	1298.477	GP
725	63337.779	27044.903	1298.447	GP
726	63336.2	27043.573	1298.414	GP
727	63334.907	27046.662	1298.51	GP
728	63362.181	27062.583	1298.309	GP
729	63363.581	27062.444	1298.143	GP
730	63365.079	27064.159	1297.809	GP
731	63361.463	27059.982	1298.333	GP
732	63359.849	27058.508	1298.477	GP
733	63359.686	27055.587	1298.552	GP
734	63355.76	27052.292	1298.603	GP
735	63355.383	27050.836	1298.749	GP
736	63354.009	27050.109	1298.79	GP
737	63351.951	27045.444	1298.752	GP
738	63352.112	27043.804	1298.576	GP
739	63356.054	27041.366	1298.044	GP
740	63385.677	27058.797	1293.008	GP
741	63399.184	27037.875	1293	B C
750	63216.059	26937.095	1304.442	B C
751	63205.395	26934.034	1304.958	B C
752	63200.558	26949.026	1305.178	B C
753	63201.262	26949.772	1305.156	B C
754	63201.65	26949.978	1305.141	B C
755	63196.679	26965.336	1305.212	B C
756	63212.224	26956.553	1305.905	READING
757	63214.327	26957.073	1305.876	READING
758	63210.528	26962.542	1306.046	READING
759	63208.036	26971.12	1306.184	READING
760	63204.943	26981.367	1306.259	READING
761	63203.026	26987.638	1306.356	READING
762	63204.757	26988.272	1306.365	READING
763	63201.703	26995.041	1307.536	F ENCE
764	63194.349	26994.106	1307.47	F ENCE
765	63189.429	26993.381	1307.277	F ENCE
766	63177.345	26990.916	1307.191	F ENCE
767	63172.948	26988.808	1306.897	F ENCE
768	63170.66	26987.762	1306.992	F ENCE
769	63168.147	26986.101	1306.858	GP
770	63168.228	26985.676	1306.735	GP

1	63148.959	26865.979	1311.734	CP1
771	63168.775	26984.913	1306.585	GP
772	63169.051	26984.295	1306.464	GP
773	63170.002	26983.173	1306.324	GP
774	63173.941	26982.01	1306.374	GP
775	63177.817	26978.114	1306.152	GP
776	63184.064	26974.519	1306.241	GP
777	63184.998	26978.558	1306.297	GP
778	63190.368	26981.122	1306.372	GP
779	63193.745	26980.372	1306.261	GP
780	63190.67	26978.691	1306.312	GP
781	63192.373	26972.885	1306.267	GP
782	63196.055	26974.904	1306.256	GP
783	63193.094	26971.923	1306.435	GP
784	63198.431	26971.599	1306.232	GP
785	63192.338	26970.888	1306.307	GP
786	63198.048	26968.915	1305.953	GP
787	63200.416	26964.012	1305.785	GP
788	63196.739	26966.532	1305.321	GP
789	63200.141	26963.773	1305.638	GP
790	63194.177	26967.256	1305.708	GP
791	63199.062	26963.284	1305.473	GP
792	63201.559	26960.331	1305.786	GP
793	63203.695	26957.063	1305.871	GP
794	63206.614	26954.681	1305.82	GP
795	63201.302	26954.806	1305.322	GP
796	63207.349	26952.381	1305.651	GP
797	63206.746	26949.07	1305.618	GP
798	63205.058	26950.873	1305.723	GP
799	63203.13	26948.94	1305.34	GP
800	63204.073	26946.923	1305.294	GP
801	63205.716	26944.095	1305.329	GP
802	63208.825	26941.114	1305.23	GP
803	63206.528	26940.22	1305.21	GP
804	63211.259	26939.5	1305.171	GP
805	63209.034	26937.438	1305.011	GP
806	63212.562	26937.738	1305.008	GP
807	63211.37	26936.049	1304.664	GP
808	63215.157	26939.042	1305.058	GP
809	63214.149	26936.691	1304.542	GP
810	63217.785	26940.277	1305.296	GP
811	63218.452	26943.347	1305.515	GP
812	63220.265	26942.308	1304.819	GP
813	63219.837	26939.944	1304.803	GP
814	63221.808	26941.737	1304.73	GP
815	63217.974	26937.125	1304.528	GP
816	63219.397	26936.874	1304.719	GP
817	63221.422	26938.48	1304.5	GP
818	63219.257	26938.209	1304.602	GP
819	63223.597	26939.417	1304.704	GP

1	63148.959	26865.979	1311.734	CP1
820	63225.096	26938.688	1304.725	GP
821	63222.359	26942.728	1304.911	GP
822	63216.918	26952.544	1305.634	GP
823	63220.091	26945.837	1305.509	GP
824	63217.672	26953.932	1305.626	GP
825	63218.33	26955.848	1305.728	GP
826	63215.804	26956.418	1305.757	GP
827	63212.664	26955.775	1305.828	GP
828	63212.192	26951.22	1305.637	SITING ARA
829	63209.291	26950.592	1305.639	SITING ARA
830	63212.898	26948.335	1305.5	SITING ARA
831	63209.942	26947.661	1305.545	SITING ARA
832	63213.708	26945.431	1305.496	SITING ARA
833	63210.78	26944.767	1305.495	SITING ARA
834	63214.338	26942.331	1305.548	SITING ARA
835	63211.37	26941.768	1305.433	SITING ARA
836	63220.744	26951.017	1305.748	CTE EN
837	63219.415	26955.498	1305.829	CTE EN
838	63219.764	26949.33	1305.656	CTE EN
839	63220.888	26944.845	1305.335	CTE EN
840	63222.299	26950.075	1305.801	CTE EN
841	63204.737	26951.817	1305.856	TRE E
842	63203.045	26958.846	1305.995	TRE E
843	63201.64	26980.99	1306.3	TRE E
844	63203.914	26991.631	1307.025	TRE E
845	63187.623	26988.952	1307.018	TRE E
846	63178.117	26986.81	1306.597	TRE E
847	63173.962	26986.68	1306.526	TRE E
848	63173.978	26979.064	1306.271	TRE E
849	63196.369	26983.712	1306.336	TRE E
850	63188.379	26981.588	1306.401	TRE E
851	63204.557	26998.133	1307.484	ROAD
852	63204.369	27003.629	1307.757	ROAD
853	63199.997	26997.946	1307.467	ROAD
854	63194.382	27002.491	1307.646	ROAD
855	63190.573	27002.203	1307.589	ROAD
856	63191.168	26997.019	1307.354	ROAD
857	63188.437	27002.156	1307.539	ROAD
858	63185.064	26996.768	1307.297	ROAD
859	63186.153	27002.25	1307.504	ROAD
860	63180.434	26995.999	1307.282	ROAD
861	63182.65	27002.951	1307.484	ROAD
862	63180.462	26995.979	1307.274	ROAD
863	63179.263	27004.453	1307.47	ROAD
864	63176.484	26995.031	1307.262	ROAD
865	63174.918	27006.603	1307.538	ROAD
866	63169.73	27008.582	1307.596	ROAD
867	63168.676	27010.726	1307.616	ROAD
868	63173.331	26994.128	1307.244	ROAD

1	63148.959	26865.979	1311.734	CP1
869	63162.494	27012.177	1307.636	ROAD
870	63170.084	26992.881	1307.176	ROAD
871	63156.371	27003.865	1307.369	ROAD
872	63167.103	26991.937	1307.104	ROAD
873	63156.417	27004.311	1307.411	ROAD
874	63158.442	27003.651	1307.412	ROAD
875	63165.314	26990.495	1307.063	ROAD
876	63160.776	27002.188	1307.389	ROAD
877	63162.927	26988.57	1307.029	ROAD
878	63162.809	27000.052	1307.349	ROAD
879	63163.541	26998.638	1307.319	ROAD
880	63164.368	26996.355	1307.309	ROAD
881	63162.443	26993.807	1307.154	ROAD
882	63159.236	26991.259	1307.04	ROAD
883	63157.229	26991.742	1307.173	ROAD
884	63154.816	26993.534	1307.249	S TAT UE
885	63154.77	26993.513	1307.237	S TAT UE
886	63155.993	26992.029	1307.204	S TAT UE
890	63250.82	26925.684	1307.369	S TAT UE
891	63251.849	26927.662	1307.323	S TAT UE
892	63253.134	26927.342	1307.304	S TAT UE
893	63253.292	26927.328	1307.329	S TAT UE
894	63253.035	26927.101	1307.825	S TAT UE
895	63251.479	26926.684	1307.826	S TAT UE
896	63251.666	26922.828	1307.455	S TAT UE
897	63252.662	26922.313	1307.84	S TAT UE
898	63252.007	26922.79	1307.46	S TAT UE
899	63254.308	26922.91	1307.832	S TAT UE
900	63253.024	26922.084	1307.475	S TAT UE
901	63253.577	26921.78	1307.462	S TAT UE
902	63254.677	26922.703	1307.44	S TAT UE
903	63254.159	26922.148	1307.442	S TAT UE
904	63255.441	26923.966	1307.465	S TAT UE
905	63253.907	26919.549	1307.581	POLE
906	63254.985	26917.667	1307.44	POLE
907	63241.053	26930.033	1307.405	POLE
908	63275.804	26907.705	1307.574	POLE
909	63300.746	26897.833	1307.551	POLE
910	63321.309	26878.653	1308.228	POLE
911	63351.308	26851.172	1308.16	ROAD
912	63354.854	26847.916	1308.325	ROAD
913	63350.457	26845.104	1308.479	ROAD
914	63346.292	26848.016	1308.352	ROAD
915	63346.748	26843.029	1308.556	ROAD
916	63342.437	26847.689	1308.384	ROAD
917	63337.66	26850.406	1308.447	ROAD
918	63333.609	26854.503	1308.429	ROAD
919	63330.276	26861.518	1308.248	ROAD
920	63326.419	26868.111	1308.117	ROAD

1	63148.959	26865.979	1311.734	CP1
921	63321.846	26873.979	1308.012	ROAD
922	63317.263	26878.598	1307.914	ROAD
923	63311.754	26883.755	1307.811	ROAD
924	63303.969	26877.551	1307.865	ROAD
925	63296.494	26884.75	1307.749	ROAD
926	63299.38	26891.41	1307.666	ROAD
927	63290.259	26890.104	1307.687	ROAD
928	63291.818	26898.032	1307.556	ROAD
929	63283.597	26893.551	1307.723	ROAD
930	63278.47	26903.84	1307.635	ROAD
931	63266.592	26899.831	1307.766	ROAD
932	63262.268	26911.961	1307.548	ROAD
933	63244.242	26907.817	1307.761	ROAD
934	63253.436	26904.996	1307.77	ROAD
935	63341.197	26851.67	1308.397	GP
936	63340.228	26853	1308.524	GP
937	63338.415	26853.535	1308.674	GP
938	63337.09	26853.102	1308.489	GP
939	63335.506	26853.66	1308.412	GP
940	63334.889	26856.638	1308.323	GP
941	63334.333	26858.914	1308.305	GP
942	63334.394	26861.259	1308.2	GP
943	63334.741	26862.689	1308.172	GP
944	63333.877	26863.539	1308.086	GP
945	63333.351	26864.885	1308.119	GP
946	63332.256	26864.883	1308.081	GP
947	63331.882	26867.249	1308.004	GP
948	63329.263	26867.644	1308.083	GP
949	63329.19	26871.208	1308.046	GP
950	63325.67	26872.166	1307.972	GP
951	63324.053	26876.278	1307.87	GP
952	63318.89	26878.524	1307.916	GP
953	63317.942	26883.994	1307.768	GP
954	63311.424	26889.311	1307.719	GP
955	63306.747	26890.138	1307.665	GP
956	63304.297	26895.456	1307.64	GP
957	63299.428	26894.781	1307.597	GP
958	63296.424	26899.25	1307.626	GP
959	63290.894	26898.609	1307.599	GP
960	63286.758	26903.302	1307.533	GP
961	63280.36	26904.114	1307.622	GP
962	63276.43	26909.605	1307.744	GP
963	63270.869	26909.818	1307.585	GP
964	63265.634	26915.84	1307.612	GP
965	63260.37	26915.235	1307.474	GP
966	63256.945	26920.135	1307.438	GP
967	63252.302	26920.09	1307.441	GP
968	63249.831	26920.2	1307.461	GP
969	63249.685	26922.967	1307.41	GP

1	63148.959	26865.979	1311.734	CP1
970	63212.667	26955.192	1305.365	B C
971	63212.677	26954.457	1305.171	B C
972	63264.941	26971.26	1305.369	B C
973	63265.314	26970.661	1305.136	B C
980	63312.121	27063.067	1300.689	TRE
981	63311.556	27070.362	1300.568	TRE
982	63285.141	27086.95	1300.489	TRE
983	63281.301	27072.052	1301.314	TRE
984	63282.373	27072.258	1301.336	GP
985	63287.057	27082.399	1300.435	GP
986	63284.044	27072.475	1301.142	GP
987	63289.785	27078.399	1300.497	GP
988	63286.472	27073.603	1300.726	GP
989	63292.106	27080.1	1300.335	GP
990	63290.09	27073.902	1300.65	GP
991	63292.909	27073.179	1300.603	GP
992	63295.308	27071.658	1300.549	GP
993	63291.847	27070.304	1300.866	GP
994	63295.352	27071.729	1300.522	GP
995	63298.515	27070.022	1300.561	GP
996	63293.92	27070.124	1300.637	GP
997	63299.624	27067.72	1300.686	GP
998	63290.726	27068.592	1300.489	GP
999	63299.658	27063.957	1300.654	GP
1000	63293.022	27068.929	1300.299	GP
1001	63300.927	27056.614	1300.908	GP
1002	63299.294	27073.68	1300.383	GP
1003	63302.601	27042.699	1301.832	GP
1004	63307.287	27052.135	1301.387	GP
1005	63303.033	27036.567	1302.17	GP
1006	63307.832	27045.675	1301.678	GP
1007	63308.775	27064.047	1300.598	GP
1008	63310.827	27062.711	1301.081	GP
1009	63312.047	27064.554	1300.327	GP
1010	63310.694	27067.509	1300.279	GP
1011	63309.435	27069.895	1300.509	GP
1012	63309.161	27072.358	1299.88	GP
1013	63307.566	27074.473	1299.845	GP
1014	63305.378	27072.394	1300.38	GP
1015	63305.567	27076.825	1300.159	GP
1016	63305.644	27084.193	1300.089	GP
1017	63304.841	27080.052	1300.011	GP
1018	63314.149	27079.301	1300.007	B C
1019	63308.551	27078.412	1300.101	B C
1020	63288.678	27076.261	1300.615	B C
1021	63286.104	27080.927	1300.645	B C
1022	63284.425	27073.917	1300.787	B C
1023	63288.302	27098.567	1299.4	B C
1024	63286.233	27095.985	1300.024	B C

1	63148.959	26865.979	1311.734	CP1
1025	63282.74	27091.446	1300.044	B C
1026	63283.65	27090.632	1300.063	B C
1027	63283.842	27090.499	1299.899	B C
1028	63284.061	27090.238	1299.75	S TAIRS
1029	63284.31	27090.153	1299.577	S TAIRS
1030	63283.543	27090.66	1300.046	S TAIRS
1031	63282.719	27091.398	1300.052	S TAIRS
1032	63287.121	27095.253	1299.886	S TAIRS
1033	63287.9	27101.613	1299.417	B C
1034	63287.128	27107.54	1299.381	B C
1035	63286.386	27113.505	1299.386	B C
1036	63286.042	27116.459	1299.41	B C
1037	63285.621	27119.443	1299.387	B C
1038	63285.195	27122.774	1299.381	B C
1039	63284.487	27128.763	1299.46	B C
1040	63284.092	27132.5	1299.391	B C
1041	63283.697	27135.783	1299.34	B C
1042	63283.316	27138.808	1299.32	B C
1043	63282.865	27141.798	1299.313	B C
1044	63282.493	27144.821	1299.293	B C
1045	63282.102	27147.736	1299.274	B C
1046	63281.73	27150.726	1299.292	B C
1047	63281.365	27153.704	1299.274	B C
1048	63280.993	27156.691	1299.268	B C
1049	63280.608	27159.671	1299.273	B C
1050	63280.299	27162.717	1299.264	B C
1051	63303.926	27084.906	1299.983	GP
1052	63305.54	27090.865	1299.801	GP
1053	63302.611	27098.512	1299.732	GP
1054	63304.326	27099.341	1299.611	GP
1055	63302.268	27104.934	1299.591	GP
1056	63299.628	27119.952	1299.375	ROAD
1057	63295.63	27108.813	1299.638	ROAD
1058	63300.788	27109.288	1299.576	ROAD
1059	63303.351	27083.858	1300.097	ROAD
1060	63298.971	27083.427	1300.181	ROAD
1061	63300.696	27069.021	1300.48	ROAD
1062	63295.844	27061.379	1300.976	ROAD
1063	63295.983	27065.962	1300.864	ROAD
1064	63287.588	27059.272	1301.368	ROAD
1065	63286.775	27062.386	1301.371	ROAD
1066	63306.391	27062.911	1300.738	ROAD
1067	63306.736	27053.168	1301.213	ROAD
1068	63312.487	27058.322	1300.346	ROAD
1069	63311.785	27052.642	1300.275	ROAD
1070	63319.004	27050.957	1299.874	ROAD
1071	63315.969	27049.669	1299.904	ROAD
1072	63322.546	27044.392	1299.623	ROAD
1073	63319.513	27043.724	1299.626	ROAD

1	63148.959	26865.979	1311.734	CP1
1074	63325.572	27040.618	1299.373	ROAD
1075	63323.497	27038.066	1299.369	ROAD
1076	63329.401	27037.547	1299.211	ROAD
1077	63329.097	27033.716	1299.203	ROAD
1078	63334.707	27035.785	1299.03	ROAD
1079	63335.016	27031.993	1299.086	ROAD
1080	63342.063	27035.163	1298.909	ROAD
1081	63343.118	27031.666	1298.915	ROAD
1082	63350.278	27035.32	1298.614	ROAD
1083	63350.526	27031.181	1298.657	ROAD
1084	63355.561	27037.087	1297.991	ROAD
1085	63355.46	27031.333	1298.669	ROAD
1086	63345.588	27036.262	1299.085	GP
1087	63341.274	27040.186	1298.682	GP
1088	63351.542	27050.203	1298.711	GP
1089	63344.967	27052.301	1298.602	GP
1090	63354.329	27057.343	1298.516	GP
1091	63344.245	27057.482	1298.578	GP
1092	63336.301	27046.716	1298.599	GP
1093	63334.125	27056.233	1298.686	GP
1094	63329.968	27048.333	1298.625	GP
1095	63327.082	27059.201	1298.74	GP
1096	63325.614	27052.543	1298.735	GP
1097	63323.833	27061.558	1299.116	GP
1098	63320.179	27057.549	1299.368	GP
1099	63320.454	27063.989	1299.384	GP
1100	63314.651	27060.042	1300.16	GP
1101	63315.556	27064.191	1299.648	GP
1102	63311.409	27061.573	1300.883	GP
1103	63311.217	27065.188	1300.38	GP

ANNEX A3

OBSERVATION AND CALCULATION

(ROAD ALIGNMENT)

Observer: Group 25
Recorder: Group 25
Instrument: Total Station

Tribhuvan University Institute of Engineering
Pulchowk Campus
Department of Civil Engineering
Survey Instruction Committee

Road Alignment

LINE/LEG		IP TO IP	IP STATION	CHAINAGE OF IP	DEFLECTION ANGLE				BEARI	RADIUS OF	TANGENT	LENGTH OF	APEX	CHAINAGE	CHAINAGE	CHAINAGE	REMARKS
FROM IP	TO IP	DISTANCE		STATION					DEGRE E	CURVE	LENGTH	CURVE	DISTANCE	OF BC	OF MC	OF EC	
IP0		0	IP0	0													
IP0	IP1	33.652	IP1	33.652	+	6	2	58	50.18	200	10.568	21.117	0.279	23.084	33.643	44.202	
IP1	IP2	46.313	IP2	79.947	+	33	10	11	56.23	25	7.446	14.473	1.085	72.501	79.738	86.975	
IP2	A	20.864	A	100.393					89.40								
A	B	70.115	B	170.508					89.40								
B	IP3	21.812	IP3	192.32	+	66	34	3	89.40	20	13.129	23.236	3.925	179.191	190.809	202.427	
IP3	IP4	43.114	IP4	232.412	+	19	8	22	155.96	145	24.446	48.437	2.046	207.966	232.185	256.404	
IP4	IP5	41.106	IP5	273.064	-	21	9	16	175.10	60	11.204	22.153	1.037	261.86	272.937	284.014	
IP5	IP6	39.038	IP6	311.848	-	8	47	1	153.95	60	4.608	9.198	0.177	307.24	311.839	316.438	
IP6	IP7	41.077	IP7	352.907	-	50	4	17	145.17	45	21.018	39.326	4.666	331.889	351.552	371.215	
IP7	IP8	42.88	IP8	393.077	-	29	28	45	95.09	60	15.785	30.871	2.042	377.292	392.728	408.164	
IP8	IP9	44.484	IP9	436.863	-	6	53	6	65.62	250	15.039	30.041	0.452	421.824	436.845	451.866	
IP9	IP10	59.581	IP10	496.408	+	18	28	13	58.73	100	16.259	32.237	1.313	480.149	496.268	512.387	
IP10	IP11	42.299	IP11	538.427	-	24	12	43	77.20	90	19.304	38.032	2.047	519.123	538.139	557.155	
IP11	IP12	43.044	IP12	580.895	+	16	44	34	52.99	80	11.773	23.377	0.862	569.122	580.811	592.5	
IP12	IP13	30.7	IP13	611.427	+	28	27	33	69.73	25	6.34	12.418	0.791	605.087	611.296	617.505	
IP13	IP14	20.518	IP14	631.683	-	60	7	19	98.19	15	8.682	15.74	2.331	623.001	630.871	638.741	
IP14	IP15	22.928	IP15	652.987	+	12	49	35	38.07	75	8.43	16.79	0.472	644.557	652.952	661.347	
IP15	IP16	55.838	IP16	708.755	+	38	21	25	50.90	20	6.956	13.389	1.175	701.799	708.494	715.189	
IP16	A	12.696	A	720.929					89.25								
A	B	7.884	B	728.813					89.25								
B	IP17	11.358	IP17	740.171					89.25								

Tribhuvan University
Institute of Engineering
Central Campus, Pulchowk
Department of Civil Engineering

Fly Levelling

Forward Loop (SBM2 to IP0)

RL of SBM = 1311.599

Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	Stadia Interval S1+S2	Hz. Distance(m) S x 100	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
SBM2	0.965	0.94	0.916	0.94	0.049								1311.599	0.049	4.9	SBM2
1	0.702	0.661	0.599	0.654	0.103	1.794	1.766	1.741	1.767	0.053		0.827	1310.772	0.156	15.6	
IP0					0	1.978	1.925	1.871	1.925	0.107		1.271	1309.501	0.107	10.7	IP0
Sum				1.594					3.692		0	2.098			31.2	

Arithmetic Check:	
$\lvert\sum BS-\sum FS\rvert=$	2.098
$\lvert\sum Rise-\sum Fall\rvert=$	2.098
|Last RL - First RL| =	2.098
Distance(k1)=	31.2

Backward Loop (IP0 to SBM2)

Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	Stadia Interval S1+S2	Hz.Distance (m) S x 100	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
IP0	1.719	1.664	1.608	1.664	0.111								1309.501	0.111	11.1	IP0
1	1.375	1.36	1.345	1.36	0.03	0.449	0.398	0.347	0.398	0.102	1.266		1310.767	0.132	13.2	
SBM2					0	0.541	0.524	0.506	0.524	0.035	0.836		1311.603	0.035	3.5	SBM2
Sum				3.024					0.922		2.102	0			27.8	

Arithmetic Check:	
$\lvert\sum BS-\sum FS\rvert=$	2.102
$\lvert\sum Rise-\sum Fall\rvert=$	2.102
|Last RL - First RL| =	2.102
Distance(k1)=	27.8

Error Computation:	
Error of Closure (m) (mm)=	-4
Loop Distance in (km) =	0.059
Permissible Error = $24\sqrt{k}$ =	5.83
RL of IP0 (m) =	1309.499

Tribhuvan University Institute of Engineering
Pulchowk Campus
Department of Civil Engineering Survey Instruction Committee

Levelling for L- Section and X - Section

Observer : Group 25
Recorder : Group 25
Instrument : Auto-level

RL of IP0	1309.499

RL of A	1303.517

Pt.	Lt. Offset	C.L Chainage	Rt.Offset	BS	IS	FS	HI	RL	Correction	Corr. Elevation	Remarks
IP0		0		1.139			1310.638	1309.499	0	1309.499	
	5				0.875			1309.763	0.07	1309.833	
	10				0.737			1309.901	0.07	1309.971	
			5		1.972			1308.666	0.07	1308.736	
			10		2.334			1308.304	0.07	1308.374	
		15			1.373			1309.265	0.07	1309.335	
			5		1.676			1308.962	0.07	1309.032	
	5				0.884			1309.754	0.07	1309.824	
			10		2.024			1308.614	0.07	1308.684	
	10				0.811			1309.827	0.07	1309.897	
BC1		23.084			1.614			1309.024	0.07	1309.094	
	5				1.737			1308.901	0.07	1308.971	
	10				1.569			1309.069	0.07	1309.139	
			5		1.972			1308.666	0.07	1308.736	
			10		2.125			1308.513	0.07	1308.583	
MC1		33.643			**1.991**			1308.647	0.07	1308.717	
	5				1.999			1308.639	0.07	1308.709	
	10				1.994			1308.644	0.07	1308.714	
			5		2.111			1308.527	0.07	1308.597	
			10		2.095			1308.543	0.07	1308.613	
		33.66		0.69		1.991	1309.337	1308.647	0.141	1308.788	

EC1		44.202			1.052			1308.285	0.141	1308.426	
	5				1.069			1308.268	0.141	1308.409	
	10				0.864			1308.473	0.141	1308.614	
			5		0.994			1308.343	0.141	1308.484	
			10		1.09			1308.247	0.141	1308.388	
		45			1.136			1308.201	0.141	1308.342	
			5		2.7			1306.637	0.141	1306.778	
			10		2.692			1306.645	0.141	1306.786	
	5				2.702			1306.635	0.141	1306.776	
	10				2.639			1306.698	0.141	1306.839	
		60		1.065		2.687	1307.715	1306.65	0.211	1306.861	
BC2		72.501			1.202			1306.513	0.211	1306.724	
	5				1.158			1306.557	0.211	1306.768	
			5		1.386			1306.329	0.211	1306.54	
	10				0.985			1306.73	0.211	1306.941	
			10		1.42			1306.295	0.211	1306.506	
MC2		79.738			1.717			1305.998	0.211	1306.209	
			5		2.013			1305.702	0.211	1305.913	
	5				1.147			1306.568	0.211	1306.779	
			10		2.53			1305.185	0.211	1305.396	
	10				1.065			1306.65	0.211	1306.861	
EC2		86.975			2.582			1305.133	0.211	1305.344	
	5				2.595			1305.12	0.211	1305.331	
			5		2.274			1305.441	0.211	1305.652	
	10				2.549			1305.166	0.211	1305.377	
			10		2.1			1305.615	0.211	1305.826	
		90		1.189		2.669	1306.235	1305.046	0.282	1305.328	
A		100.393				3		1303.235	0.282	1303.517	
				4.083		**10.347**					
			RISE/FALL	**6.264**							
			Error	**-0.2815**							
			Permisible								
Pt	Lt.Offset	C.L Chainage	Rt. Offset	BS	IS	FS	HI	REDUCE	CORRECTI	CORR. ELEVAT	REMARK
B		170.508		2.574			1306.757	1304.183	0	1304.183	
BC3		179.191			0.613			1306.144	-0.013	1306.131	
	5				0.645			1306.112	-0.013	1306.099	
	10				0.58			1306.177	-0.013	1306.164	

			5		1.313			1305.444	-0.013	1305.431	
			10		1.491			1305.266	-0.013	1305.253	
TP				2.452		0.643	1308.566	1306.114	-0.013	1306.101	
MC3		190.809		2.654		1.143	1310.077	1307.423	-0.027	1307.396	
	5				1.822			1308.255	-0.027	1308.228	
	10				0.701			1309.376	-0.027	1309.349	
			5		3.606			1306.471	-0.027	1306.444	
			10		4.591			1305.486	-0.027	1305.459	
EC3		202.427			1.941			1308.136	-0.027	1308.109	
			5		3.154			1306.923	-0.027	1306.896	
	5				1.056			1309.021	-0.027	1308.994	
			10		4.19			1305.887	-0.027	1305.86	
				1.395		1.178	1310.294	1308.899	-0.04	1308.859	
	10				0.73			1309.564	-0.04	1309.524	
BC4		207.966			2.098			1308.196	-0.04	1308.156	
	5				1.252			1309.042	-0.04	1309.002	
			5		3.285			1307.009	-0.04	1306.969	
	10				0.605			1309.689	-0.04	1309.649	
			10		4.3			1305.994	-0.04	1305.954	
MC4		232.185			2.795			1307.499	-0.04	1307.459	
			5		3.769			1306.525	-0.04	1306.485	
	5				2.244			1308.05	-0.04	1308.01	
			10		4.952			1305.342	-0.04	1305.302	
	10				1.665			1308.629	-0.04	1308.589	
TP				0.969		1.972	1309.291	1308.322	-0.054	1308.268	
EC4		256.404			3.46			1305.831	-0.054	1305.777	
			5		4.356			1304.935	-0.054	1304.881	
	5				2.648			1306.643	-0.054	1306.589	
			10		5.098			1304.193	-0.054	1304.139	
	10				1.843			1307.448	-0.054	1307.394	
TP				1.36		2.634	1308.017	1306.657	-0.067	1306.59	
BC5		261.86			2.151			1305.866	-0.067	1305.799	
			5		2.93			1305.087	-0.067	1305.02	
			10		3.358			1304.659	-0.067	1304.592	
	5				1.45			1306.567	-0.067	1306.5	
	10				0.291			1307.726	-0.067	1307.659	
MC5		272.937			1.98			1306.037	-0.067	1305.97	
	5				0.559			1307.458	-0.067	1307.391	
			5		2.611			1305.406	-0.067	1305.339	
	10				0.437			1307.58	-0.067	1307.513	
			10		4.05			1303.967	-0.067	1303.9	
EC5		284.014			1.154			1306.863	-0.067	1306.796	

	5				0.895			1307.122	-0.067	1307.055	
			5		1.957			1306.06	-0.067	1305.993	
	10				0.878			1307.139	-0.067	1307.072	
			10		3.14			1304.877	-0.067	1304.81	
		285			1.117			1306.9	-0.067	1306.833	
TP				1.253		1.056	1308.214	1306.961	-0.08	1306.881	
BC6		290.37			1.416			1306.798	-0.08	1306.718	
			5		1.575			1306.639	-0.08	1306.559	
	5				0.67			1307.544	-0.08	1307.464	
	10				0.473			1307.741	-0.08	1307.661	
			10		3.01			1305.204	-0.08	1305.124	
MC6		294.969			1.508			1306.706	-0.08	1306.626	
			5		0.734			1307.48	-0.08	1307.4	
			10		0.423			1307.791	-0.08	1307.711	
	5				2.028			1306.186	-0.08	1306.106	
	10				3.012			1305.202	-0.08	1305.122	
EC6		299.568			1.481			1306.733	-0.08	1306.653	
			5		2.074			1306.14	-0.08	1306.06	
	5				1.024			1307.19	-0.08	1307.11	
			10		3.014			1305.2	-0.08	1305.12	
	10				0.765			1307.449	-0.08	1307.369	
		300			1.498			1306.716	-0.08	1306.636	
		315			0.723			1307.491	-0.08	1307.411	
BC7		315.019			0.715			1307.499	-0.08	1307.419	
	5				0.282			1307.932	-0.08	1307.852	
	10				0.481			1307.733	-0.08	1307.653	
			5		1.272			1306.942	-0.08	1306.862	
			10		1.91			1306.304	-0.08	1306.224	
TP				1.692		1.36	1308.546	1306.854	-0.094	1306.76	
MC7		334.682			1.999			1306.547	-0.094	1306.453	
			5		2.08			1306.466	-0.094	1306.372	
	5				1.35			1307.196	-0.094	1307.102	
			10		2.402			1306.144	-0.094	1306.05	
	10				0.109			1308.437	-0.094	1308.343	
EC7		354.345			2.405			1306.141	-0.094	1306.047	
	5				0.896			1307.65	-0.094	1307.556	
			5		2.94			1305.606	-0.094	1305.512	
			10		3.679			1304.867	-0.094	1304.773	
TP				1.459		1.05	1308.955	1307.496	-0.107	1307.389	
	10				1.074			1307.881	-0.107	1307.774	
		360			3.22			1305.735	-0.107	1305.628	
BC8		360.422			3.208			1305.747	-0.107	1305.64	
			5		4.254			1304.701	-0.107	1304.594	
	5				1.317			1307.638	-0.107	1307.531	

			10		3.017			1305.938	-0.107	1305.831	
	10				1.09			1307.865	-0.107	1307.758	
MC8		375.858			3.89			1305.065	-0.107	1304.958	
	5				3.262			1305.693	-0.107	1305.586	
	10				2.303			1306.652	-0.107	1306.545	
			5		5.042			1303.913	-0.107	1303.806	
			10		6.11			1302.845	-0.107	1302.738	
TP				0.677		1.962	1307.67	1306.993	-0.107	1306.886	
EC8		391.294			3.84			1303.83	-0.107	1303.723	
	5				2.285			1305.385	-0.107	1305.278	
			5		1.28			1306.39	-0.107	1306.283	
			10		4.726			1302.944	-0.107	1302.837	
	10				1.995			1305.675	-0.107	1305.568	
BC9		404.954			3.584			1304.086	-0.107	1303.979	
			5		4.336			1303.334	-0.107	1303.227	
	5				3.526			1304.144	-0.107	1304.037	
			10		4.84			1302.83	-0.107	1302.723	
	10				3.032			1304.638	-0.107	1304.531	
MC9		419.975			4.112			1303.558	-0.107	1303.451	
	5				3.473			1304.197	-0.107	1304.09	
			5		5.115			1302.555	-0.107	1302.448	
	10				3.525			1304.145	-0.107	1304.038	
			10		5.895			1301.775	-0.107	1301.668	
EC9		434.996			4.73			1302.94	-0.107	1302.833	
	5				4.175			1303.495	-0.107	1303.388	
			5		5.8			1301.87	-0.107	1301.763	
			10		5.85			1301.82	-0.107	1301.713	
	10				3.495			1304.175	-0.107	1304.068	
TP				1.005		1.613	1307.062	1306.057	-0.134	1305.923	
				0.678		1.879	1305.861	1305.183	-0.147	1305.036	
BC10		463.279			3.663			1302.198	-0.147	1302.051	
			5		3.875			1301.986	-0.147	1301.839	
			10		3.945			1301.916	-0.147	1301.769	
	5				1.837			1304.024	-0.147	1303.877	
	10				1.852			1304.009	-0.147	1303.862	
MC10		479.398			2.403			1303.458	-0.147	1303.311	
			5		3.349			1302.512	-0.147	1302.365	
			10		3.85			1302.011	-0.147	1301.864	
	5				2.306			1303.555	-0.147	1303.408	
	10				1.905			1303.956	-0.147	1303.809	
EC10		495.517			2.383			1303.478	-0.147	1303.331	
	5				2.034			1303.827	-0.147	1303.68	
	10				1.13			1304.731	-0.147	1304.584	
			5		2.928			1302.933	-0.147	1302.786	

			10		3.81			1302.051	-0.147	1301.904	
TP				1.49		1.225	1306.126	1304.636	-0.161	1304.475	
BC11		502.253			1.485			1304.641	-0.161	1304.48	
			5		2.34			1303.786	-0.161	1303.625	
			10		3.158			1302.968	-0.161	1302.807	
	5				1.514			1304.612	-0.161	1304.451	
	10				0.75			1305.376	-0.161	1305.215	
MC11		521.269			2.333			1303.793	-0.161	1303.632	
			5		2.635			1303.491	-0.161	1303.33	
			10		3.46			1302.666	-0.161	1302.505	
	5				2.302			1303.824	-0.161	1303.663	
	10				1.292			1304.834	-0.161	1304.673	
EC11		540.285			3.532			1302.594	-0.161	1302.433	
	5				2.599			1303.527	-0.161	1303.366	
	10				1.738			1304.388	-0.161	1304.227	
			5		5.494			1300.632	-0.161	1300.471	
			10		6.04			1300.086	-0.161	1299.925	
BC12		552.252			3.04			1303.086	-0.161	1302.925	
	5				2.276			1303.85	-0.161	1303.689	
	10				1.18			1304.946	-0.161	1304.785	
			5		4.05			1302.076	-0.161	1301.915	
			10		5.447			1300.679	-0.161	1300.518	
MC12		563.941			4.088			1302.038	-0.161	1301.877	
TP				1.664		1.988	1305.802	1304.138	-0.174	1303.964	
MC12	5	563.941			3.138			1302.664	-0.174	1302.49	
	10				2.048			1303.754	-0.174	1303.58	
			5		4.823			1300.979	-0.174	1300.805	
			10		6.083			1299.719	-0.174	1299.545	
EC12		575.63			4.977			1300.825	-0.174	1300.651	
	5				3.705			1302.097	-0.174	1301.923	
	10				2.827			1302.975	-0.174	1302.801	
			5		5.405			1300.397	-0.174	1300.223	
			10		5.451			1300.351	-0.174	1300.177	
TP				1.095		1.883	1305.014	1303.919	-0.188	1303.731	
TP				0.67		1.849	1303.835	1303.165	-0.188	1302.977	
BC13		588.217			3.758			1300.077	-0.188	1299.889	
			5		4			1299.835	-0.188	1299.647	
			10		4.036			1299.799	-0.188	1299.611	
	5				2.755			1301.08	-0.188	1300.892	
	10				2.066			1301.769	-0.188	1301.581	
MC13		594.426			3.485			1300.35	-0.188	1300.162	
			5		3.766			1300.069	-0.188	1299.881	
			10		4.012			1299.823	-0.188	1299.635	
	5				3.462			1300.373	-0.188	1300.185	

	10				2.475			1301.36	-0.188	1301.172	
EC13		600.635			3.76			1300.075	-0.188	1299.887	
			5		3.94			1299.895	-0.188	1299.707	
			10		4.551			1299.284	-0.188	1299.096	
	5				3.456			1300.379	-0.188	1300.191	
	10				2.76			1301.075	-0.188	1300.887	
TP				0.613		1.59	1302.858	1302.245	-0.214	1302.031	
BC14		606.131			2.477			1300.381	-0.214	1300.167	
			5		3.566			1299.292	-0.214	1299.078	
			10		3.91			1298.948	-0.214	1298.734	
	5				2.094			1300.764	-0.214	1300.55	
	10				1.331			1301.527	-0.214	1301.313	
MC14		614.001			3.568			1299.29	-0.214	1299.076	
			5		4.217			1298.641	-0.214	1298.427	
			10		5.647			1297.211	-0.214	1296.997	
	5				3.092			1299.766	-0.214	1299.552	
	10				1.754			1301.104	-0.214	1300.89	
EC14		621.871			3.995			1298.863	-0.214	1298.649	
			5		4.936			1297.922	-0.214	1297.708	
			10		5.667			1297.191	-0.214	1296.977	
	5				3.169			1299.689	-0.214	1299.475	
	10				2.083			1300.775	-0.214	1300.561	
BC15		627.687			4.115			1298.743	-0.214	1298.529	
			5		4.298			1298.56	-0.214	1298.346	
			10		6.449			1296.409	-0.214	1296.195	
	5				3.372			1299.486	-0.214	1299.272	
	10				2.673			1300.185	-0.214	1299.971	
TP				1.008		2.598	1301.268	1300.26	-0.228	1300.032	
MC15		636.082			3.058			1298.21	-0.228	1297.982	
	5				2.433			1298.835	-0.228	1298.607	
	10				1.473			1299.795	-0.228	1299.567	
			5		3.95			1297.318	-0.228	1297.09	
			10		4.436			1296.832	-0.228	1296.604	
EC15		644.477			4.239			1297.029	-0.228	1296.801	
	5				2.914			1298.354	-0.228	1298.126	
	10				2.094			1299.174	-0.228	1298.946	
			5		5.189			1296.079	-0.228	1295.851	
			10		5.585			1295.683	-0.228	1295.455	
TP				1.023		2.725	1299.566	1298.543	-0.241	1298.302	
		660			3.352			1296.214	-0.241	1295.973	
	5				3.029			1296.537	-0.241	1296.296	
	10				2.132			1297.434	-0.241	1297.193	
			5		3.968			1295.598	-0.241	1295.357	
			10		4.325			1295.241	-0.241	1295	

		675			2.857			1296.709	-0.241	1296.468	
	5				1.708			1297.858	-0.241	1297.617	
	10				0.718			1298.848	-0.241	1298.607	
			5		3.346			1296.22	-0.241	1295.979	
			10		3.868			1295.698	-0.241	1295.457	
TP				2.318		2.067	1299.817	1297.499	-0.255	1297.244	
BC16		684.929			0.927			1298.89	-0.255	1298.635	
	5				2.61			1297.207	-0.255	1296.952	
	10				0.526			1299.291	-0.255	1299.036	
			5		1.766			1298.051	-0.255	1297.796	
			10		3.021			1296.796	-0.255	1296.541	
MC16		691.624			0.86			1298.957	-0.255	1298.702	
	5				0.958			1298.859	-0.255	1298.604	
	10				0.744			1299.073	-0.255	1298.818	
			5		1.532			1298.285	-0.255	1298.03	
			10		1.997			1297.82	-0.255	1297.565	
EC16		698.319			2.04			1297.777	-0.255	1297.522	
	5				1.71			1298.107	-0.255	1297.852	
			5		2.452			1297.365	-0.255	1297.11	
	10				1.724			1298.093	-0.255	1297.838	
			10		2.457			1297.36	-0.255	1297.105	
TP				1.745		2.62	1298.942	1297.197	-0.268	1296.929	
A		704.059			1.8			1297.142	-0.268	1296.874	
			5		2.016			1296.926	-0.268	1296.658	
			10		2.07			1296.872	-0.268	1296.604	
			15		2.417			1296.525	-0.268	1296.257	
			20		2.289			1296.653	-0.268	1296.385	
			25		2.827			1296.115	-0.268	1295.847	
			30		2.955			1295.987	-0.268	1295.719	
	5				1.673			1297.269	-0.268	1297.001	
	10				1.629			1297.313	-0.268	1297.045	
	15				1.444			1297.498	-0.268	1297.23	
	20				1.42			1297.522	-0.268	1297.254	
	25				1.409			1297.533	-0.268	1297.265	
	30				1.343			1297.599	-0.268	1297.331	
	35				1.348			1297.594	-0.268	1297.326	
	40				0.7			1298.242	-0.268	1297.974	
	45				0.849			1298.093	-0.268	1297.825	
IP17		723.301				1.512		1297.43	-0.281	1297.149	
	5				1.388			1297.554	-0.281	1297.273	
	10				1.459			1297.483	-0.281	1297.202	
			5		1.451			1297.491	-0.281	1297.21	
			10		1.443			1297.499	-0.281	1297.218	
				29.794		**36.547**					

RISE/FALL	6.753
ERROR	0.563

	LOOP FROM IP17 TO B						
STATION	BS	FS	Rise	Fall	Elevation	Correction	Elevation
B	2.574				**1304.18**	0	1304.183
	2.452	0.643	1.931		1306.114	-0.013	1306.101
	2.654	1.143	1.309		1307.423	-0.027	1307.396
	1.395	1.178	1.476		1308.899	-0.04	1308.859
	0.969	1.972		0.577	1308.322	-0.054	1308.268
	1.36	2.634		1.665	1306.657	-0.067	1306.59
	1.253	1.056	0.304		1306.961	-0.08	1306.881
	1.692	1.36		0.107	1306.854	-0.094	1306.76
	1.459	1.05	0.642		1307.496	-0.107	1307.389
	0.677	1.962		0.503	1306.993	-0.121	1306.872
	1.005	1.613		0.936	1306.057	-0.134	1305.923
	0.678	1.879		0.874	1305.183	-0.147	1305.036
	1.49	1.225		0.547	1304.636	-0.161	1304.475
	1.664	1.988		0.498	1304.138	-0.174	1303.964
	1.095	1.883		0.219	1303.919	-0.188	1303.731
	0.67	1.849		0.754	1303.165	-0.201	1302.964
	0.613	1.59		0.92	1302.245	-0.214	1302.031
	1.008	2.598		1.985	1300.26	-0.228	1300.032
	1.023	2.725		1.717	1298.543	-0.241	1298.302
	2.318	2.067		1.044	1297.499	-0.255	1297.244
	1.745	2.62		0.302	1297.197	-0.268	1296.929
IP17	2.014	1.512	0.233		1297.43	-0.281	1297.149
	2.831	1.493	0.521		1297.951	-0.295	1297.656
	1.979	1.245	1.586		1299.537	-0.308	1299.229
	2.052	1.118	0.861		1300.398	-0.322	1300.076
	2.343	0.81	1.242		1301.64	-0.335	1301.305
	1.977	1.135	1.208		1302.848	-0.349	1302.499
	1.065	0.615	1.362		1304.21	-0.362	1303.848

	2.094	0.636	0.429		1304.639	-0.375	1304.264
	2.288	1.106	0.988		1305.627	-0.389	1305.238
	2.132	1.197	1.091		1306.718	-0.402	1306.316
	1.669	1.061	1.071		1307.789	-0.416	1307.373
	2.054	0.907	0.762		1308.551	-0.429	1308.122
	2.62	1.049	1.005		1309.556	-0.442	1309.114
	2.252	0.645	1.975		1311.531	-0.456	1311.075
	1.736	0.84	1.412		1312.943	-0.469	1312.474
	0.752	1.309	0.427		1313.37	-0.483	1312.887
	0.746	2.046		1.294	1312.076	-0.496	1311.58
	1.31	2.111		1.365	1310.711	-0.509	1310.202
	0.879	2.116		0.806	1309.905	-0.523	1309.382
	0.749	2.834		1.955	1307.95	-0.536	1307.414
	0.854	2.417		1.668	1306.282	-0.55	1305.732
B		2.39		1.536	1304.746	-0.563	1304.183
	66.19	**65.627**	**21.835**	**21.27**			
	Arithmetic check(e_1)						
		Error			**0.563**		

ANNEX A4

OBSERVATION AND CALCULATION

(BRIDGE SITE SURVEY)

Tribhuvan University
Institute of Engineering
Central Campus, Pulchowk
Department of Civil Engineering

Calculation of Base Triangles

Inst St and HI	Sighted To	Face	Set I					Set II					Mean Horizontal Angle	Remarks
			HCR Observation			Face Angle	Mean	HCR Observation			Face Angle	Mean		
			D	M	S			D	M	S				
A	D	L	0	0	0	32° 21' 10"	∠DAB	90	0	0	32° 21' 28"	∠DAB	∠DAB	
		R	180	0	3			269	59	43				
	B	L	32	21	10	32° 21' 15"	32° 21' 13"	122	21	28	32° 21' 30"	32° 21' 29"	32° 21' 21"	
		R	212	21	18		∠BAC	302	21	13		∠BAC	∠BAC	
	C	L	117	4	0	84° 42' 50"		207	4	26	84° 42' 58"			
		R	297	4	28	84° 43' 10"	84° 43' 0"	27	4	12	84° 42' 59"	84° 42' 58"	84° 42' 59"	
B	C	L	0	0	0	32° 13' 5"	∠CBA	90	0	0	32° 12' 49"	∠CBA	∠CBA	
		R	180	0	43			269	59	46				
	A	L	32	13	5	32° 13' 17"	32° 13' 11"	122	12	49	32° 13' 3"	32° 12' 56"	32° 13' 3"	
		R	212	14	0		∠ABD	302	12	49		∠ABD	∠ABD	
	D	L	109	58	45	77° 45' 40"		199	57	40	77° 44' 51"			
		R	289	59	45	77° 45' 45"	77° 45' 43"	19	57	44	77° 44' 55"	77° 44' 53"	77° 45' 19"	
C	A	L	0	0	0	63° 3' 36"	∠ACB	90	0	0	63° 3' 28"	∠ACB	∠ACB	
		R	179	59	48			269	59	40				
	B	L	63	3	36	63° 3' 51"		153	3	28	63° 3' 44"			
		R	243	3	39		63° 3' 44"	333	3	24		63° 3' 36"	63° 3' 40"	
D	B	L	0	0	0	69° 54' 0"	∠BDA	90	0	0	69° 54' 10"	∠BDA	∠BDA	
		R	179	59	54			270	0	17				
	A	L	69	54	0	69° 54' 8"		159	54	10	69° 53' 53"			
		R	249	54	2		69° 54' 4"	339	54	10		69° 54' 2"	69° 54' 3"	

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Bridge Site Triangulation

Inst St and HI	Sighted To	Face	HCR Observation			Face Angle	Mean	Remarks
			D	M	S			
D	A	L	0	0	0	50° 45' 7"	50° 45' 4"	∠ADE
	E	L	50	45	7			
	A	R	180	0	6	50° 45' 0"		
	E	R	230	45	6			
D	E	L	0	0	0	64° 36' 9"	64° 36' 8"	∠EDF
	F	L	64	36	9			
	E	R	179	59	59	64° 36' 8"		
	F	R	244	36	7			
A	E	L	0	0	0	59° 1' 17"	59° 1' 16"	∠EAD
	D	L	59	1	17			
	E	R	179	59	57	59° 1' 16"		
	D	R	239	1	13			
E	F	L	0	0	0	34° 48' 12"	34° 48' 7"	∠FED
	D	L	34	48	12			
	F	R	180	0	4	34° 48' 1"		
	D	R	214	48	5			
E	D	L	0	0	0	70° 14' 45"	70° 14' 51"	∠DEA
	A	L	70	14	45			
	D	R	179	59	53	70° 14' 58"		
	A	R	250	14	51			

E	G	L	0	0	0	68° 49' 28"	68° 49' 22"	∠GEF
	F	L	68	49	28			
	G	R	180	0	17	68° 49' 15"		
	F	R	248	49	32			
G	F	L	0	0	0	69° 18' 25"	69° 18' 26"	∠FGE
	E	L	69	18	25			
	F	R	180	0	3	69° 18' 26"		
	E	R	249	18	29			
F	E	L	0	0	0	41° 51' 52"	41° 51' 43"	∠EFG
	G	L	41	51	52			
	E	R	180	0	14	41° 51' 35"		
	G	R	221	51	49			
F	D	L	0	0	0	80° 35' 22"	80° 35' 22"	∠DFE
	E	L	80	35	22			
	D	R	180	0	23	80° 35' 23"		
	E	R	260	35	46			

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Fly Levelling

Forward Loop (SBM2 to Rover 1)

RL of SBM 2 =	1311.599

Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	Stadia Interval S1+S2	Hz. Distance(m) S x 100	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
SBM2	0.664	0.65	0.638	0.651	0.026								1311.599	0.026	2.6	SBM2
1	0.774	0.759	0.734	0.756	0.04	1.47	1.453	1.436	1.453	0.034		0.802	1310.797	0.074	7.4	
2	0.698	0.675	0.654	0.676	0.044	1.95	1.928	1.905	1.928	0.045		1.172	1309.625	0.089	8.9	
3	1.318	1.286	1.254	1.286	0.064	1.744	1.72	1.695	1.72	0.049		1.044	1308.581	0.113	11.3	
4	0.69	0.65	0.61	0.65	0.08	1.896	1.865	1.834	1.865	0.062		0.579	1308.002	0.142	14.2	
5	1.294	1.238	1.182	1.238	0.112	1.87	1.823	1.785	1.826	0.085		1.176	1306.826	0.197	19.7	
Rover 1					0	1.43	1.37	1.31	1.37	0.12		0.132	1306.694	0.12	12	Rover 1
Sum				5.257					10.162		0	4.905			76.1	

Arithmetic Check:	
$\lvert\sum BS-\sum FS\rvert=$	4.905
$\lvert\sum Rise-\sum Fall\rvert=$	4.905
|Last RL - First RL| =	4.095
Distance(k1)=	76.1

Backward Loop (Rover 1 to SBM 2)

Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	Stadia Interval S1+S2	Hz.Distance (m) S x 100	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
Rover 1	1.43	1.37	1.31	1.37	0.12								1306.694	0.12	12	Rover 1
1	1.704	1.664	1.624	1.664	0.08	1.294	1.236	1.182	1.237	0.112	0.133		1306.827	0.192	19.2	
2	2.029	1.992	1.959	1.993	0.07	0.735	0.696	0.65	0.694	0.085	0.97		1307.797	0.155	15.5	
3	1.713	1.689	1.663	1.688	0.05	1.255	1.219	1.182	1.219	0.073	0.774		1308.571	0.123	12.3	
4	1.994	1.976	1.958	1.976	0.036	0.807	0.783	0.759	0.783	0.048	0.905		1309.476	0.084	8.4	
5	1.597	1.588	1.58	1.588	0.017	0.666	0.65	0.635	0.65	0.031	1.326		1310.802	0.048	4.8	
SBM2					0	0.795	0.781	0.777	0.784	0.018	0.804		1311.606	0.018	1.8	SBM2
Sum				10.279					5.367		4.912	0			74	

Arithmetic Check:	
$\lvert\sum BS-\sum FS\rvert=$	4.912
$\lvert\sum Rise-\sum Fall\rvert=$	4.912
|Last RL - First RL| =	4.912
Distance(k1)=	74

Error Computation:	
Error of Closure (m) (mm)=	-7
Loop Distance in (km) =	0.1501
Permissible Error = $24\sqrt{k}$ =	9.298
RL of SBM (m) =	1306.6905

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Fly Levelling

Forward Loop (Rover 1 to A)

RL of Rover 1 =	1306.691

Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	Stadia Interval S1+S2	Hz. Distance(m) S x 100	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
Rover 1	0.9	0.85	0.8	0.85	0.1								1306.691	0.1	10	SBM2
1	0.704	0.686	0.665	0.685	0.039	1.933	1.885	1.837	1.885	0.096		1.035	1305.656	0.135	13.5	
2	0.625	0.61	0.594	0.61	0.031	1.567	1.543	1.518	1.543	0.049		0.858	1304.798	0.08	8	
3	0.877	0.856	0.836	0.856	0.041	1.718	1.702	1.688	1.703	0.03		1.093	1303.705	0.071	7.1	
A					0	1.627	1.608	1.59	1.608	0.037		0.752	1302.953	0.037	3.7	
Sum				3.001					6.739		0	3.738			42.3	

Arithmetic Check:	
|∑BS-∑FS|=	3.738
|∑Rise-∑Fall|=	3.738
|Last RL - First RL| =	3.738
Distance(k1)=	42.3

Backward Loop (A to Rover 1)

Stations Chainage	BS			Mean BS	S1	FS			Mean FS	S2	Rise +	Fall -	Elevation	Stadia Interval S1+S2	Hz.Distance (m) S x 100	Remarks
	Top	Mid	Bot		(T-B)	Top	Mid	Bot		(T-B)						
A	1.636	1.616	1.596	1.616	0.04								1302.953	0.04	4	Rover 1
1	1.662	1.649	1.634	1.648	0.028	0.885	0.868	0.85	0.868	0.035	0.748		1303.701	0.063	6.3	
2	1.705	1.676	1.649	1.677	0.056	0.675	0.66	0.644	0.66	0.031	0.988		1304.689	0.087	8.7	
3	2.028	1.978	1.935	1.98	0.093	0.774	0.747	0.72	0.747	0.054	0.93		1305.619	0.147	14.7	
Rover 1					0	0.958	0.911	0.864	0.911	0.094	1.069		1306.688	0.094	9.4	
Sum				6.921					3.186		3.735	0			43.1	

Arithmetic Check:	
|∑BS-∑FS|=	3.735
|∑Rise-∑Fall|=	3.735
|Last RL - First RL| =	3.735
Distance(k1)=	43.1

Error Computation:	
Error of Closure (m) (mm)=	3
Loop Distance in (km) =	0.0854
Permissible Error = 24√k =	7.014
RL of SBM (m) =	1302.9545

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RECIPROCAL LEVELLING

First Set (When Instrument is nearer to A):

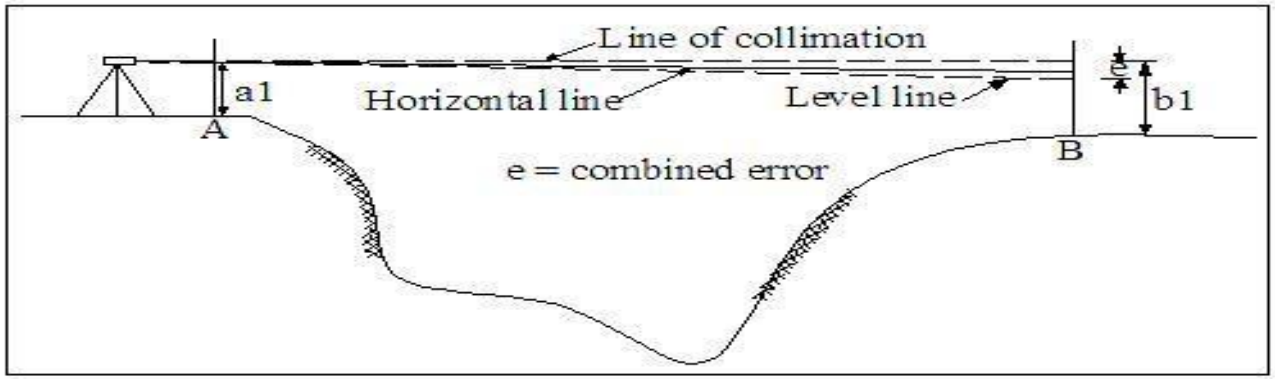
Instrument At	Sighted to	Staff Reading			Mean = (T+M+B) / 3	True Height Difference (h1) = Mean A - Mean B	Remarks
		Top	Middle	Bottom			
Near to A	A	1.452	1.435	1.419	1.435	0.665	
	B	1.14	0.77	0.4	0.77		

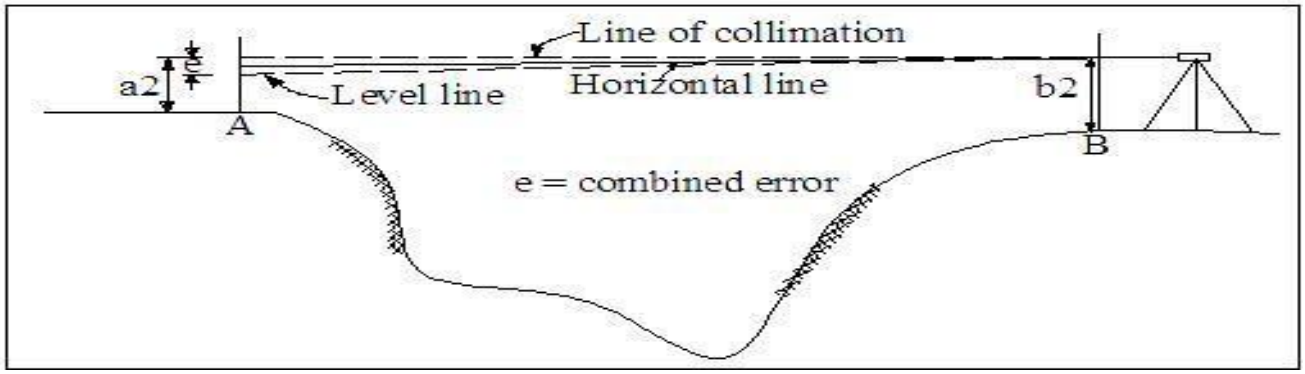
Second Set (When Instrument is nearer to B):

Instrument At	Sighted to	Staff Reading			Mean = (T+M+B) / 3	True Height Difference (h1) = Mean B - Mean A	Remarks
		Top	Middle	Bottom			
Near to B	B	2.07	1.725	1.38	1.725	0.667	
	A	1.071	1.058	1.044	1.058		

Error Computation:	
Error of Closure = h1 - h2 (mm)=	-2
Bridge Axis Length (m) =	70.108
Permissible Error (mm) = 24√k =	9.361

RL Computation	
RL of A (m) =	1302.955
Height difference (mm) =	666
RL of B (m) =	1303.621


Line of collimation
a1
Horizontal line
Level line
e
b1
A
B
e = combined error


Line of collimation
e
a2
Level line
Horizontal line
b2
A
B
e = combined error

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Bridge Site Detailing

SN	Easting	Northing	RL	Remarks
1	27323.197	63046.365	1303.004	A
2	27319.642	63116.325	1303.702	B
3	27364.65	63052.355	1300.244	C
4	27281.097	63105.886	1304.206	D
5	27263.42	63041.855	1302.591	E
6	27220.087	63043.283	1300.955	G
7	27243.273	63099.206	1304.182	F
100	27215.062	63020.692	1306.392	BKT
101	27215.004	63020.747	1306.395	BKT
102	27208.938	63020.435	1306.499	BKT
103	27205.782	63020.162	1306.607	BKT
104	27216.051	63021.675	1306.251	CONC.B
105	27217.308	63020.434	1306.285	CONC.B
106	27217.251	63020.352	1306.253	CONC.B
107	27216.087	63020.484	1306.356	CONC.B
108	27231.139	63059.611	1299.106	CULVERT
109	27233.578	63059.164	1298.88	CULVERT
110	27232.451	63065.457	1299.966	CULVERT
111	27234.938	63065.709	1300.059	CULVERT
112	27236.923	63074.315	1299.388	CULVERT
113	27234.6	63074.778	1299.412	CULVERT
114	27217.046	63020.201	1306.309	DRAIN
115	27216.493	63020.102	1306.233	DRAIN
116	27216.529	63018.64	1306.333	DRAIN
117	27217.191	63018.558	1306.481	DRAIN
118	27329.306	63047.926	1302.898	GP
119	27334.051	63048.027	1302.603	GP
120	27333.98	63050.446	1302.085	GP
121	27334.125	63053.596	1301.216	GP
122	27333.348	63056.251	1300.438	GP
123	27338.998	63048.81	1302.462	GP
124	27339.142	63049.987	1302.15	GP
125	27339.275	63051.555	1301.7	GP
126	27339.133	63053.543	1301.045	GP
127	27345.471	63049.169	1302.021	GP
128	27346.182	63052.45	1301.189	GP
129	27328.332	63059.089	1299.75	GP
130	27347.124	63057.26	1300.091	GP
131	27323.071	63062.216	1298.962	GP
132	27348.031	63062.349	1298.459	GP
133	27323.792	63066.833	1297.556	GP
134	27354.388	63050.971	1300.63	GP

135	27323.422	63070.106	1296.603	GP
136	27356.068	63061.048	1298.914	GP
137	27323.164	63073.604	1296.199	GP
138	27356.813	63063.337	1297.784	GP
139	27323.721	63080.477	1295.647	GP
140	27360.557	63062.804	1297.562	GP
141	27337.329	63075.519	1295.902	GP
142	27361.145	63056.386	1299.079	GP
143	27340.168	63071.979	1296.809	GP
144	27360.462	63050.942	1300.116	GP
145	27344.326	63068.728	1297.348	GP
146	27346.955	63068.305	1297.15	GP
147	27366.17	63055.554	1299.31	GP
148	27353.999	63060.973	1299.084	GP
149	27357.266	63061.826	1298.476	GP
150	27354.95	63067.241	1297.317	GP
151	27351.255	63073.972	1295.55	GP
152	27354.259	63077.266	1295.097	GP
153	27369.842	63053.49	1299.492	GP
154	27356.242	63079.265	1294.189	GP
155	27368.17	63043.729	1301.423	GP
156	27345.197	63079.375	1294.565	GP
157	27375.521	63041.931	1300.461	GP
158	27375.607	63046.88	1300.131	GP
159	27375.994	63053.282	1298.487	GP
160	27344.362	63078.576	1293.878	GP
161	27327.189	63043.931	1303.507	GP
162	27332.797	63037.452	1304.298	GP
163	27337.045	63033.93	1305.094	GP
164	27347.33	63032.98	1305.661	GP
165	27348.984	63026.536	1305.911	GP
166	27352.099	63021.086	1306.566	GP
167	27357.11	63015.848	1307.125	GP
168	27362.476	63007.212	1307.849	GP
169	27377.828	63056.838	1297.562	GP
170	27384.207	63061.929	1295.816	GP
171	27325.504	63028.865	1305.768	GP
172	27388.626	63055.991	1296.278	GP
173	27324.789	63032.463	1305.032	GP
174	27323.739	63037.539	1304.556	GP
175	27370.526	63031.111	1303.284	GP
176	27326.364	63016.736	1306.236	GP
177	27357.034	63030.549	1305.075	GP
178	27350.459	63033.797	1305.279	GP
179	27344.714	63036.045	1304.173	GP
180	27313.612	63030.823	1304.929	GP
181	27346.747	63042.346	1303.011	GP
182	27310.184	63026.437	1305.789	GP
183	27307.197	63019.482	1306.163	GP
184	27317.523	63054.183	1300.839	GP

185	27302.731	63033.212	1304.881	GP
186	27314.603	63059.719	1299.837	GP
187	27300.167	63039.32	1303.605	GP
188	27311.79	63064.8	1298.257	GP
189	27300.719	63043.046	1302.168	GP
190	27308.866	63067.326	1297.127	GP
191	27304.098	63056.842	1299.76	GP
192	27305.666	63071.17	1296.16	GP
193	27313.5	63054.546	1300.606	GP
194	27294.489	63063.536	1297.346	GP
195	27317.805	63066.666	1297.684	GP
196	27320.746	63072.633	1296.249	GP
197	27319.235	63043.012	1303.293	GP
198	27294.673	63041.435	1301.516	GP
199	27313.981	63010.982	1306.678	GP
200	27291.202	63036.849	1302.853	GP
201	27293.107	63045.278	1300.456	GP
202	27288.993	63031.597	1304.013	GP
203	27291.809	63054.821	1298.749	GP
204	27290.942	63037.46	1302.552	GP
205	27291.385	63064.618	1297.07	GP
206	27286.257	63041.973	1301.488	GP
207	27286.741	63056.17	1298.509	GP
208	27290.045	63043.625	1299.985	GP
209	27263.457	63041.83	1302.497	GP
210	27322.001	63051.418	1301.782	GP
211	27352.965	63089.081	1295.134	GP
212	27359.586	63089.86	1294.892	GP
213	27366.553	63088.106	1294.657	GP
214	27360.089	63082.337	1294.805	GP
215	27355.881	63077.95	1294.985	GP
216	27348.205	63076.385	1295.435	GP
217	27279.645	63040.677	1301.66	GP
218	27263.412	63041.855	1302.418	GP
219	27279.56	63034.593	1303.205	GP
220	27280.569	63044.001	1300.953	GP
221	27220.087	63043.283	1300.955	GP
222	27220.187	63038.74	1301.38	GP
223	27219.685	63030.821	1303.836	GP
224	27219.667	63030.848	1303.836	GP
225	27219.057	63026.544	1305.726	GP
226	27219.07	63026.549	1305.725	GP
227	27226.581	63021.634	1305.785	GP
228	27226.56	63021.667	1305.785	GP
229	27243.273	63099.206	1304.182	GP
230	27264.909	63042.556	1302.364	GP
231	27230.439	63036.757	1302.741	GP
232	27231.847	63031.526	1303.85	GP
233	27231.849	63031.527	1303.85	GP
234	27231.129	63027.882	1305.464	GP

235	27231.13	63027.882	1305.464	GP
236	27236.544	63025.009	1305.465	GP
237	27236.533	63024.997	1305.451	GP
238	27237.877	63019.474	1305.843	GP
239	27237.155	63015.939	1306.062	GP
240	27231.296	63010.425	1306.399	GP
241	27234.058	63008.833	1306.481	GP
242	27228.762	63010.302	1306.641	GP
243	27215.712	63025.07	1305.749	GP
244	27209.542	63024.3	1305.981	GP
245	27215.084	63020.764	1306.086	GP
246	27208.973	63020.312	1306.167	GP
247	27202.949	63020.677	1306.236	GP
248	27197.683	63020.311	1306.222	GP
249	27205.285	63030.517	1305.495	GP
250	27211.772	63029.672	1305.245	GP
251	27265.129	63038.423	1303.093	GP
252	27267.549	63036.609	1303.815	GP
253	27268.202	63033.202	1304.429	GP
254	27245.516	63028.21	1305.483	GP
255	27270.403	63028.522	1305.234	GP
256	27249.031	63023.09	1305.492	GP
257	27267.755	63022.255	1305.727	GP
258	27249.76	63018.786	1305.722	GP
259	27254.363	63026.966	1305.646	GP
260	27258.28	63023.445	1305.536	GP
261	27254.384	63031.032	1304.799	GP
262	27256.381	63033.994	1304.178	GP
263	27263.016	63029.076	1305.013	GP
264	27259.61	63035.583	1304.351	GP
265	27264.773	63036.453	1303.725	GP
266	27259.318	63037.144	1303.349	GP
267	27271.007	63047.72	1300.616	GP
268	27257.427	63038.486	1302.536	GP
269	27270.71	63052.568	1299.472	GP
270	27256.761	63041.662	1302.171	GP
271	27272.854	63055.721	1298.893	GP
272	27258.811	63047.058	1301.061	GP
273	27277.15	63059.02	1298.067	GP
274	27265.126	63049.563	1300.461	GP
275	27276.889	63062.381	1297.335	GP
276	27265.584	63051.815	1299.747	GP
277	27280.175	63067.871	1297.135	GP
278	27261.417	63050.429	1300.307	GP
279	27287.363	63069.183	1296.811	GP
280	27262.674	63053.588	1299.697	GP
281	27258.129	63052.348	1299.634	GP
282	27221.129	63040.246	1301.393	GP
283	27223.383	63036.784	1301.821	GP
284	27223.568	63034.291	1302.431	GP

285	27219.273	63033.009	1303.172	GP
286	27219.272	63033	1303.174	GP
287	27219.691	63047.786	1300.242	GP
288	27219.636	63051.251	1299.618	GP
289	27220.319	63054.581	1298.915	GP
290	27218.664	63059.139	1299.454	GP
291	27211.023	63056.278	1299.668	GP
292	27197.181	63060.149	1300.15	GP
293	27188.402	63041.613	1301.622	GP
294	27192.76	63041.399	1301.813	GP
295	27194.718	63038.547	1303.217	GP
296	27195.925	63036.759	1304.28	GP
297	27196.39	63034.6	1305.305	GP
298	27202.577	63029.494	1306.036	GP
299	27211.733	63028.881	1306.063	GP
300	27214.061	63030.357	1305.171	GP
301	27217.957	63026.387	1305.98	GP
302	27224.015	63023.822	1305.616	GP
303	27226.192	63025.963	1305.387	GP
304	27229.11	63032.117	1303.764	GP
305	27228.104	63035.476	1302.668	GP
306	27229.672	63040.822	1302.036	GP
307	27229.83	63043.348	1301.121	GP
308	27224.816	63043.055	1300.323	GP
309	27224.425	63044.947	1300.167	GP
310	27227.077	63046.09	1299.889	GP
311	27226.261	63040.715	1300.393	GP
312	27226.029	63036.891	1301.365	GP
313	27223.996	63037.654	1301.486	GP
314	27227.869	63037.37	1301.958	GP
315	27164.113	63064.119	1304.923	GP
316	27179.009	63035.624	1303.995	GP
317	27161.18	63065.427	1305.925	GP
318	27181.275	63032.346	1305.537	GP
319	27160.331	63054.43	1304.341	GP
320	27169.058	63058.466	1301.291	GP
321	27220.085	63023.497	1306.768	GP
322	27215.021	63020.694	1306.405	GP
323	27217.94	63025.734	1306.063	GP
324	27197.76	63035.202	1304.771	GP
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326	27191.296	63033.668	1305.807	GP
327	27185.26	63032.377	1306.043	GP
328	27182.753	63033.146	1305.523	GP
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331	27170.617	63041.175	1301.885	GP
332	27166.362	63037.61	1303.047	GP
333	27168.269	63031.28	1305.421	GP
334	27219.432	63051.71	1299.534	GP

335	27199.983	63031.909	1305.774	GP
336	27205.041	63029.208	1306.126	GP
337	27215.38	63028.013	1305.678	GP
338	27214.877	63030.164	1305.054	GP
339	27217.841	63027.231	1305.685	GP
340	27218.538	63029.714	1304.792	GP
341	27217.198	63029.35	1304.613	GP
342	27216.853	63028.131	1304.973	GP
343	27221.713	63029.104	1304.216	GP
344	27220.568	63031.239	1303.639	GP
345	27319.158	63120.299	1304.661	GP
346	27323.334	63122.02	1304.679	GP
347	27328.076	63124.558	1304.512	GP
348	27325.514	63130.163	1305.644	GP
349	27329.873	63134.684	1305.884	GP
350	27328.46	63134.423	1305.384	GP
351	27328.208	63131.243	1304.467	GP
352	27327.84	63124.396	1304.489	GP
353	27335.971	63119.599	1302.761	GP
354	27332.199	63114.486	1302.068	GP
355	27335.51	63112.231	1300.977	GP
356	27335.048	63110.719	1300.149	GP
357	27334.468	63103.702	1299.126	GP
358	27328.409	63103.711	1299.434	GP
359	27325.199	63110.06	1301.012	GP
360	27321.072	63114.961	1303.167	GP
361	27316.789	63132.287	1306.884	GP
362	27320.226	63136.405	1307.331	GP
363	27312.809	63137.658	1308.179	GP
364	27308.16	63145.934	1309.795	GP
365	27297.772	63142.005	1309.779	GP
366	27295.412	63134.023	1308.618	GP
367	27290.548	63125.947	1307.457	GP
368	27286.697	63120.18	1306.567	GP
369	27285.564	63111.56	1305.148	GP
370	27288.483	63104.014	1303.044	GP
371	27295.848	63103.989	1302.189	GP
372	27303.057	63102.092	1301.167	GP
373	27295.352	63092.99	1299.193	GP
374	27311.731	63098.173	1299.223	GP
375	27291.681	63091.571	1299.151	GP
376	27304.225	63088.376	1296.899	GP
377	27302.985	63082.814	1296.322	GP
378	27311.272	63078.501	1295.884	GP
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381	27318.517	63083.586	1295.131	GP
382	27320.526	63083.345	1295.365	GP
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385	27345.212	63086.815	1295.304	GP
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391	27372.105	63104.462	1298.757	GP
392	27321.823	63082.459	1295.251	GP
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395	27370.395	63120.33	1301.524	GP
396	27369.976	63129.233	1302.16	GP
397	27359.187	63122.405	1302.108	GP
398	27346.255	63120.233	1301.827	GP
399	27352.622	63112.43	1300.617	GP
400	27342.905	63119.008	1300.412	GP
401	27340.465	63124.762	1302.492	GP
402	27335.271	63126.985	1303.751	GP
403	27330.895	63126.683	1303.312	GP
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406	27312.553	63138.036	1308.203	GP
407	27309.215	63141.487	1309.096	GP
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409	27246.727	63117.69	1307.124	GP
410	27257.765	63113.737	1306.244	GP
411	27262.425	63118.126	1306.995	GP
412	27270.704	63121.893	1307.431	GP
413	27247.433	63098.51	1303.689	GP
414	27245.143	63093.458	1303.359	GP
415	27243.041	63093.007	1303.304	GP
416	27242.275	63091.175	1303.082	GP
417	27241.773	63093.217	1303.361	GP
418	27247.452	63111.419	1305.873	GP
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420	27243.349	63116.383	1306.954	GP
421	27243.987	63119.175	1307.434	GP
422	27244.762	63123.708	1308.217	GP
423	27244.568	63125.545	1308.582	GP
424	27248.507	63121.448	1307.787	GP
425	27249.912	63118.992	1307.344	GP
426	27254.417	63114.334	1306.382	GP
427	27252.624	63110.015	1305.423	GP
428	27255.085	63120.466	1307.602	GP
429	27248.852	63106.3	1305.067	GP
430	27261.114	63115.68	1306.49	GP
431	27257.405	63123.341	1308.089	GP
432	27263.101	63113.454	1306.123	GP
433	27267.758	63116.766	1306.488	GP
434	27245.61	63093.875	1303.341	GP

435	27244.878	63091.485	1303.101	GP
436	27246.454	63090.733	1302.753	GP
437	27247.397	63090.364	1302.533	GP
438	27247.701	63090.056	1302.264	GP
439	27249.096	63087.362	1301.731	GP
440	27250.08	63085.951	1301.078	GP
441	27249.49	63084.666	1300.804	GP
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443	27246.546	63098.845	1303.712	GP
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445	27273.405	63129.033	1308.636	GP
446	27269.758	63134.367	1309.569	GP
447	27309.43	63148.753	1310.245	TRE
448	27314.333	63141.573	1308.644	TRE
449	27323.288	63140.732	1307.719	TRE
450	27322.869	63129.701	1305.892	TRE
451	27312.794	63125.996	1306.074	TRE
452	27310.977	63123.392	1305.757	TRE
453	27307.36	63127.47	1306.778	TRE
454	27300.234	63126.504	1307.202	TRE
455	27296.946	63137.583	1308.985	TRE
456	27291.736	63125.998	1307.467	TRE
457	27286.749	63120.225	1306.624	TRE
458	27281.137	63131.609	1308.77	TRE
459	27287.006	63113.019	1305.457	TRE
460	27275.97	63123.964	1307.652	TRE
461	27286.911	63107.586	1304.37	TRE
462	27274.708	63102.3	1303.428	TRE
463	27261.337	63121.013	1307.648	TRE
464	27270.895	63113.482	1306.158	TRE
465	27255.306	63117.316	1307.068	TRE
466	27261.784	63111.711	1305.804	TRE
467	27260.983	63127.211	1308.723	TRE
468	27246.659	63117.627	1307.147	TRE
469	27330.621	63115.767	1302.576	TRE
470	27333.596	63127.386	1304.352	TRE
471	27331.365	63112.484	1301.614	TRE
472	27344.151	63129.411	1304.007	TRE
473	27331.472	63103.666	1299.594	TRE
474	27331.279	63098.132	1297.988	TRE
475	27350.534	63137.596	1304.882	TRE
476	27333.909	63100.763	1298.634	TRE
477	27357.322	63146.031	1305.781	TRE
478	27336.544	63101.064	1298.404	TRE
479	27364.281	63146.036	1304.763	TRE
480	27339.951	63095.876	1296.681	TRE
481	27370.14	63132.293	1302.416	TRE
482	27352.16	63098.961	1296.235	TRE
483	27356.925	63111.533	1300.147	TRE
484	27247.197	63116.907	1306.944	TRE

Survey camp 2082 (31)

ANNEX-B:

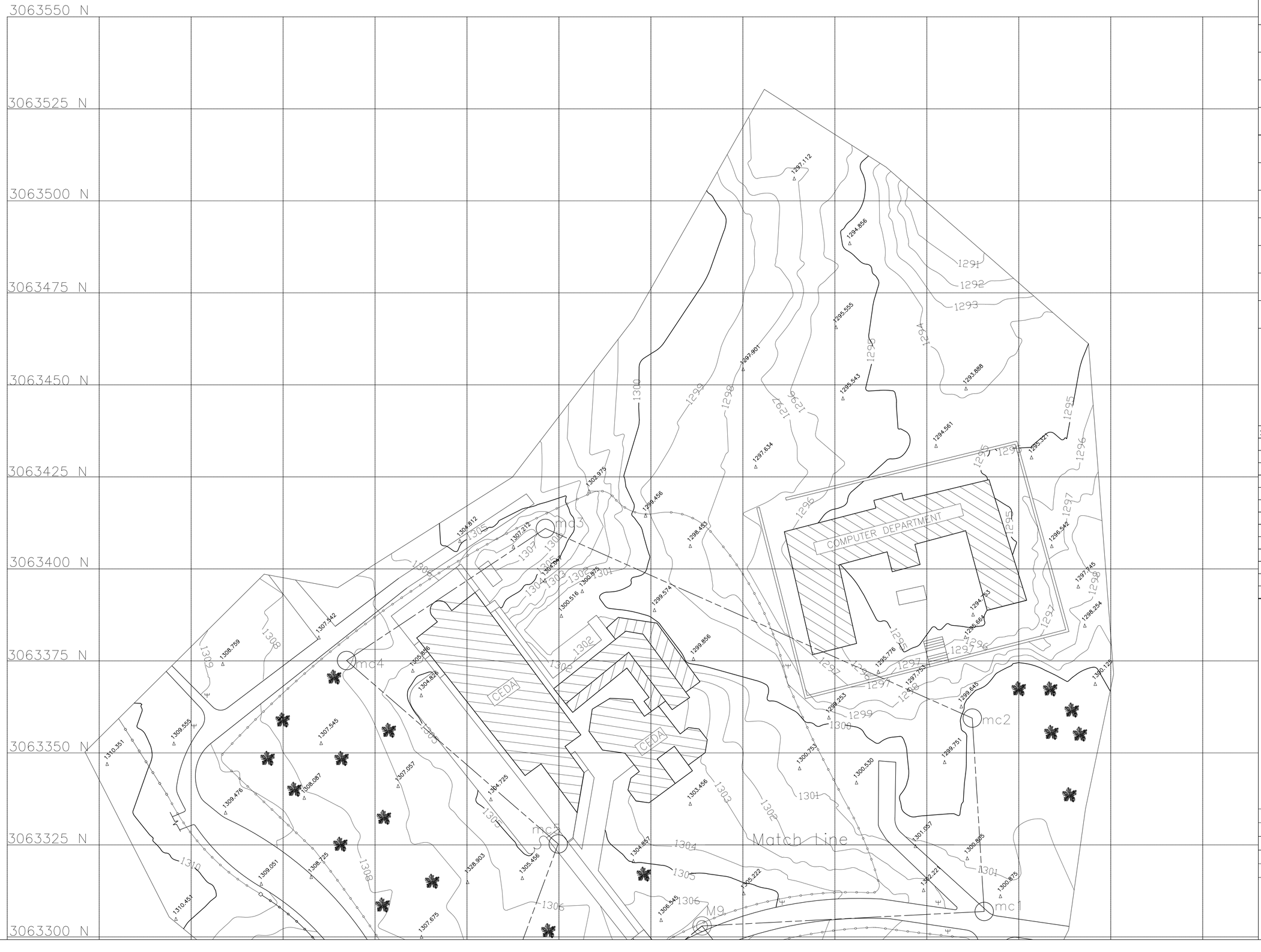
Maps, Drawings, and Graph

ANNEX-B1:

Topographic Survey Maps and Graphs

TOPOGRAPHIC MAP

SCALE:- 1:500



LEGENDS

S.N.	DESCRIPTION	SYMBOL
1.	MAJOR STATION	◎
2.	MAJOR LEG	
3.	MINOR STATION	○
4.	MINOR LEG	
5.	MAJOR CONTOUR	1305
6.	MINOR CONTOUR	1306
7.	BUILDINGS	
8.	ROAD	
9.	FENCE	
10.	SPOTHEIGHT	△
11	ELECTRIC POLE	
12	TREES	

COORDINATES

SN	Easting	Northing	RL	Remarks
1	626865.979	3063148.959	1311.734	CP1
2	626859.254	3063070.173	1311.466	CP2
3	626986.953	3063305.275	1304.232	M9
4	626921.5125	3063248.492	1307.565	M10
5	626923.814	3063150.877	1308.208	M11
6	627065.62	3063306.918	1300.632	mc1
7	627062.395	3063359.441	1298.472	mc2
8	626946.327	3063410.98	1306.276	mc3
9	626892.266	3063375.098	1306.275	mc4
10	626949.819	3063325.266	1304.14	mc5
11	627033.828	3063246.248	1303.188	ml1
12	626999.566	3063174.642	1304.155	ml2
13	626959.961	3063204.747	1306.282	ml3

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TITLE: TOPOGRAPHIC MAP
SCALE 1:500

GROUP 25

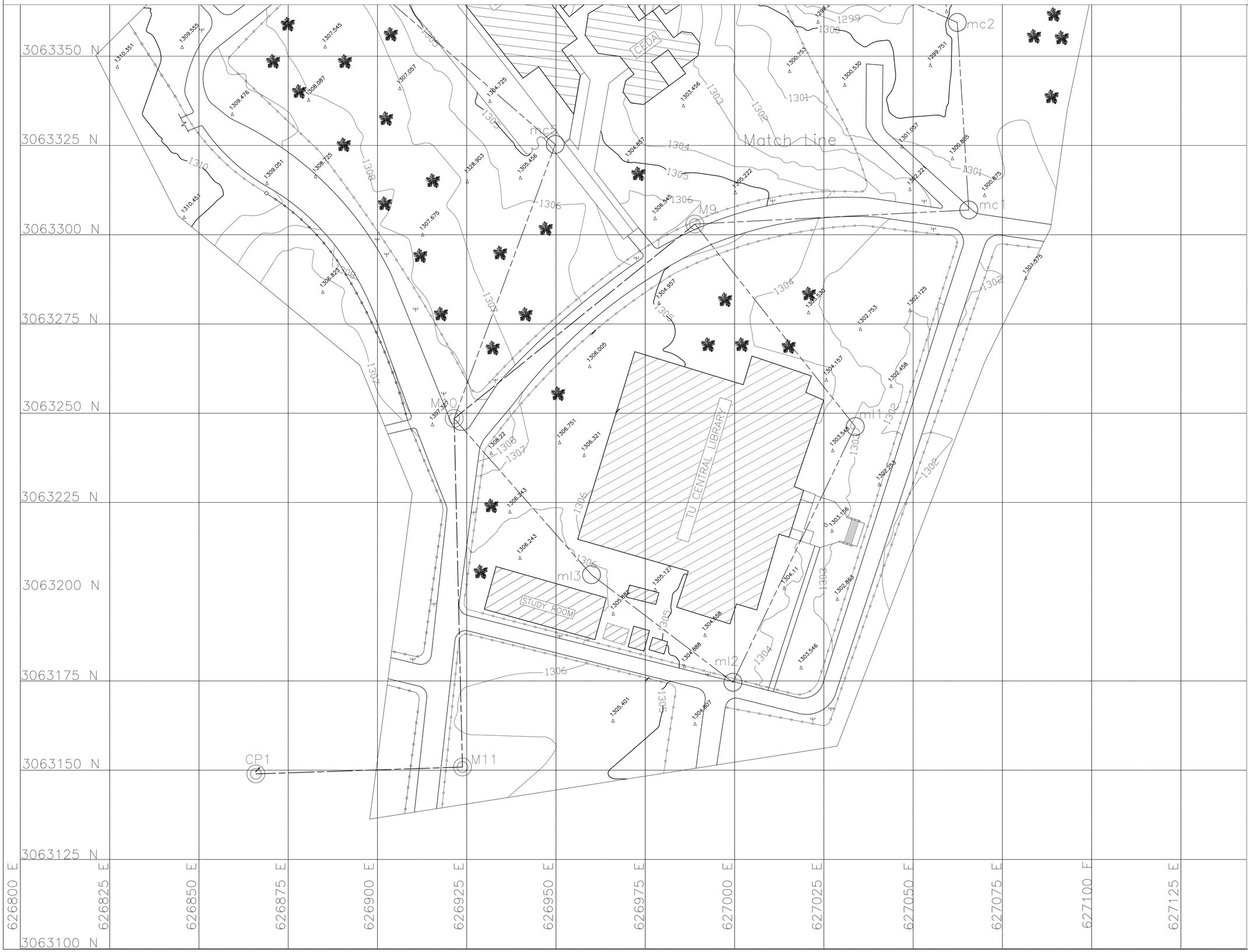
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

METER

10 5 0 15 30

TOPOGRAPHIC MAP

SCALE:- 1:500



LEGENDS

S.N.	DESCRIPTION	SYMBOL
1.	MAJOR STATION	◎
2.	MAJOR LEG	— — — — — —
3.	MINOR STATION	○
4.	MINOR LEG	— — — — — —
5.	MAJOR CONTOUR	1305
6.	MINOR CONTOUR	1306
7.	BUILDINGS	
8.	ROAD	
9.	FENCE	
10.	SPOTHEIGHT	△
11	ELECTRIC POLE	
12	TREES	

COORDINATES

SN	Easting	Northing	RL	Remarks
1	626865.979	3063149	1311.734	CP1
2	626859.254	3063070	1311.466	CP2
3	626986.953	3063305	1304.232	M9
4	626921.5125	3063248	1307.565	M10
5	626923.814	3063151	1308.208	M11
6	627065.62	3063307	1300.632	mc1
7	627062.395	3063359	1298.472	mc2
8	626946.327	3063411	1306.276	mc3
9	626892.266	3063375	1306.275	mc4
10	626949.819	3063325	1304.14	mc5
11	627033.828	3063246	1303.188	ml1
12	626999.566	3063175	1304.155	ml2
13	626959.961	3063205	1306.282	ml3

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TITLE: TOPOGRAPHIC MAP
SCALE 1:500

GROUP 25

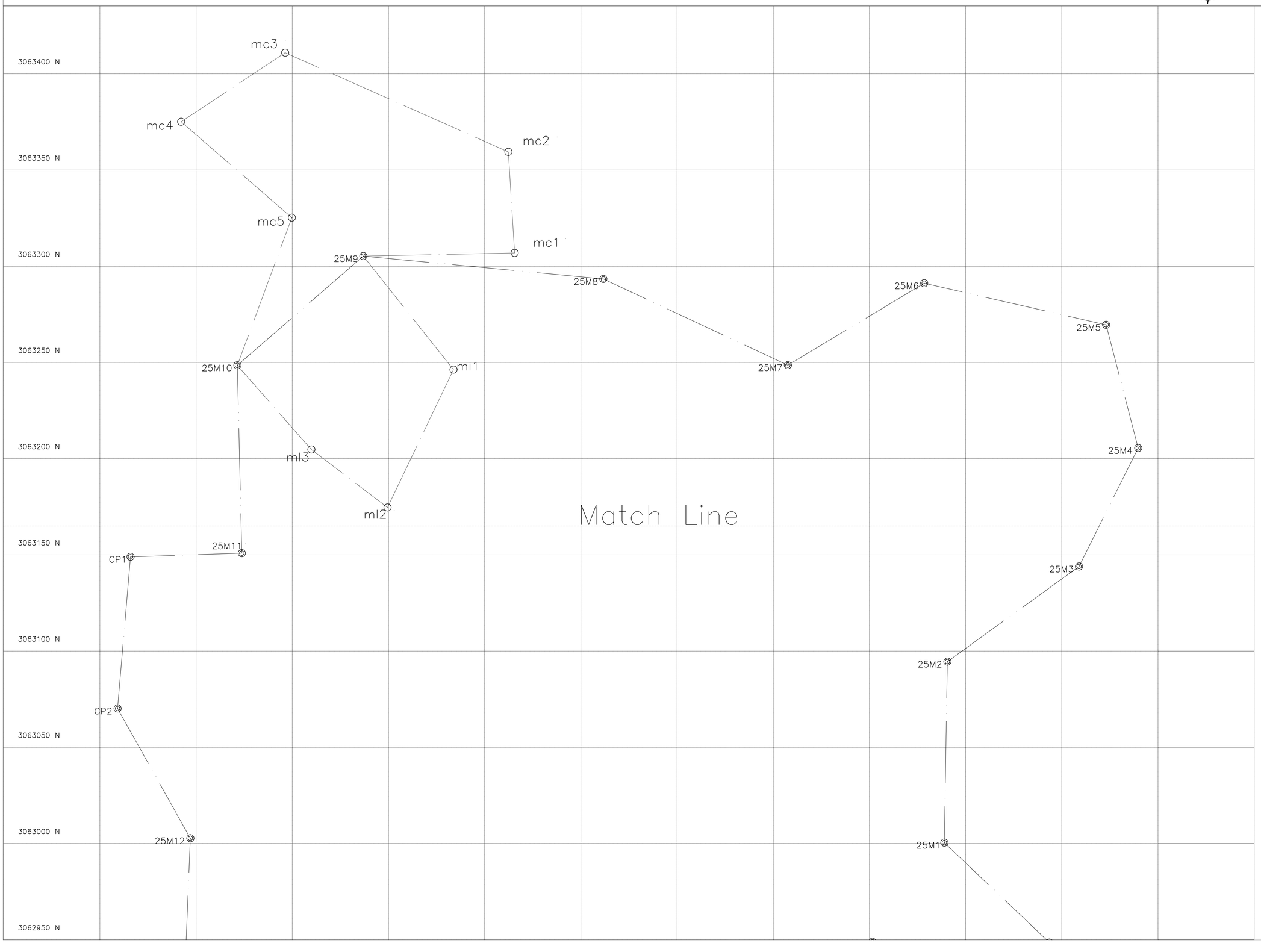
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

METER

10 5 0 15 30

MAJOR AND MINOR TRAVERSE

SCALE 1:1000



LEGENDS:

S.N.	DESCRIPTION	SYMBOL
1	MAJOR STATION	◎
2	MAJOR TRAVERSE LEG	———
3	MINOR STATION	○
4	MINOR TRAVERSE LEG	———

STATION COORDINATES (MAJOR)		
STATION	EASTING(m)	NORTHING(m)
CP1	626865.979	3063148.959
CP2	626859.243	3063070.157
M12	626897.049	3063002.716
M13	626893.311	3062910.831
M14	626957.403	3062870.718
M15	627079.861	3062875.710
M16	627156.053	3062848.774
M17	627211.569	3062889.565
M18	627251.648	3062948.944
M19	627343.603	3062948.546
M1	627288.976	3063000.444
M2	627290.564	3063094.520
M3	627359.004	3063143.938
M4	627389.754	3063205.533
M5	627373.091	3063269.557
M6	627278.514	3063291.167
M7	627207.635	3063248.580
M8	627111.736	3063293.401
M9	626986.953	3063305.275
M10	626921.513	3063248.492
M11	626923.814	3063150.877

STATION COORDINATES (MINOR)			
STATION	EASTING(m)	NORTHING(m)	ELEVATION(m)
mc1	627065.620	3063306.918	1300.632
mc2	627062.395	3063359.441	1298.472
mc3	626946.327	3063410.980	1306.276
mc4	626892.266	3063375.098	1306.275
mc5	626949.819	3063325.266	1304.140
ml1	627033.828	3063246.248	1303.188
ml2	626999.566	3063174.642	1304.155
ml3	626959.961	3063204.747	1306.282

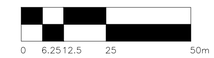
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TITLE: TOPOGRAPHIC MAP

SCALE= 1:1000

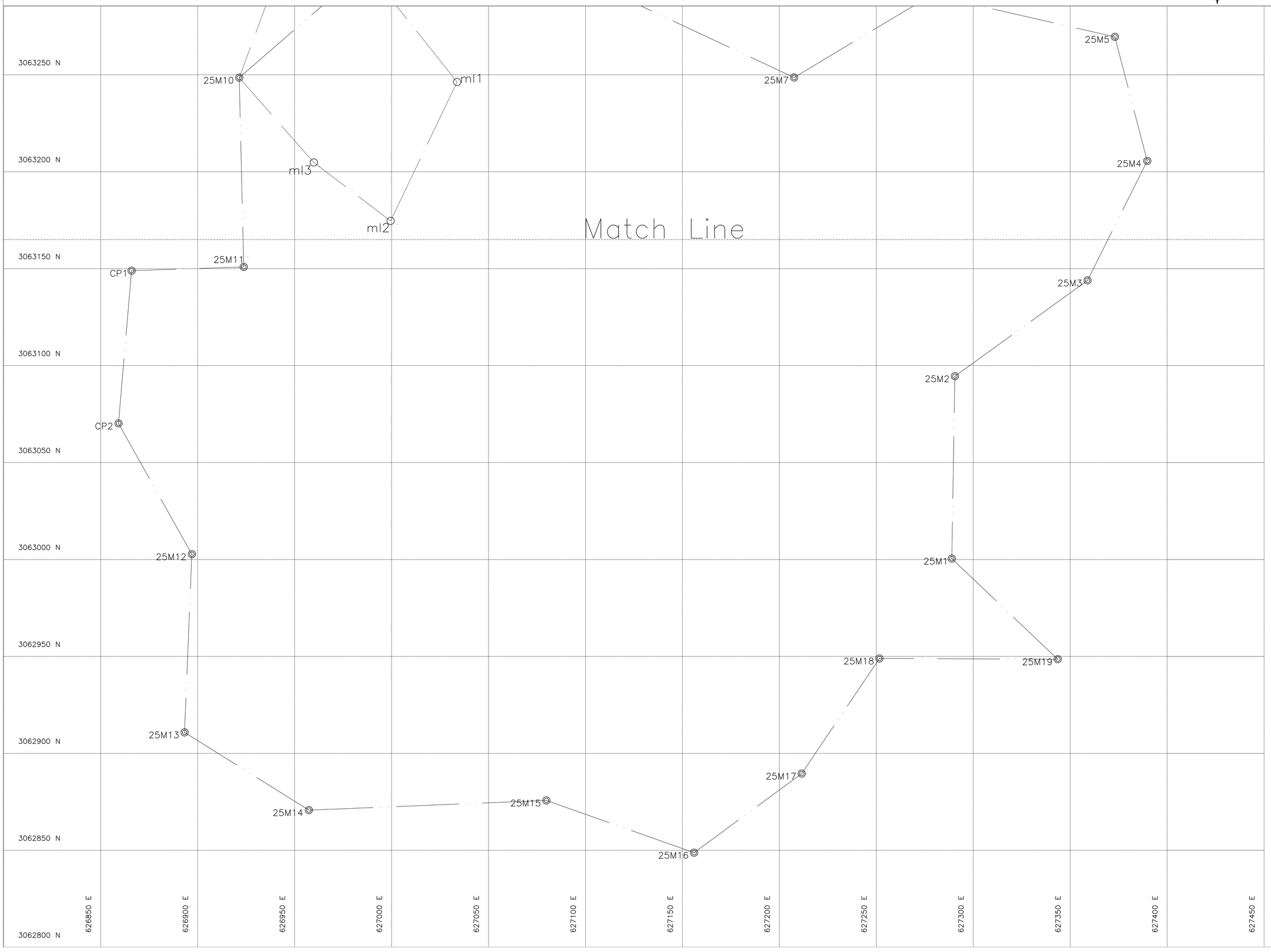
GROUP 25

PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184



MAJOR AND MINOR TRAVERSE

SCALE 1:1000



LEGENDS:

S.N.	DESCRIPTION	SYMBOL
1	MAJOR STATION	◎
2	MAJOR TRAVERSE LEG	
3	MINOR STATION	○
4	MINOR TRAVERSE LEG	

STATION COORDINATES (MAJOR)		
STATION	EASTING(m)	NORTHING(m)
CP1	626865.979	3063148.959
CP2	626859.243	3063070.157
M12	626897.049	3063002.716
M13	626893.311	3062910.831
M14	626957.403	3062870.718
M15	627079.861	3062875.710
M16	627156.053	3062848.774
M17	627211.569	3062889.565
M18	627251.648	3062948.944
M19	627343.603	3062948.546
M1	627288.976	3063000.444
M2	627290.564	3063094.520
M3	627359.004	3063143.938
M4	627389.754	3063205.533
M5	627373.091	3063269.557
M6	627278.514	3063291.167
M7	627207.635	3063248.580
M8	627111.736	3063293.401
M9	626986.953	3063305.275
M10	626921.513	3063248.492
M11	626923.814	3063150.877

STATION COORDINATES (MINOR)			
STATION	EASTING(m)	NORTHING(m)	ELEVATION(m)
mc1	627065.620	3063306.918	1300.632
mc2	627062.395	3063359.441	1298.472
mc3	626946.327	3063410.980	1306.276
mc4	626892.266	3063375.098	1306.275
mc5	626949.819	3063325.266	1304.140
ml1	627033.828	3063246.248	1303.188
ml2	626999.566	3063174.642	1304.155
ml3	626959.961	3063204.747	1306.282

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SCALE= 1:1000

GROUP 25

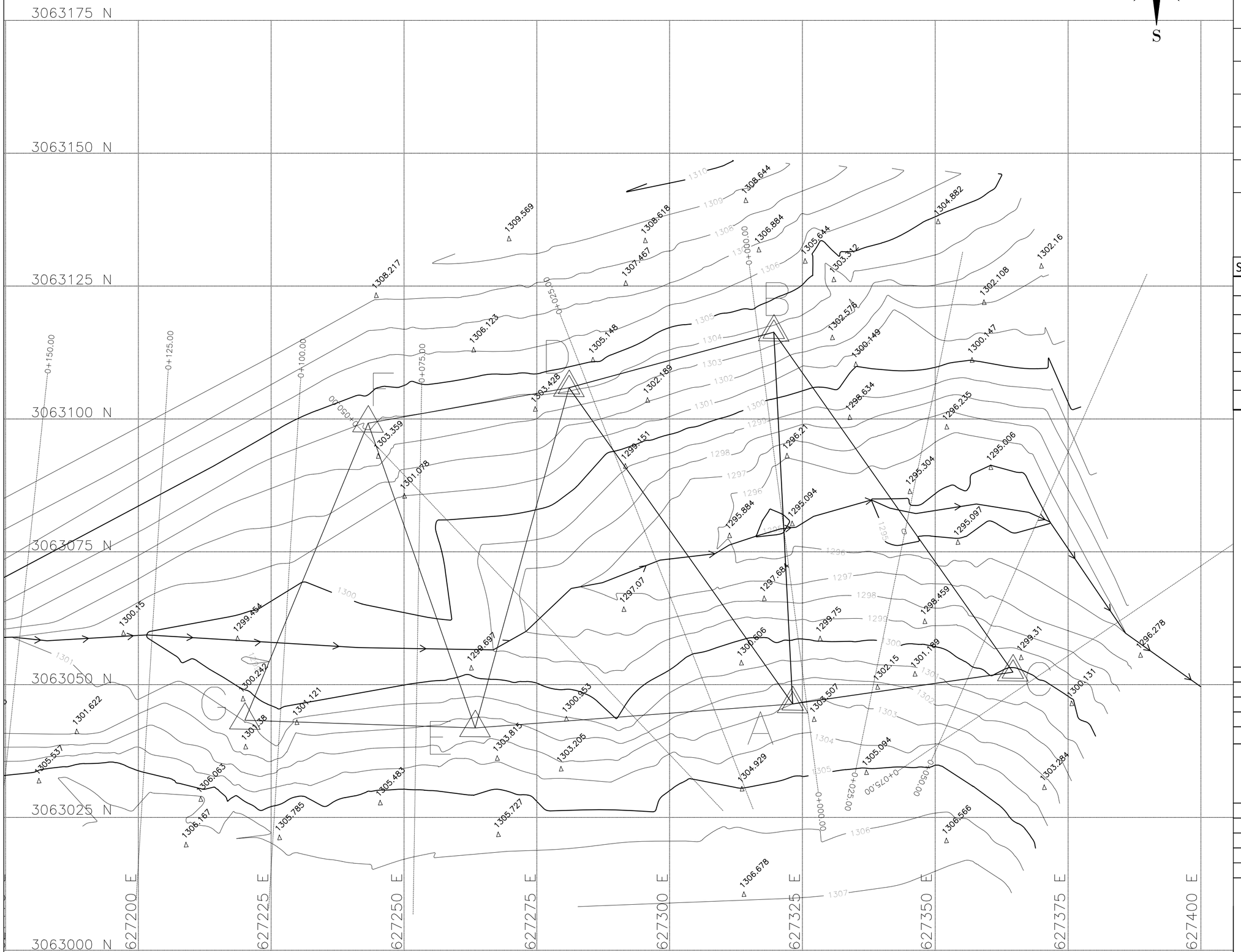
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

0 6.25 12.5 25 50m

ANNEX-B2:

Bridge Survey Maps and Graphs

TITLE: BRIDGE SITE
SCALE 1:500



LEGENDS

S.N.	DESCRIPTION	SYMBOL
1.	STATION	
2.	TRIANGULATION LINE	
3.	MAJOR CONTOUR	
4.	MINOR CONTOUR	
5.	FLOW DIRECTION	
6.	SECTION LINE	

HORIZONTAL AND VERTICAL CONTROL POINTS

COORDINATES

SN	Easting	Northing	RL	Remarks
1	27323.2	63046.37	1303.004	A
2	27319.64	63116.33	1303.702	B
3	27364.65	63052.36	1300.244	C
4	27281.1	63105.89	1304.206	D
5	27263.42	63041.86	1302.591	E
6	27220.09	63043.28	1300.955	G
7	27243.27	63099.21	1304.182	F

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TITLE: TOPOGRAPHIC MAP
BRIDGE SITE

SCALE 1:500

GROUP 25

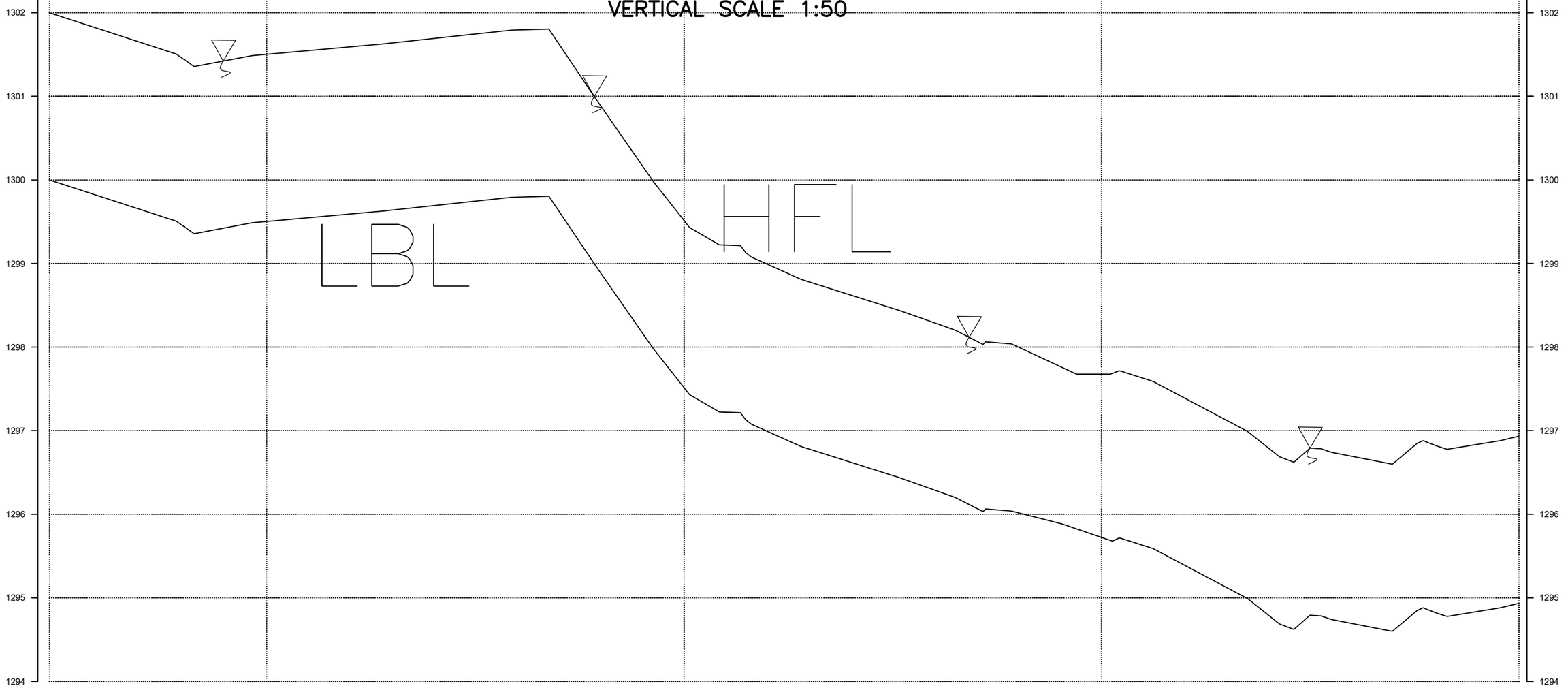
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

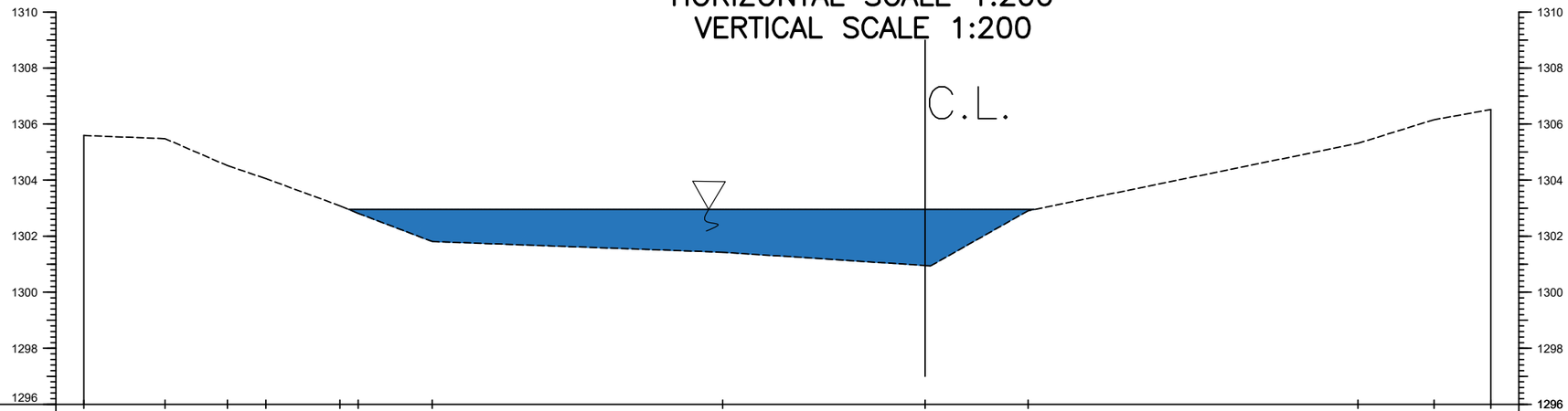
SHEET NO 1

TITLE: L–SECTION (BRIDGE SITE)
HORIZONTAL SCALE 1:500
VERTICAL SCALE 1:50



| | | |

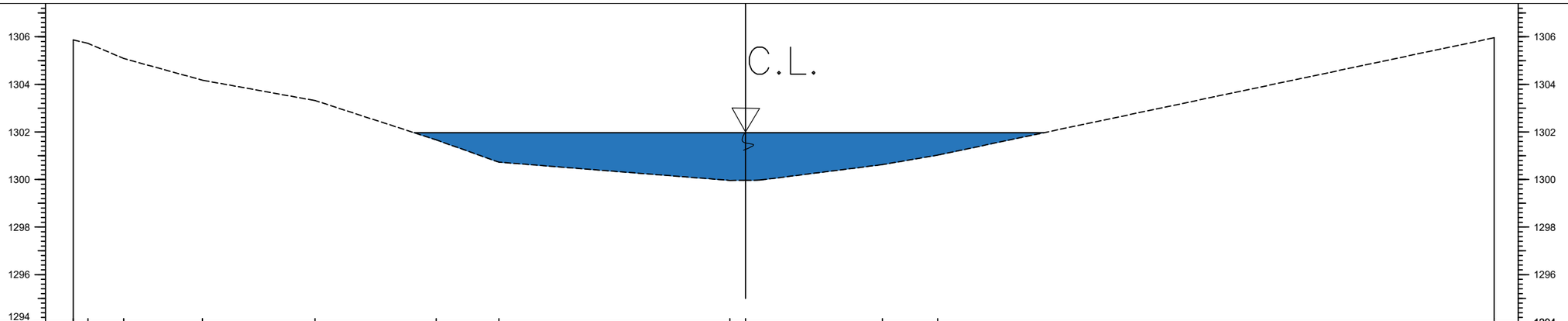
TITLE: X–SECTION OF RIVER (UPSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200



EXISTING LEVEL (m)	1305.592	1305.480	1304.517	1304.054	1303.084	1302.814	1301.808	1301.426	1300.953	1302.908	1305.313	1306.156	1306.518	
DISTANCE (m)	30.01	27.12	24.89	23.52	20.88	20.23	17.58	7.22	0.00	3.68	15.44	18.16	20.18	

-0+150.00

X–SECTION AT –0+150.00



EXISTING LEVEL (m)	1305.869	1305.723	1305.087	1304.175	1303.319	1301.666	1300.728	1299.962	1299.967	1300.626	1301.024	1305.959		
DISTANCE (m)	28.32	27.70	26.19	22.88	18.13	13.03	10.39	0.66	0.00	5.76	8.09	31.53		

-0+125.00

X–SECTION AT –0+125.00

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TITLE: X–SECTION OF RIVER (UPSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200

GROUP 25

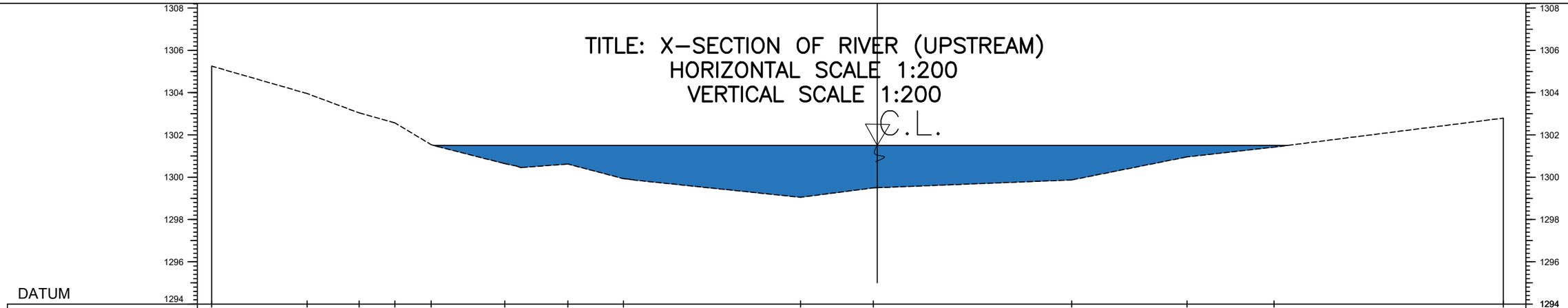
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No: 3

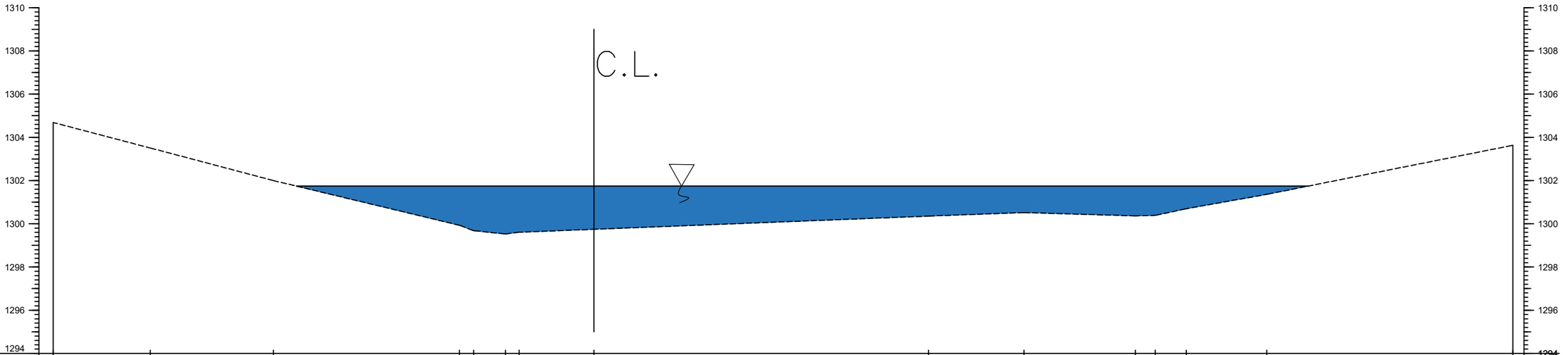
TITLE: X–SECTION OF RIVER (UPSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200



DATUM														
EXISTING LEVEL (m)	1305.254	1303.957	1303.041	1302.569	1301.540	1300.643	1300.623	1299.931	1299.049	1299.495	1299.869	1300.960	1301.421	1302.794
DISTANCE (m)	31.53	27.01	24.54	22.86	21.13	17.65	14.66	12.02	3.64	0.19	9.21	14.67	18.79	29.65

-0+100.00

X–SECTION AT –0+100.00



DATUM														
EXISTING LEVEL (m)	1304.686	1303.508	1301.998	1299.932	1299.681	1299.527	1299.608	1299.745	1300.355	1300.520	1300.365	1300.387	1300.696	1301.362	1303.633
DISTANCE (m)	25.04	20.54	14.84	6.23	5.57	4.09	3.46	0.00	15.49	19.91	25.07	25.99	27.43	31.16	42.55

-0+075.00

X–SECTION AT –0+075.00

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SURVEY INSTRUCTION COMITTEE

TITLE: X–SECTION OF RIVER (UPSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200

GROUP 25

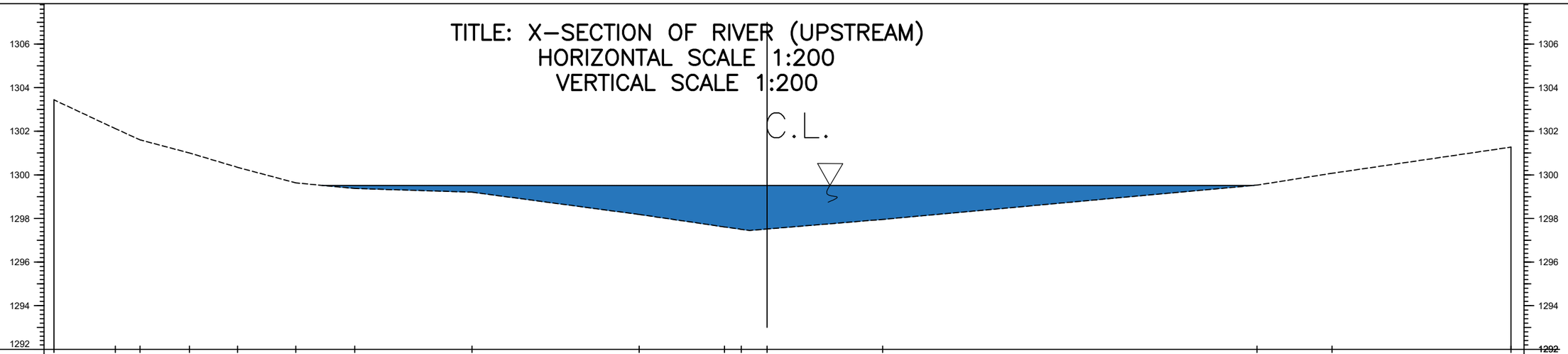
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

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REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No: 4

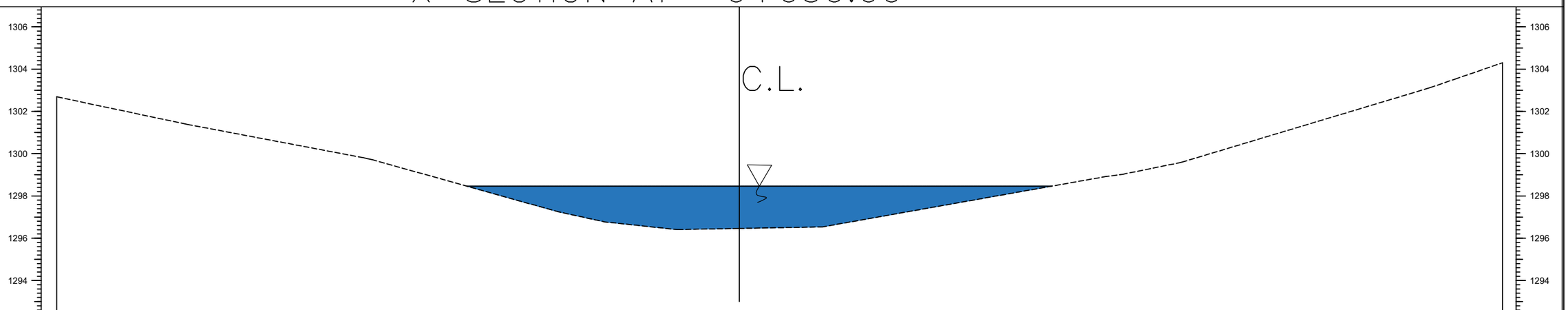
TITLE: X–SECTION OF RIVER (UPSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200



EXISTING LEVEL (m)	1303.440	1302.119	1301.603	1301.005	1300.346	1299.631	1299.376	1299.207	1298.184	1297.613	1297.502	1297.517	1297.956	1299.531	1300.071	1301.271
DISTANCE (m)	32.75	29.93	28.79	26.53	24.32	21.64	18.93	13.56	5.87	1.96	1.18	0.00	5.31	22.51	25.95	34.17

-0+050.00

X–SECTION AT –0+050.00



EXISTING LEVEL (m)	1302.691	1301.390	1299.804	1297.252	1296.773	1296.410	1296.464	1296.536	1298.919	1299.025	1299.584	1303.123	1303.560	1304.300
DISTANCE (m)	32.35	26.18	17.79	8.58	6.38	2.96	0.00	3.94	17.37	18.17	20.92	32.74	34.01	36.17

-0+025.00

X–SECTION AT –0+025.00

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TITLE: X–SECTION OF RIVER (UPSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200

GROUP 25

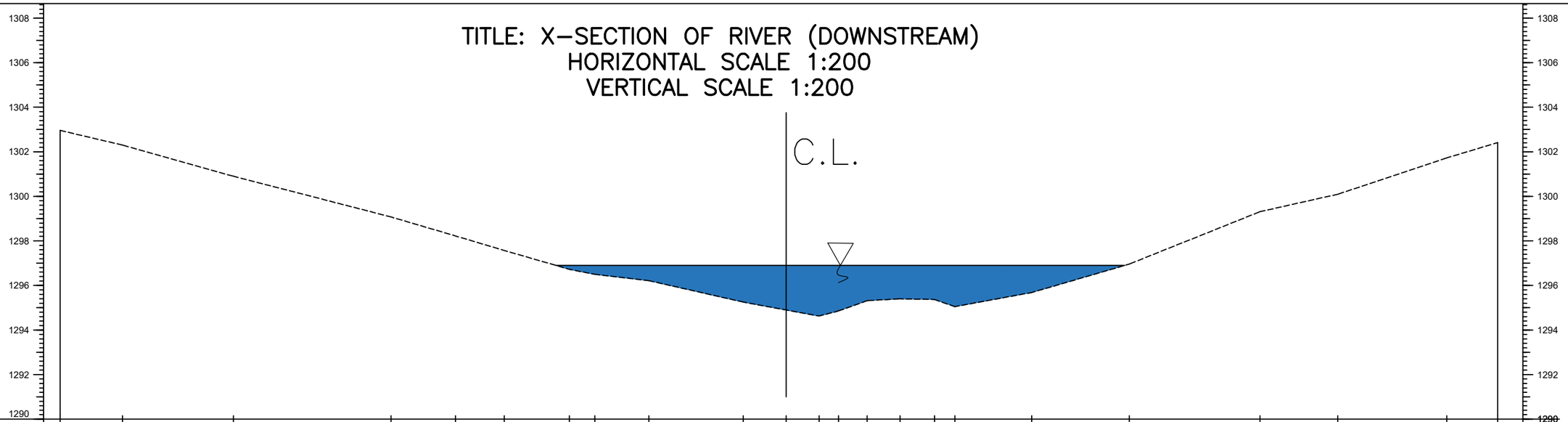
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

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SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No: 5

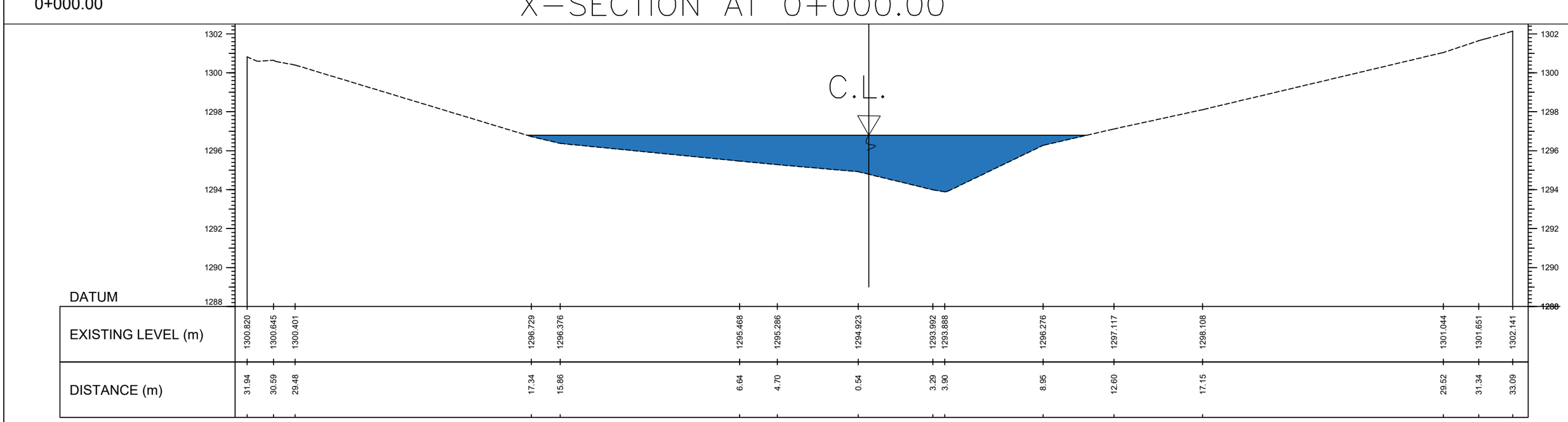
TITLE: X–SECTION OF RIVER (DOWNSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200



DATUM																					
EXISTING LEVEL (m)	1302.960	1302.300	1300.901	1299.075	1298.224	1297.569	1296.726	1296.499	1296.216	1295.254	1294.900	1294.627	1294.856	1295.319	1295.400	1295.371	1295.049	1295.684	1296.960	1299.312	1300.095	1301.727	1302.413
DISTANCE (m)	32.63	29.83	24.84	17.76	14.86	12.67	9.74	8.60	6.16	1.93	0.00	1.48	2.36	3.64	5.12	6.67	7.59	11.05	15.43	21.31	24.80	29.71	31.99

0+000.00

X–SECTION AT 0+000.00



DATUM																					
EXISTING LEVEL (m)	1300.820	1300.645	1300.401	1296.729	1296.376	1295.468	1295.286	1294.923	1293.992	1293.888	1296.276	1297.117	1298.108	1301.044	1301.651	1302.141					
DISTANCE (m)	31.94	30.59	29.48	17.34	15.86	6.64	4.70	0.54	3.29	3.90	8.95	12.60	17.15	29.52	31.34	33.09					

0+025.00

X–SECTION AT 0+025.00

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TITLE: X–SECTION OF RIVER (DOWNSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200

GROUP 25

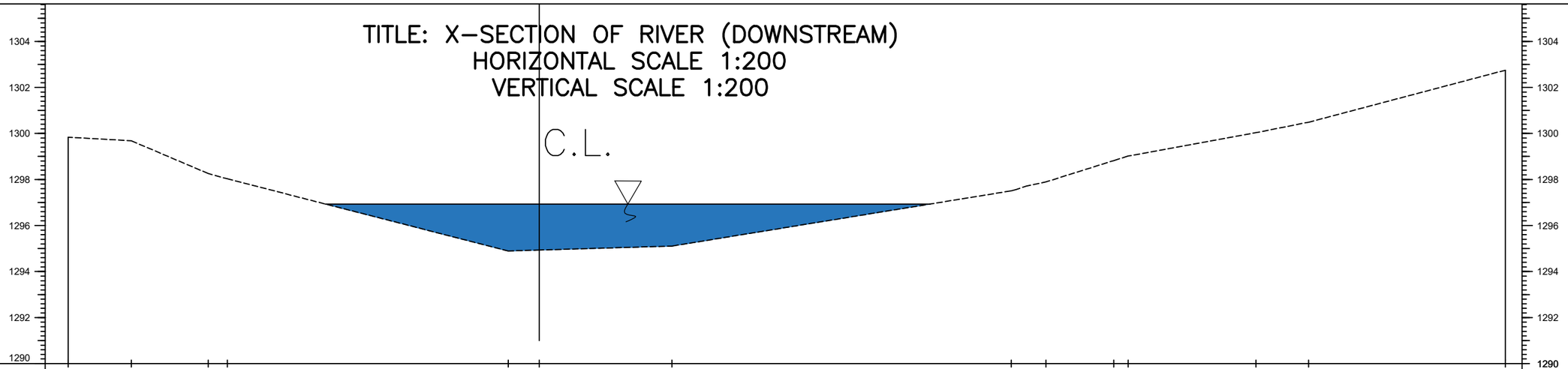
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No: 6

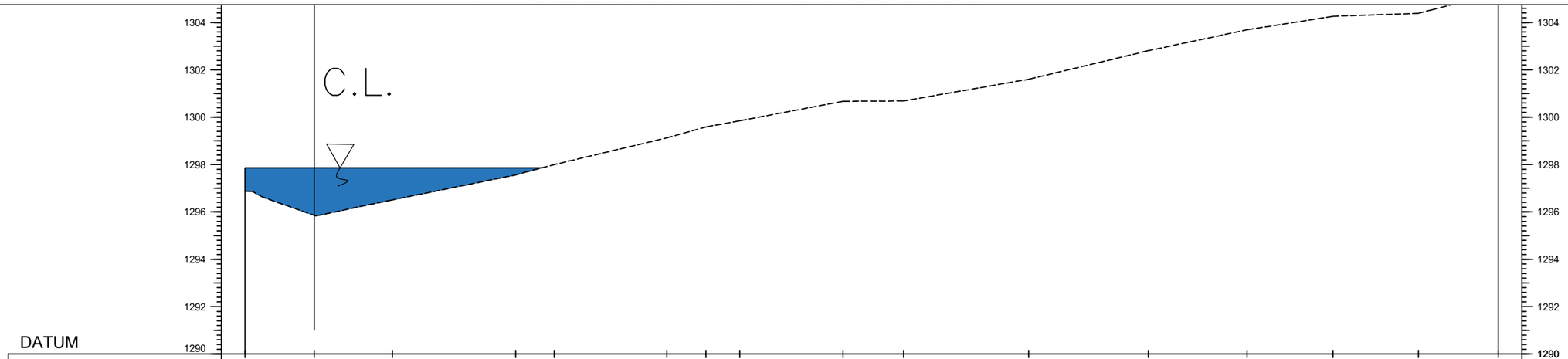
TITLE: X–SECTION OF RIVER (DOWNSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200



DATUM																							
EXISTING LEVEL (m)	1299.837	1299.682	1298.257	1298.028	1294.894	1294.934	1295.106	1297.507	1297.895	1298.822	1299.017	1300.033	1300.488	1302.747									
DISTANCE (m)	20.47	17.73	14.38	13.55	1.35	0.00	5.77	20.51	22.02	24.97	25.59	31.15	33.43	41.98									

0+050.00

X–SECTION AT 0+050.00



DATUM																							
EXISTING LEVEL (m)	1296.874	1295.856	1296.507	1297.562	1297.997	1299.125	1299.589	1299.849	1300.670	1300.684	1301.599	1302.807	1303.688	1304.261	1304.385	1305.260							
DISTANCE (m)	2.92	0.00	3.30	8.50	10.13	14.89	16.54	17.97	22.34	24.88	30.17	35.23	39.39	43.02	46.62	50.00							

0+075.00

X–SECTION AT 0+075.00

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TITLE: X–SECTION OF RIVER (DOWNSTREAM)
HORIZONTAL SCALE 1:200
VERTICAL SCALE 1:200

GROUP 25

PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PARVEEN	080BCE166
SUYOG REGMI	080BCE184

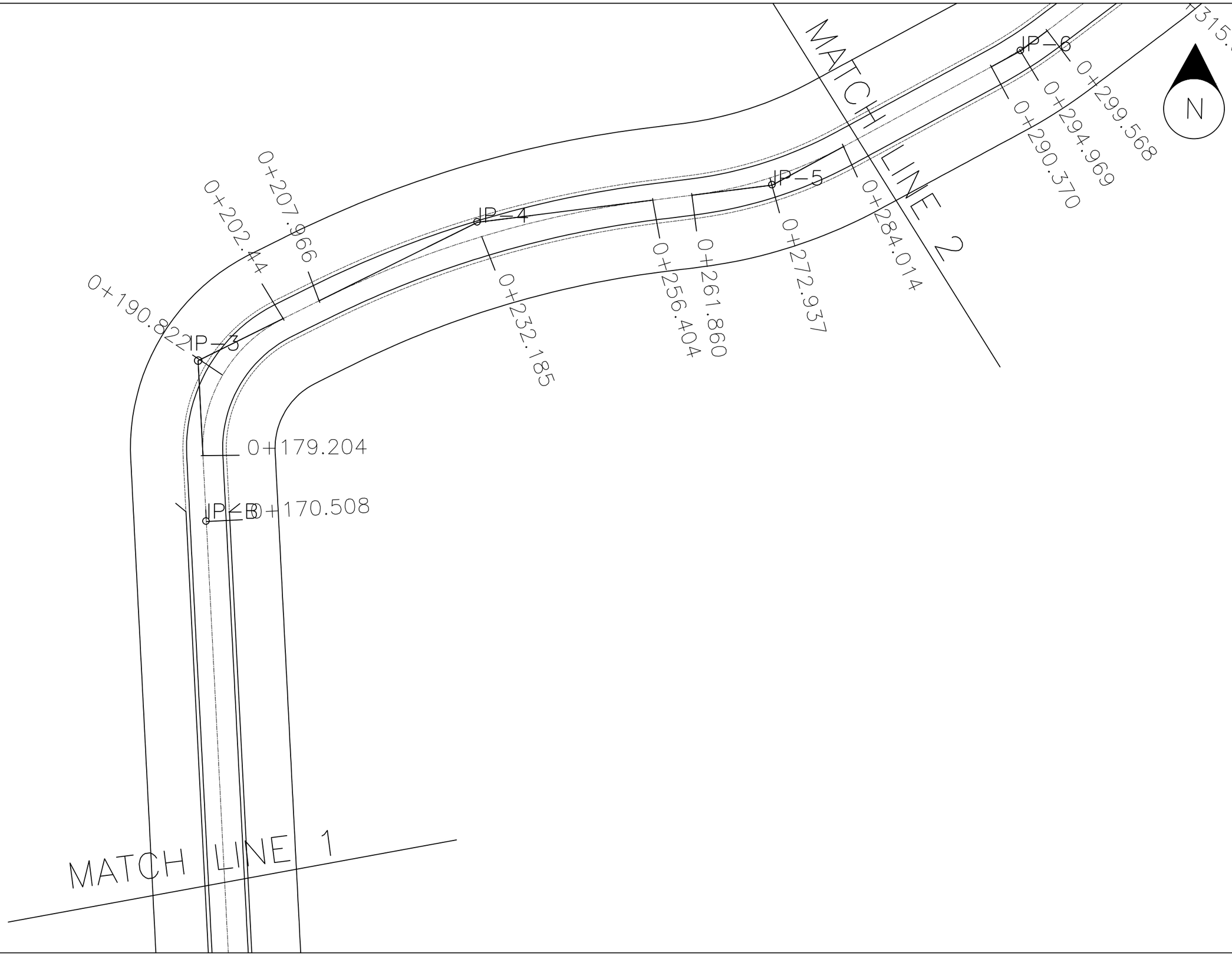
GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No: 7

ANNEX-B3:

Road Alignment Survey Maps and Graphs



LEGENDS

	CENTER LINE
	FORMATION WIDTH
	ROW
	INTESECTION POINT
	BRIDGE/CULVERT
MATCH LINE	MATCH LINE
	SIDE DRAIN

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DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE: ROAD ALIGNMENT PLAN

SCALE 1:500

GROUP 25

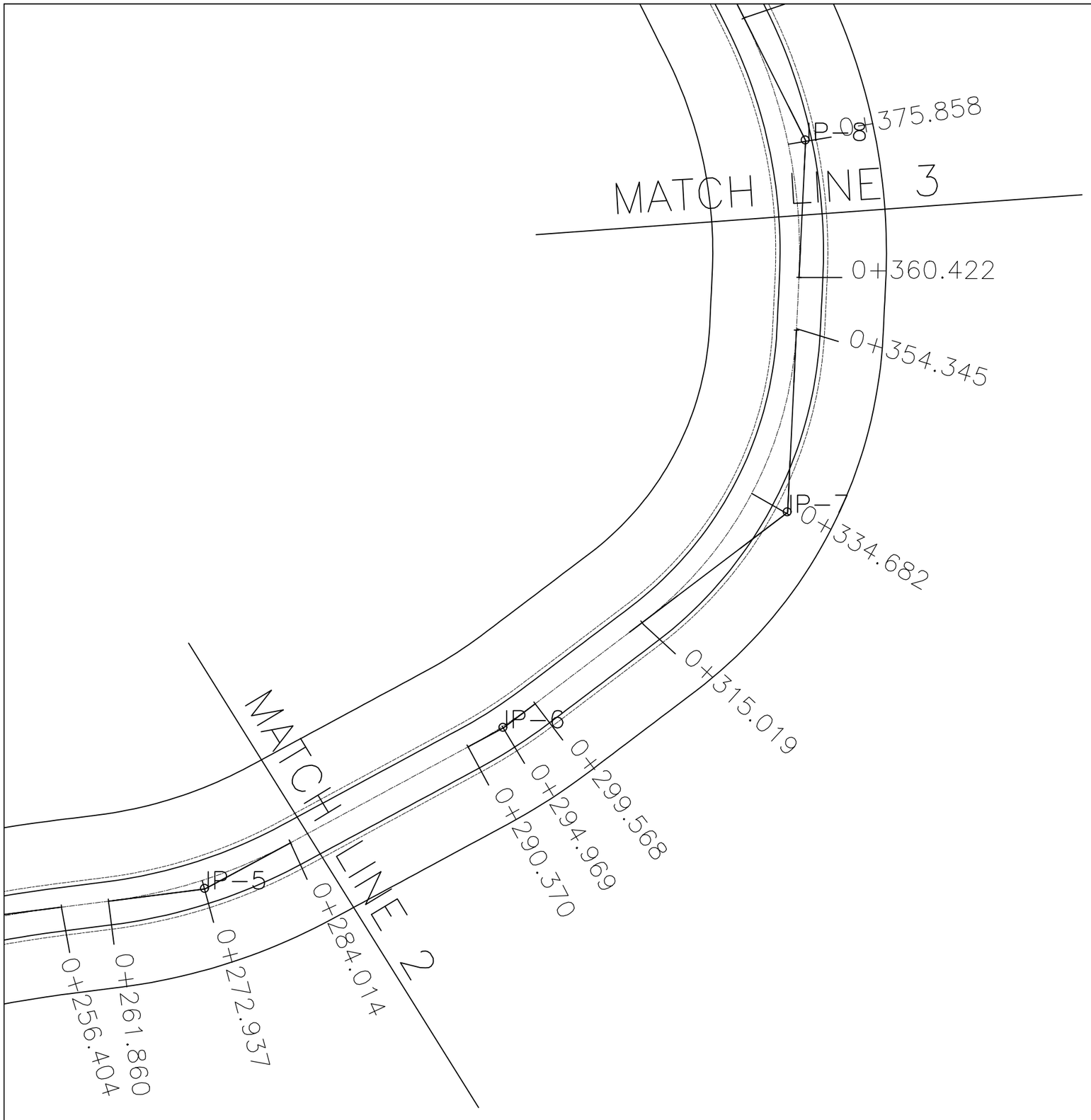
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE146
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

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REEVA AGRAWL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No:

2



N

LEGENDS

—·—·—·—	CENTER LINE
———	FORMATION WIDTH
———	ROW
○	INTESECTION POINT
	BRIDGE/CULVERT
MATCH LINE	MATCH LINE
- - - - - - - -	SIDE DRAIN

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DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE: ROAD ALIGNMENT PLAN

SCALE 1:500

GROUP 25

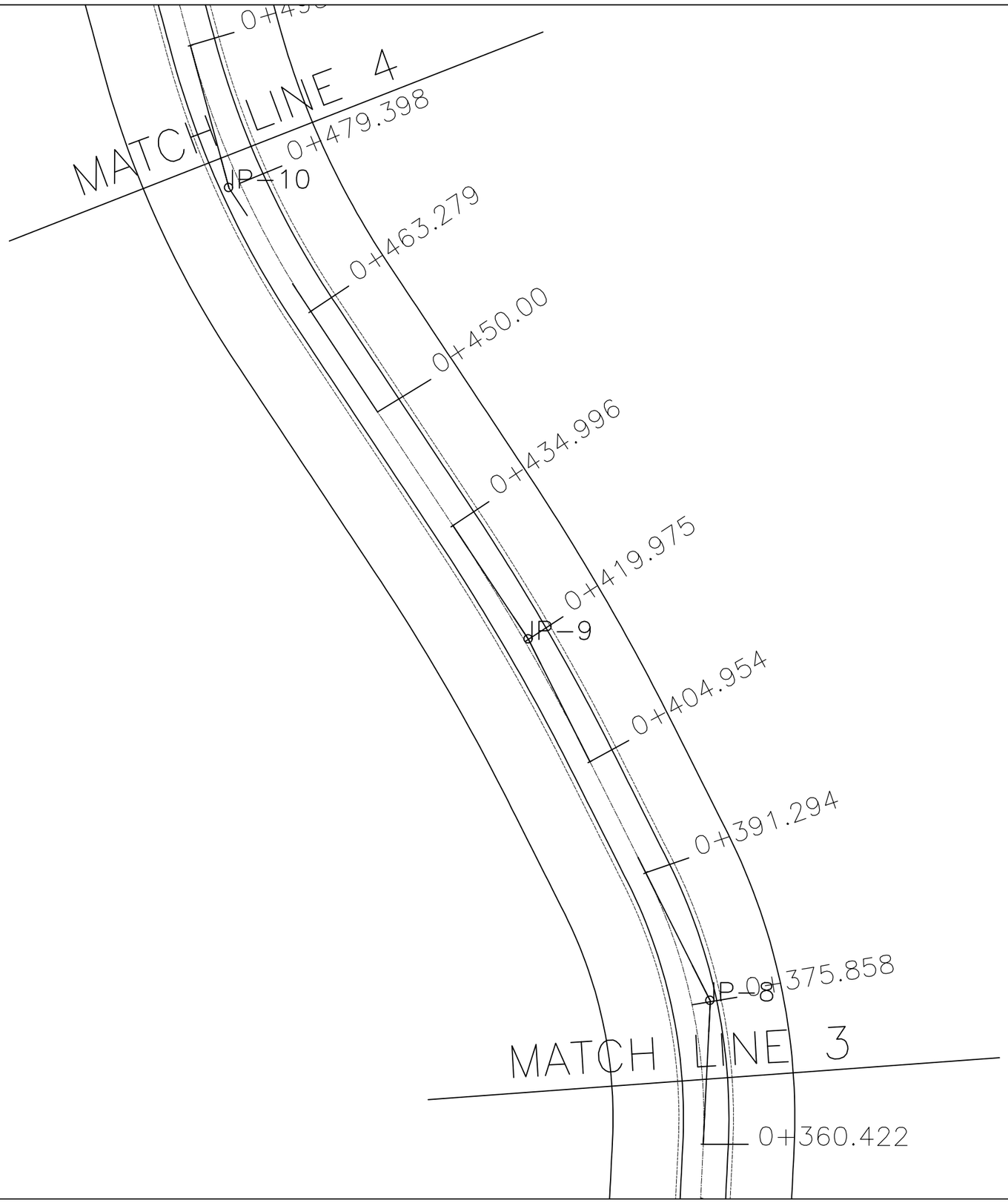
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE146
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No:

3



N

LEGENDS

—·—·—·—	CENTER LINE
———	FORMATION WIDTH
———	ROW
○	INTESECTION POINT
⌒	BRIDGE/CULVERT
MATCH LINE	MATCH LINE
- - - - - - -	SIDE DRAIN

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PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE: ROAD ALIGNMENT PLAN
SCALE 1:500

GROUP 25

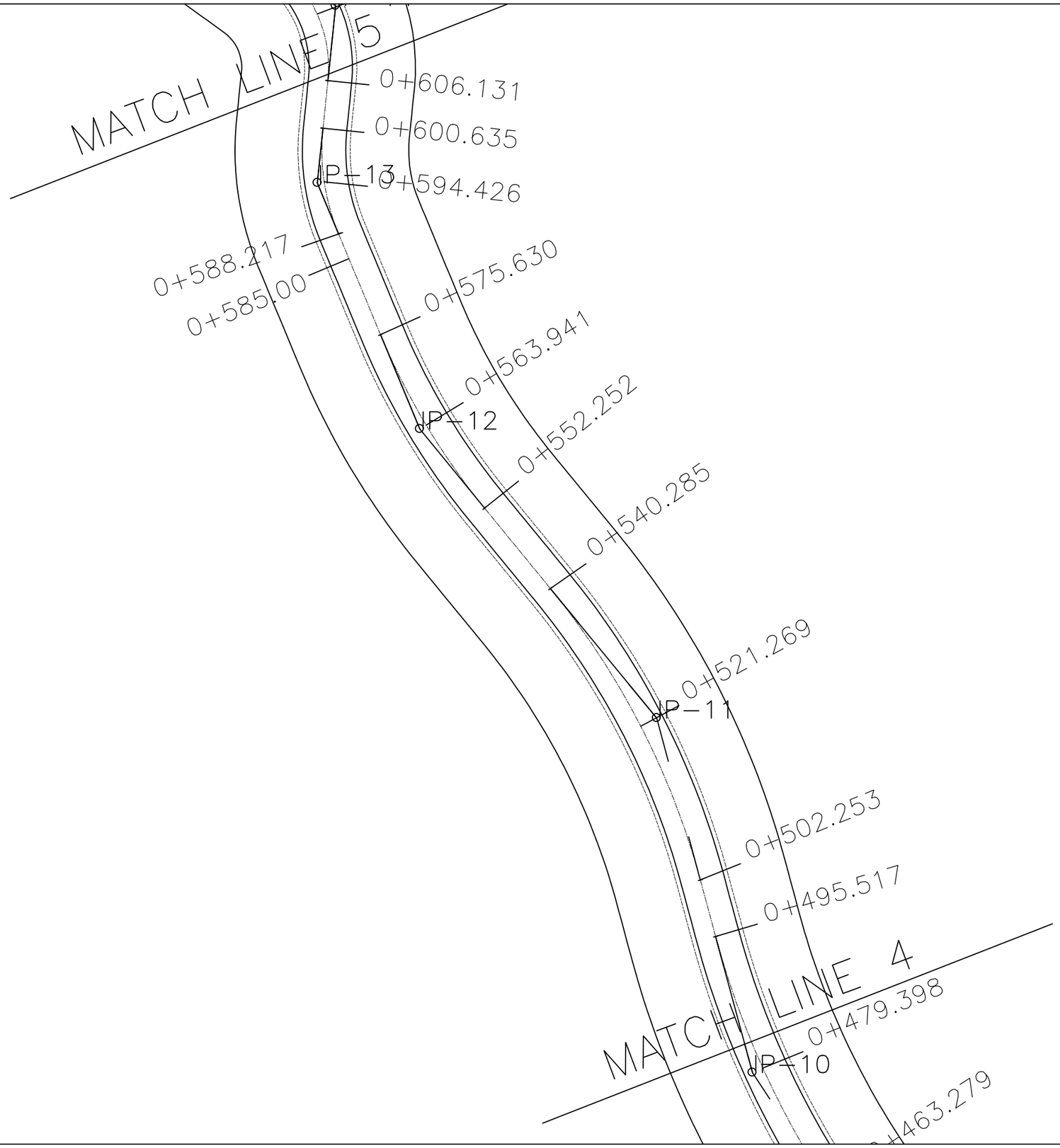
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE146
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GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No:

4



N

LEGENDS

—·—·—·—	CENTER LINE
———	FORMATION WIDTH
———	ROW
○	INTESECTION POINT
	BRIDGE/CULVERT
MATCH LINE	MATCH LINE
- - - - - - - -	SIDE DRAIN

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PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE: ROAD ALIGNMENT PLAN
SCALE 1:500

GROUP 25

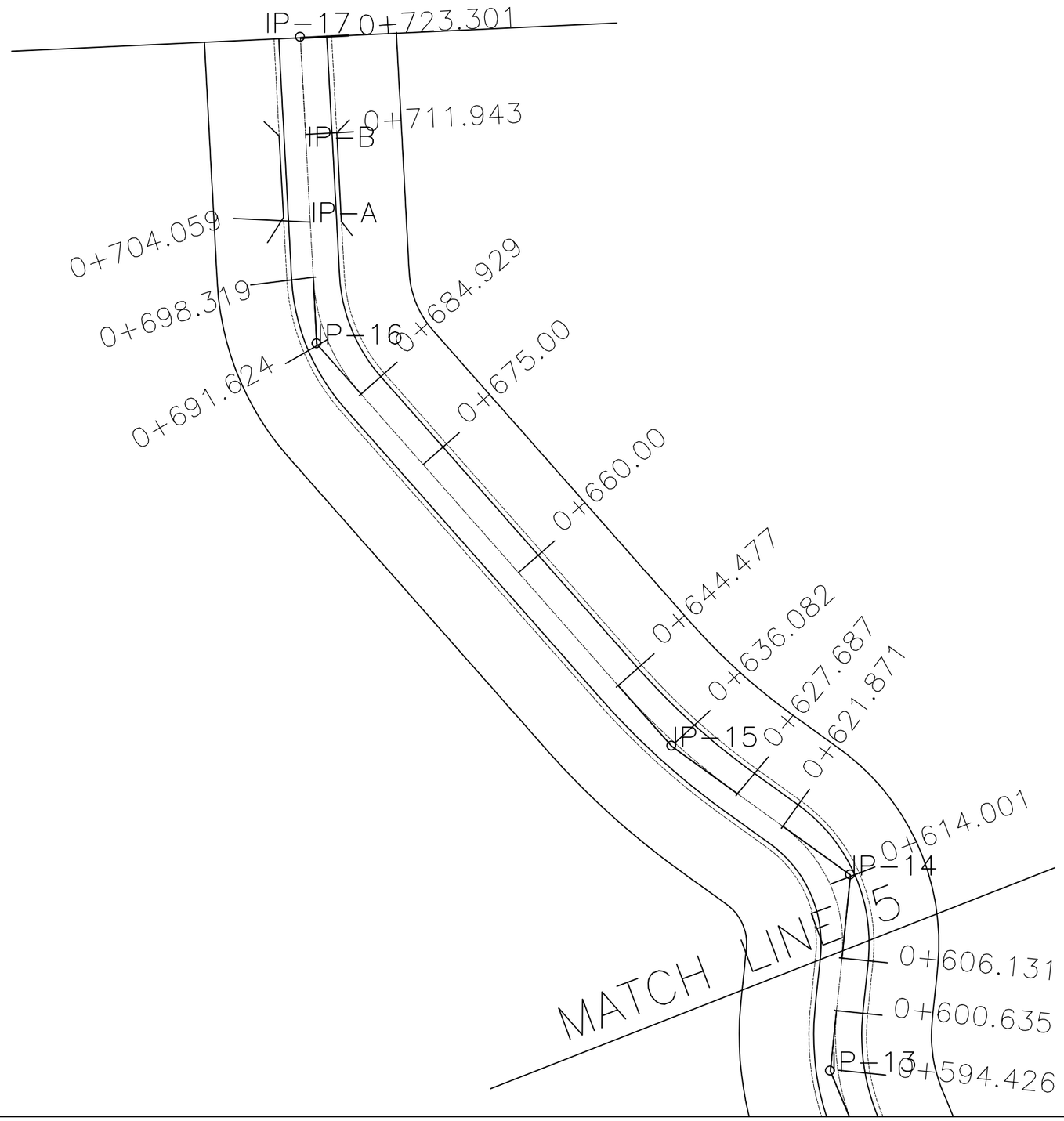
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE146
SNEHA PRAVEEN	080BCE166
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GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No:

5



N

LEGENDS

—·—·—·—	CENTER LINE
———	FORMATION WIDTH
———	ROW
○	INTESECTION POINT
	BRIDGE/CULVERT
MATCH LINE	MATCH LINE
- - - - - - - -	SIDE DRAIN

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PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE: ROAD ALIGNMENT PLAN
SCALE 1:500

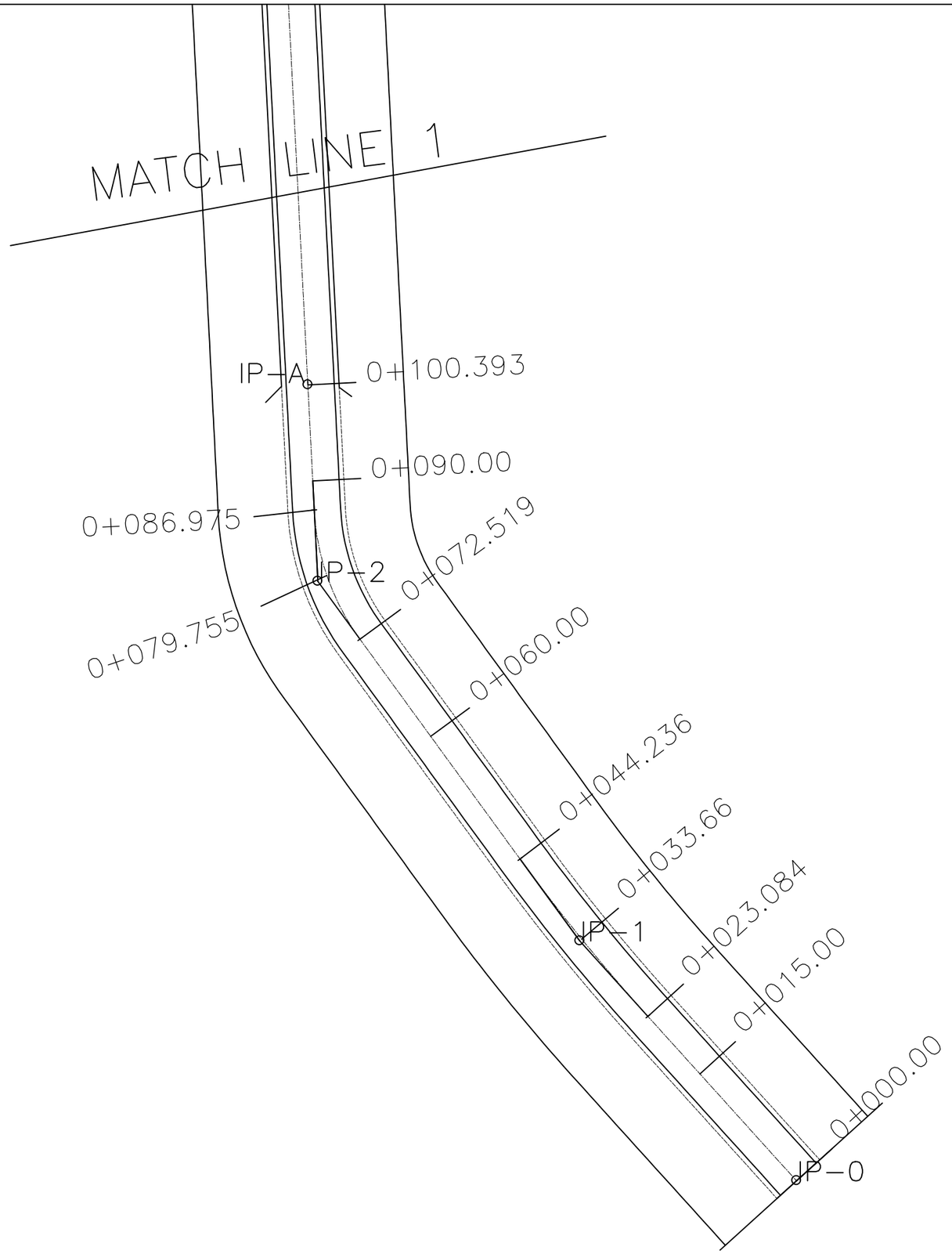
GROUP 25

PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE146
SNEHA PRAVEEN	080BCE166
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GROUP 26

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REEVA AGRAWL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

Sheet No:
6



N

LEGENDS

- CENTER LINE
- FORMATION WIDTH
- ROW
- INTESECTION POINT
- BRIDGE/CULVERT
- MATCH LINE — MATCH LINE
- SIDE DRAIN

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DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE: ROAD ALIGNMENT PLAN

SCALE 1:500

GROUP 25

PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE146
SNEHA PRAVEEN	080BCE166
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TULSI SHARMA	080BCE185

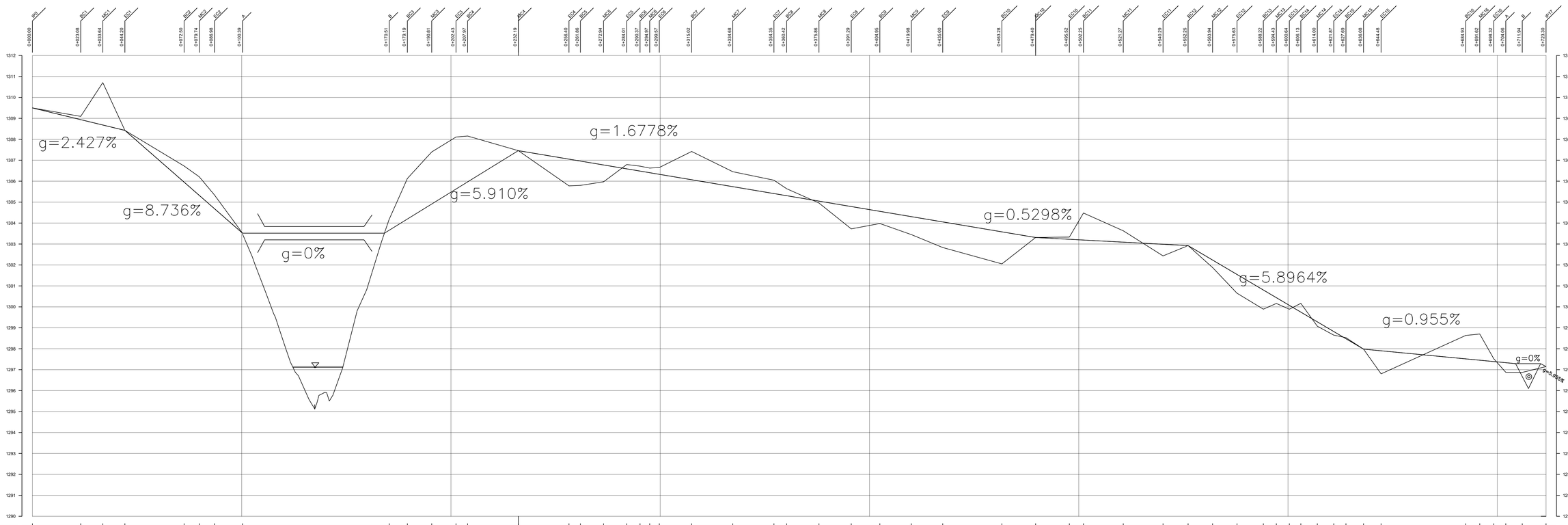
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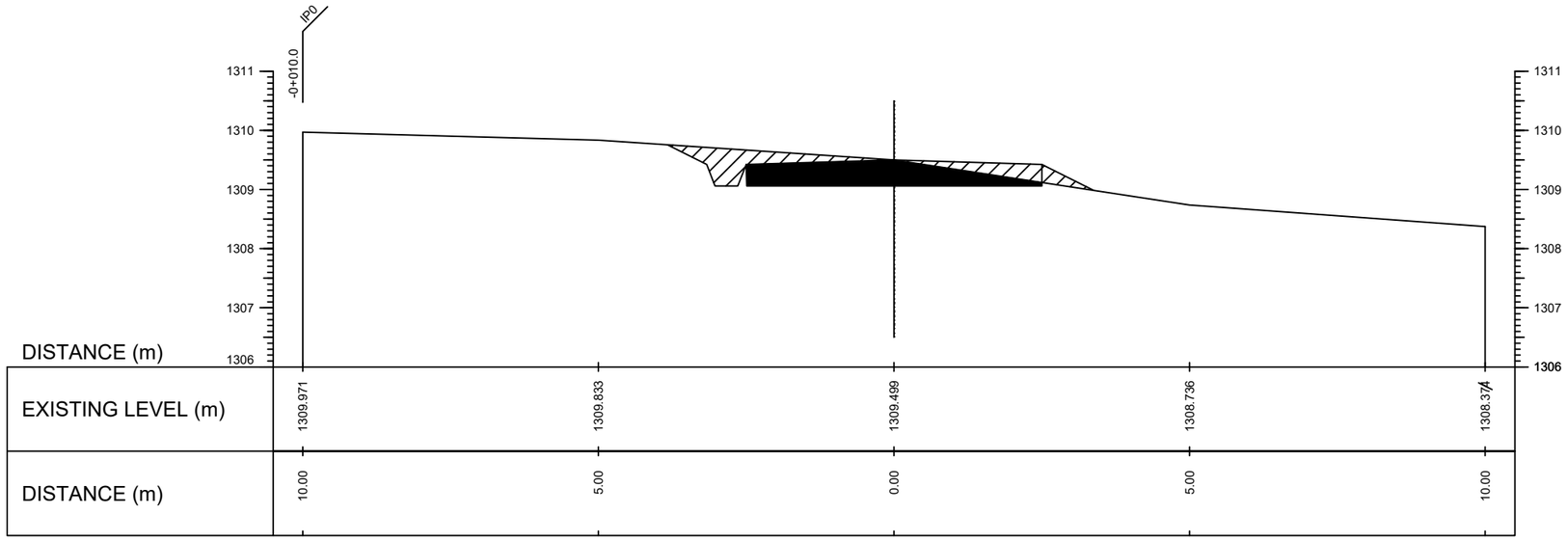
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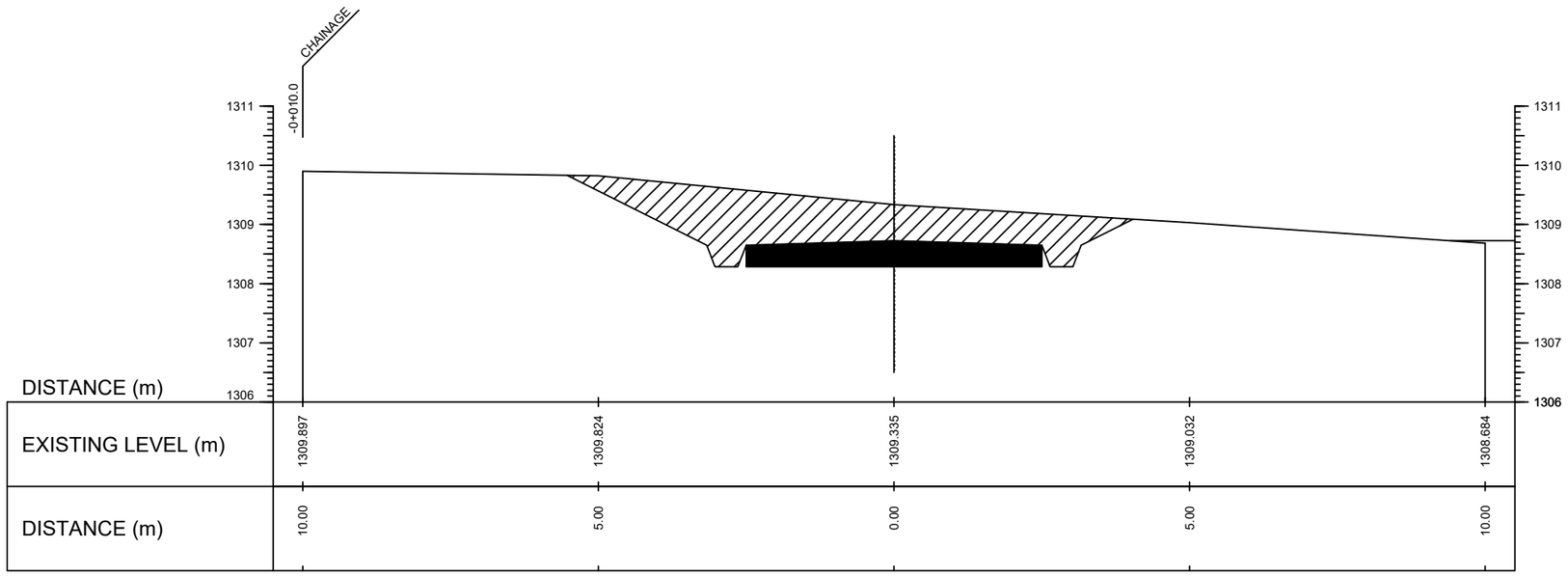
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| | | |



0+000.00



0+015.00

TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

TITLE:ROAD ALIGNMENT CROSS SECTION

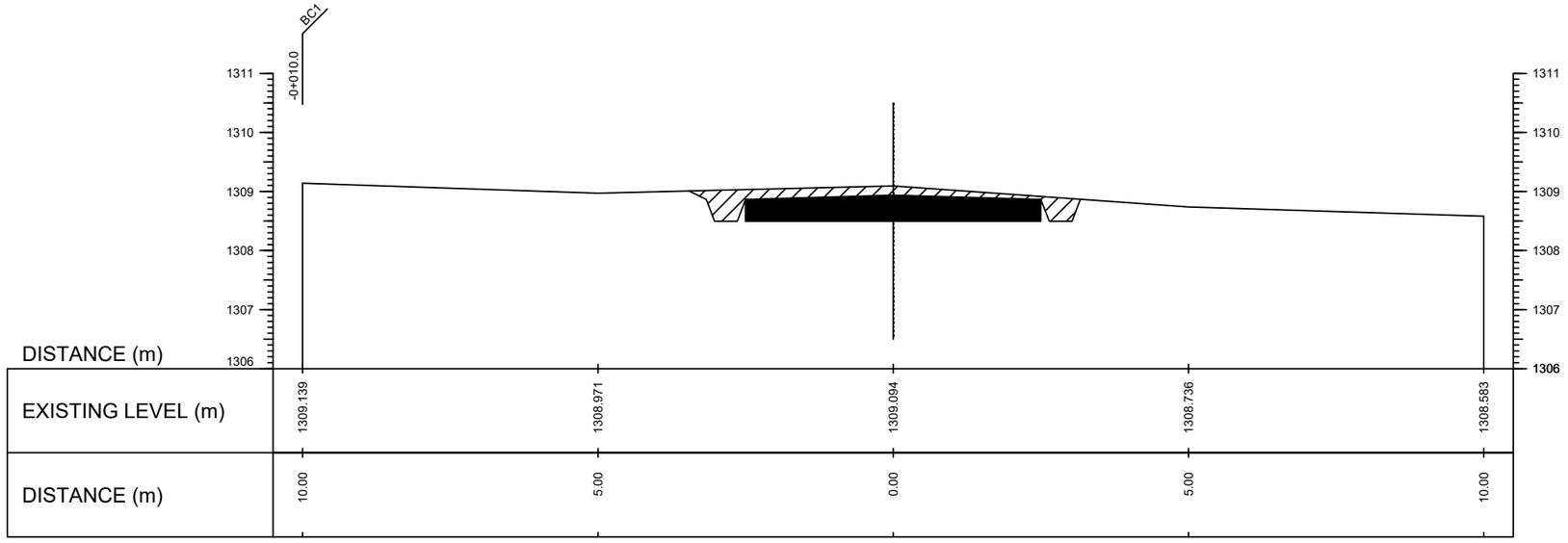
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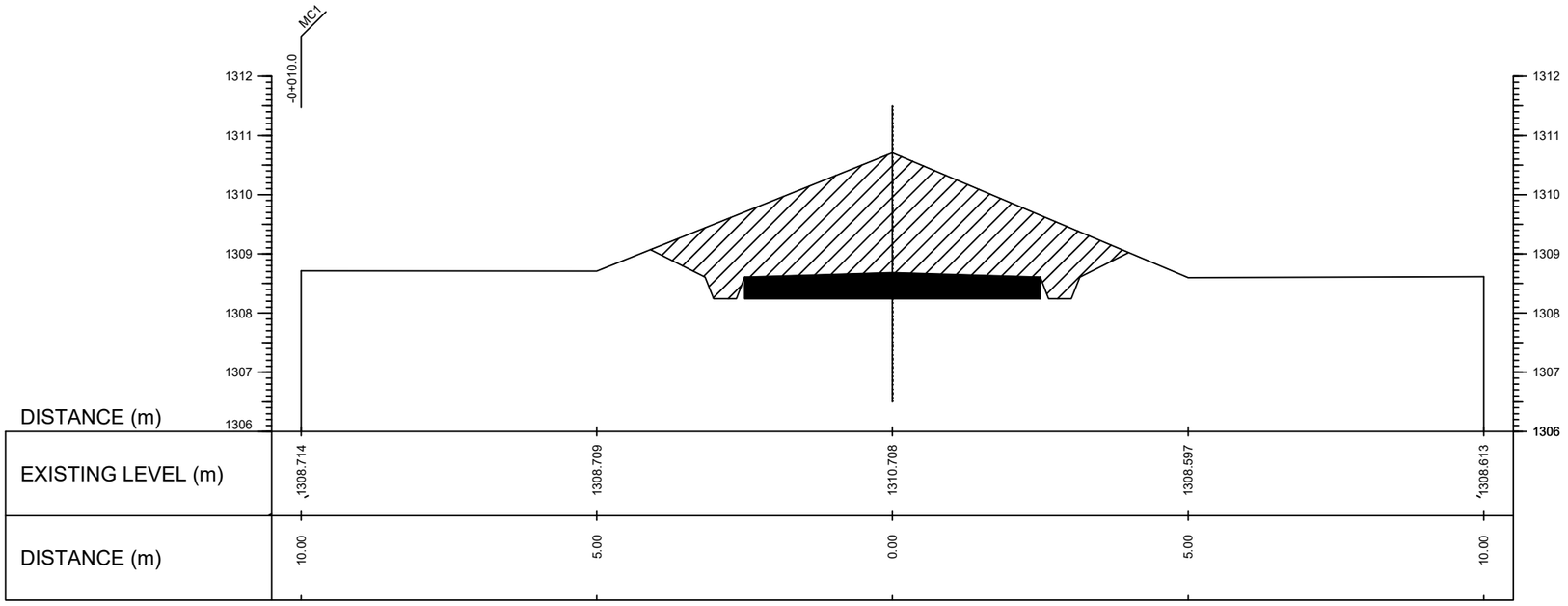
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+023.08



0+033.64

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INSTITUTE OF ENGINEERING
PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

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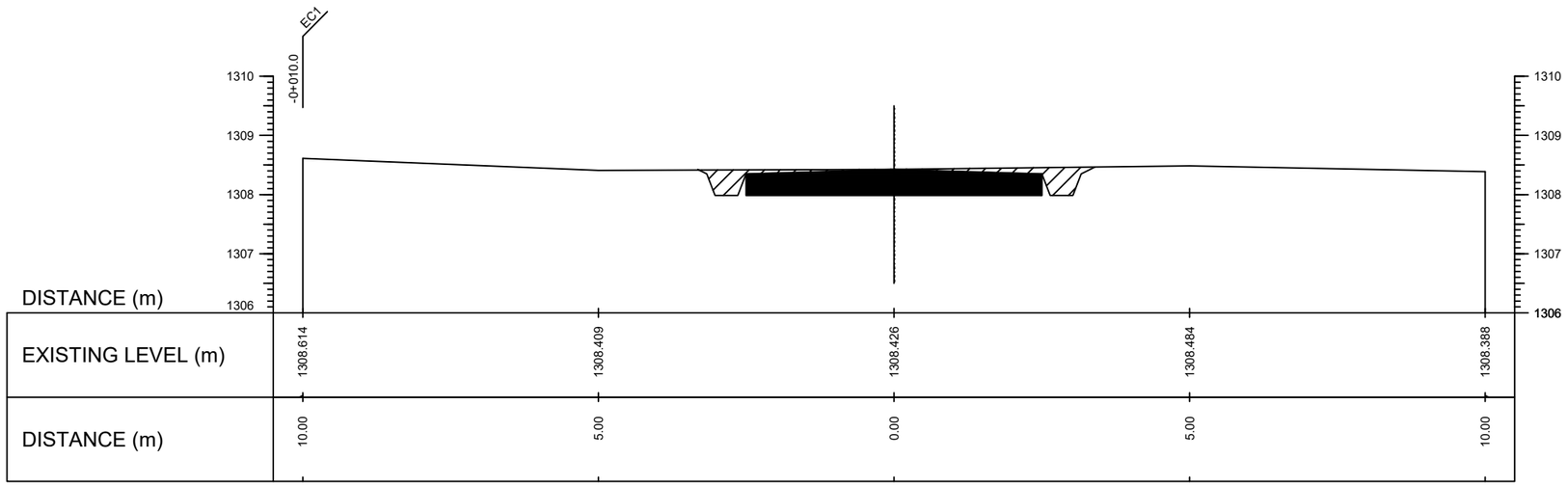
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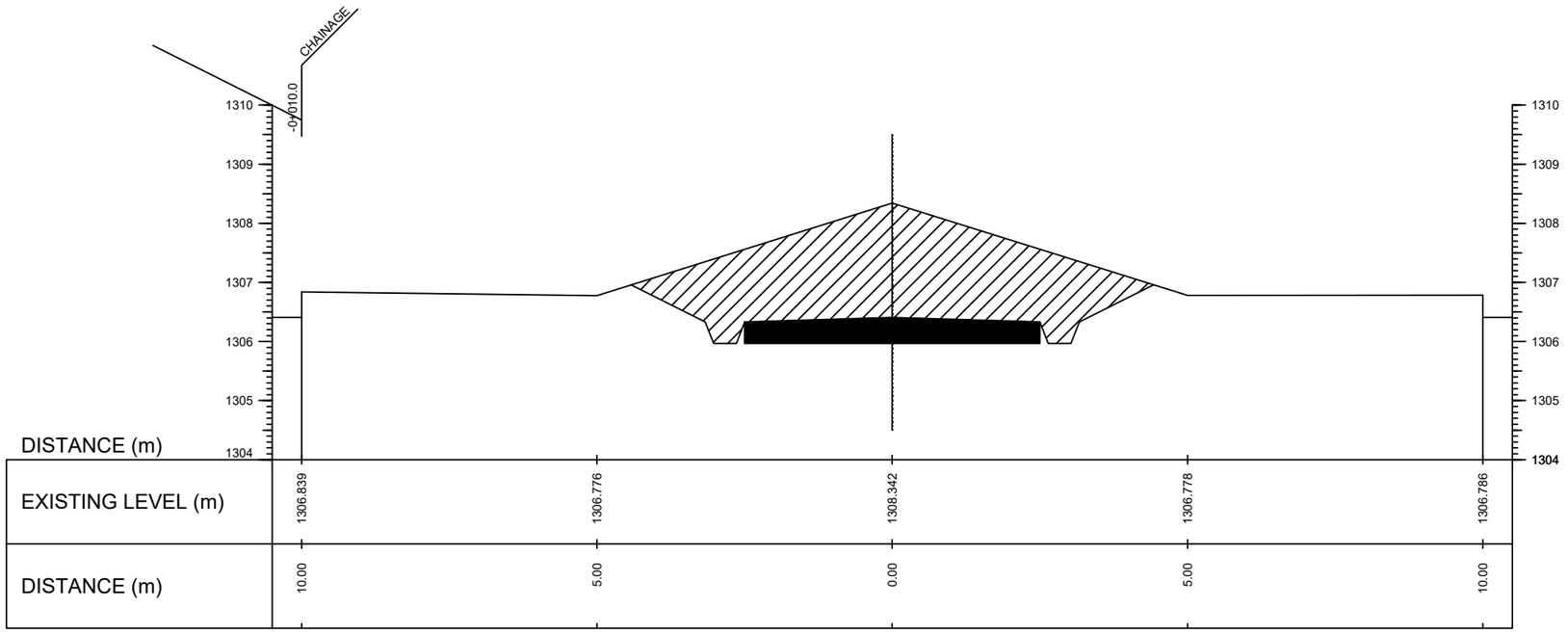
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+044.20



0+045.00

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PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

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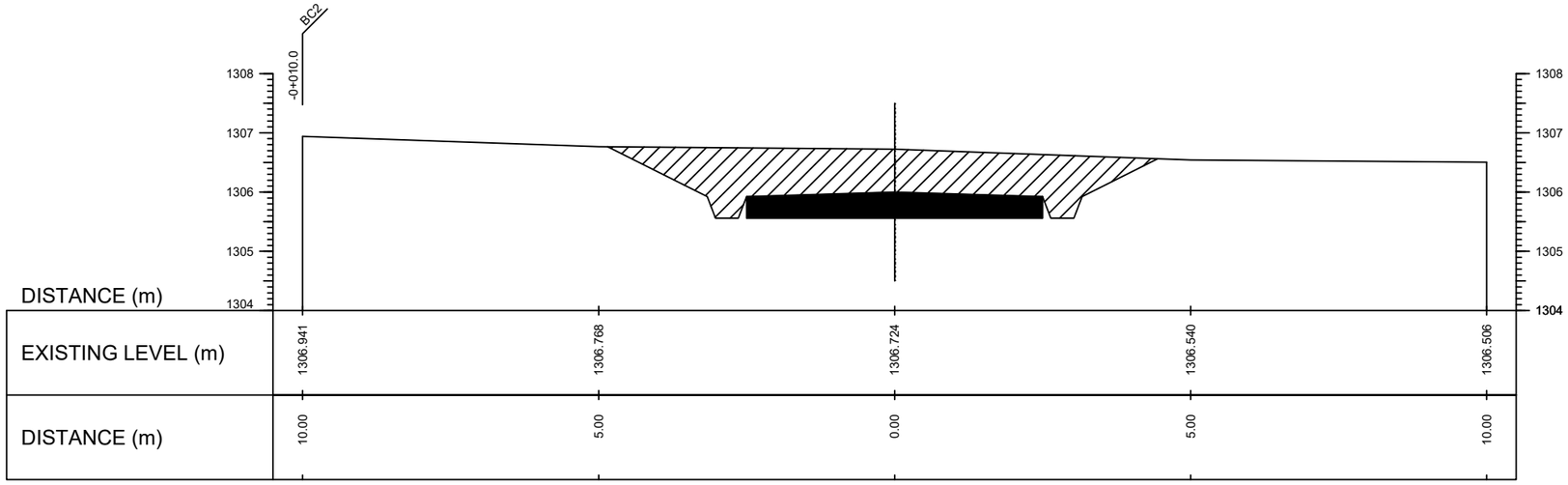
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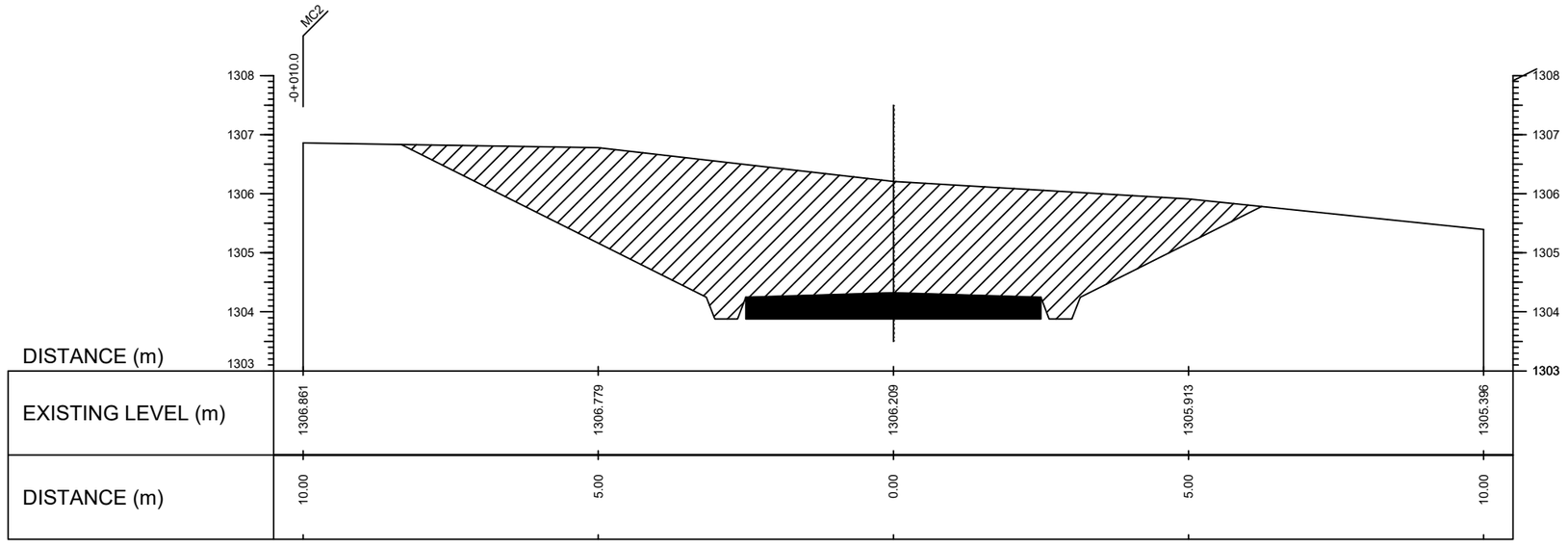
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 31

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+072.50



0+079.74

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SURVEY INSTRUCTION COMITTEE

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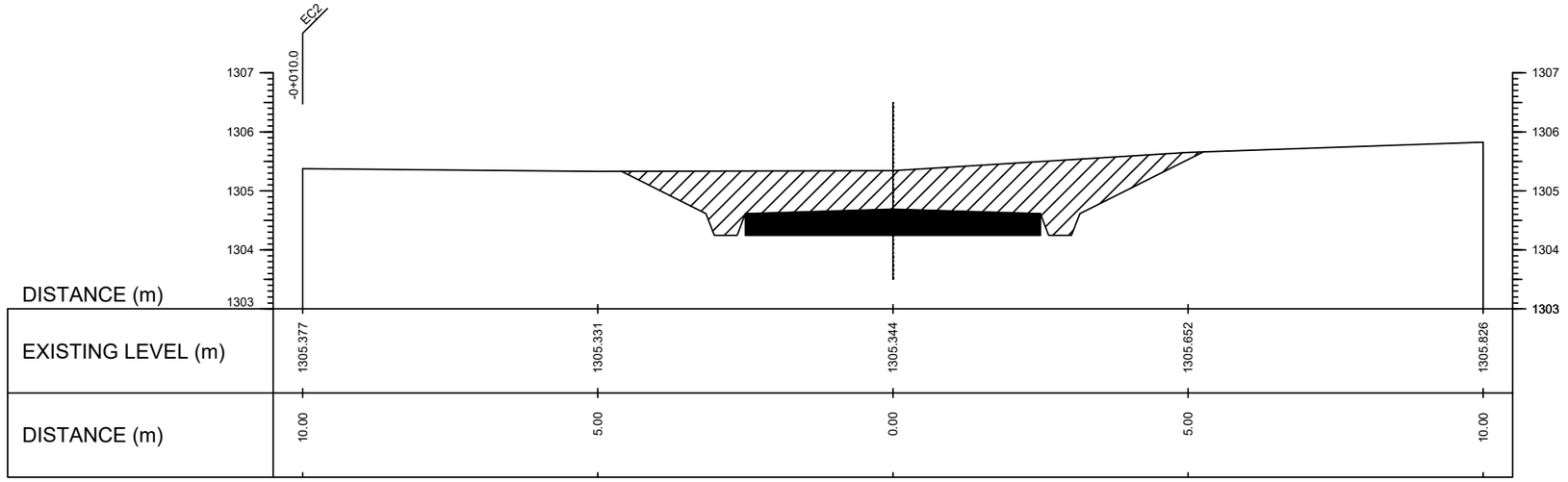
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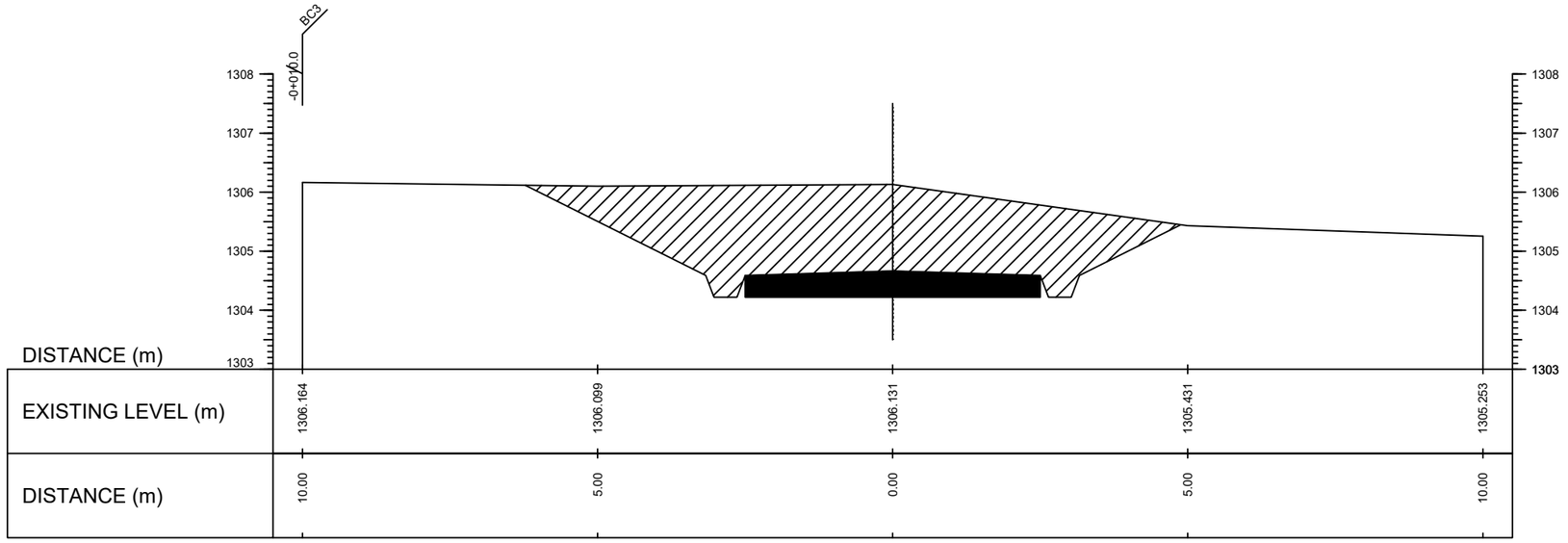
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+086.98



0+179.19

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SURVEY INSTRUCTION COMITTEE

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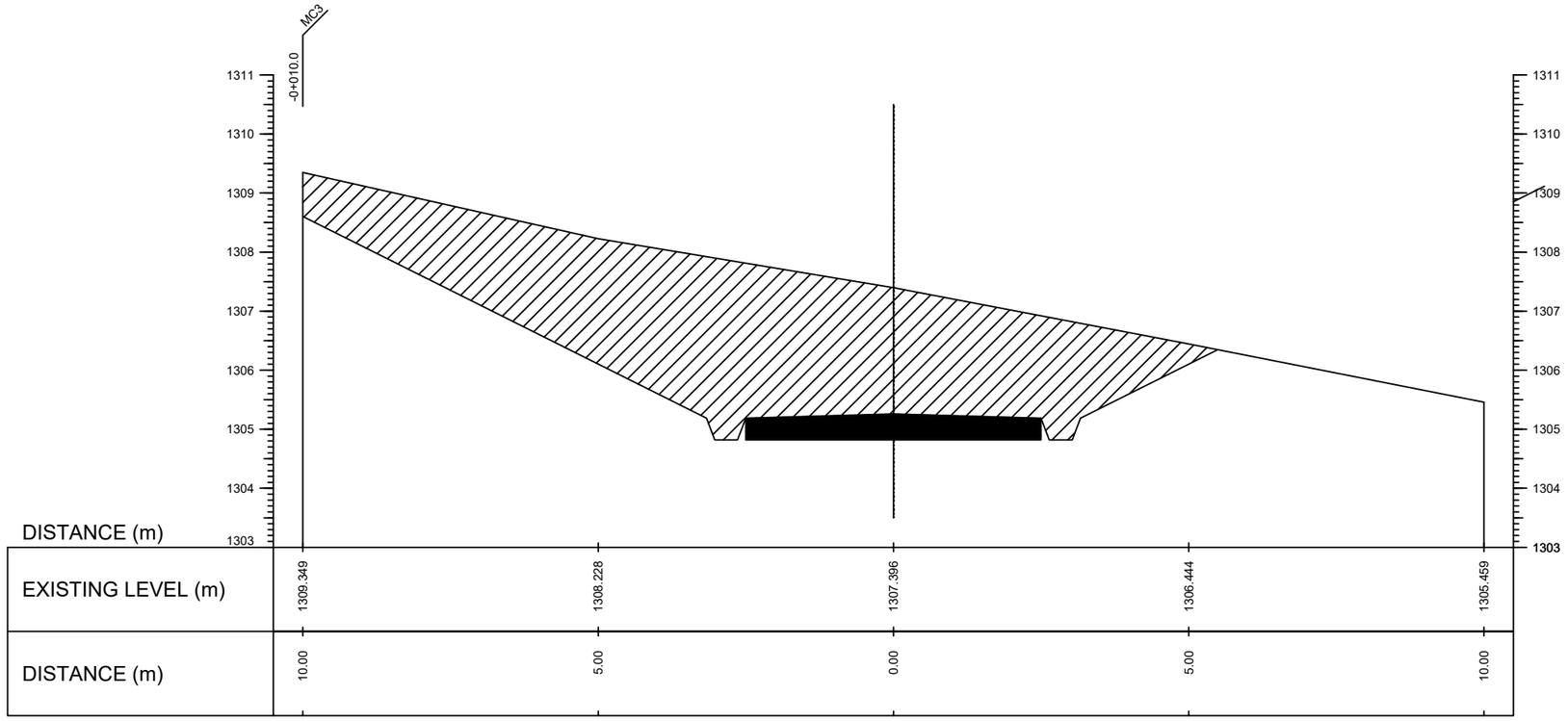
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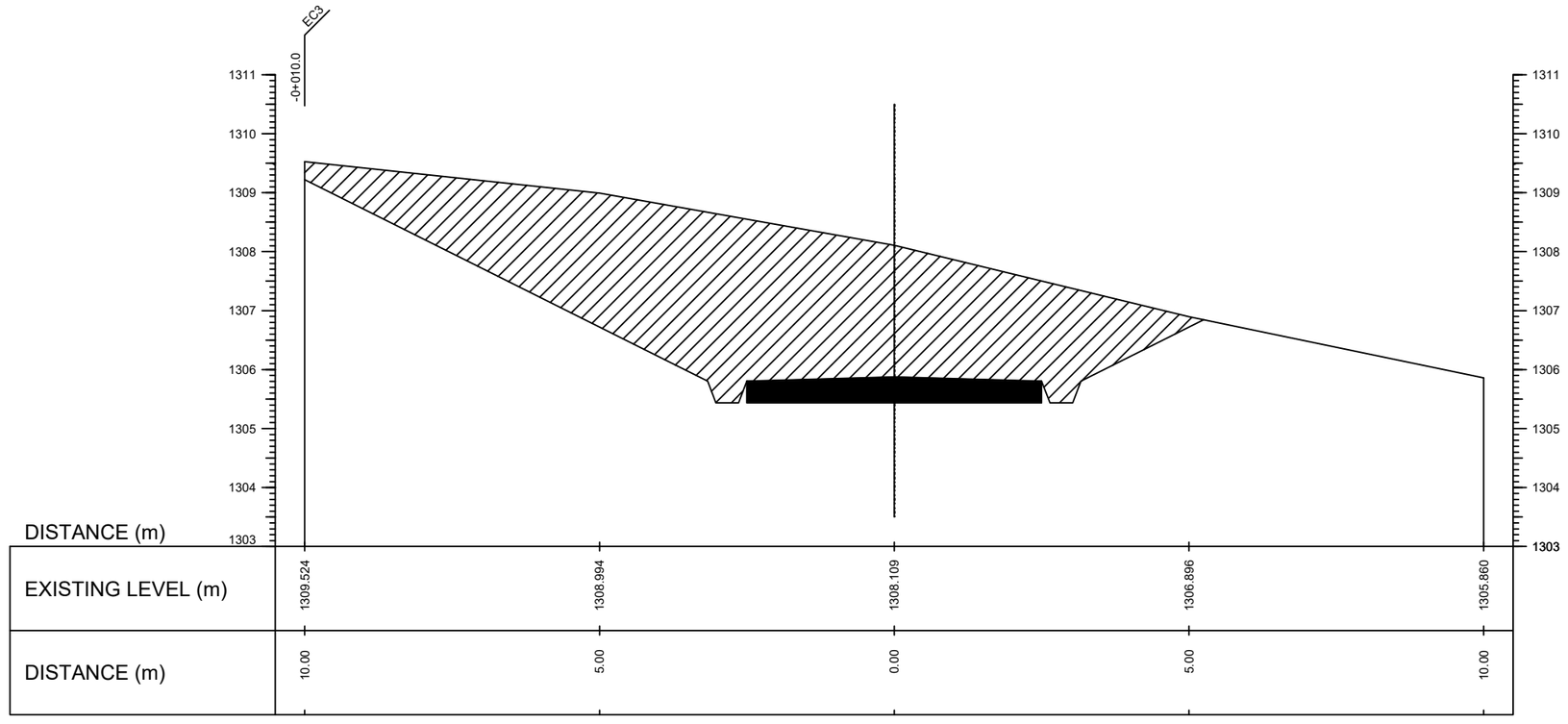
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+190.81



0+202.43

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SURVEY INSTRUCTION COMITTEE

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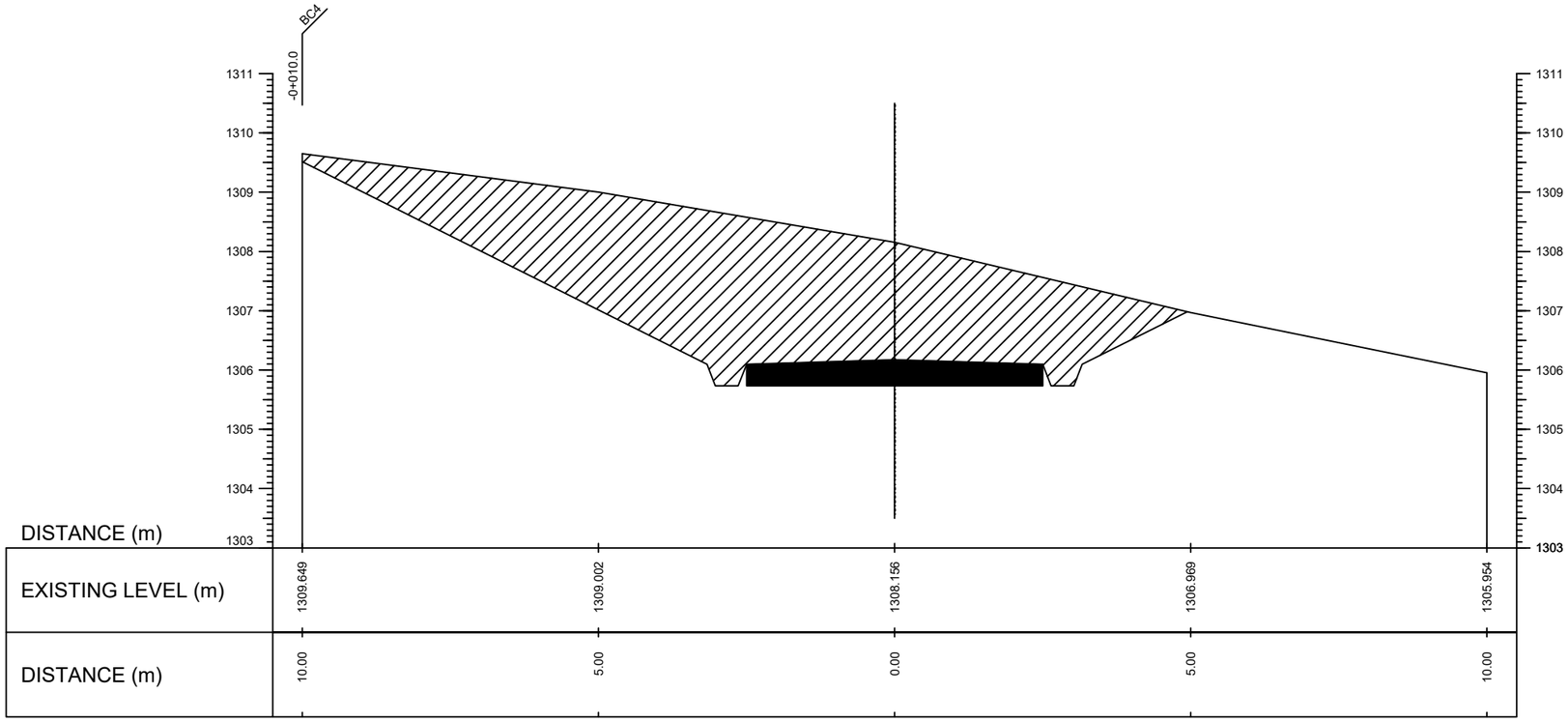
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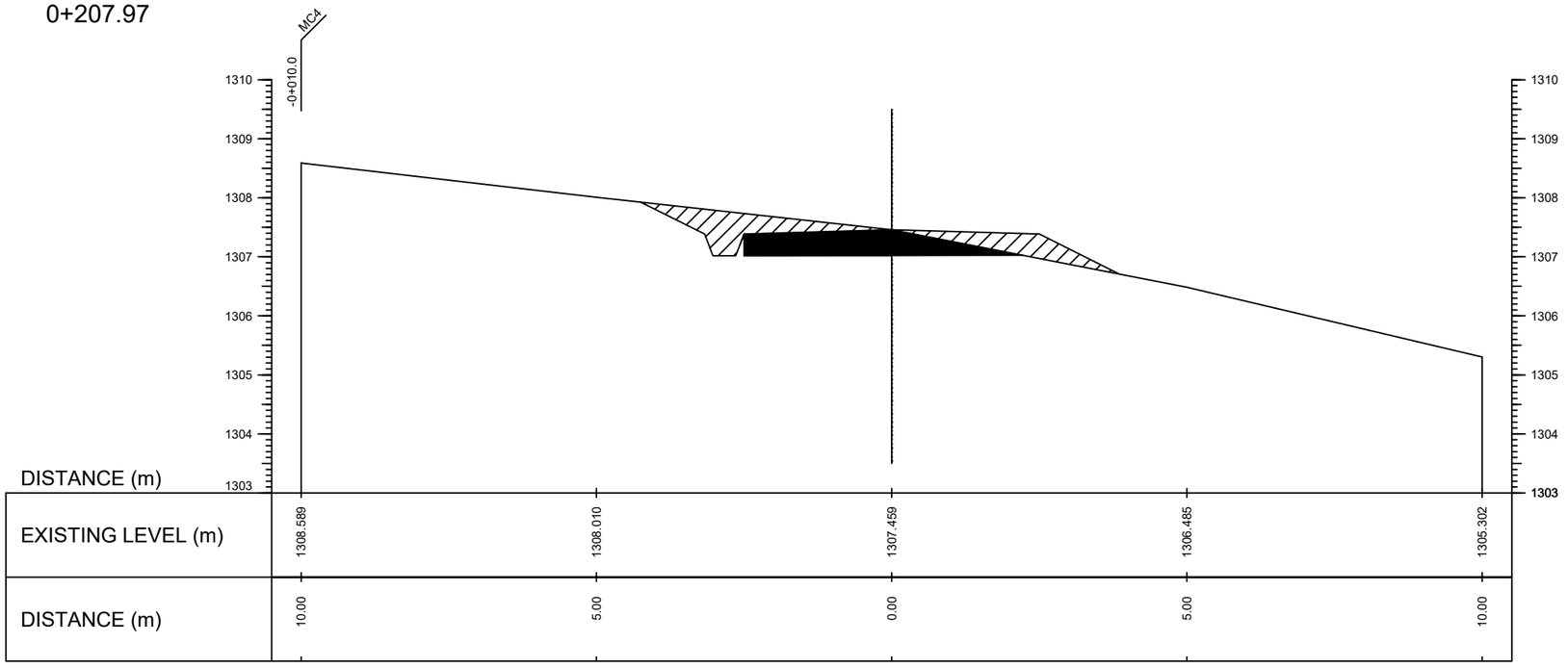
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+207.97



0+232.19

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PULCHOWK CAMPUS
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SURVEY INSTRUCTION COMITTEE

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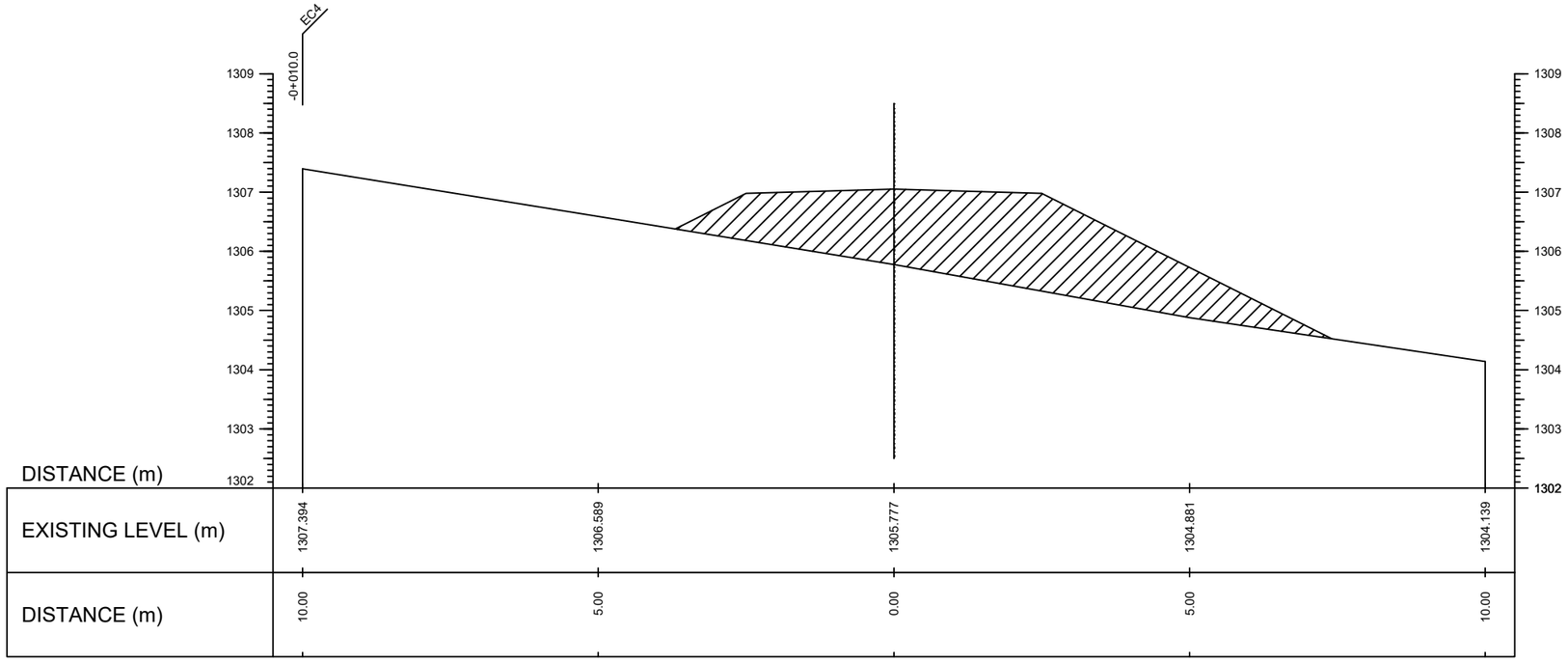
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GROUP 25

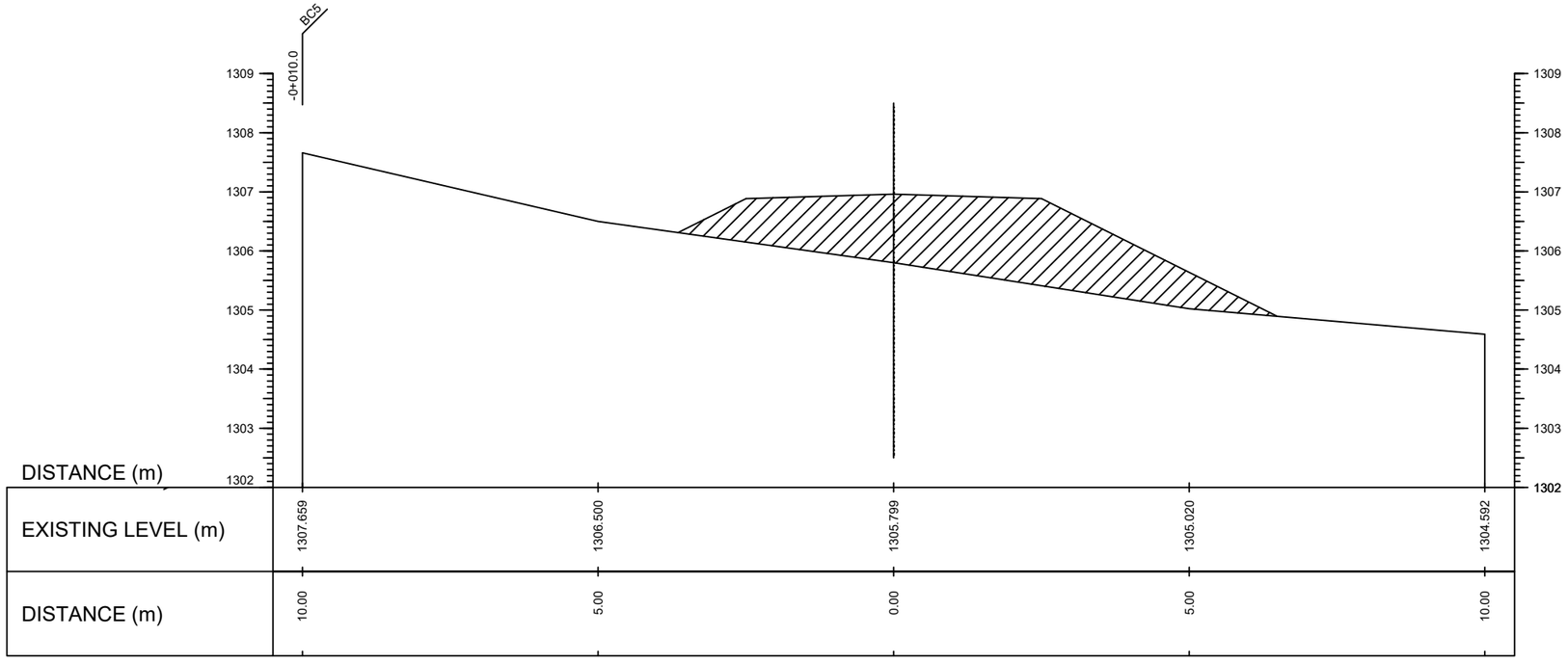
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+256.40



0+261.86

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SURVEY INSTRUCTION COMITTEE

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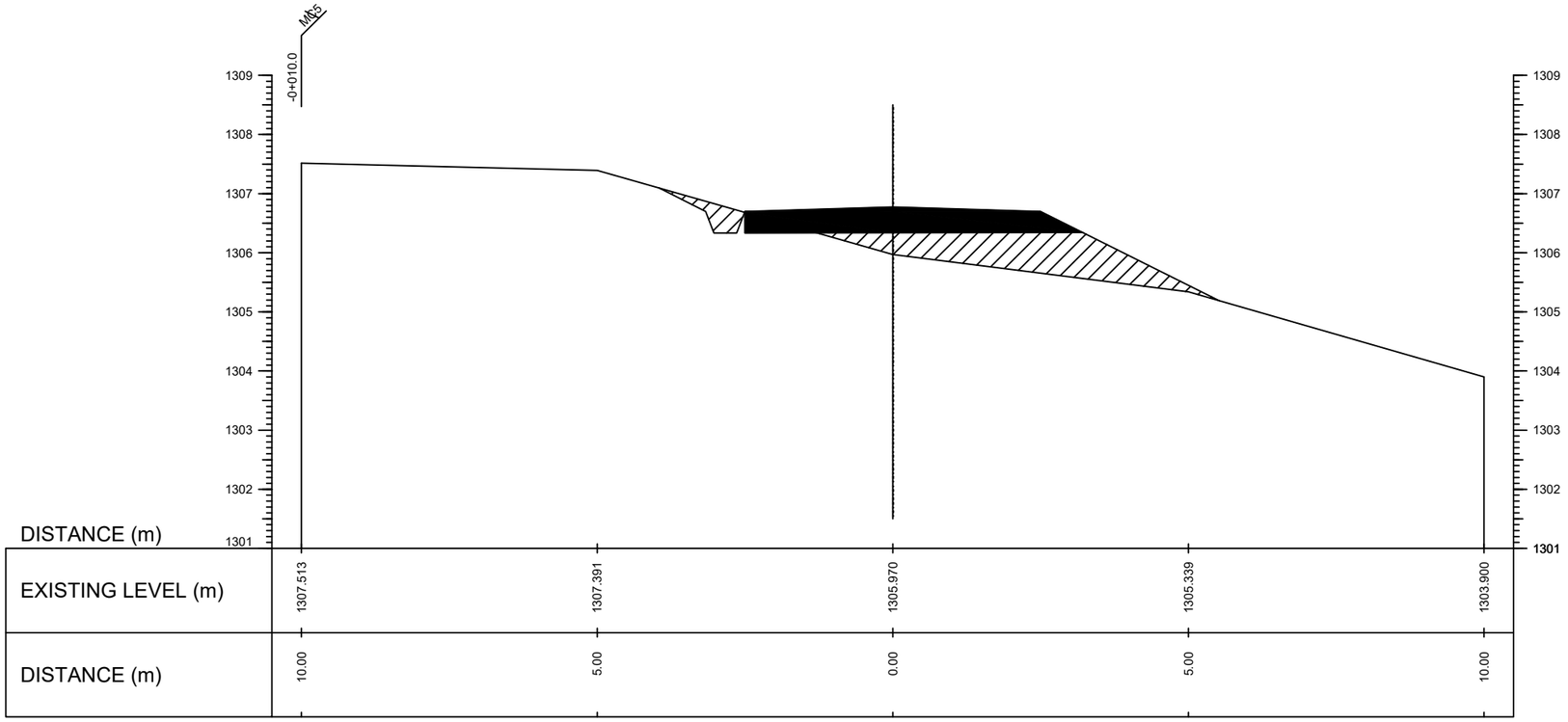
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GROUP 25

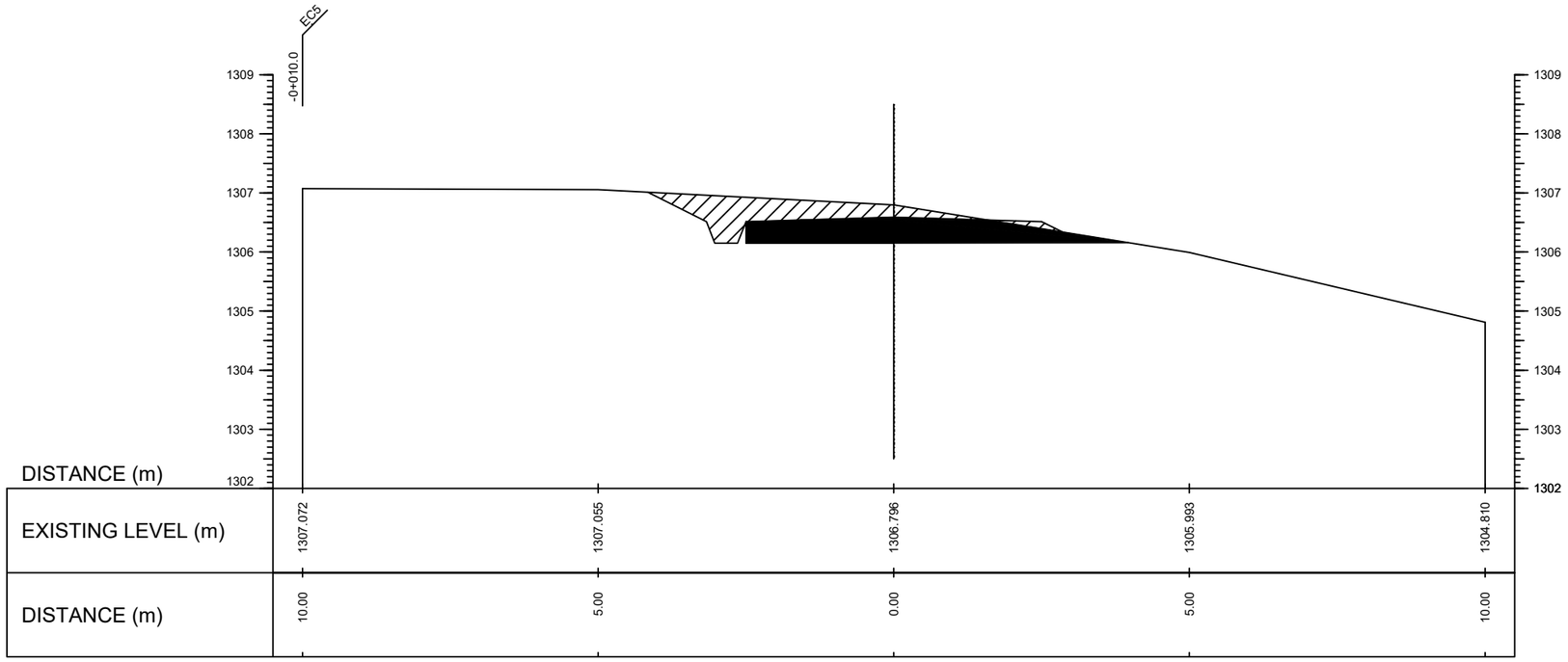
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+272.94



0+284.01

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PULCHOWK CAMPUS
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SURVEY INSTRUCTION COMITTEE

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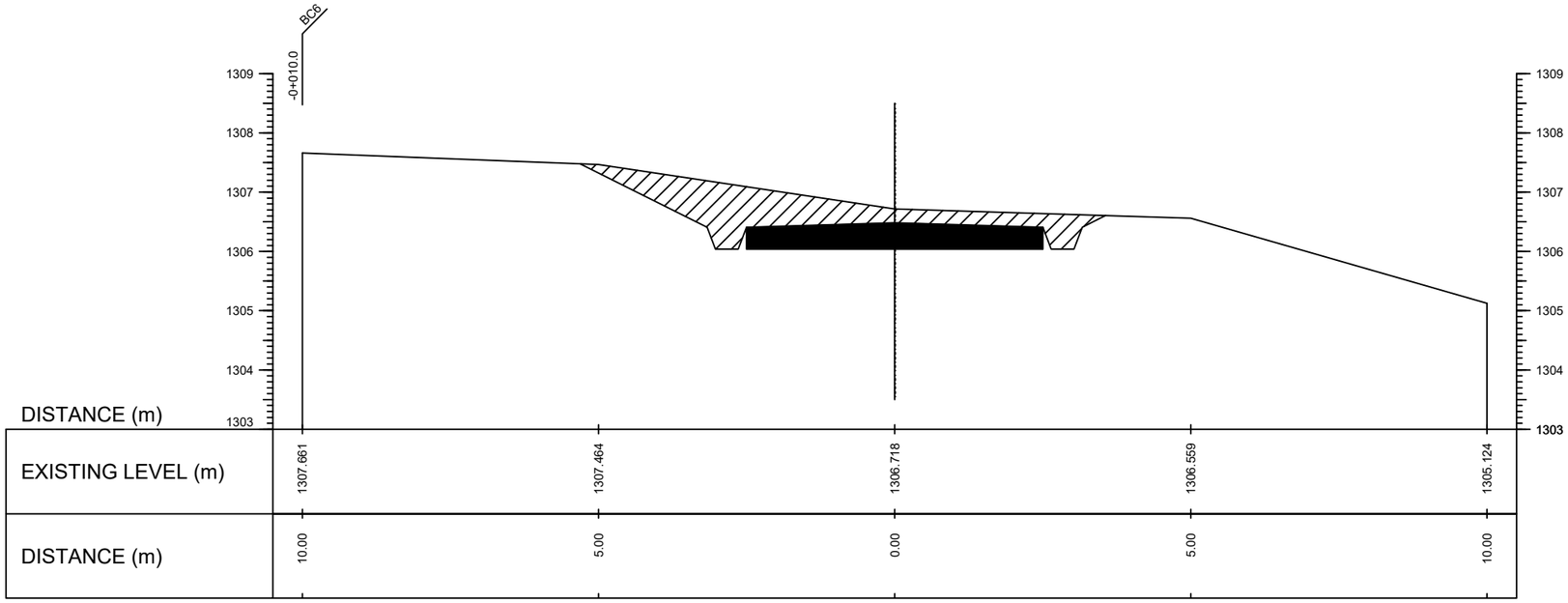
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GROUP 25

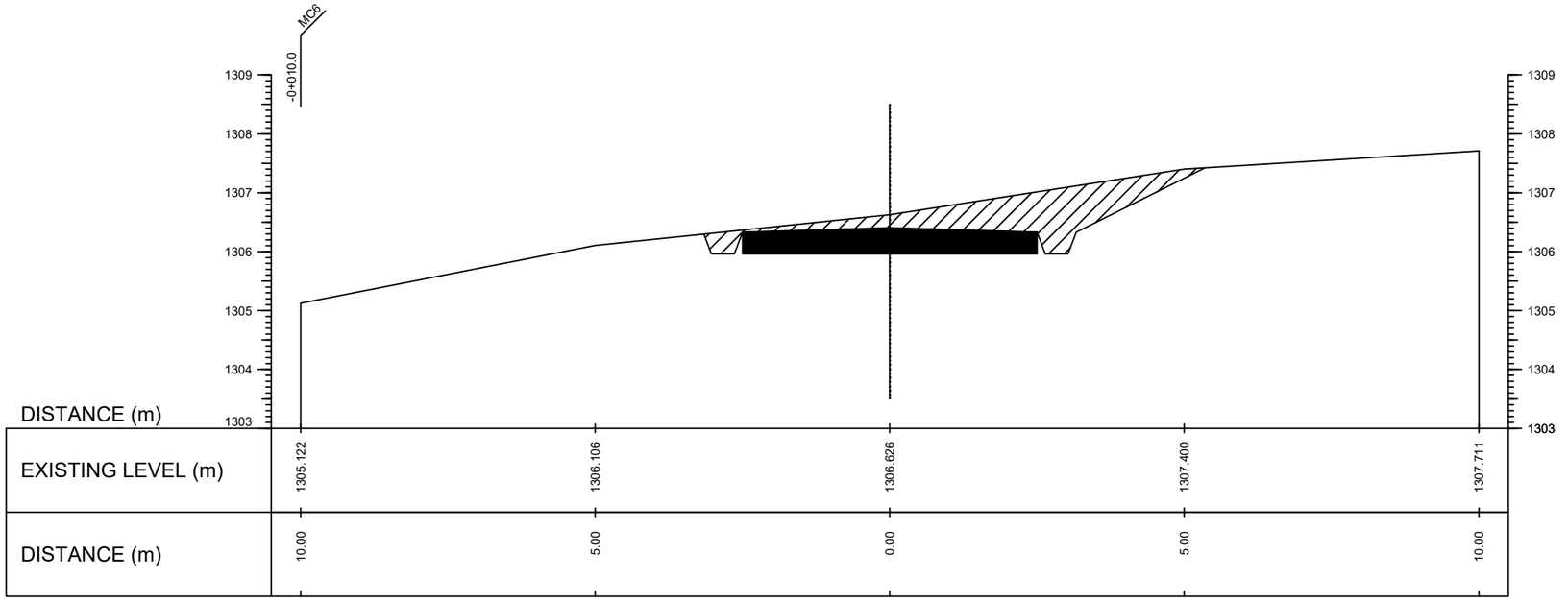
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+290.37



0+294.97

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PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

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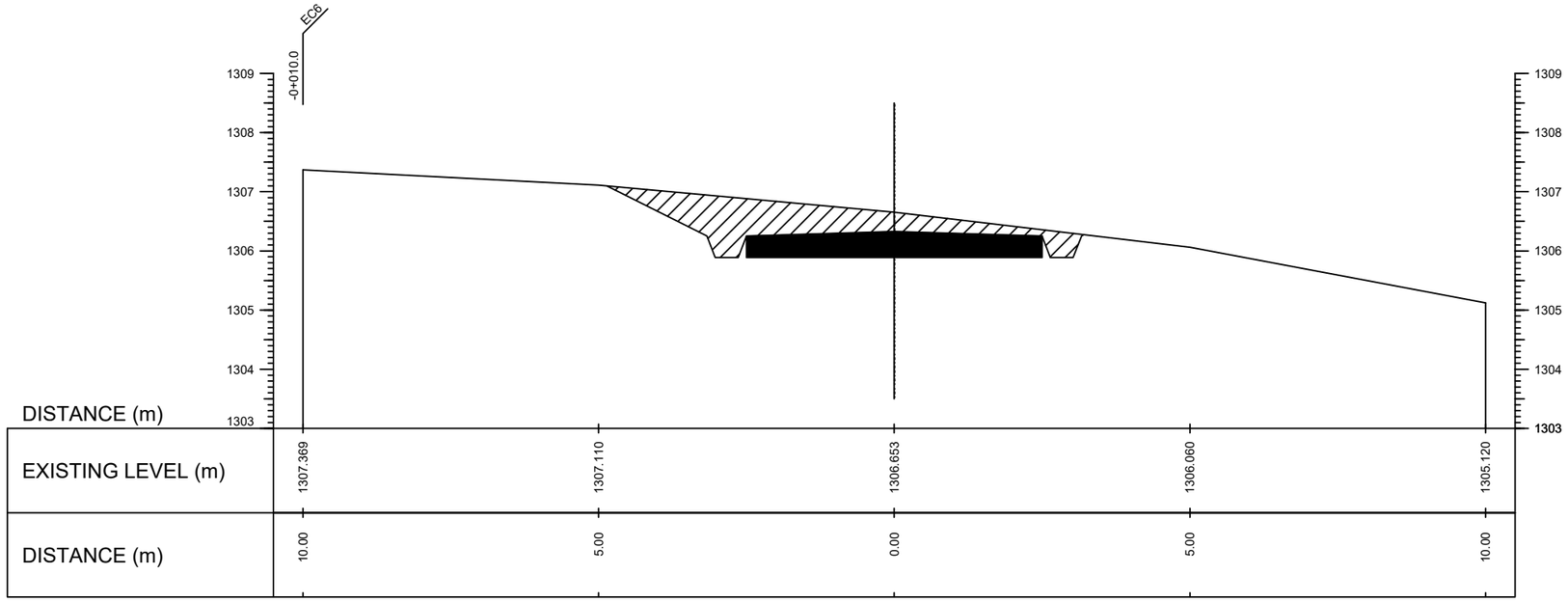
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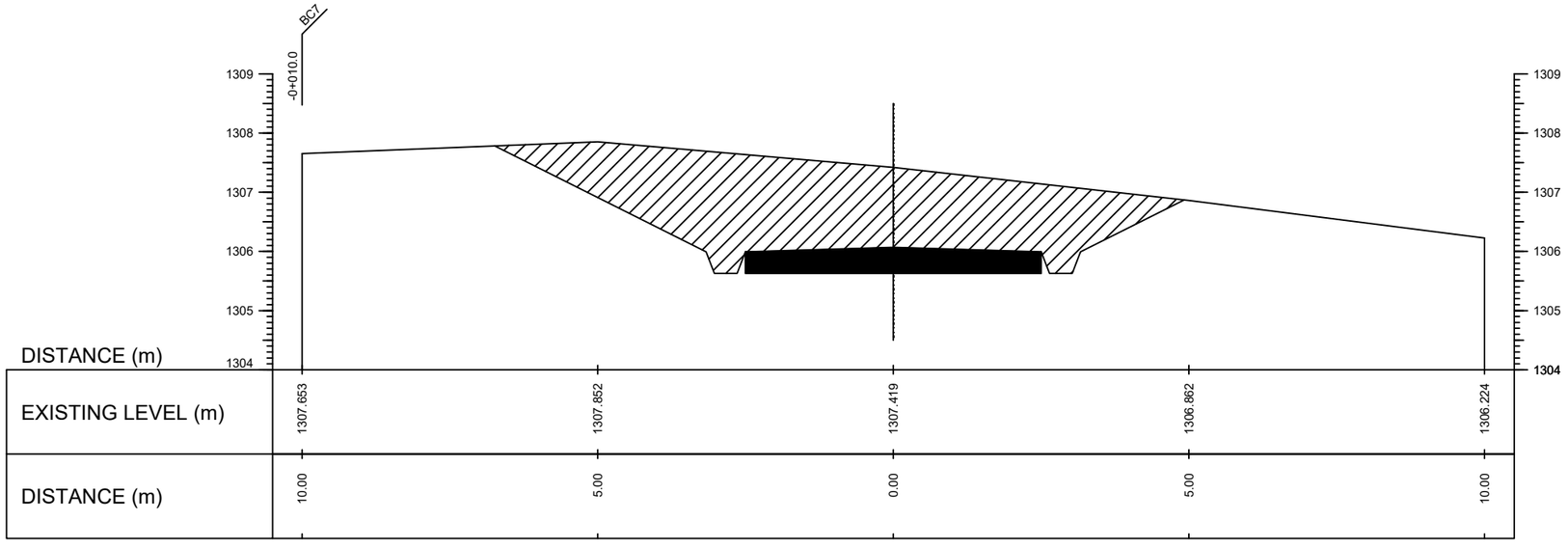
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+299.57



0+315.02

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PULCHOWK CAMPUS
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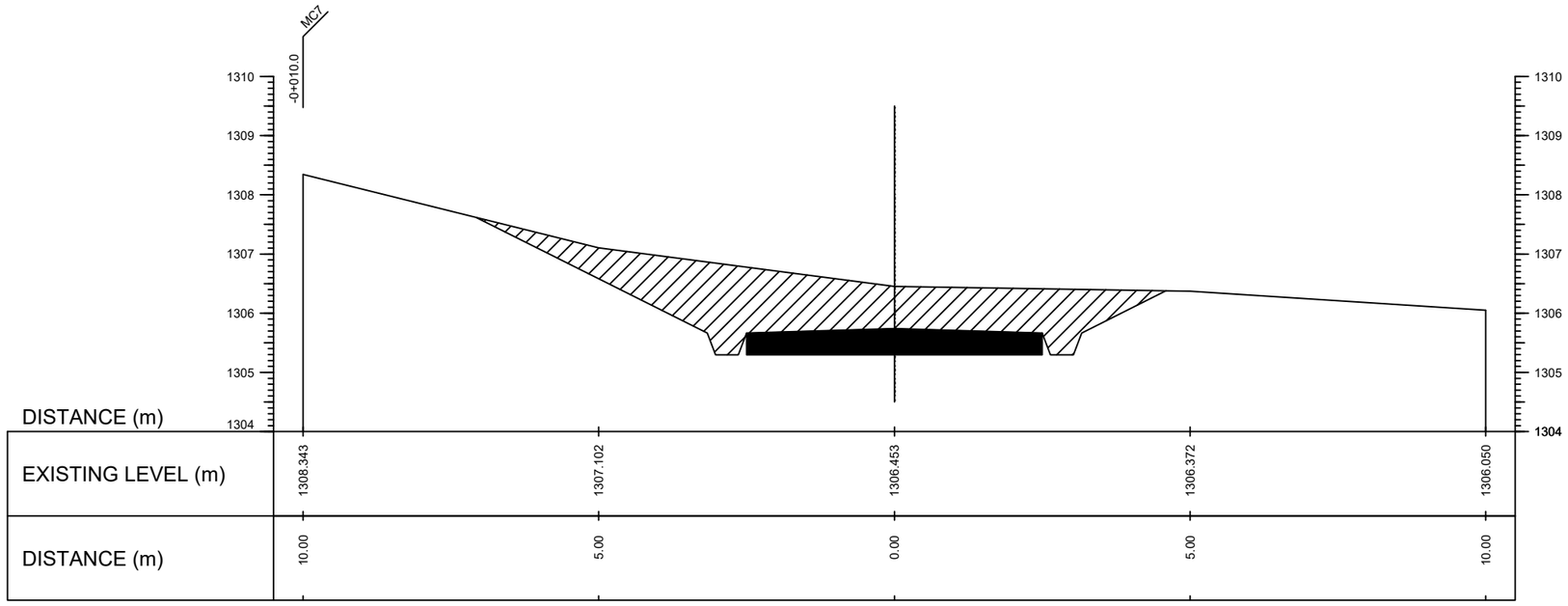
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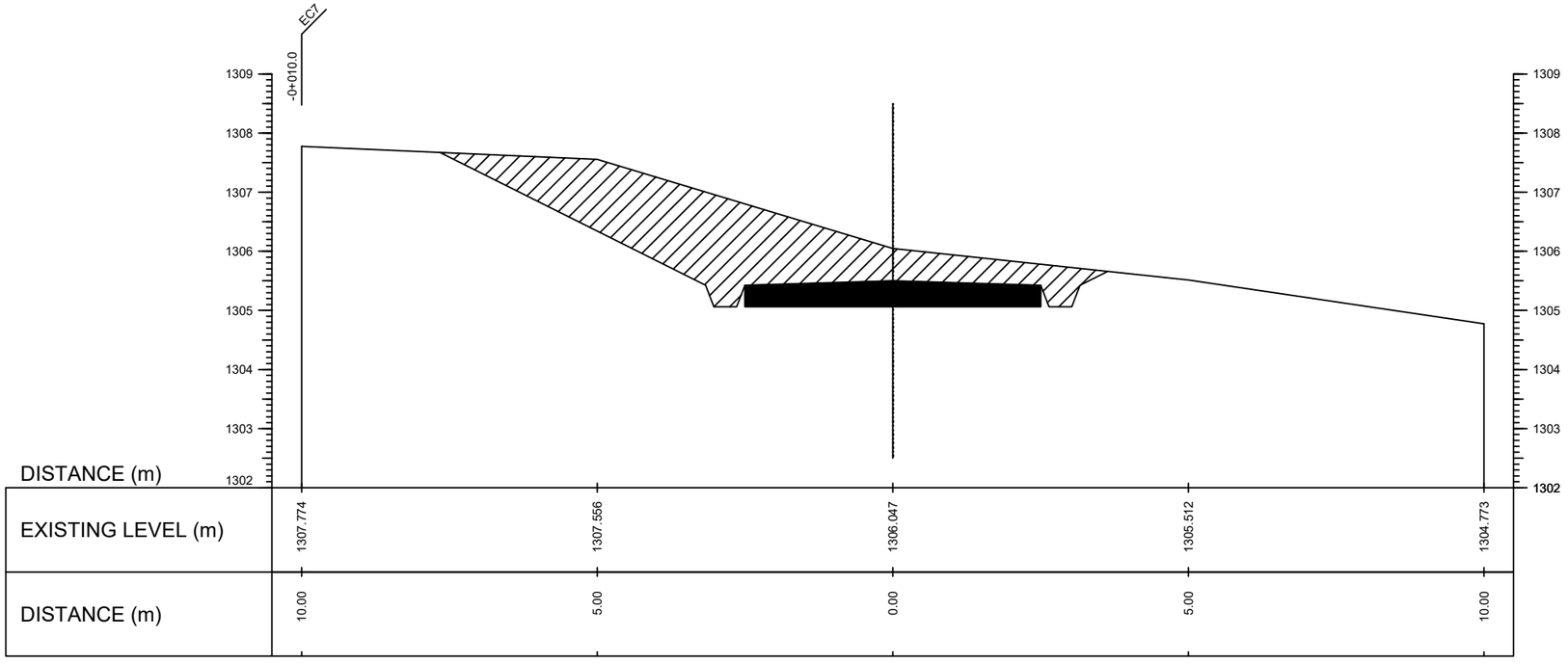
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
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SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+334.68



0+354.35

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PULCHOWK CAMPUS
DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

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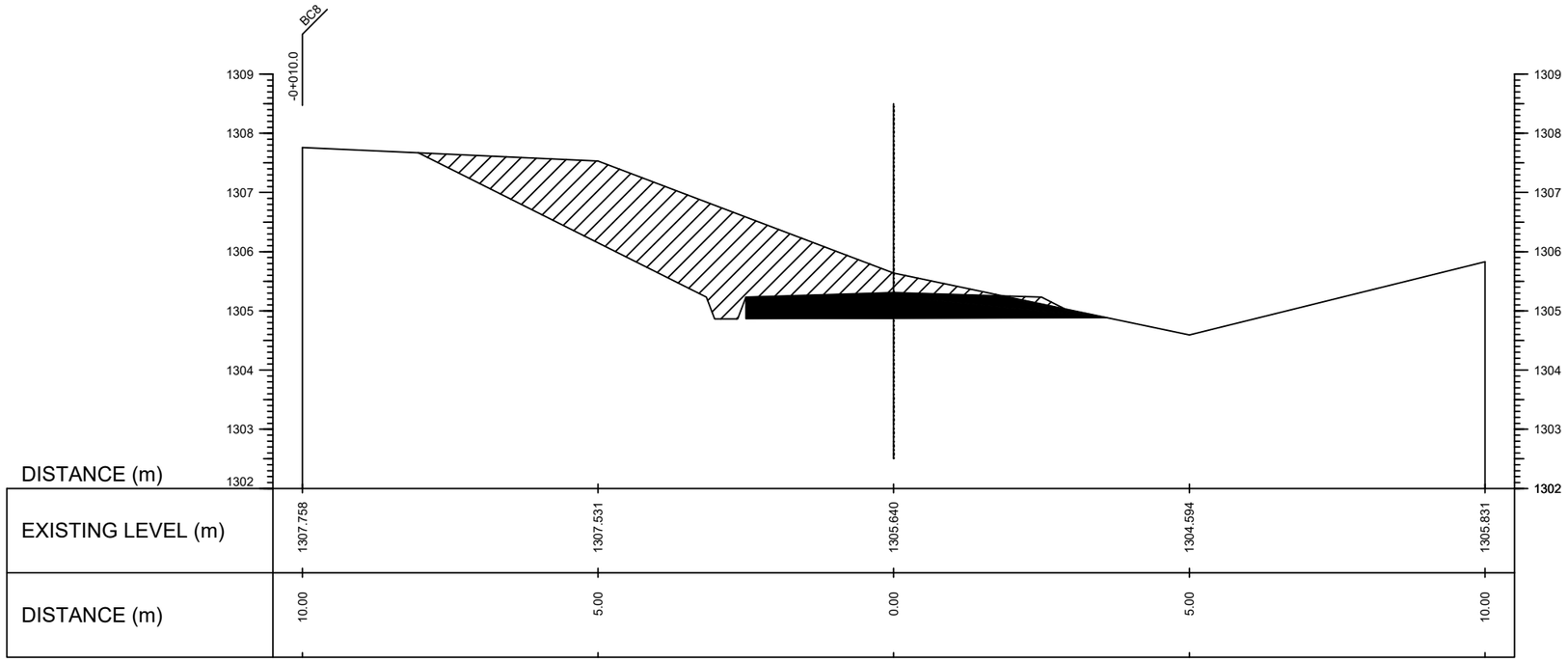
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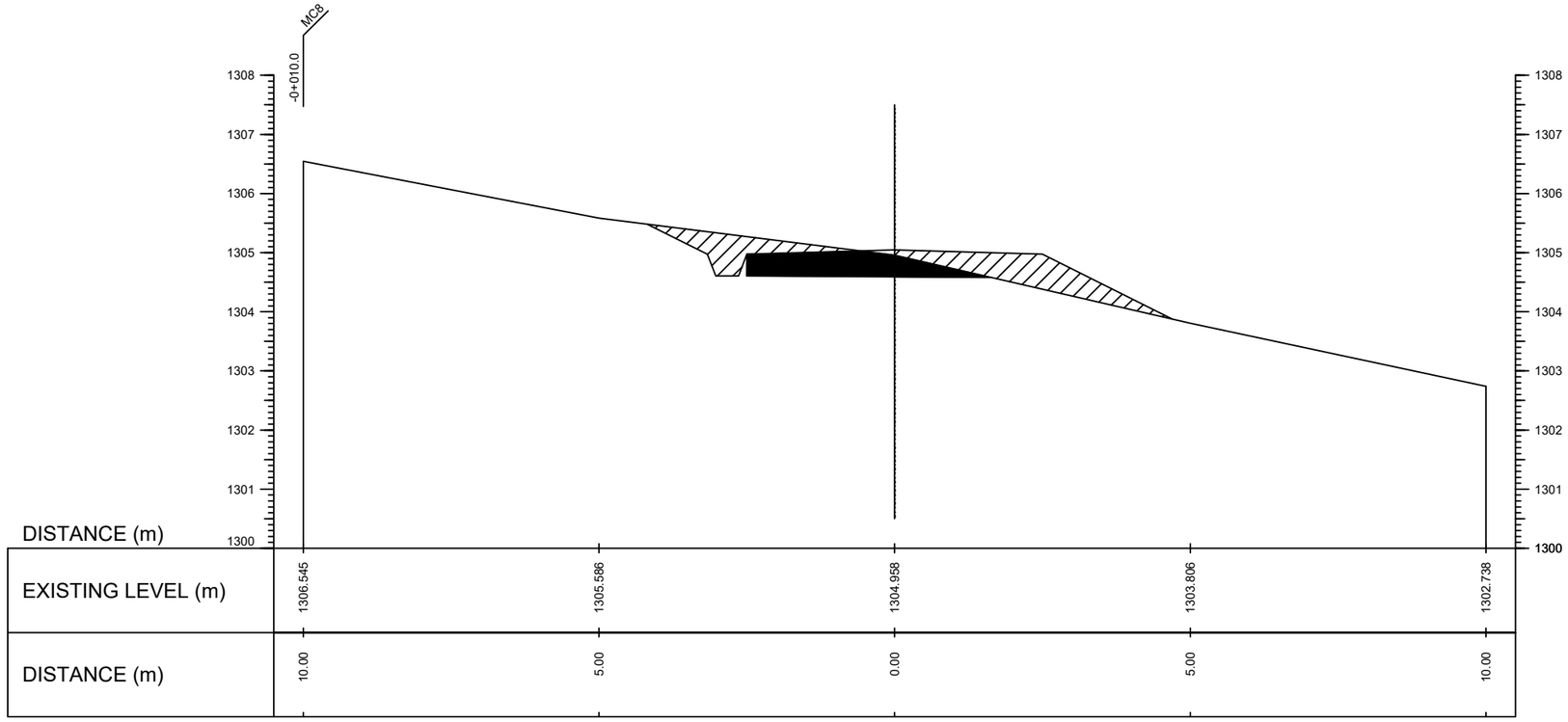
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
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SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+360.42



0+375.86

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PULCHOWK CAMPUS
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SURVEY INSTRUCTION COMITTEE

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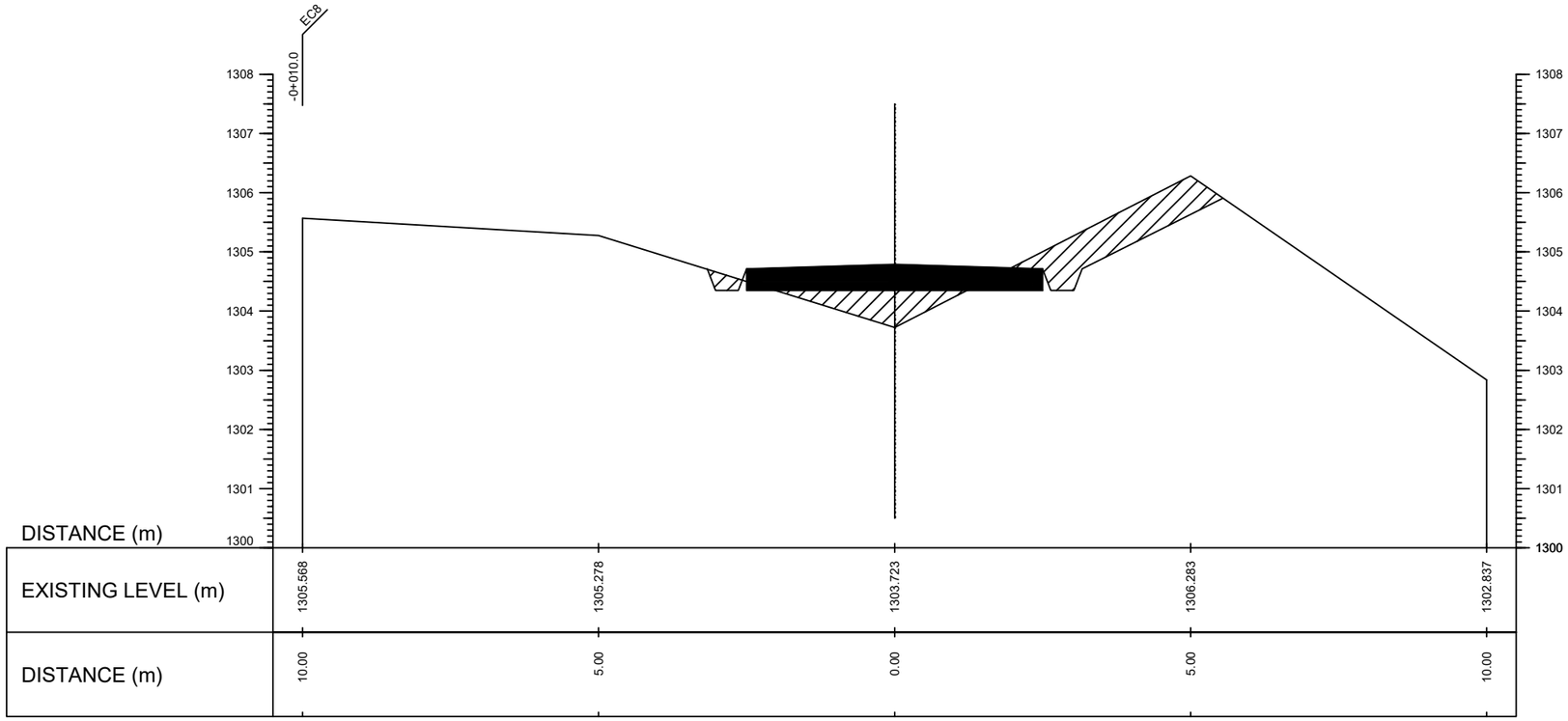
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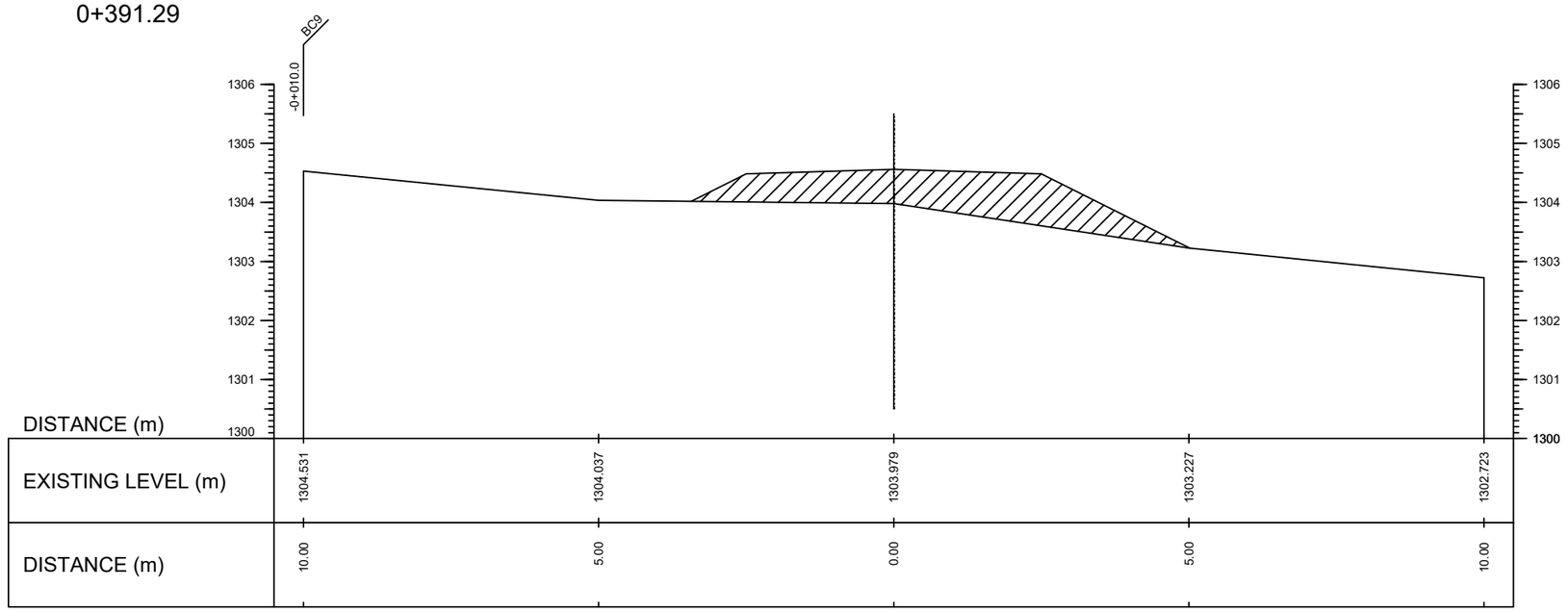
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+391.29



0+404.95

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PULCHOWK CAMPUS
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SURVEY INSTRUCTION COMITTEE

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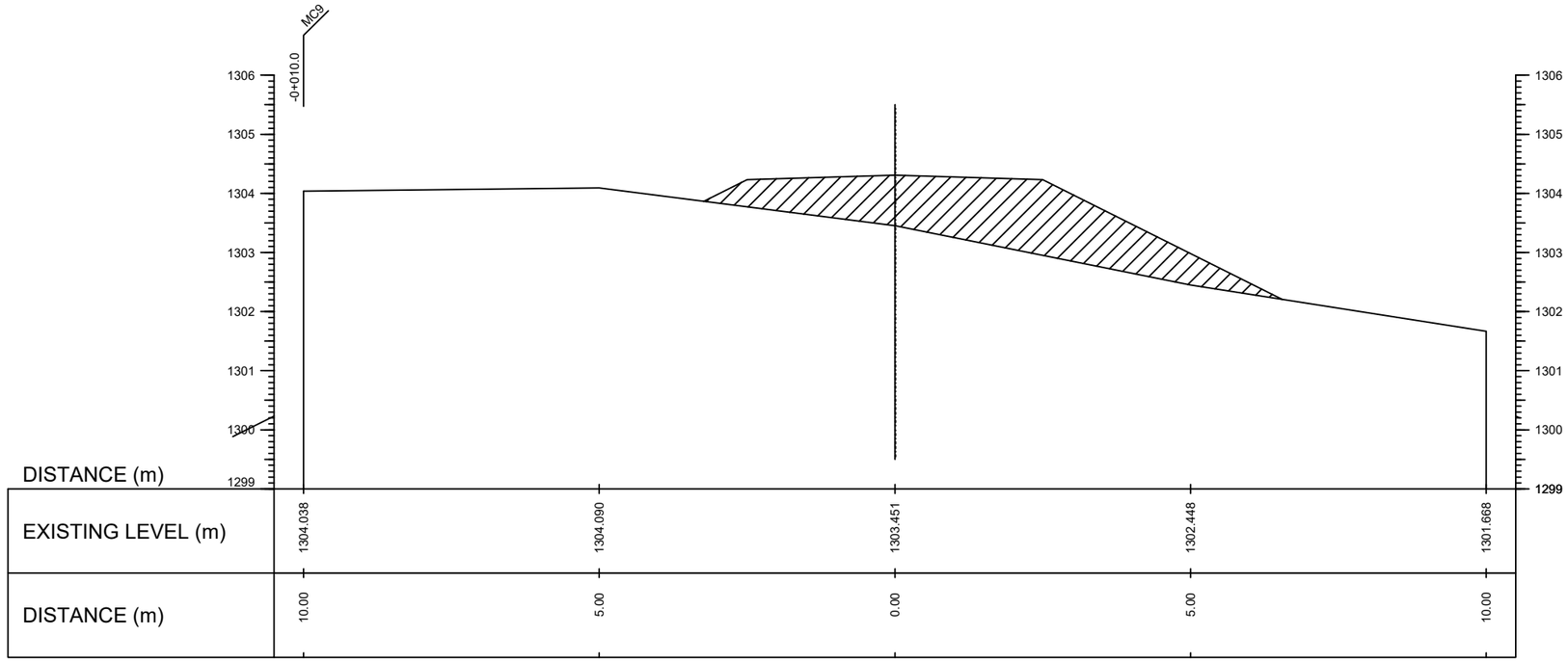
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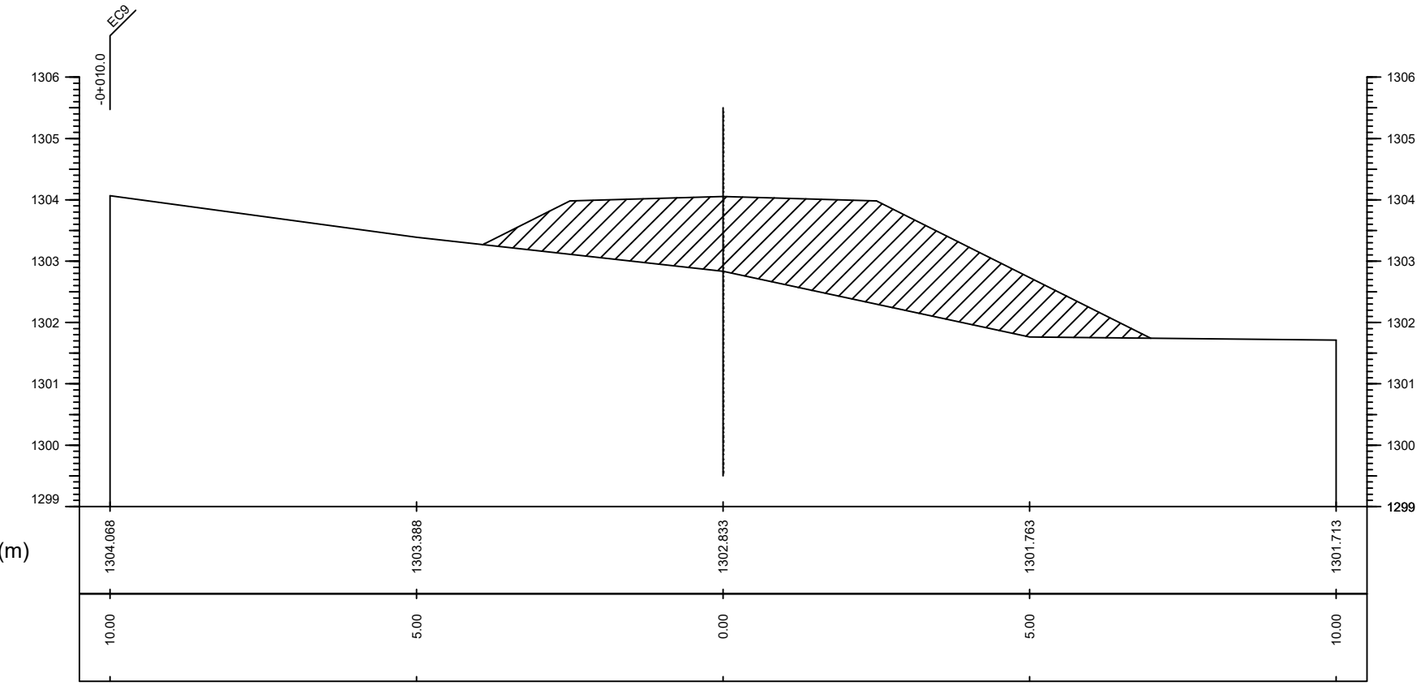
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+419.98



0+435.00

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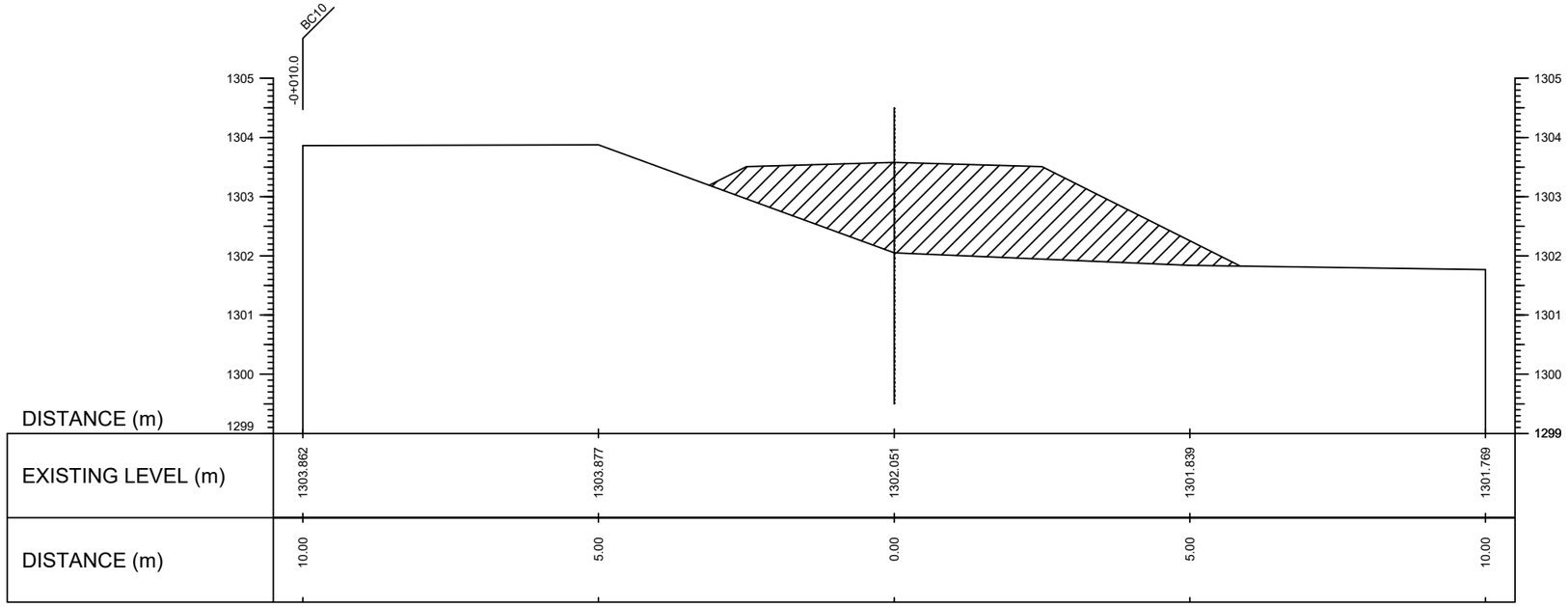
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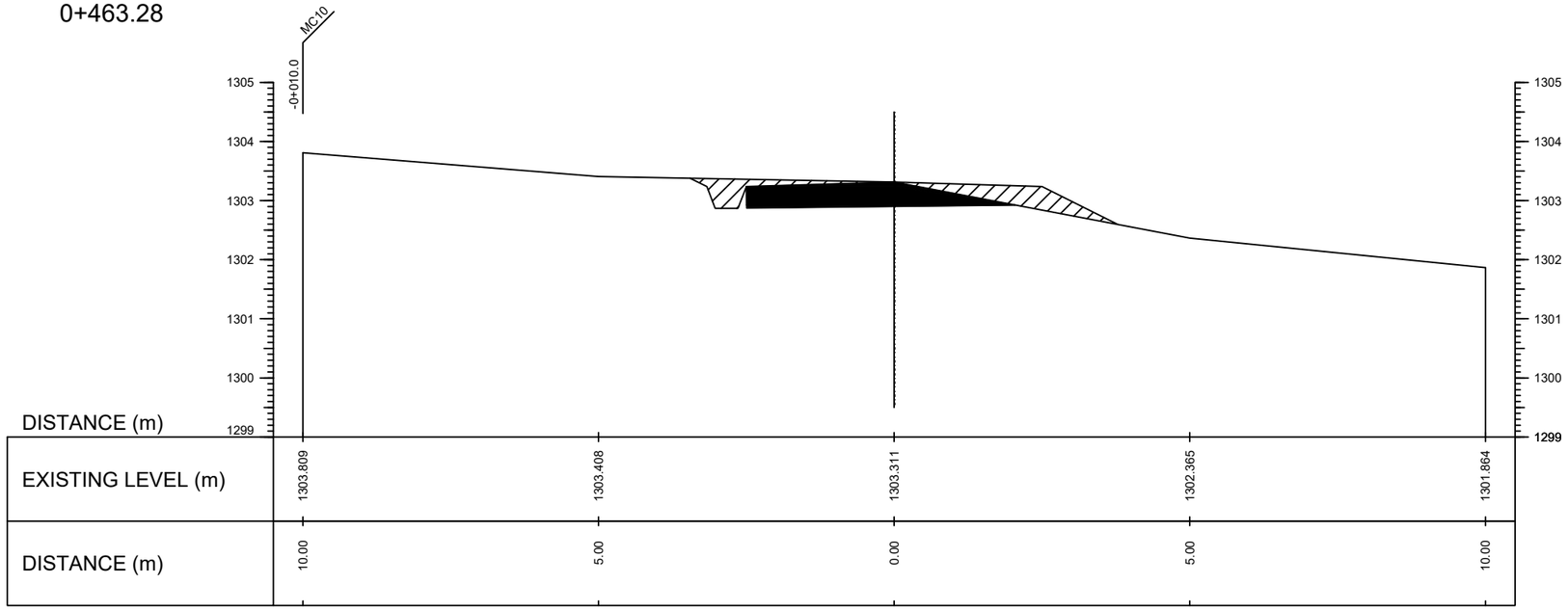
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+463.28



0+479.40

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SURVEY INSTRUCTION COMITTEE

TITLE:ROAD ALIGNMENT CROSS SECTION

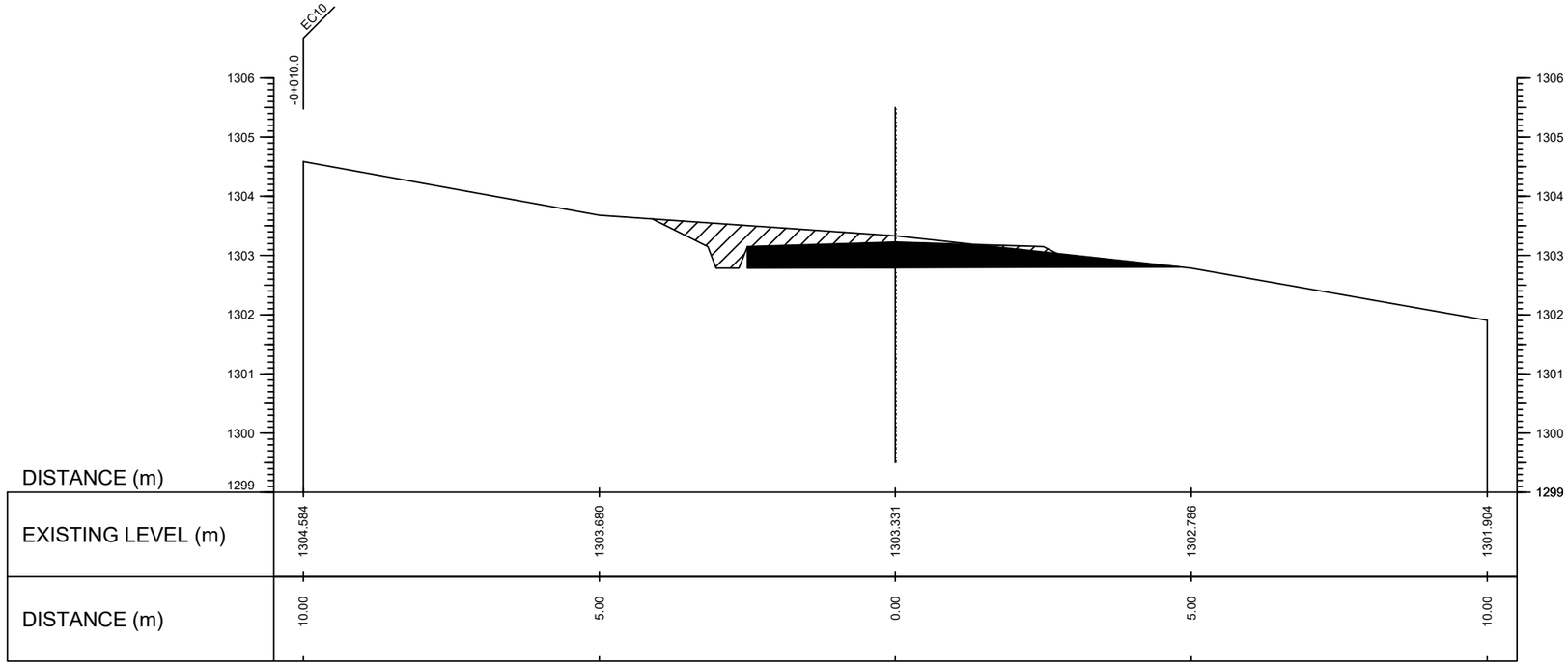
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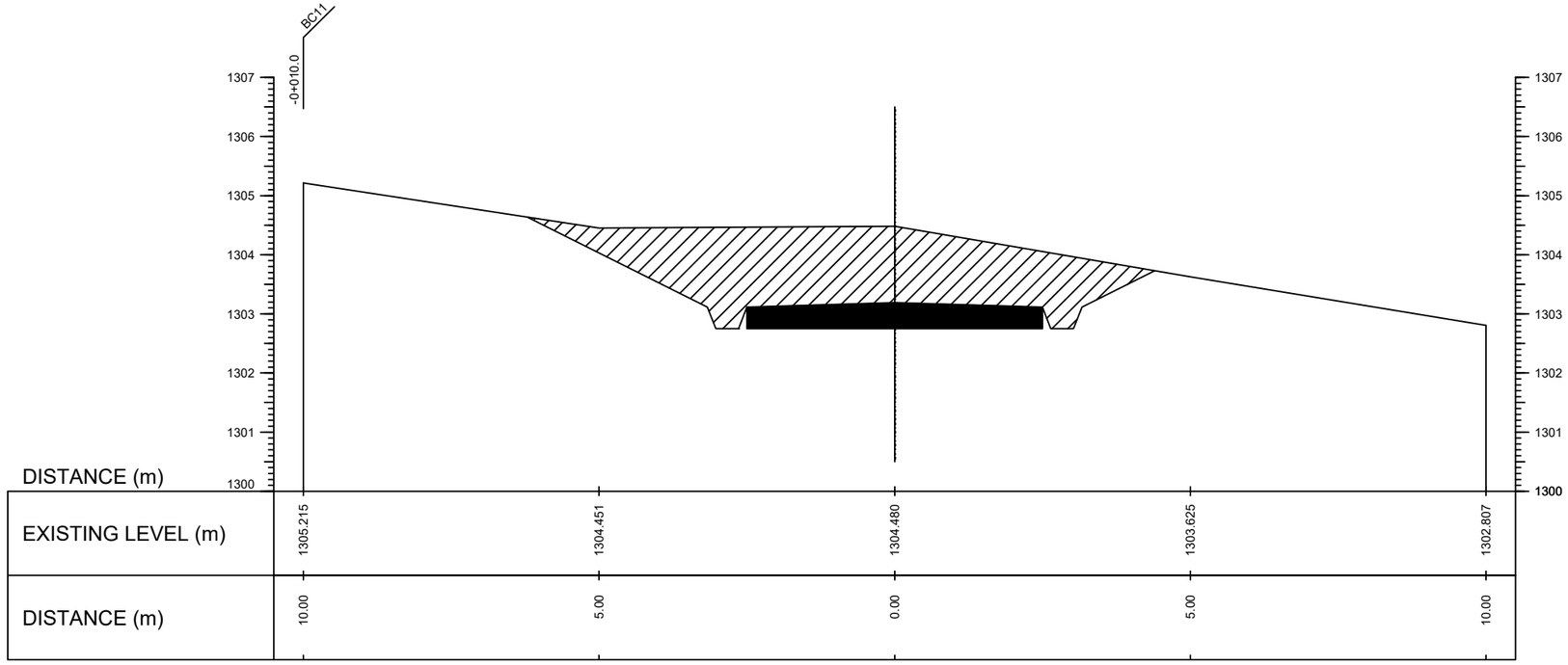
PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
SNEHA PRAVEEN	080BCE166
SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+495.52



0+502.25

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INSTITUTE OF ENGINEERING
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DEPARTMENT OF CIVIL ENGINEERING
SURVEY INSTRUCTION COMITTEE

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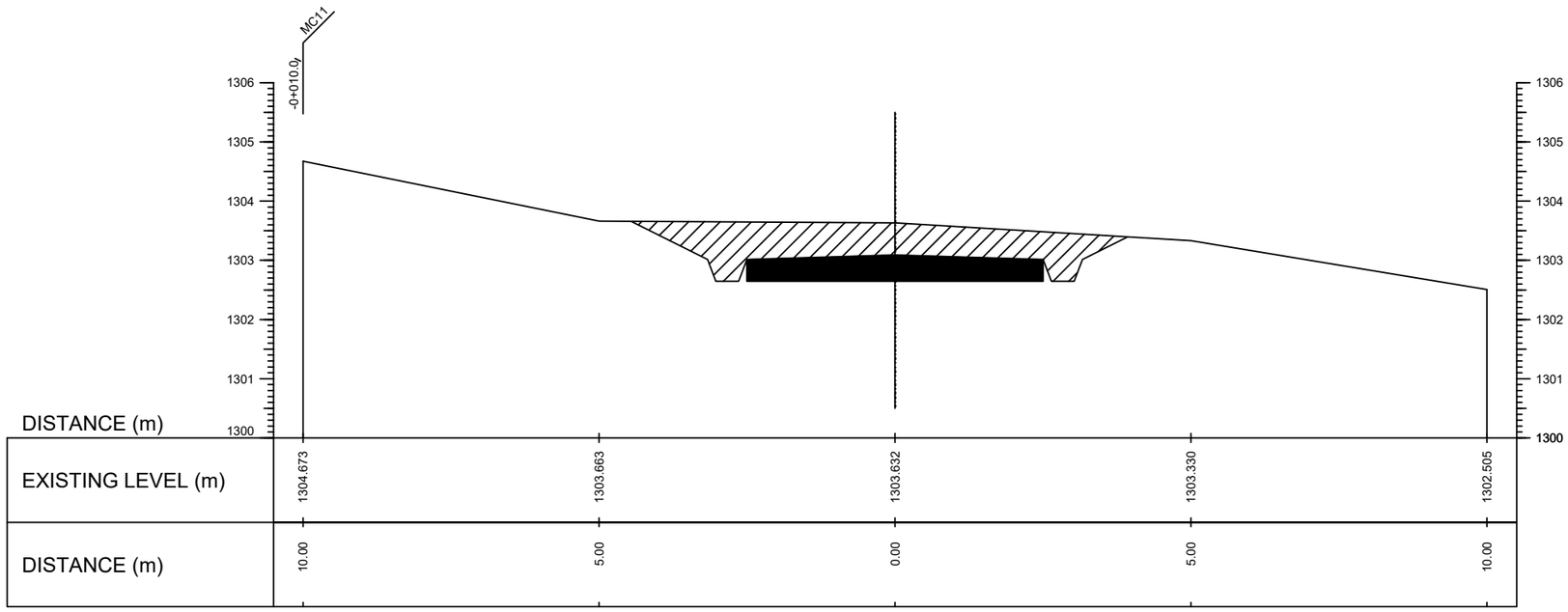
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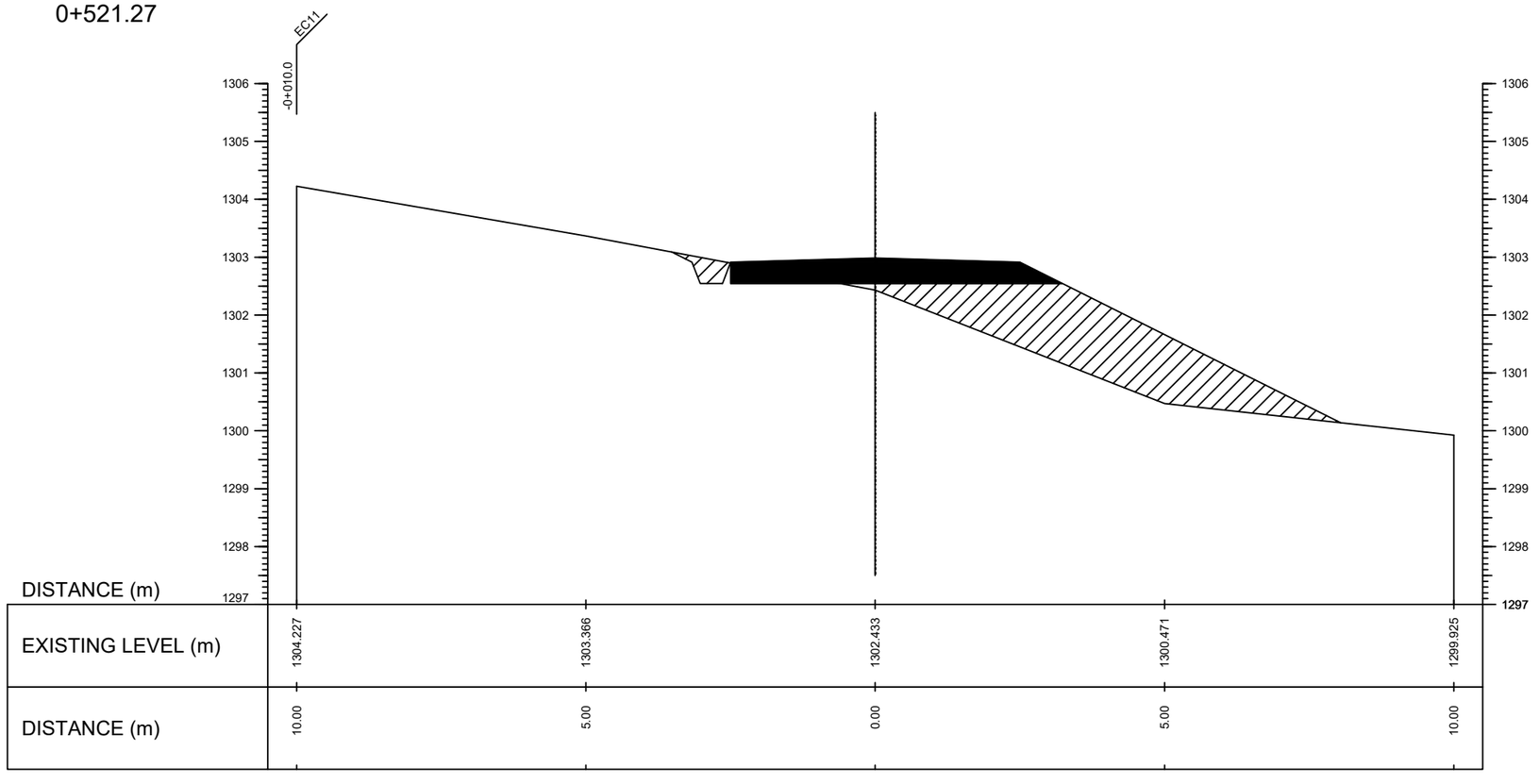
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
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SUYOG REGMI	080BCE184

GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+521.27



0+540.29

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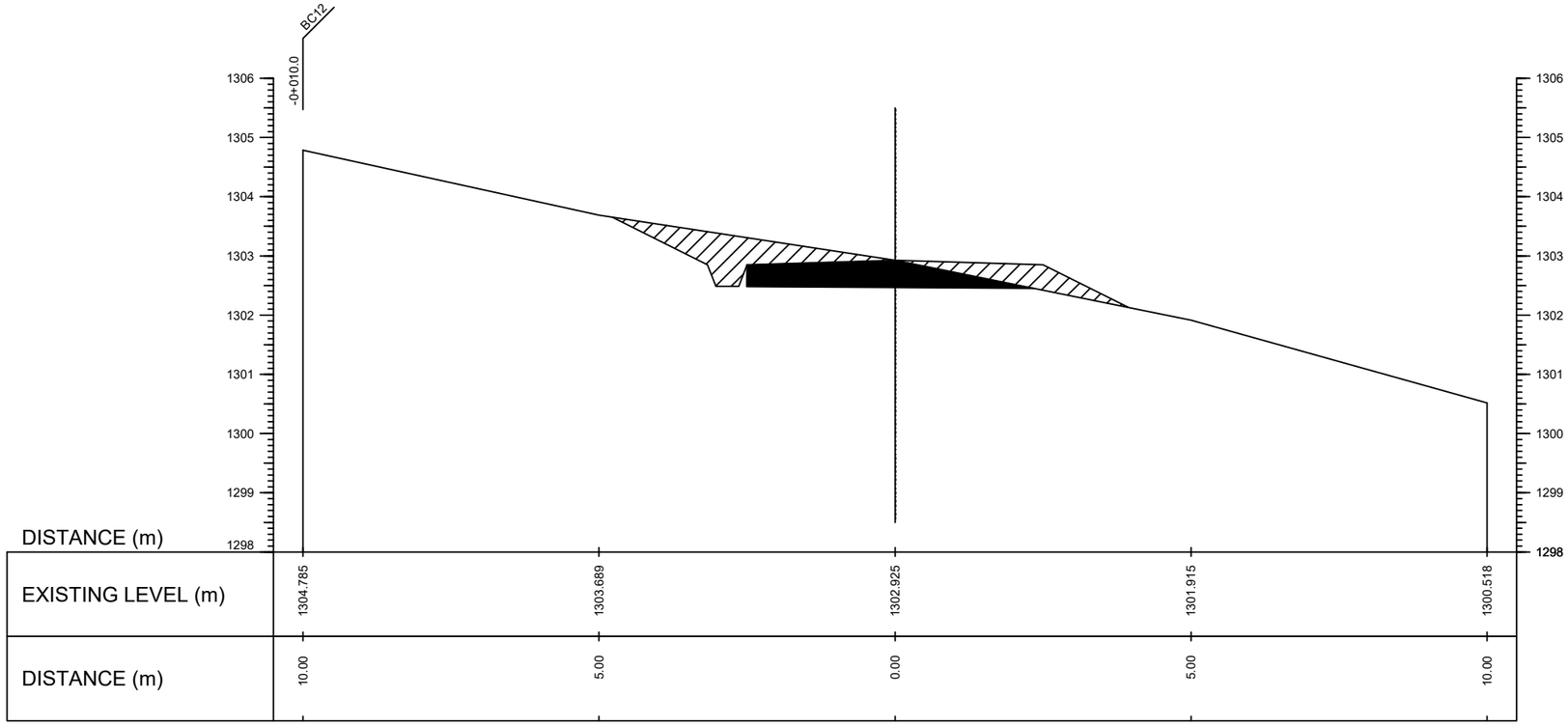
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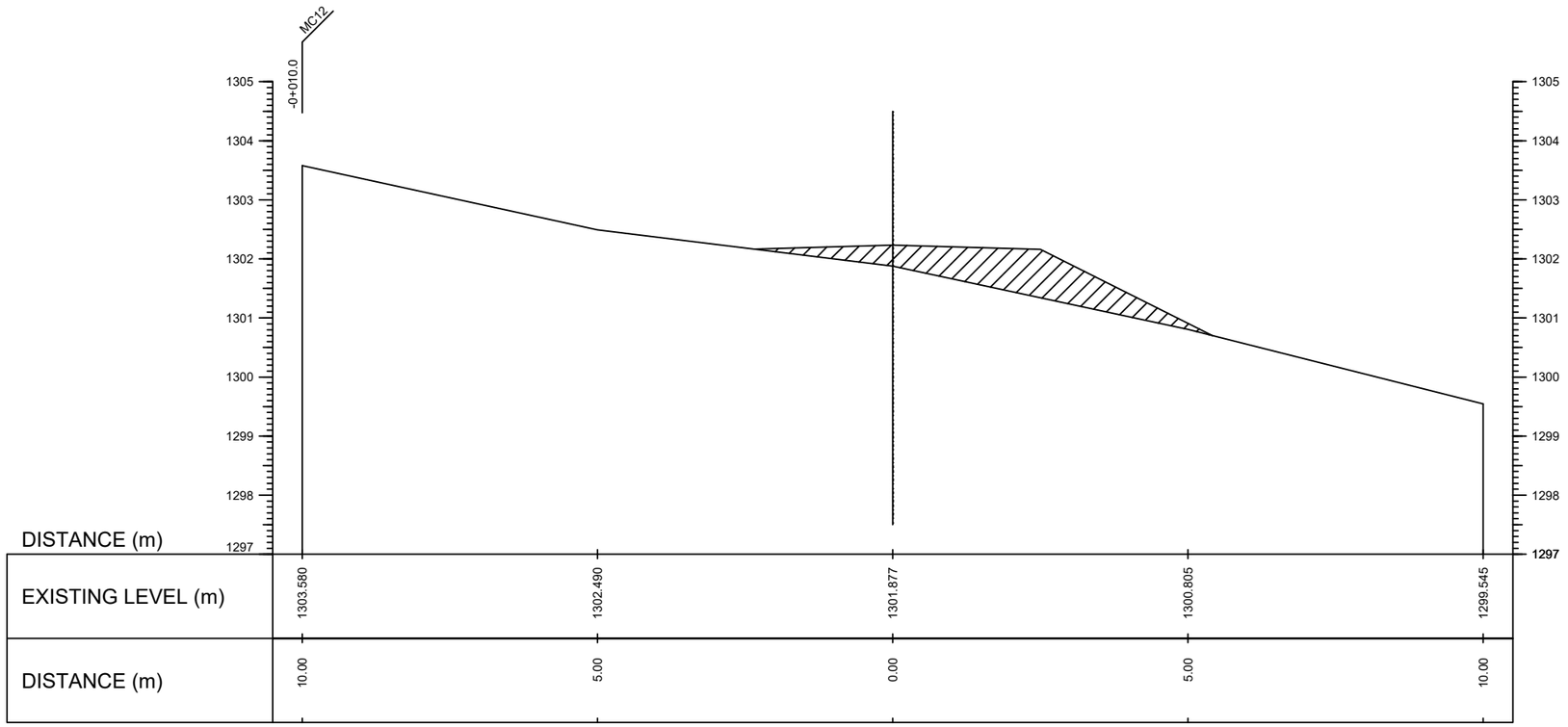
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GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+552.25



0+563.94

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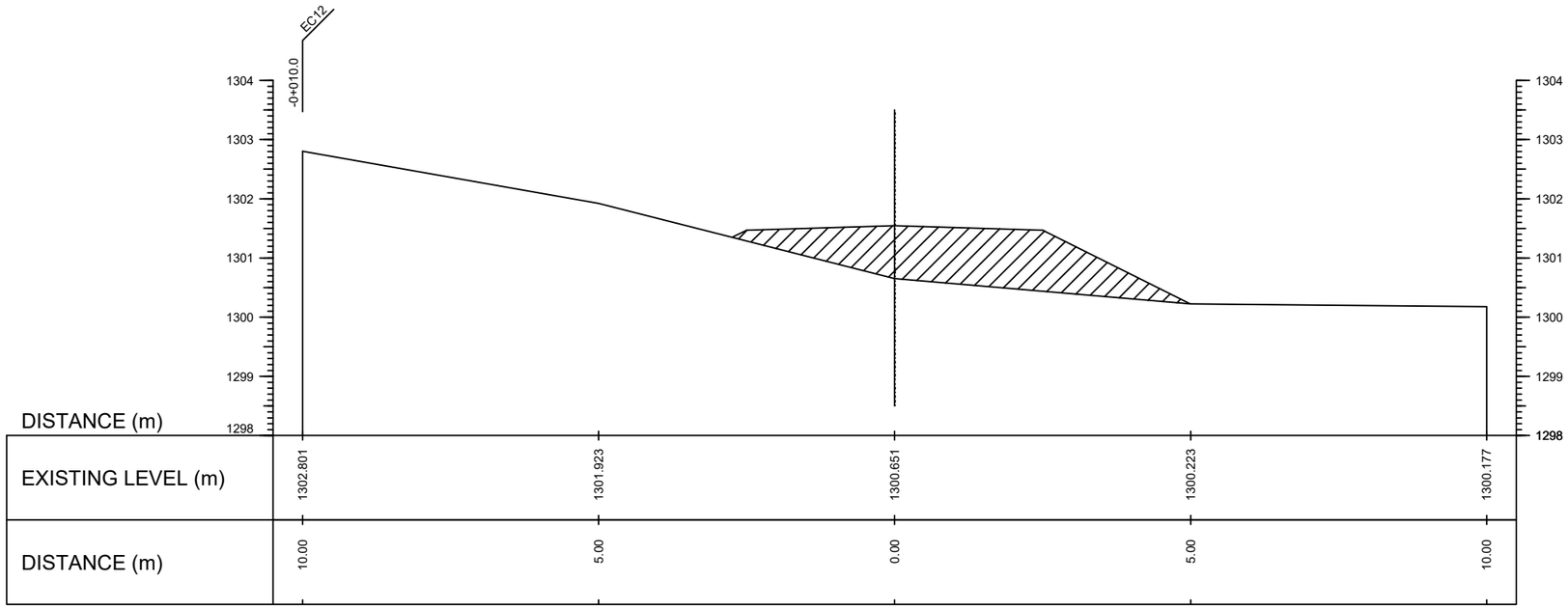
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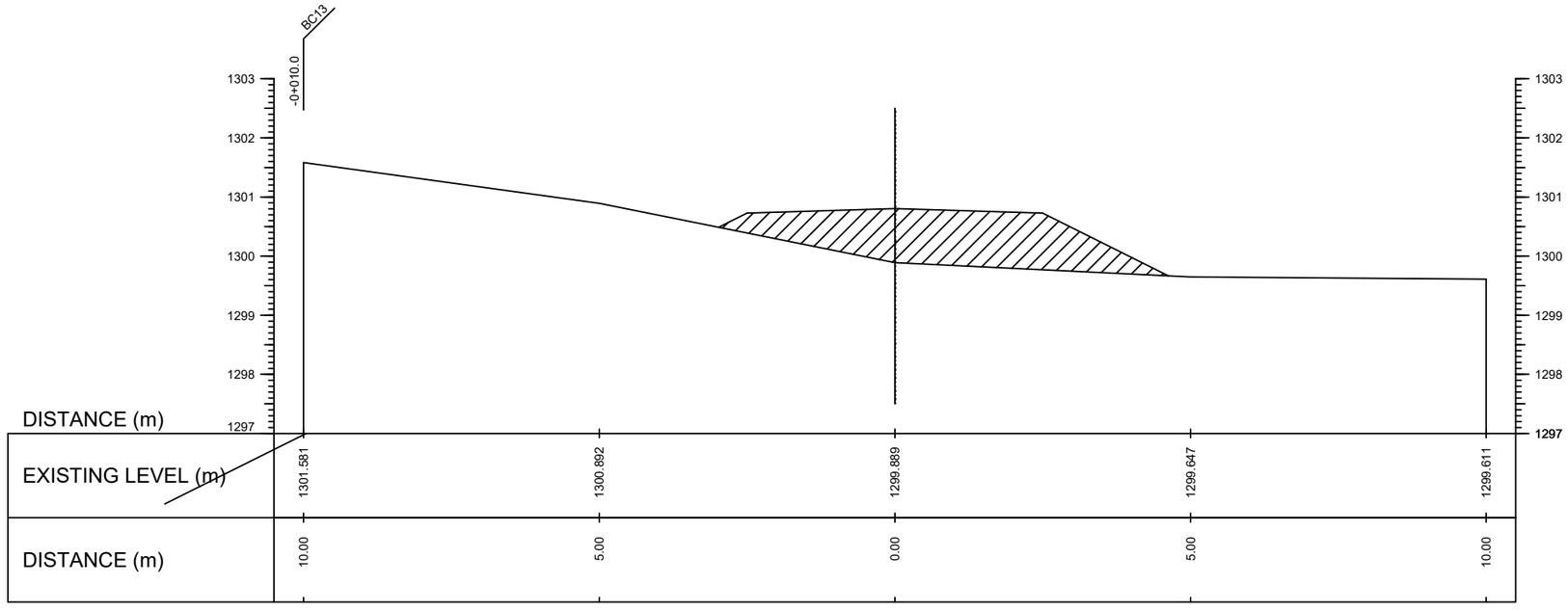
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GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+575.63



0+588.22

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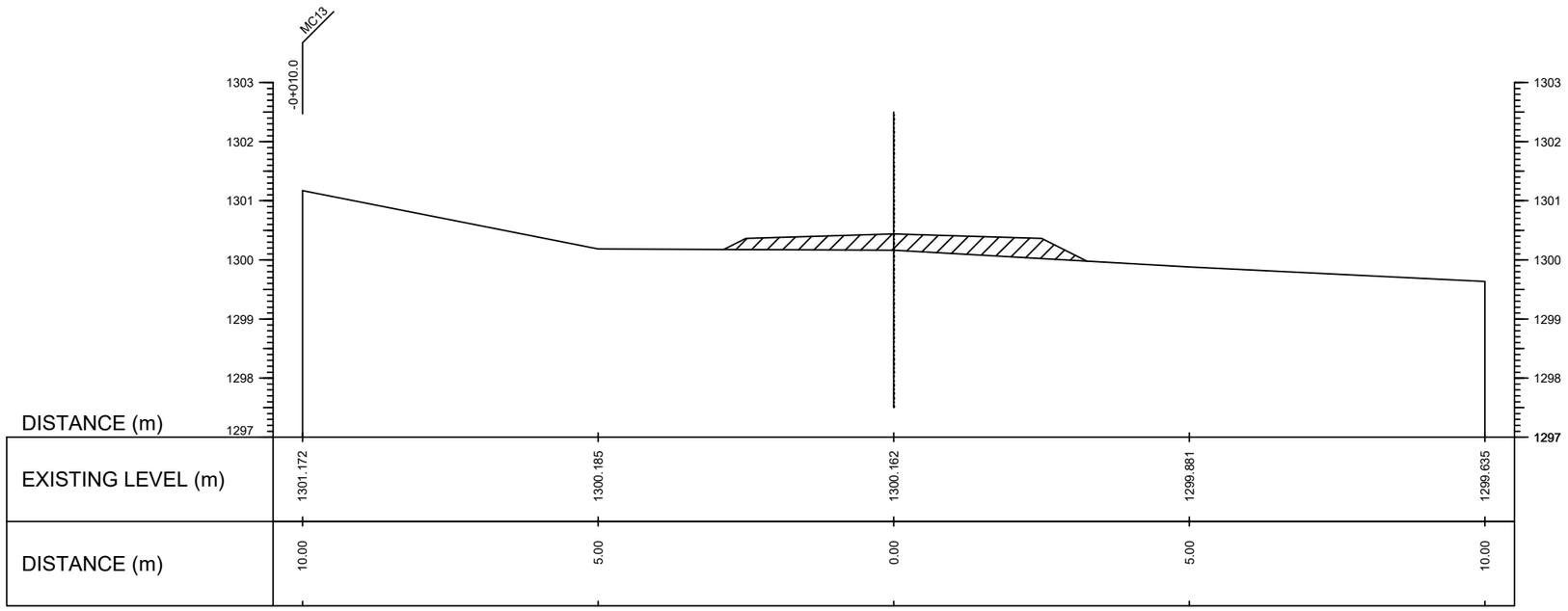
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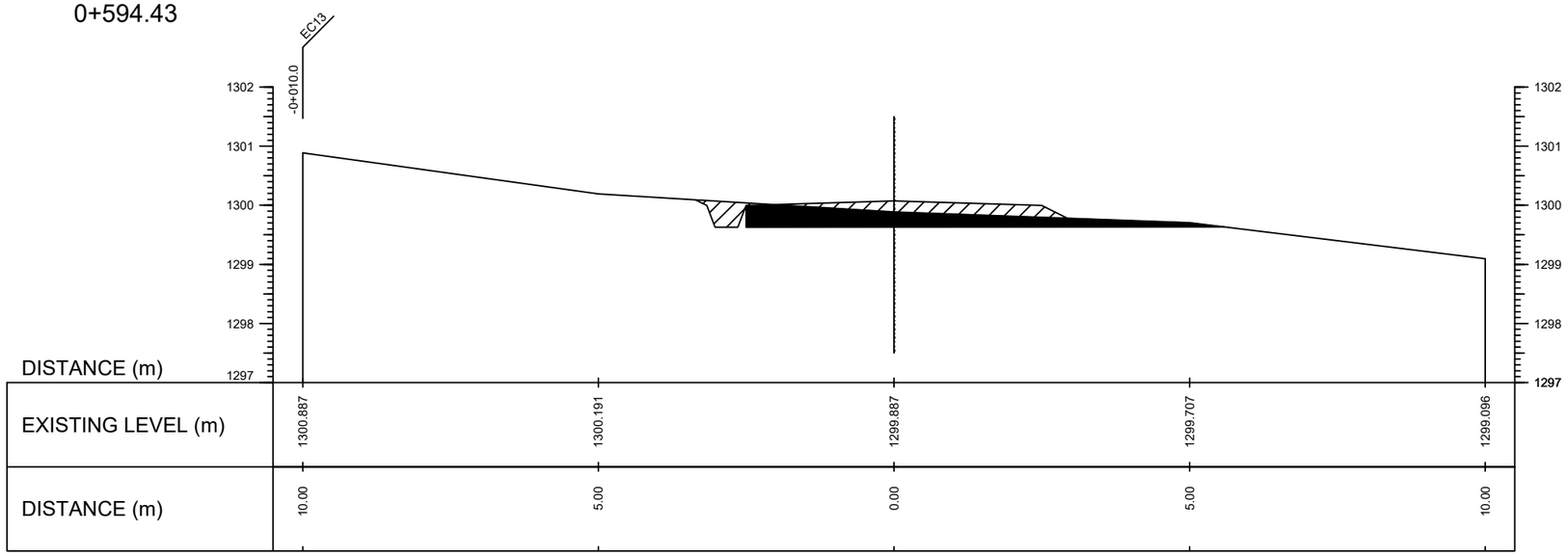
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RASABI SHARMA	080BCE120
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GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+594.43



0+600.64

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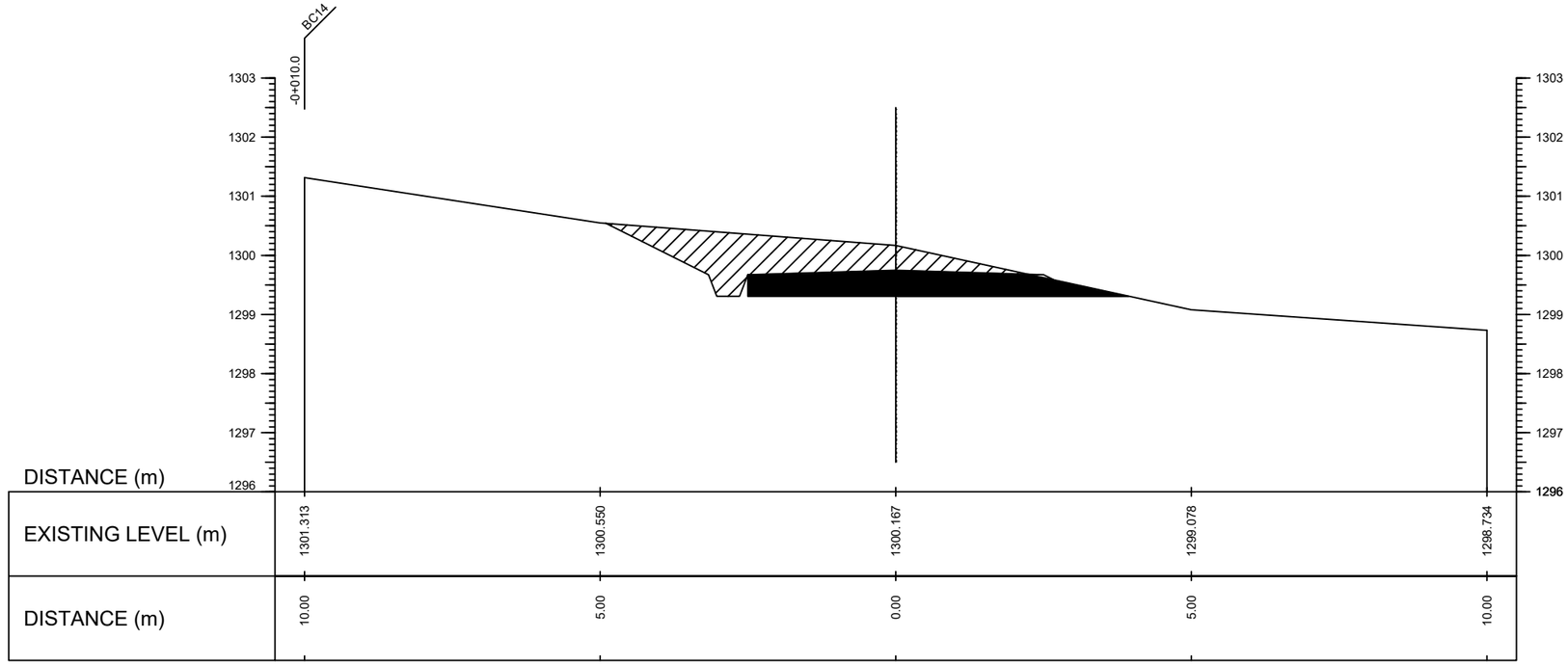
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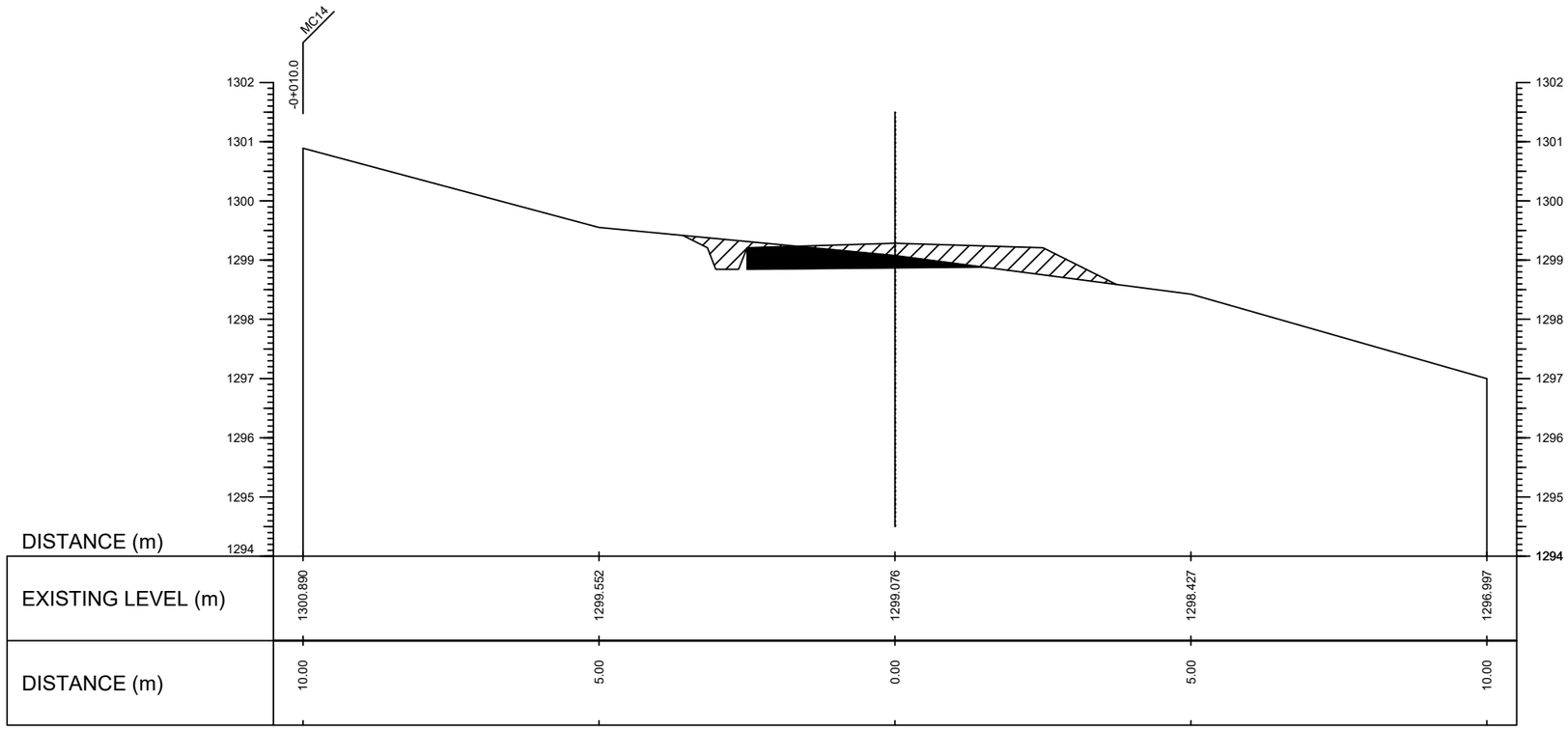
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GROUP 26

PRABESH PANDEY	080BCE099
REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+606.13



0+614.00

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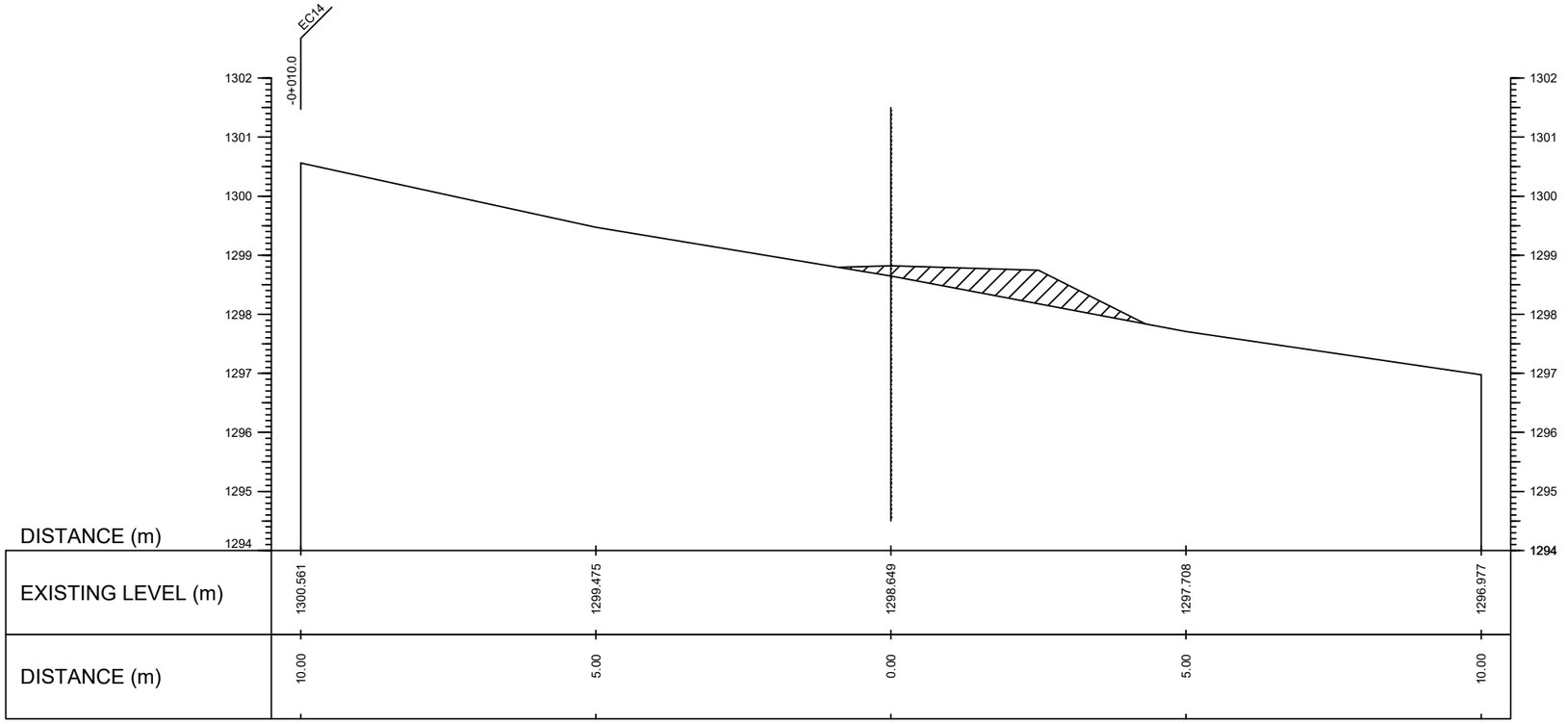
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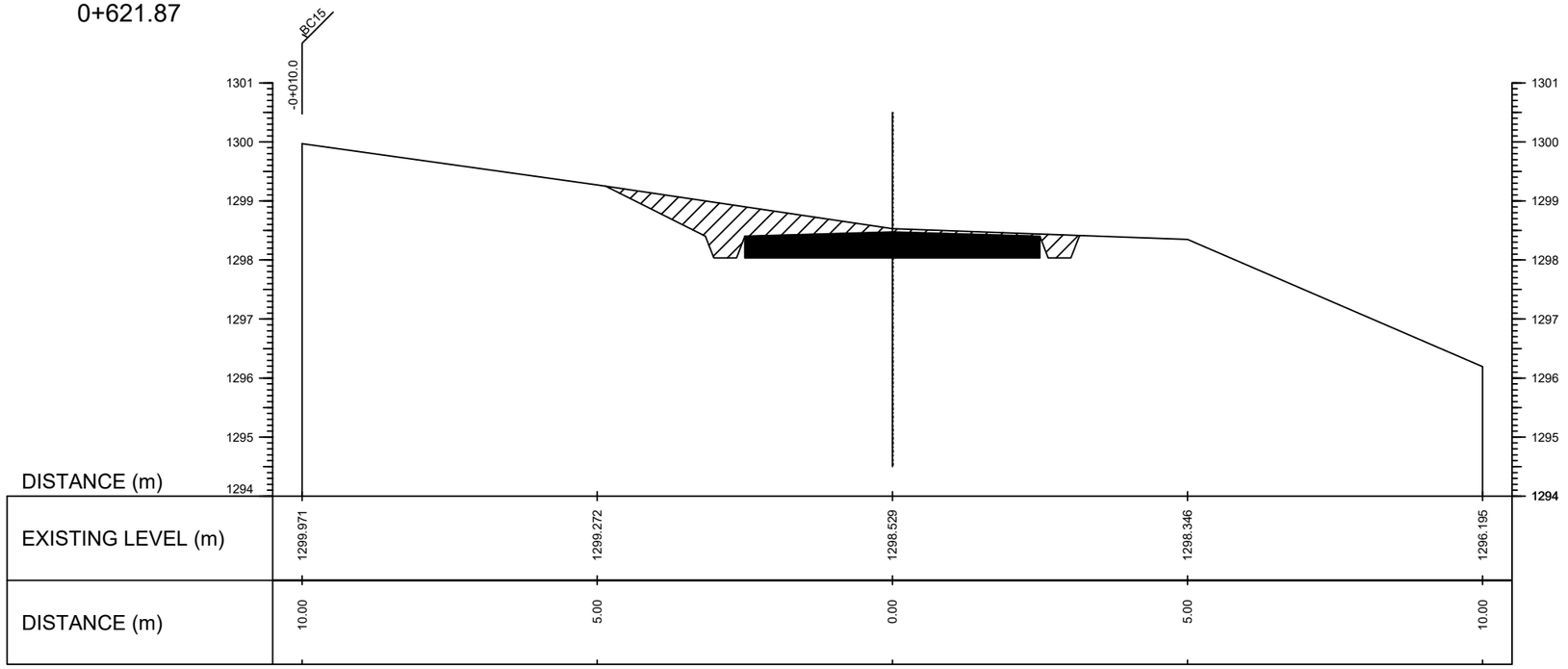
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GROUP 26

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REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+621.87



0+627.69

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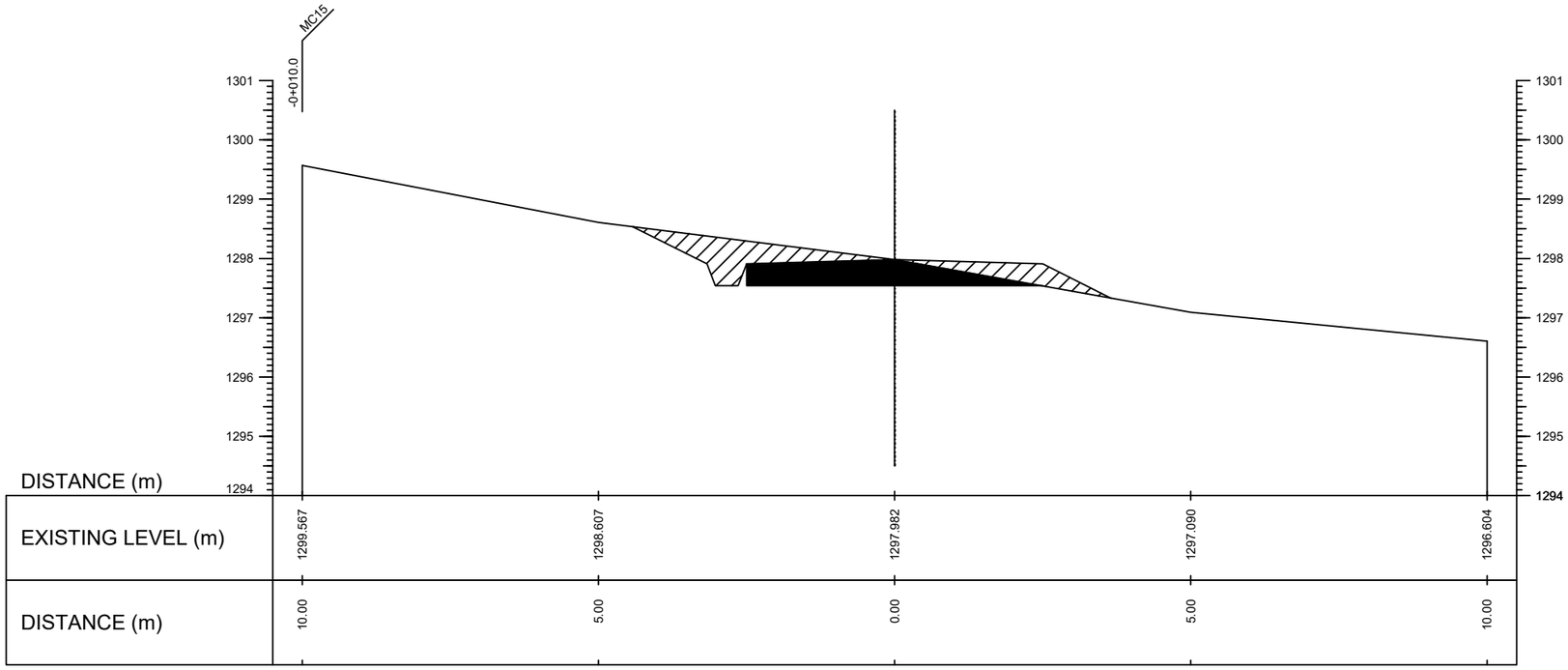
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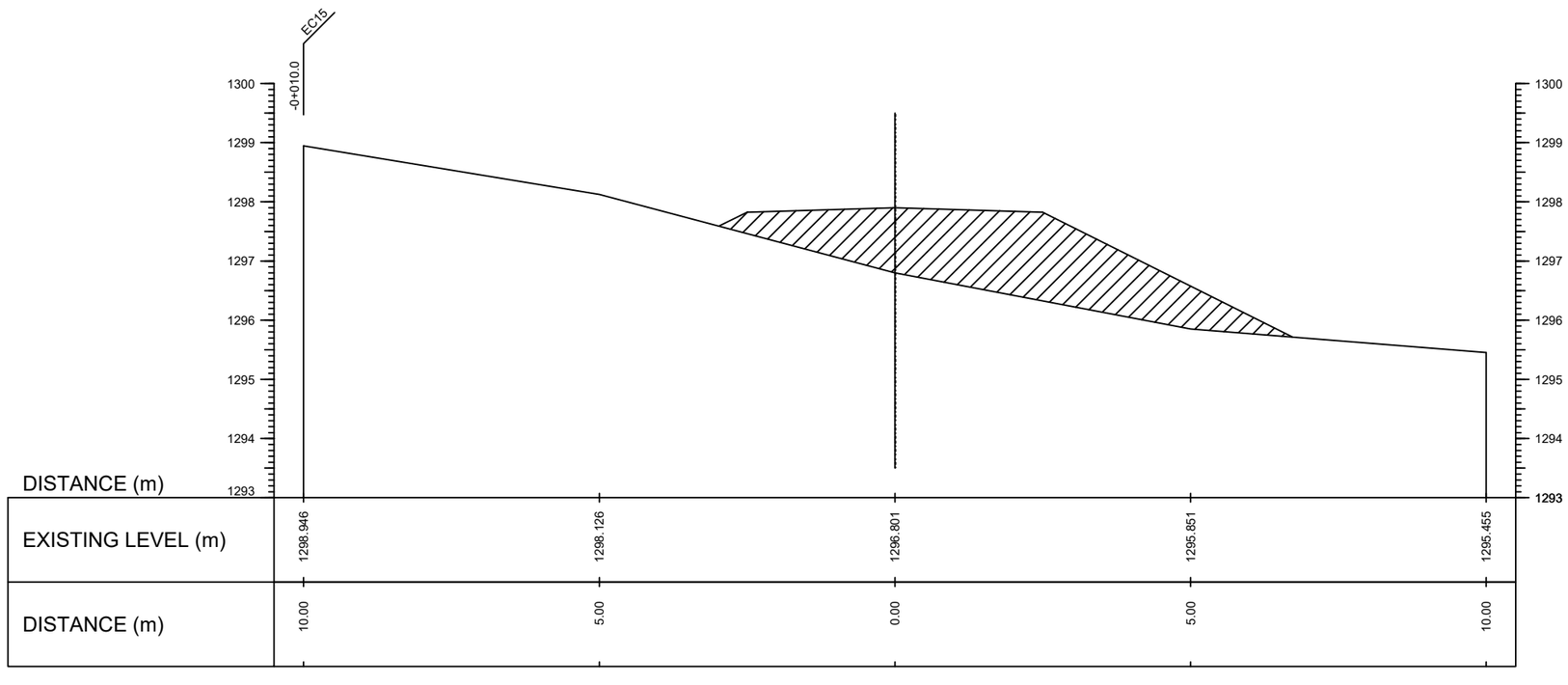
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GROUP 26

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REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+636.08



0+644.48

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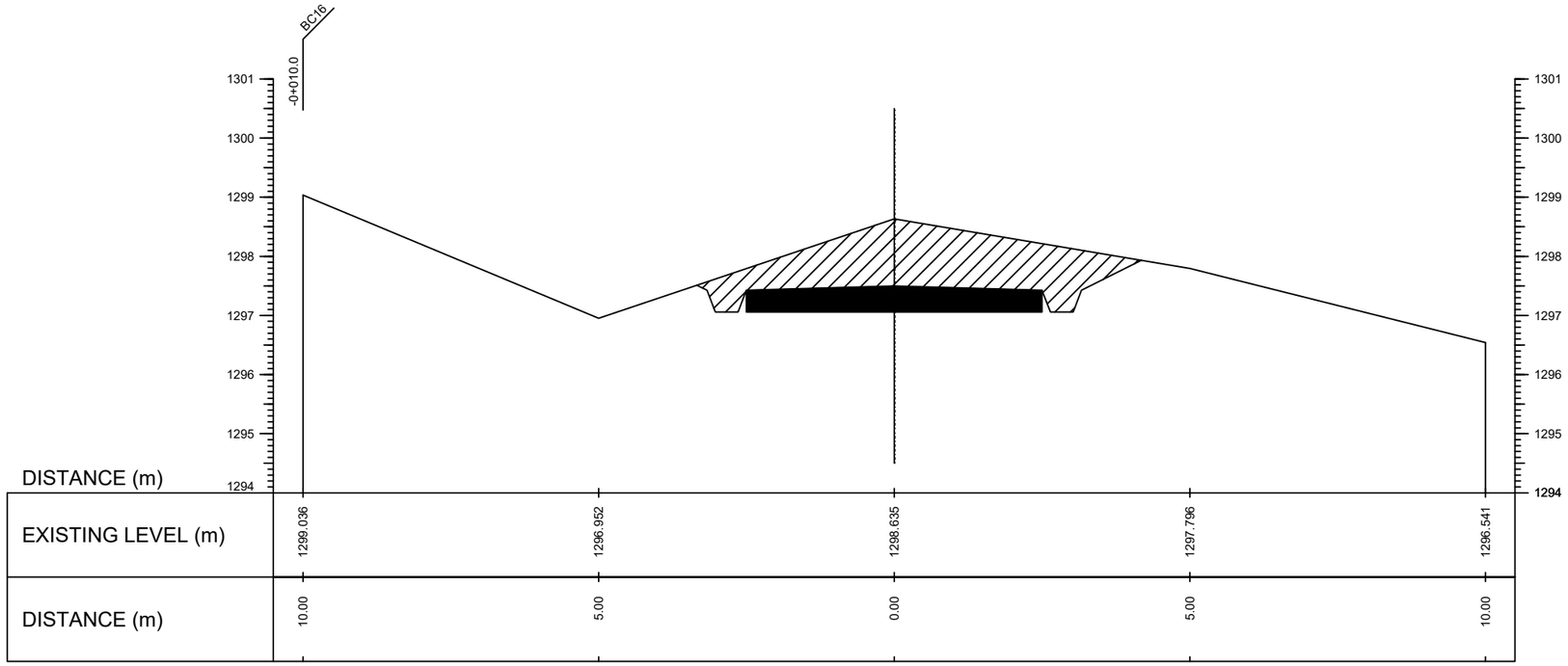
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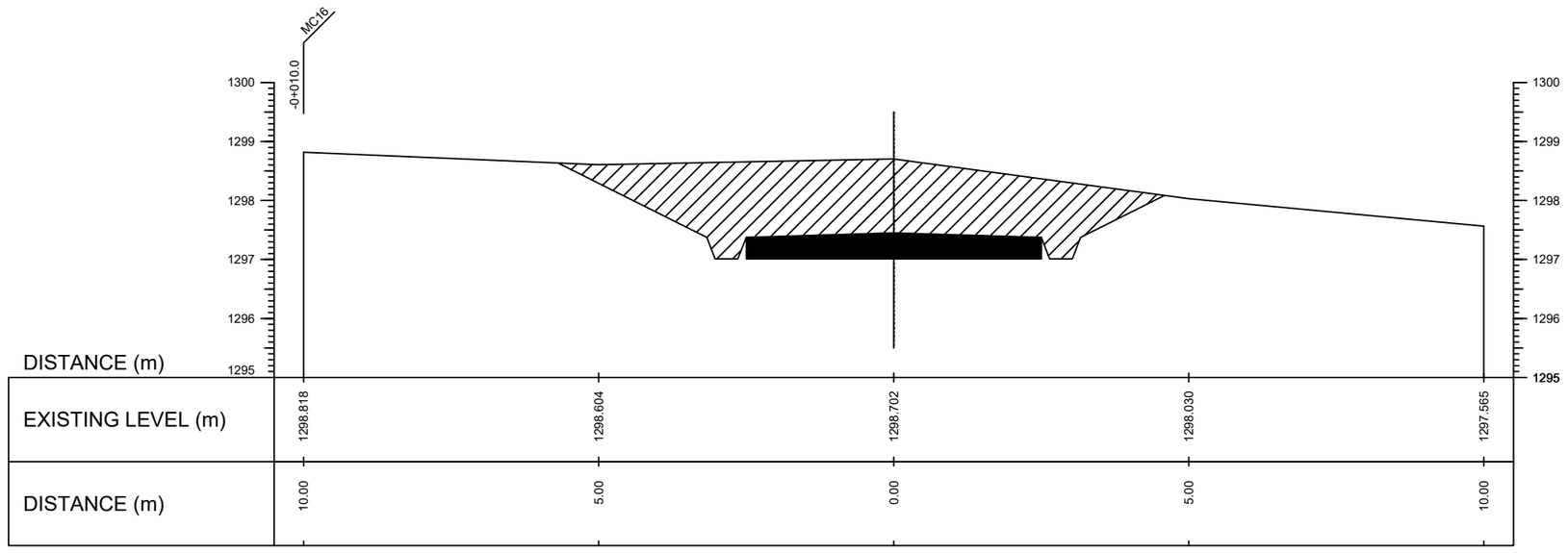
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REEVA AGRAWAL	080BCE123
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SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+684.93



0+691.62

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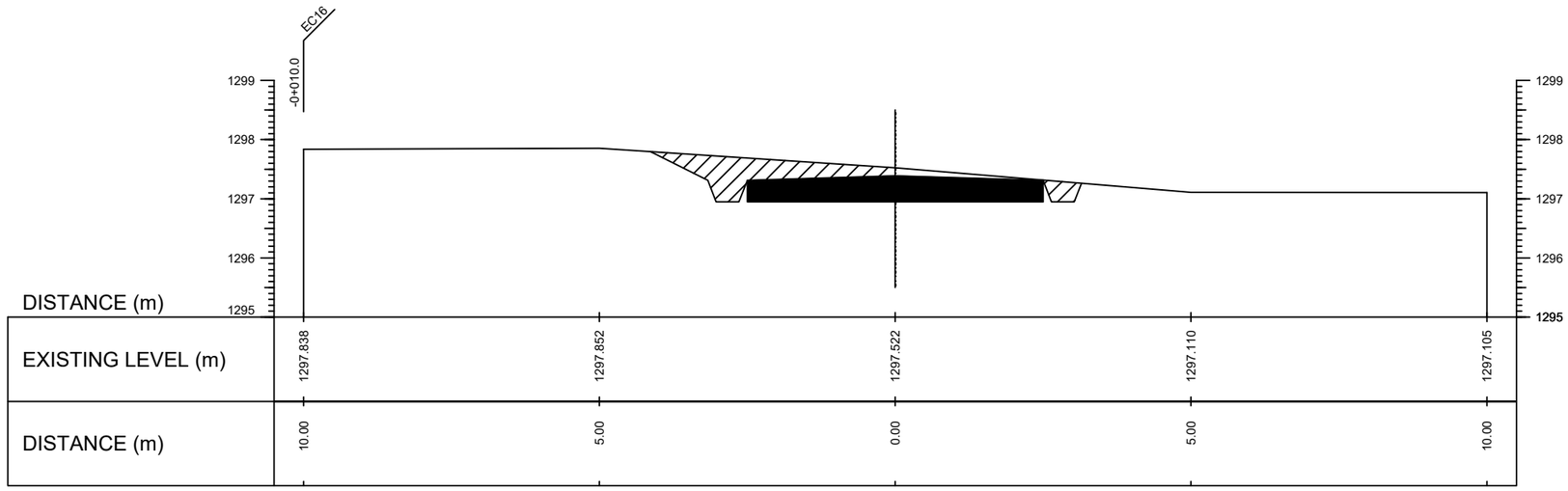
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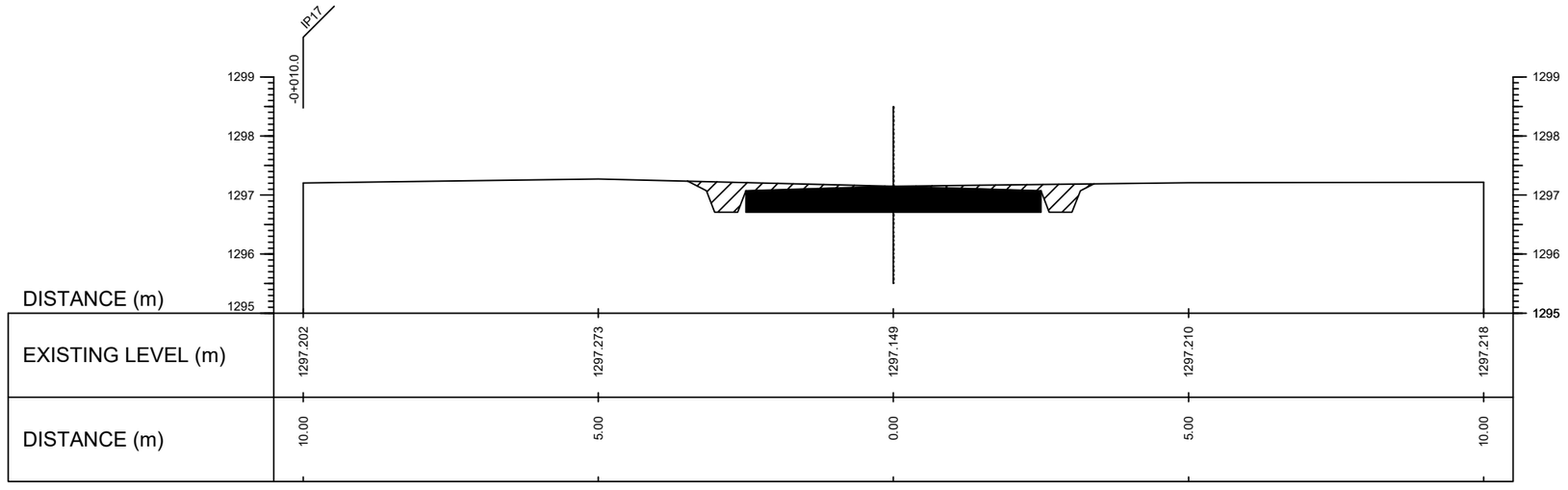
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RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
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GROUP 26

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REEVA AGRAWAL	080BCE123
SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185



0+698.32



0+723.30

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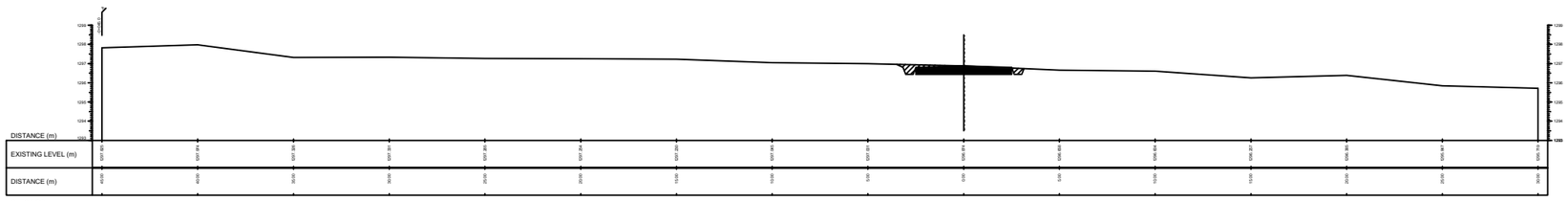
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PAWAN SHRESTHA	080BCE098
RASABI SHARMA	080BCE120
SANDIP BASNET	080BCE145
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SANDIP KANDEL	080BCE146
SRIJANA PANT	080BCE167
TULSI SHARMA	080BCE185

X-SECTION OF CULVERT



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HORIZONTAL SCALE 1:100

VERTICAL SCALE 1:100

GROUP 25

PAWAN SHRESTHA	080BCE098
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TULSI SHARMA	080BCE185

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